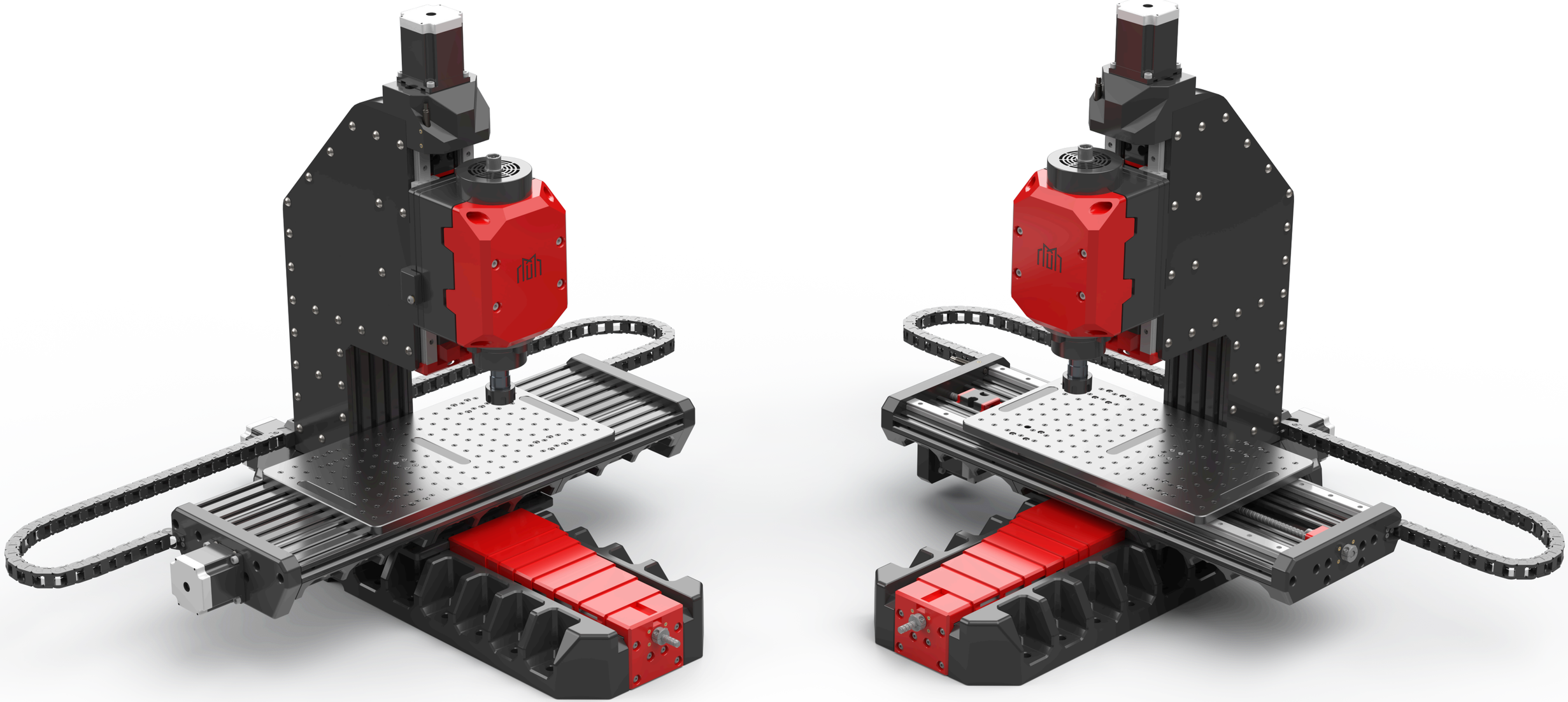
www.millennium-machines.com/



MILO V2 - ASSEMBLY INSTRUCTIONS

Version 12.12.25

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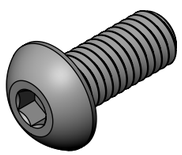
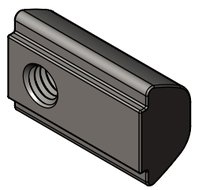
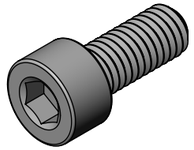


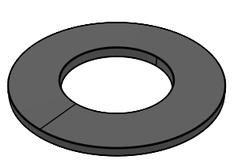
BUTTON HEAD CAP SCREW (BHCS)
Low-profile metric fastener with a domed head and hex socket drive.



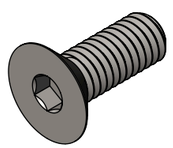
DROP IN NUT
Post-install fastener that slides into a T-slot to provide a threaded anchor point.



SOCKET HEAD CAP SCREW (SHCS)
High-strength metric fastener with a cylindrical head and hex socket drive.

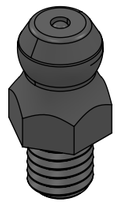


BELLEVILLE WASHER
Conical spring washer providing high load in limited space.

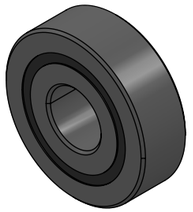


FLAT HEAD CAP SCREW (FHCS)
Countersunk fastener with a flat top and hex socket drive for a flush finish.

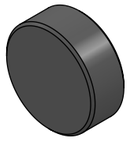
HEAT INSERT
Threaded brass insert designed to be heated and embedded into plastic parts.



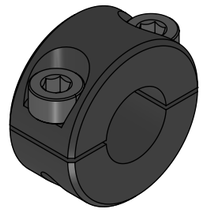
GREASE NIPPLE
Fitting used to apply lubricant into bearings or mechanical joints.



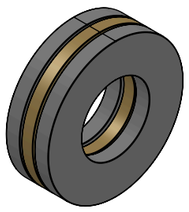
ROLLER BEARING
Rotating element that reduces friction between moving parts using steel balls.



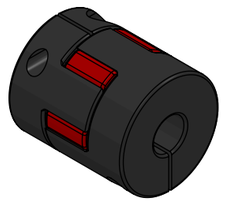
NEODYMIUM MAGNET
High-strength rare-earth magnet offering powerful magnetic force in a compact size.



DOUBLE SPLIT SHAFT COLLAR
Clamp-style collar with two splits for secure, even clamping around a shaft.



THRUST WASHER
Flat bearing surface that supports axial loads and reduces friction.



SPIDER COUPLER
Flexible shaft coupling using an elastomeric spider to absorb misalignment and vibration.

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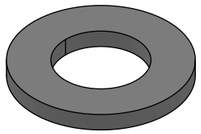
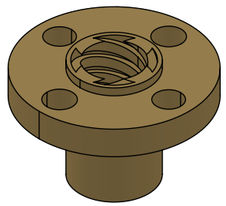
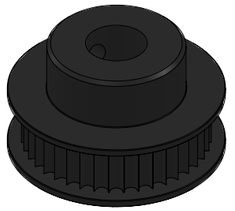


FLAT WASHER
Flat disc used to distribute load and prevent surface damage under a fastener.

DOWEL PIN
Precision cylindrical pin used for accurate alignment and positioning of components.



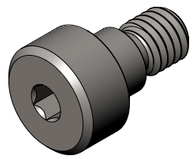
LEADSCREW NUT
Threaded brass component that converts rotary motion to linear motion on a leadscrew.



Miley Only

TIMING PULLEY
Toothed pulley designed to engage with a timing belt for precise, slip-free power transmission.

SET SCREW
Headless threaded fastener used to secure one component to a shaft through side pressure.



SHOULDER BOLT
Precision fastener with an unthreaded shoulder for guiding, pivoting, or locating parts.

Unless you're willing to experiment and almost certainly waste filament, stick with 0.2mm layers. Some parts contain overhangs that will become more difficult to print with thicker layers due to the overhang angle.

We provide a Minimum Wall Thickness for every printable part based on a 0.4 mm nozzle. If you are planning to use a different size of nozzle you must recalculate the required wall counts.

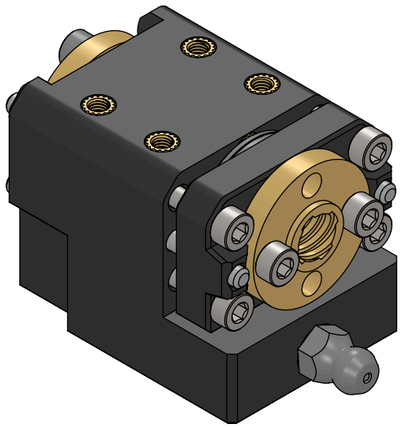
File Naming

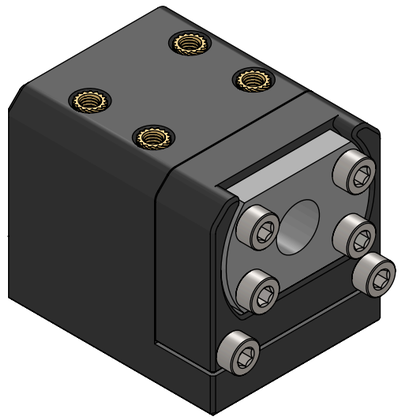
With your STL files downloaded and your printer warmed up, you may be wondering which parts to print in your favorite colors. Have no fear, we've got your back - each file is labeled with [a] in the name to indicate the print in your accent color.

Quantity Required

Example: Panel-Clip-A x8.stl If any file ends with x#, that tells you the quantity of that part required to build the mill.

Profile Name	Structural	Semi Structural	Aesthetic
Profile Code	S	SS	A
Infill Density [%]	45	25	20
Wall Count	6	4	3
Top/Bottom Layers	6	5	5
Nozzle Size [mm]	0..4	0.4	0.4





Leadscrew or Ballscrew?

Both Milo and Miley are designed to work with TR8×8 leadscrews or SFK0802 ballscrews.

That said, the standard—and the configuration we recommend—is the TR8×8 leadscrew.

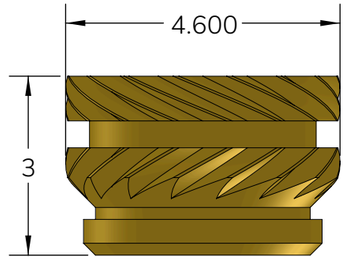
Cost aside, there are a few reasons why leadscrews tend to make more sense for small, desktop-sized CNC machines. With their 8 mm pitch, the motors don't need to spin as fast to reach practical feed rates, especially when running multi-flute end mills.

Ballscrews, like the SFK0802, have clear advantages in efficiency and backlash reduction, which is why they're popular in larger or industrial machines. However, with only a 2 mm lead, the stepper motors must spin four times faster to achieve the same travel speed as an 8 mm leadscrew. At those higher RPMs, stepper torque drops off quickly, limiting acceleration and cutting force. Unless the system uses servo motors, that efficiency advantage can easily be offset by the loss in usable torque.

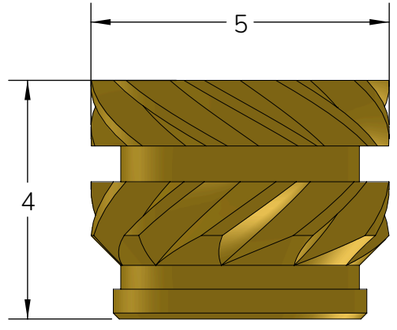
In practice, 8 mm leadscrews strike the best balance between speed, cost, and reliability. They're quiet, simple to install, and more forgiving of small alignment errors. We only recommend ballscrews if you plan to run servos or don't mind being limited to single-flute tooling.

All 3D-printed parts that use M3 heat-set inserts are supplied in two versions: one for CNC-Kitchen inserts and one for Voron inserts. Make sure to select the STL that matches the style of insert you're using.

CNC-Kitchen Specification

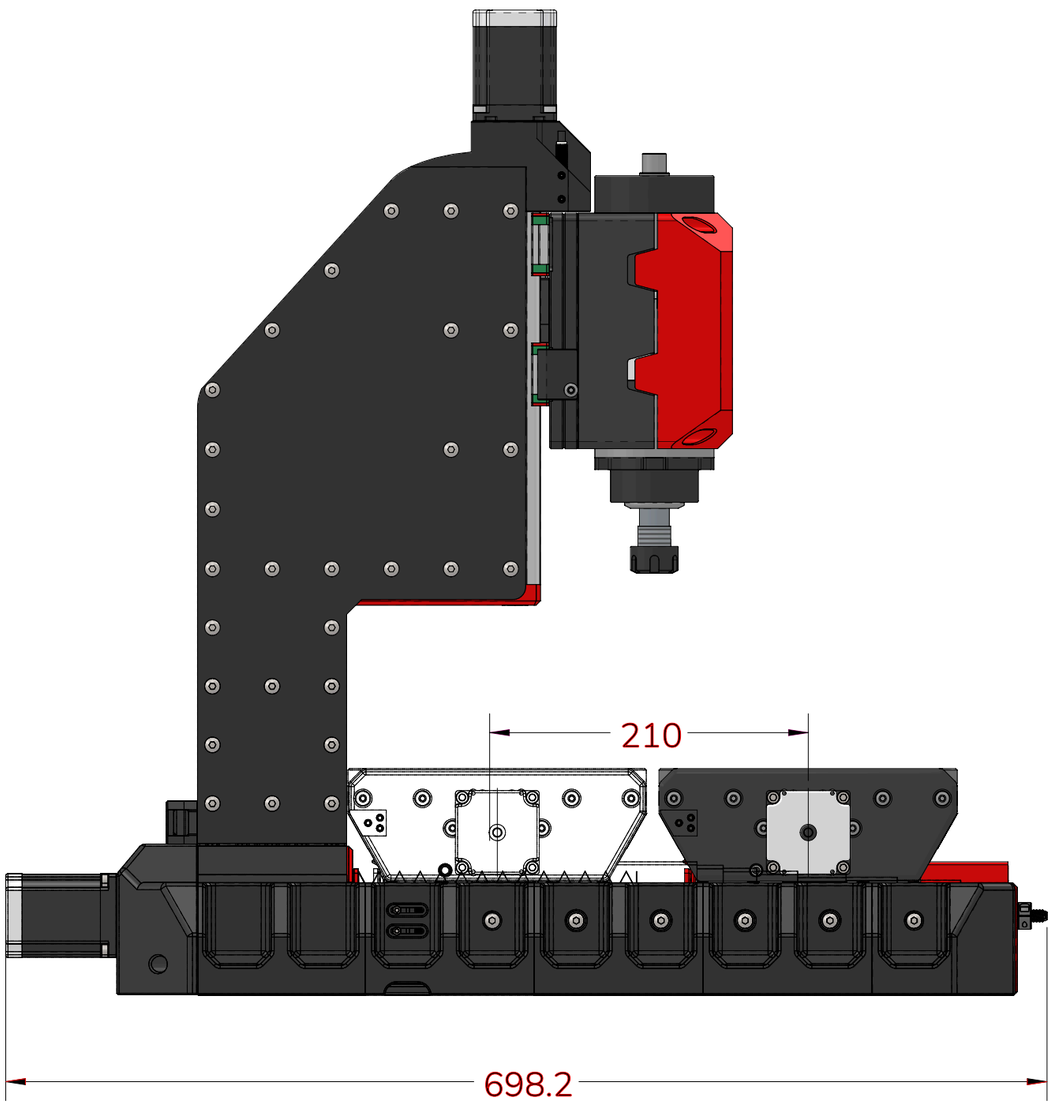


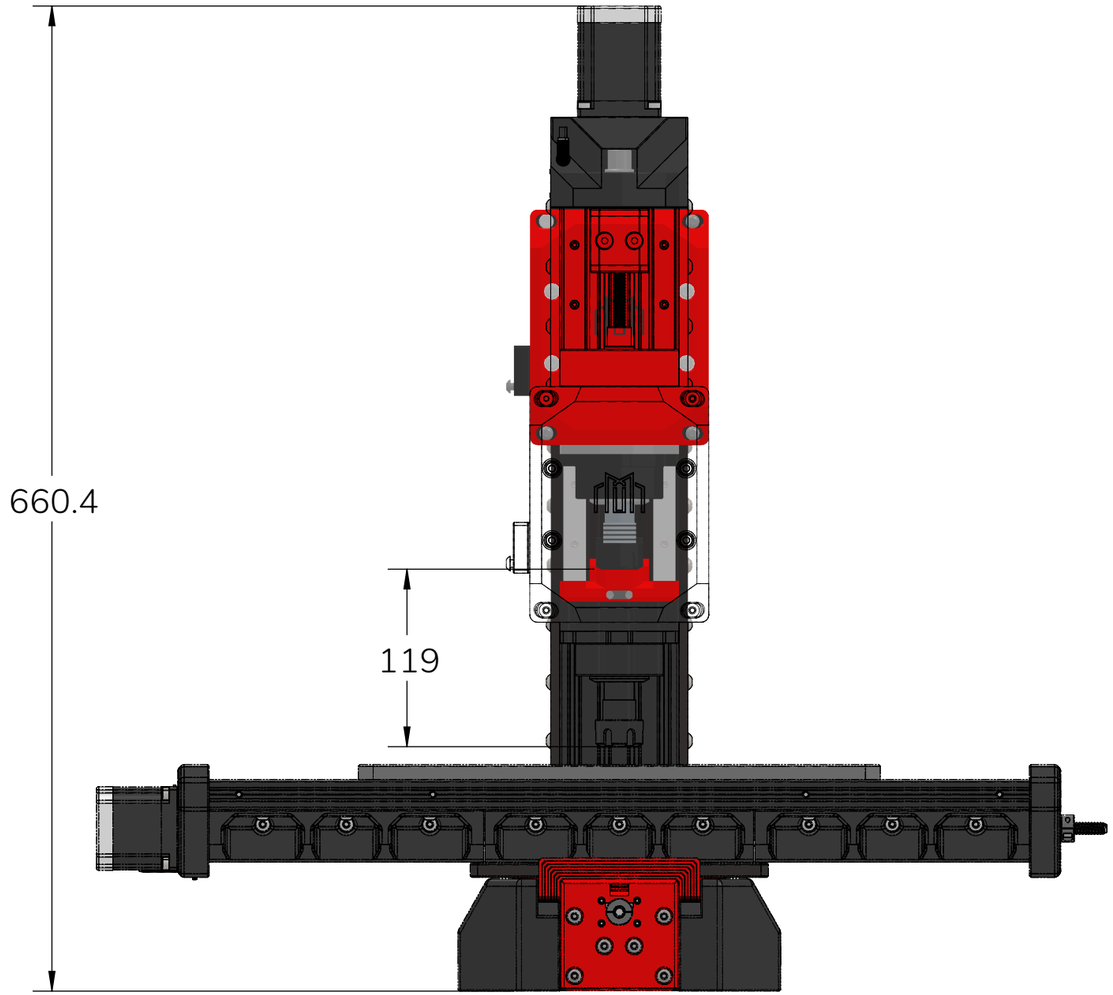
Voron Specification



INTRODUCTION

Both Milo & Miley have the same Y and Z travel





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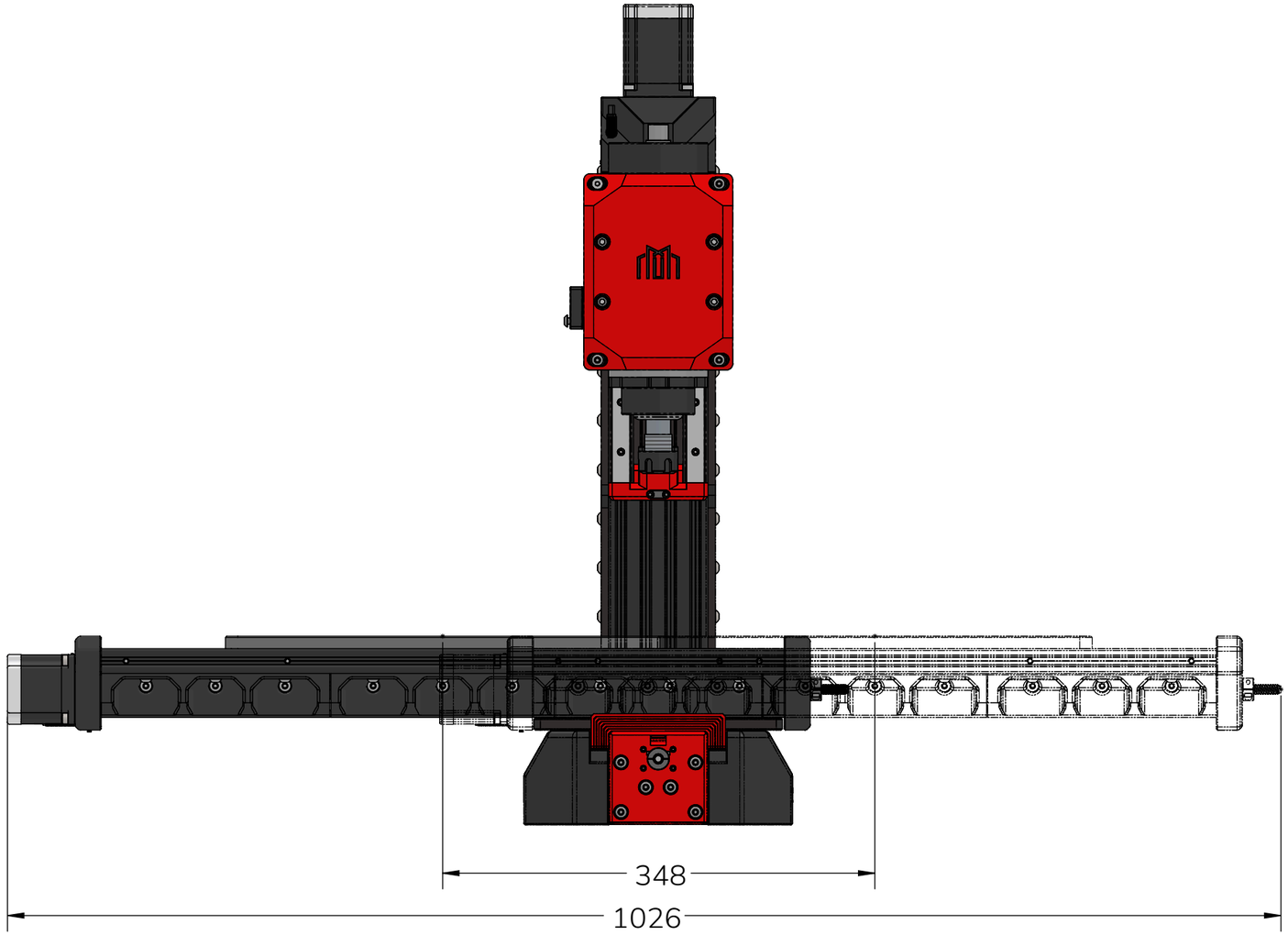
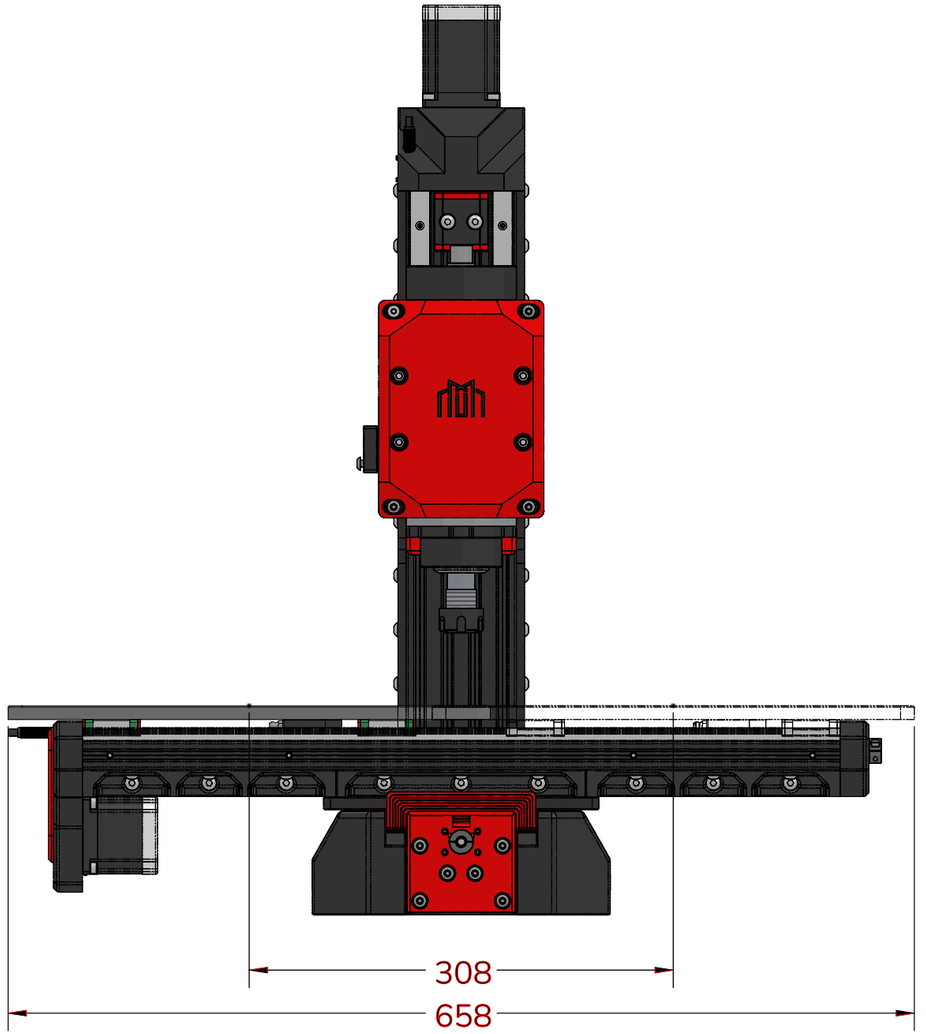
Milo



Miley



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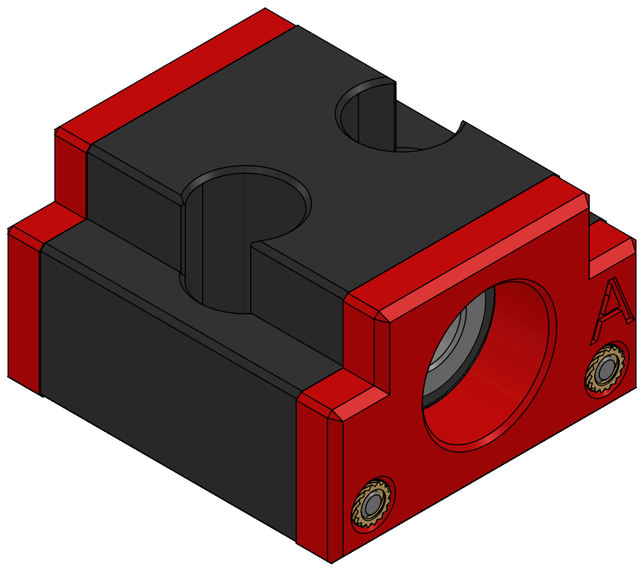
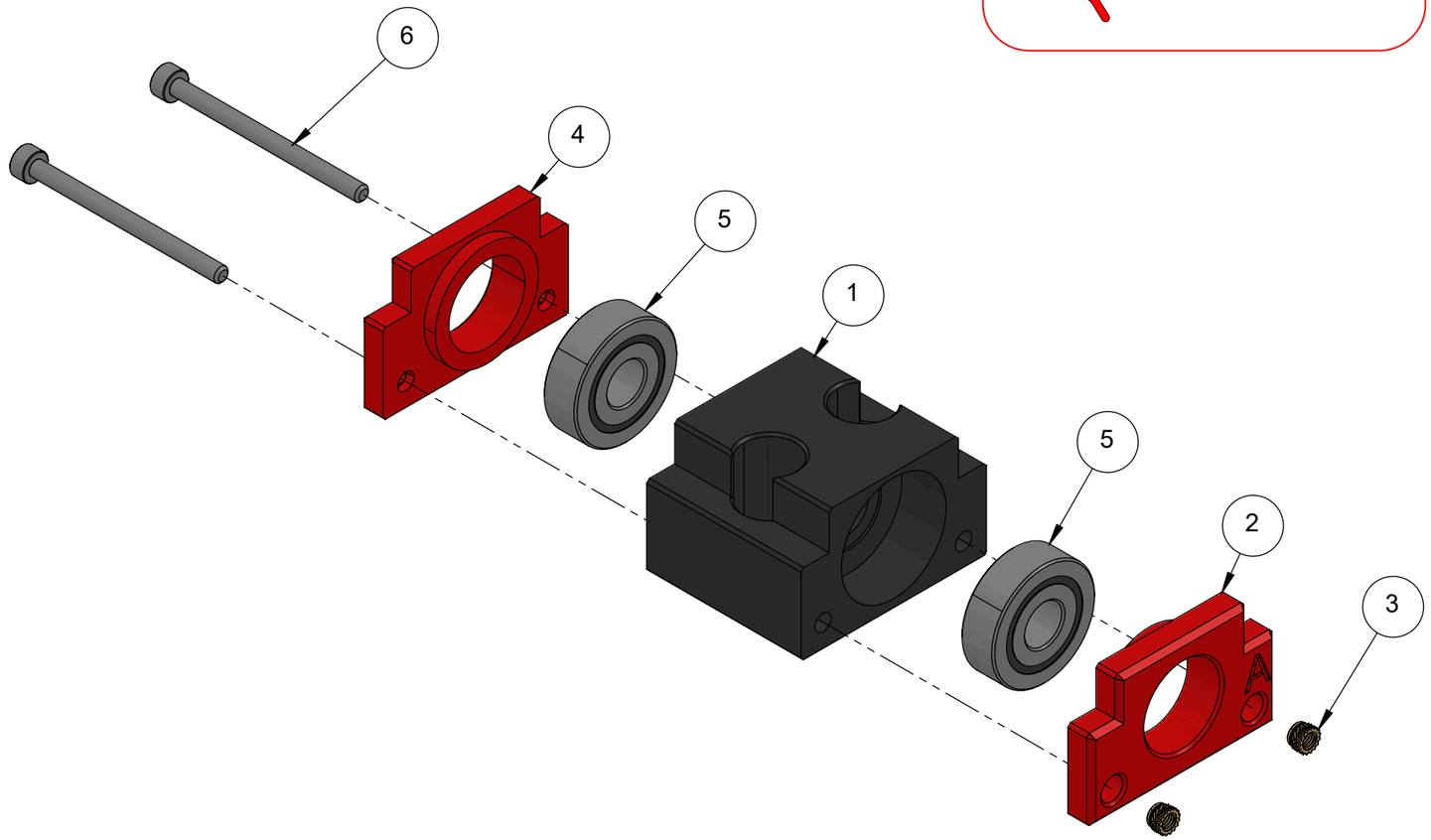
QTY 3





	PART NUMBER	QTY.
1	INTERNAL AXIS FLOATING BEARING BLOCK	1
2	FLOATING BEARING BLOCK FACEPLATE A	1
3	M3X3MM HEATSET INSERT	2
4	FLOATING BEARING BLOCK FACEPLATE B	1
5	608ZZ BALL BEARING	2
6	ISO-4762-M3X40-SHCS	2

PRE-ASSEMBLY

FLOATING BEARING BLOCK

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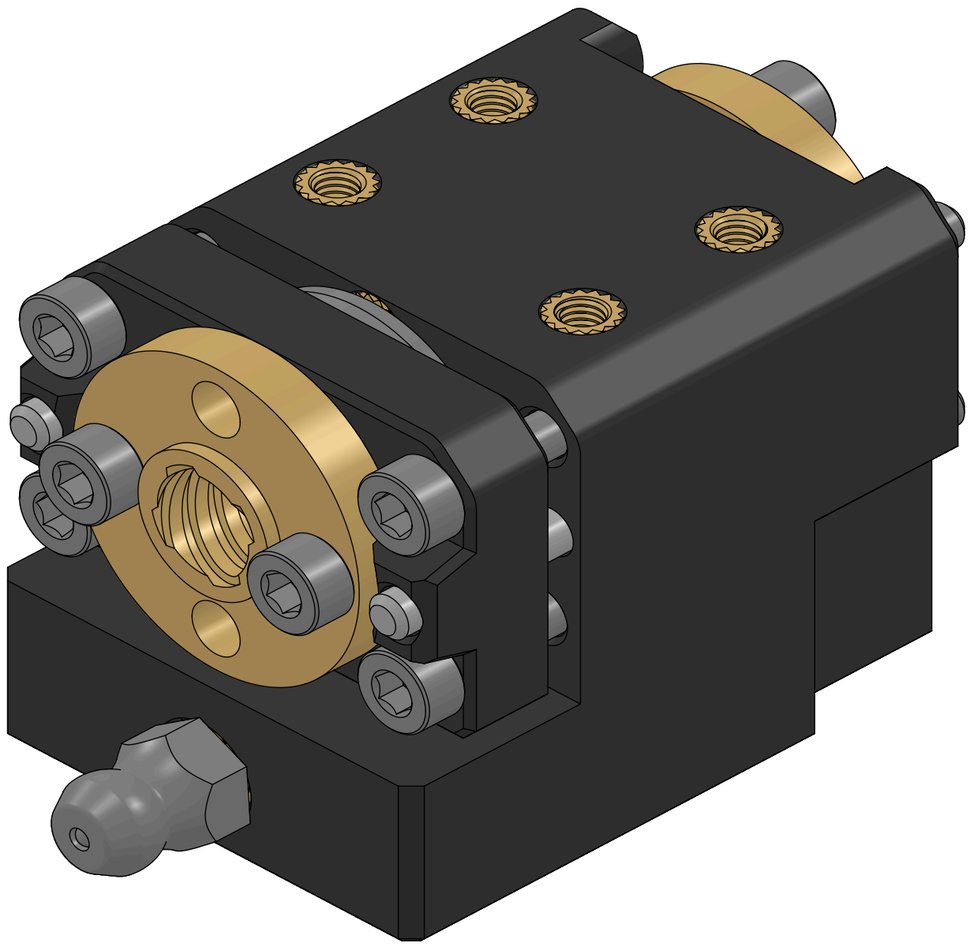

	PART NUMBER	QTY.
1	ThrED-AB-BLOCK-ASSY	1
	ThrED-AB-BLOCK	1
	M3X3MM HEATSET INSERT	10
	M5X5.8mm HEATSET INSERT	1
	TR8X8 BRASS LEADSCREW NUT	1
	ISO-4762-M3X30-SHCS	2
	ISO-2338 M3-20mm PIN	2
	DIN-71412 GREASE-NIPPLE	1
2	ThrED-COMPRESSION-PLATE-ASSEMBLY	1
	ThrED-COMPRESSION-PLATE	1
	M3X3MM HEATSET INSERT	2
	TR8X8 BRASS LEADSCREW NUT	1
	ISO-4762-M3X10-SHCS	2
3	ISO-4762-M3X40-SHCS	4
4	DIN-2093 10.2mm ID - 20mm OD BELLEVILLE WASHER	2

QTY 3

PRE-ASSEMBLY

AB BLOCK ASSEMBLY

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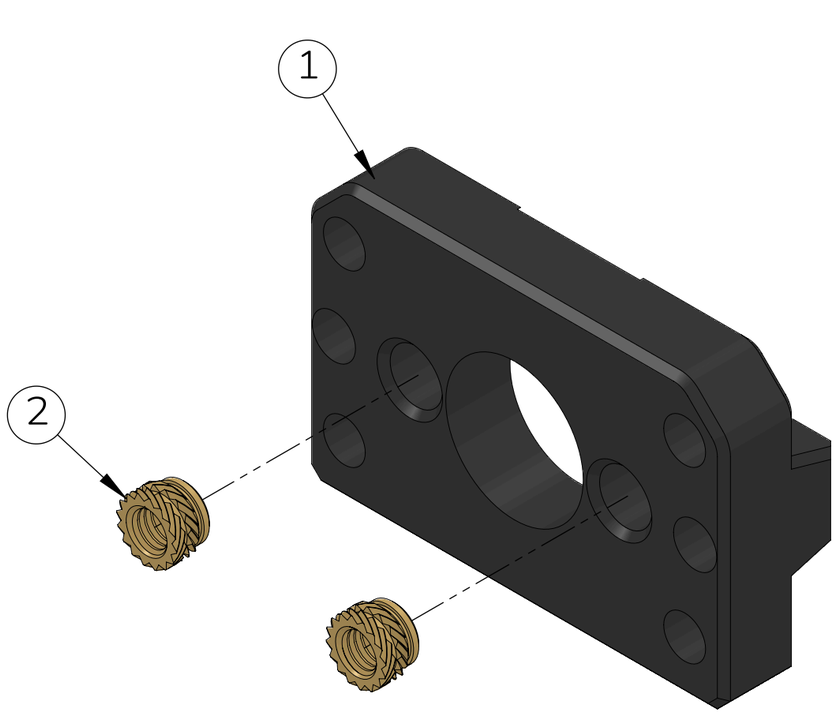
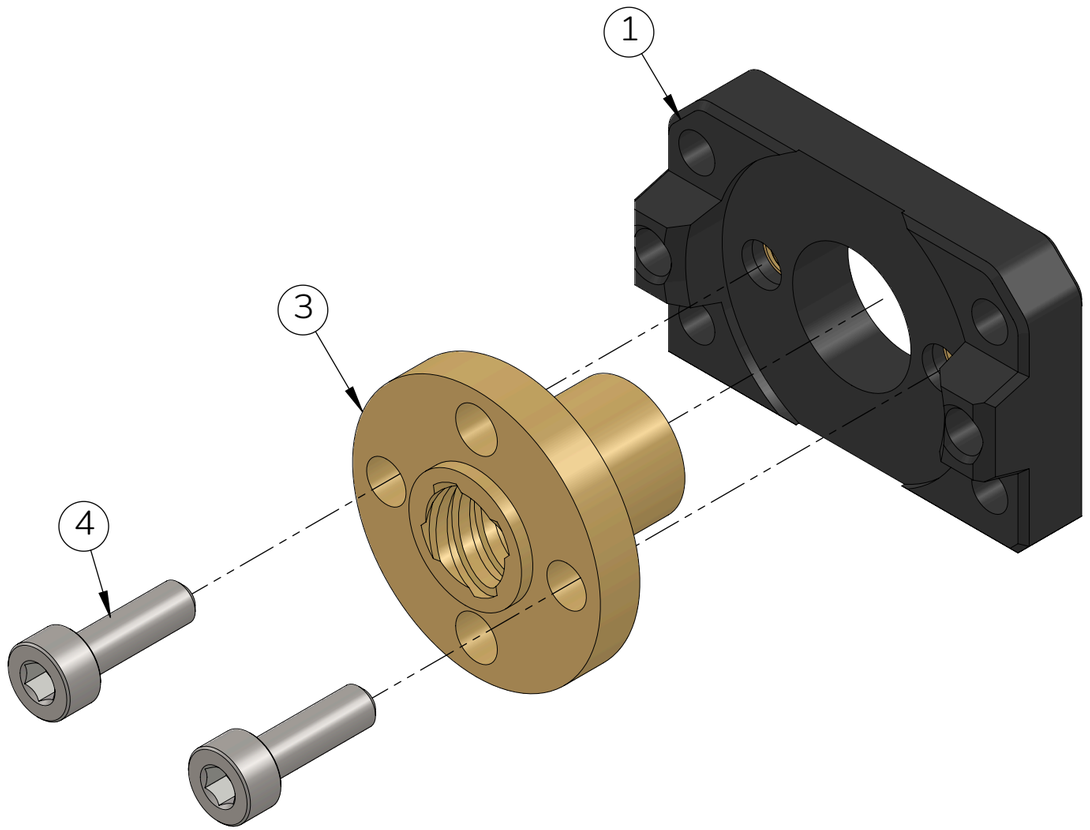
	PART NUMBER	QTY.
1	ThrED-COMPRESSION-PLATE	1
3	TR8X8 BRASS LEADSCREW NUT	1
4	ISO-4762-M3X10-SHCS	2





	PART NUMBER	QTY.
1	ThrED-COMPRESSION-PLATE	1
2	M3X3MM HEATSET INSERT	2

PRE-ASSEMBLY

AB BLOCK ASSEMBLY - COMPRESSION PLATE

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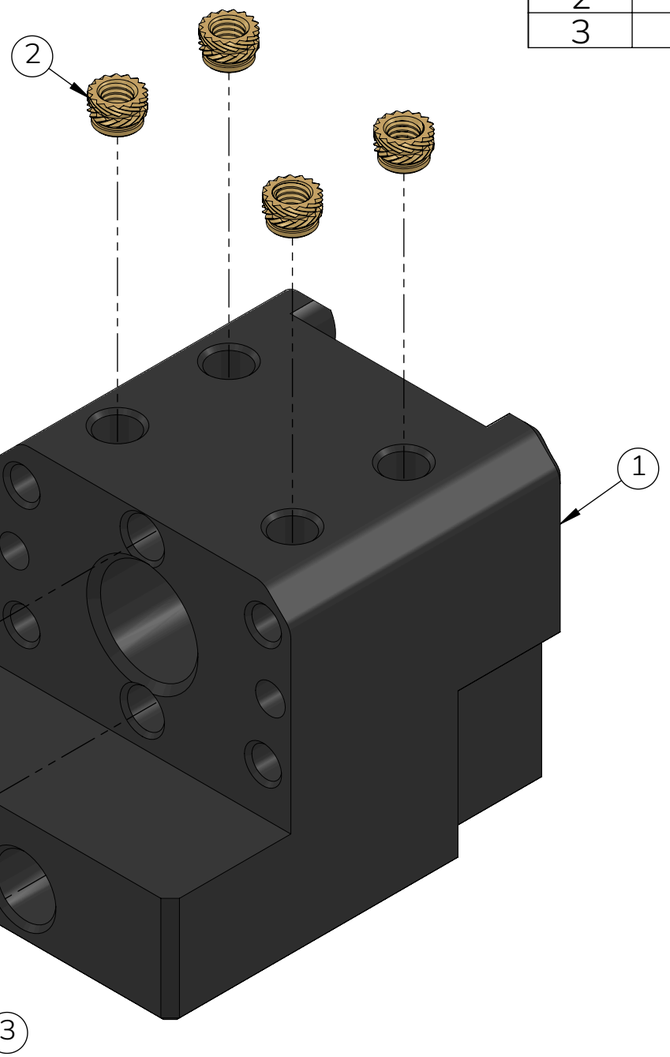
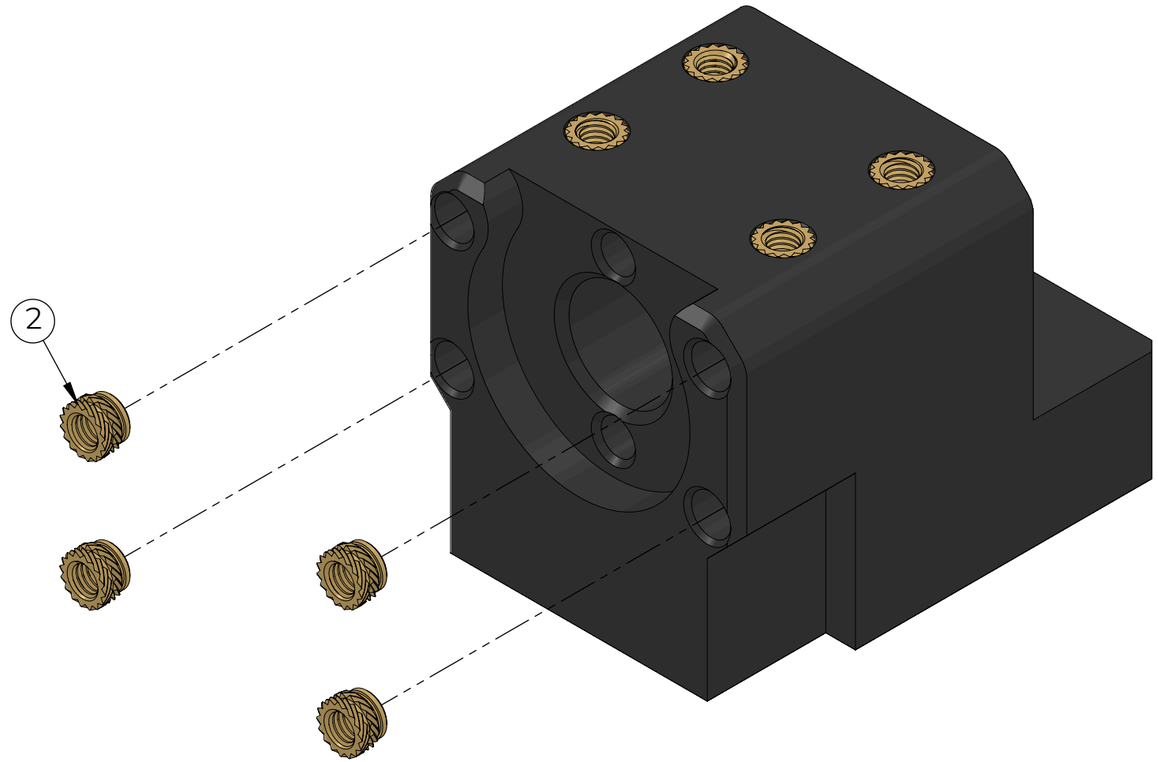
	PART NUMBER	QTY.
1	ThrED-AB-BLOCK	1
2	M3X3MM HEATSET INSERT	10
3	M5X5.8mm HEATSET INSERT	1





PRE-ASSEMBLY

AB BLOCK ASSEMBLY - HEAT INSERTS

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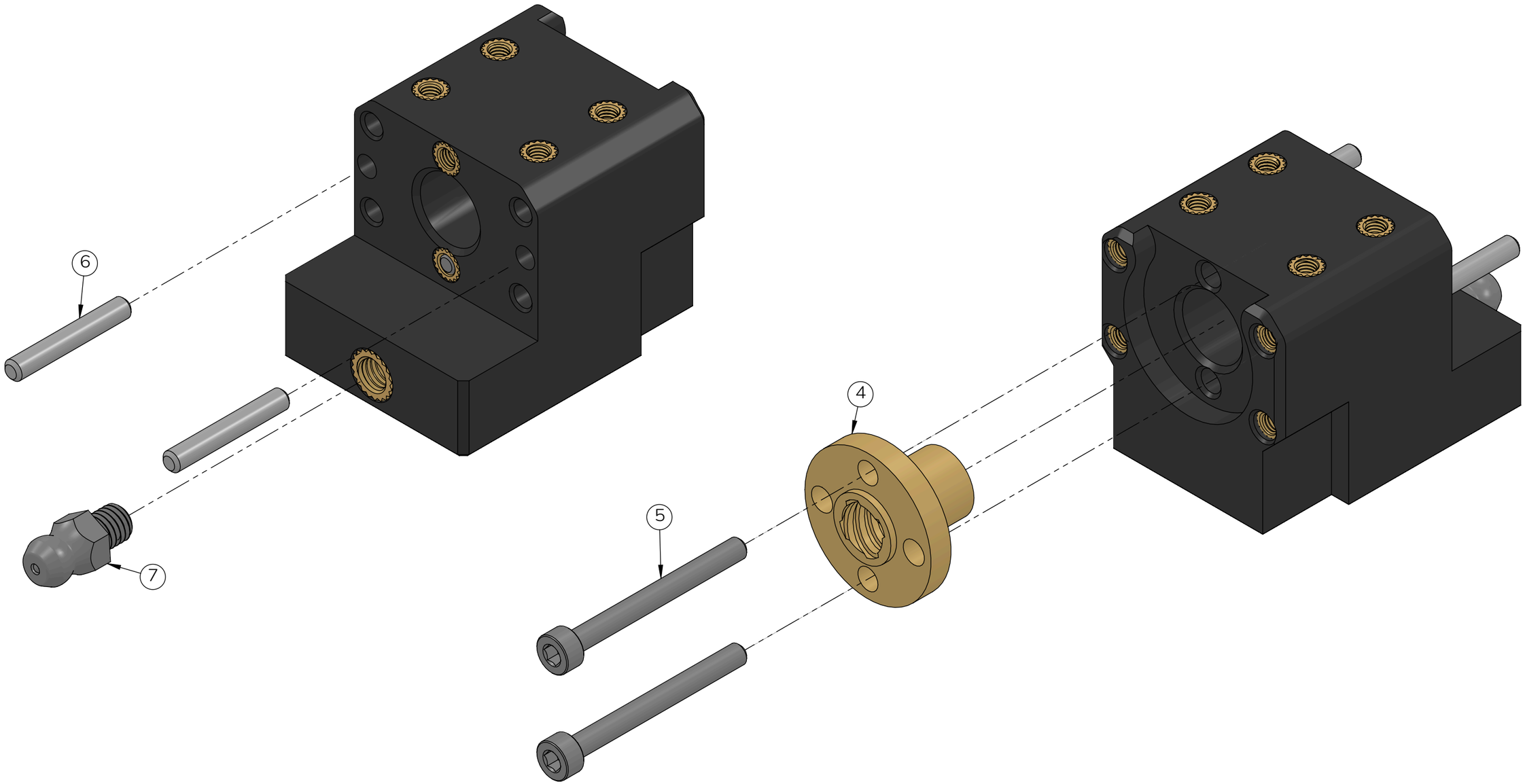
	PART NUMBER	QTY.
1	ThrED-AB-BLOCK	1
4	TR8X8 BRASS LEADSCREW NUT	1
5	ISO-4762-M3X30-SHCS	2
6	ISO-2338 M3-20mm PIN	2
7	DIN-71412 GREASE-NIPPLE	1



PRE-ASSEMBLY

AB BLOCK ASSEMBLY

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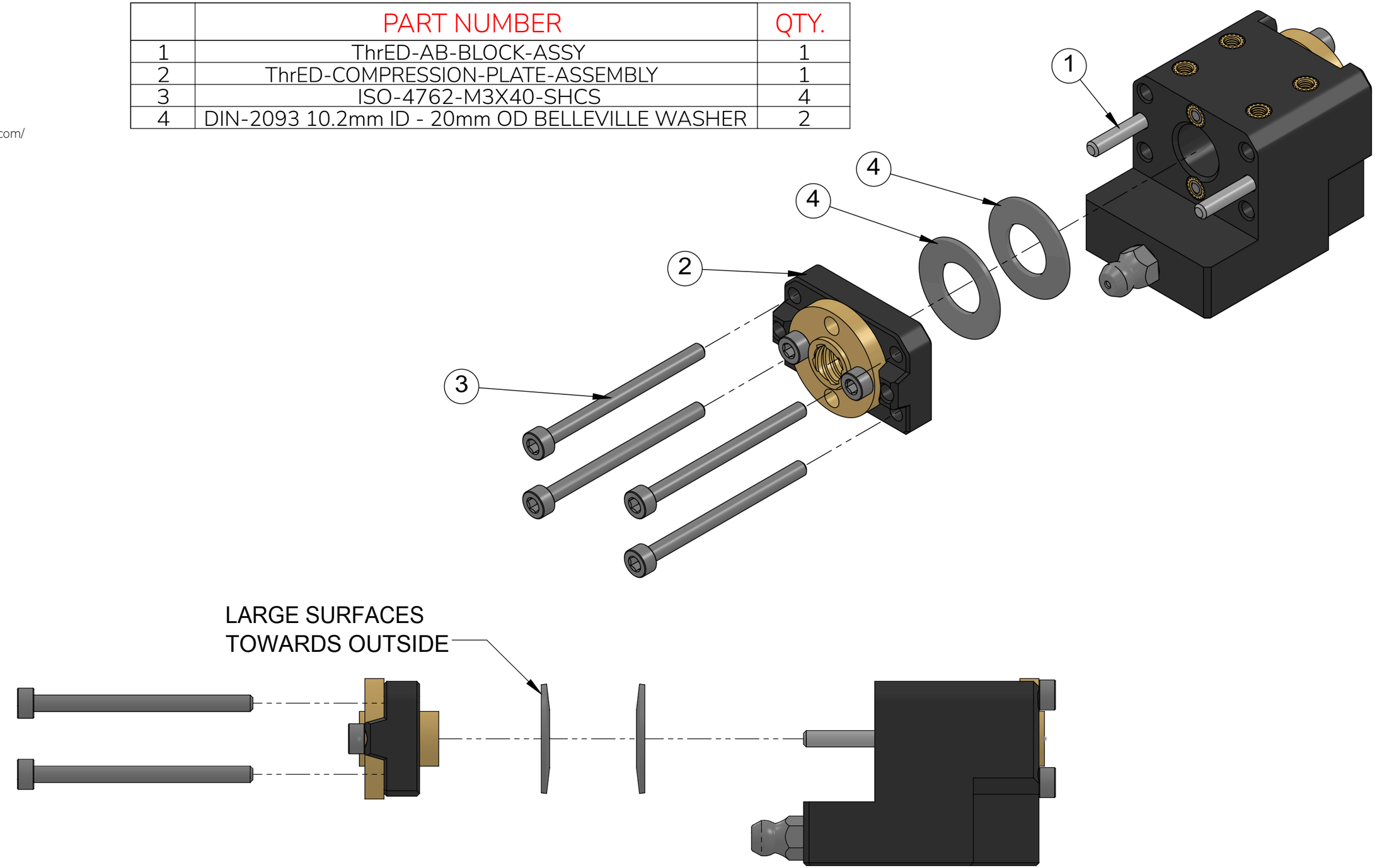
	PART NUMBER	QTY.
1	ThrED-AB-BLOCK-ASSY	1
2	ThrED-COMPRESSION-PLATE-ASSEMBLY	1
3	ISO-4762-M3X40-SHCS	4
4	DIN-2093 10.2mm ID - 20mm OD BELLEVILLE WASHER	2



PRE-ASSEMBLY

AB BLOCK ASSEMBLY

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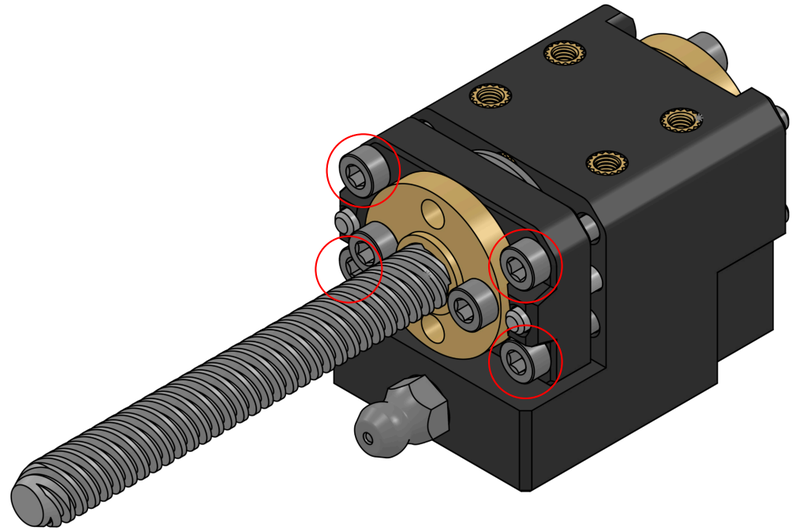
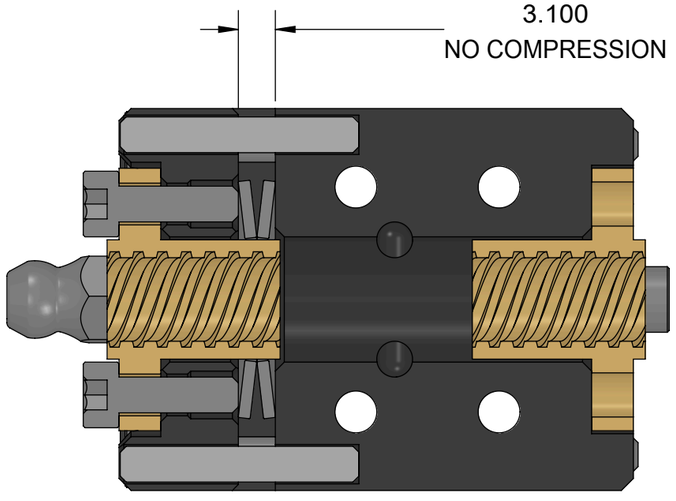
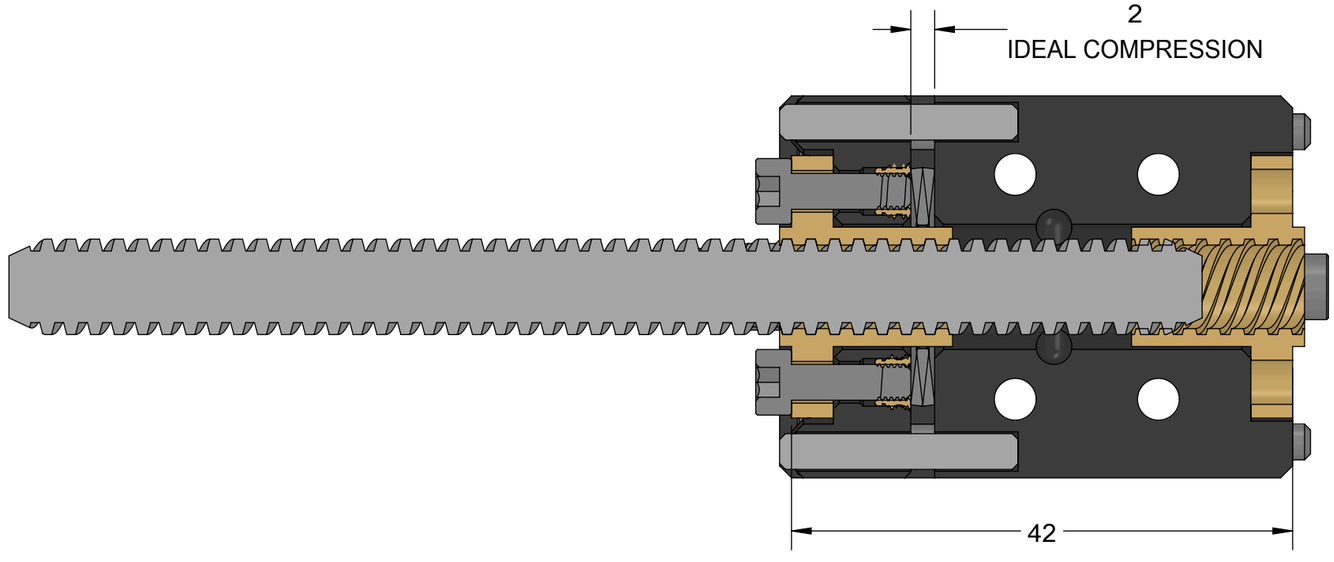
Using the (4) outside M3 bolts,
Compress the face plate slowly.
Once there is a gap around 2mm
start trying to thread in any leadscrew.
If the leadscrew doesn't thread in, keep
Slowly tightening the face plate until it does.



Once you have the leadscrew threaded into
both nuts, you should be able to remove
all 4 of the outside bolts.

If you have issues, try rotating one or both
of the nuts 90°.





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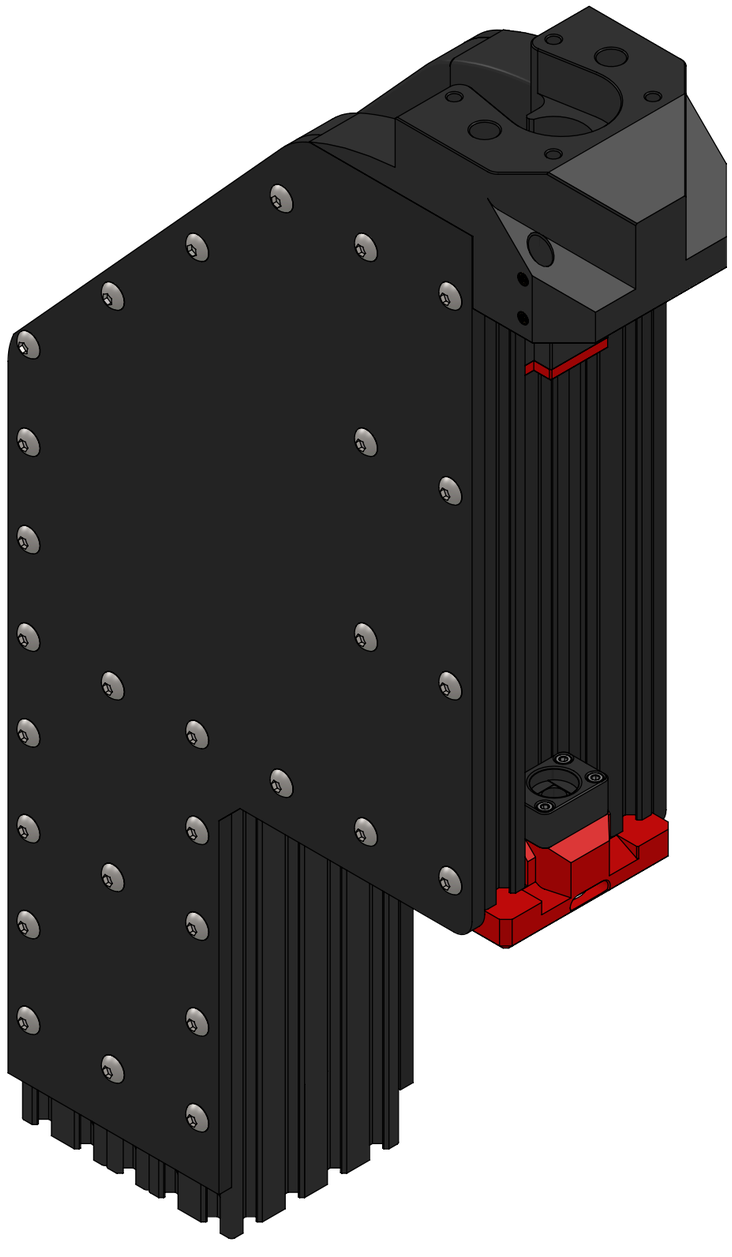

PART NUMBER	QTY.
C-BEAM-315mm	1
V-SLOT-2020X315MM	2
C-BEAM-315mm	1
C BEAM 120mm	1
C BEAM 255mm	2
FMJ V2	2
ISO-7380-M5X10 BHCS	58
M5 DROP IN NUT	52
REAR-CABLE-RUN	1
M5X5.8mm HEATSET INSERT	12
ISO-7380-M5X12 BHCS	18
COLUMN-BALLAST	2
Z-AXIS-MOTOR-MOUNT	1
M3X3MM HEATSET INSERT	8
Z-AXIS_BEARING-BLOCK	1
608ZZ BALL BEARING	3
Z-BEARING-BLOCK-INSERT	1
ISO-4762-M3X40-SHCS	6
M3 DROP IN NUT	12
INTERNAL AXIS FLOATING BEARING BLOCK	1
FLOATING BEARING BLOCK FACE-PLATE A	1
FLOATING BEARING BLOCK FACE-PLATE B	1

Z AXIS FRAME

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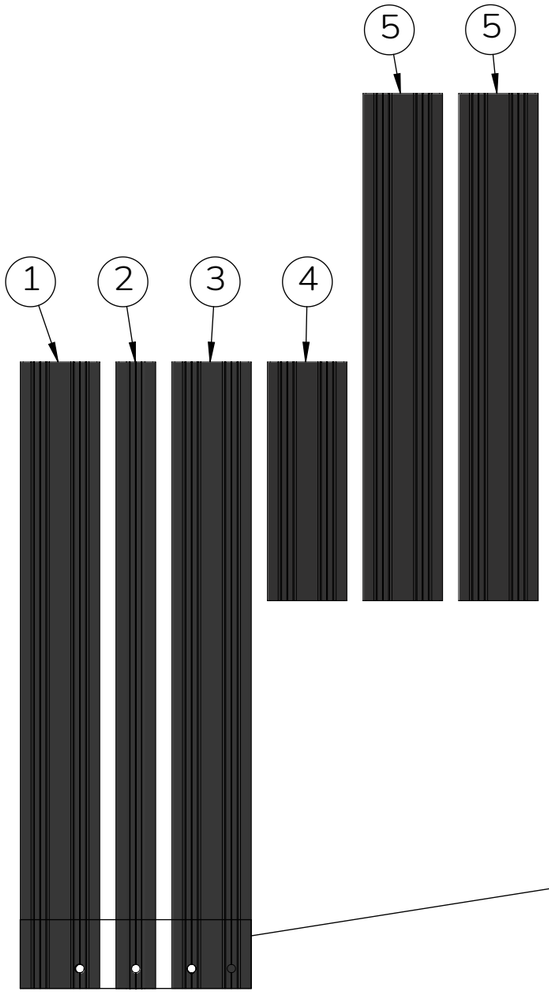
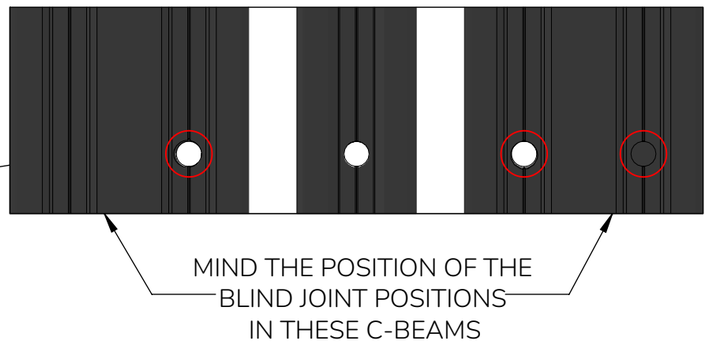
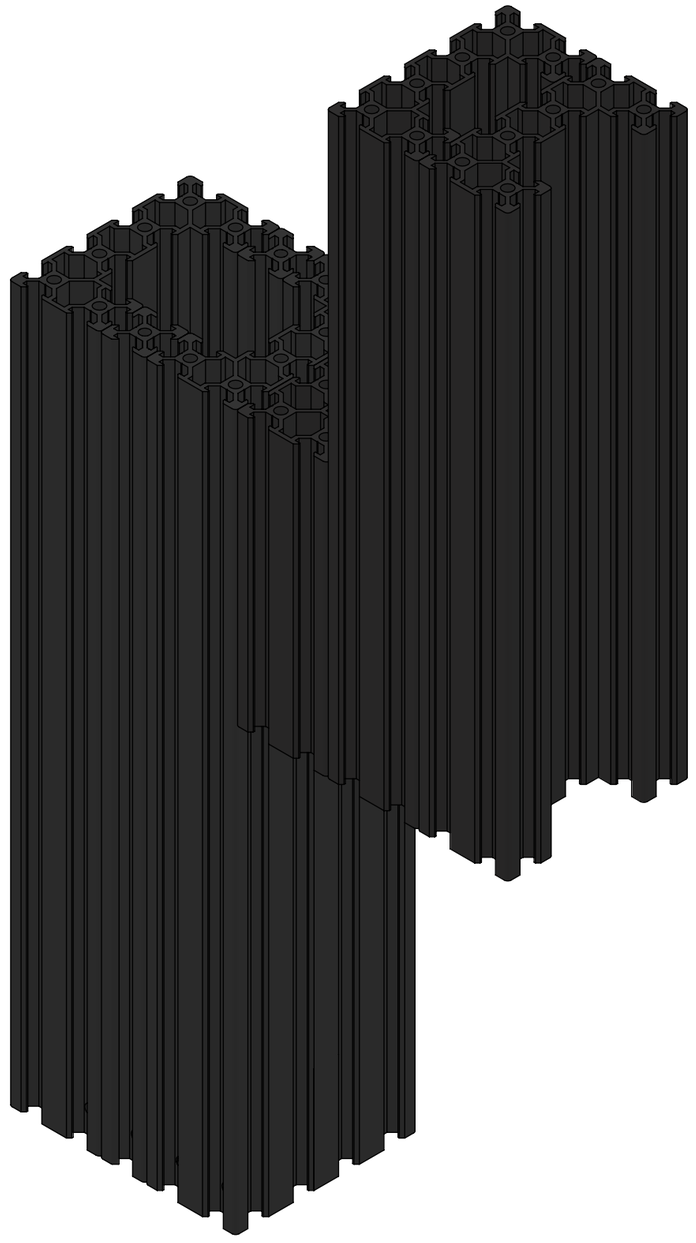


	PART NUMBER	QTY.
1	C-BEAM-315mm	1
2	V-SLOT-2020X315MM	2
3	C-BEAM-315mm	1
4	C BEAM 120mm	1
5	C BEAM 255mm	2





Z AXIS FRAME

EXTRUSIONS

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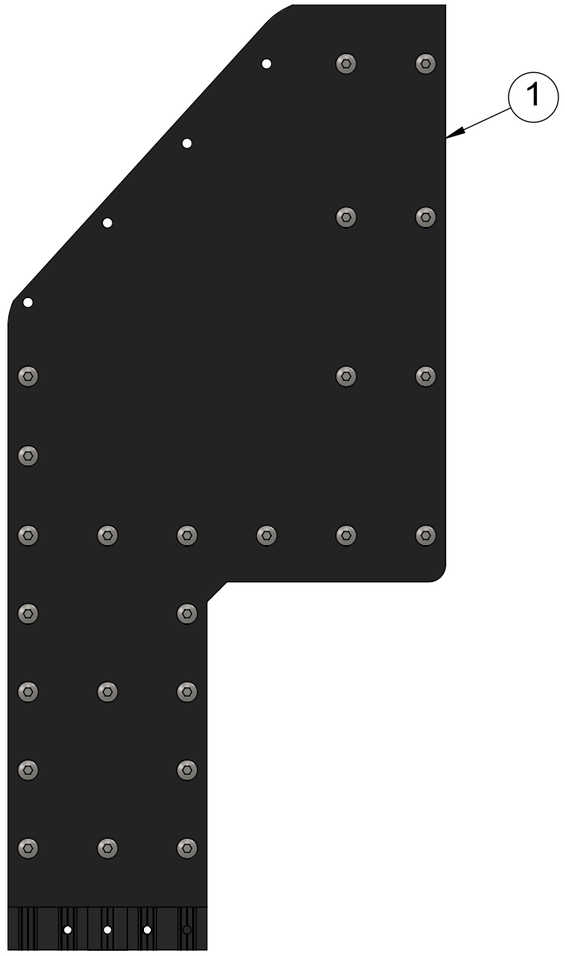
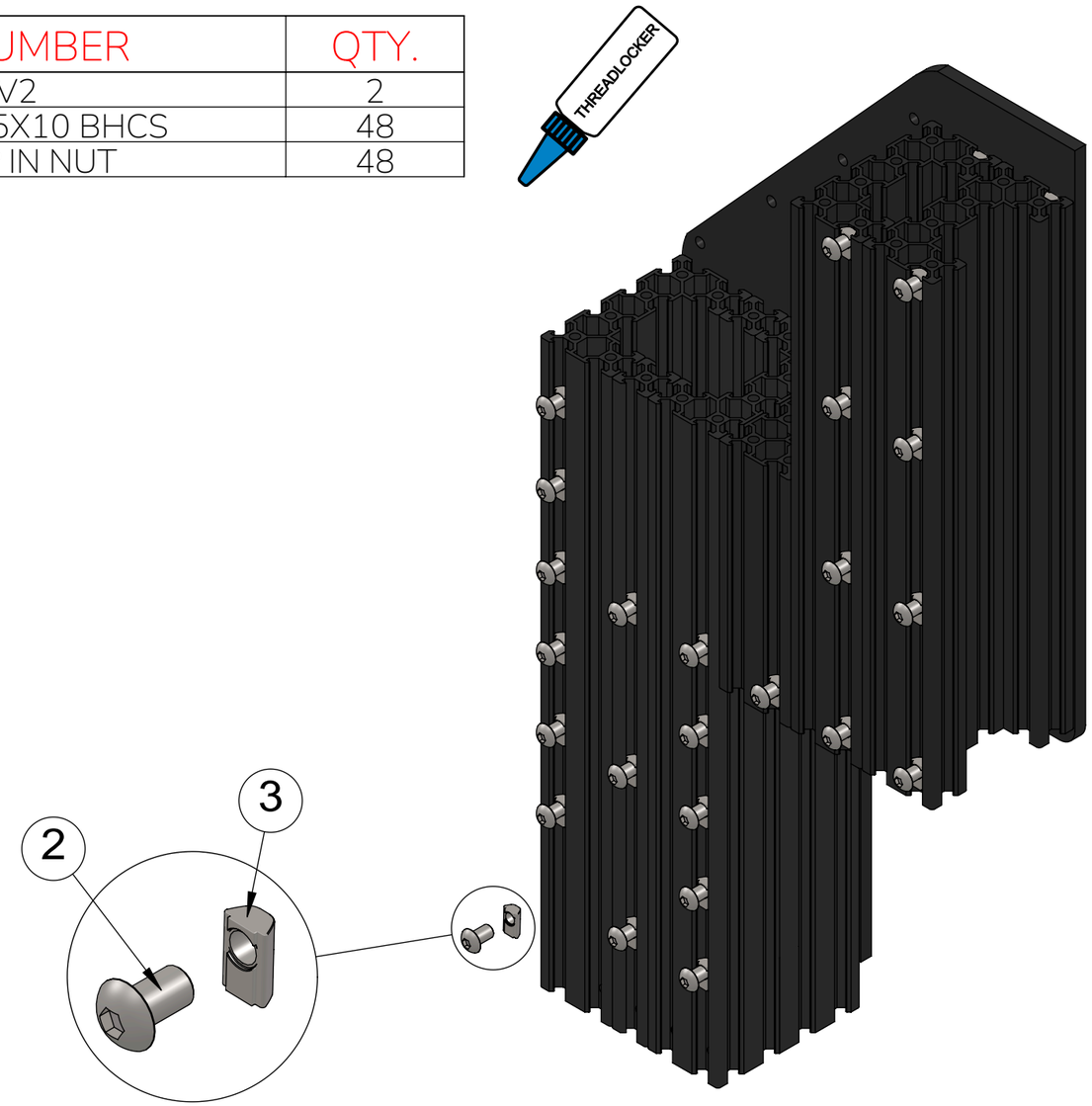
	PART NUMBER	QTY.
1	FMJ V2	2
2	ISO-7380-M5X10 BHCS	48
3	M5 DROP IN NUT	48





Z AXIS FRAME

FMJ PLATES

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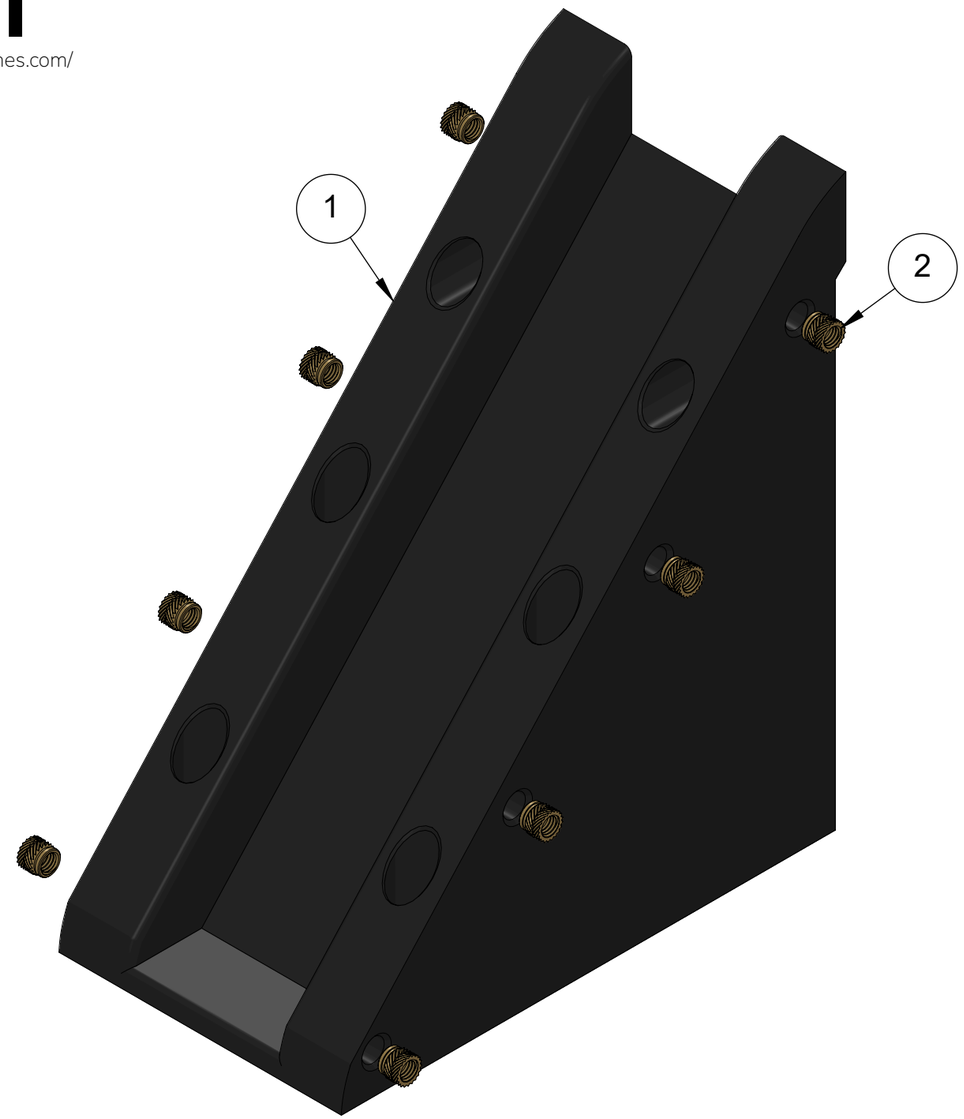

	PART NUMBER	QTY.
1	REAR-CABLE-RUN	1
2	M5X5.8mm-HEATSET	8

Z AXIS FRAME

REAR CABLE RUN

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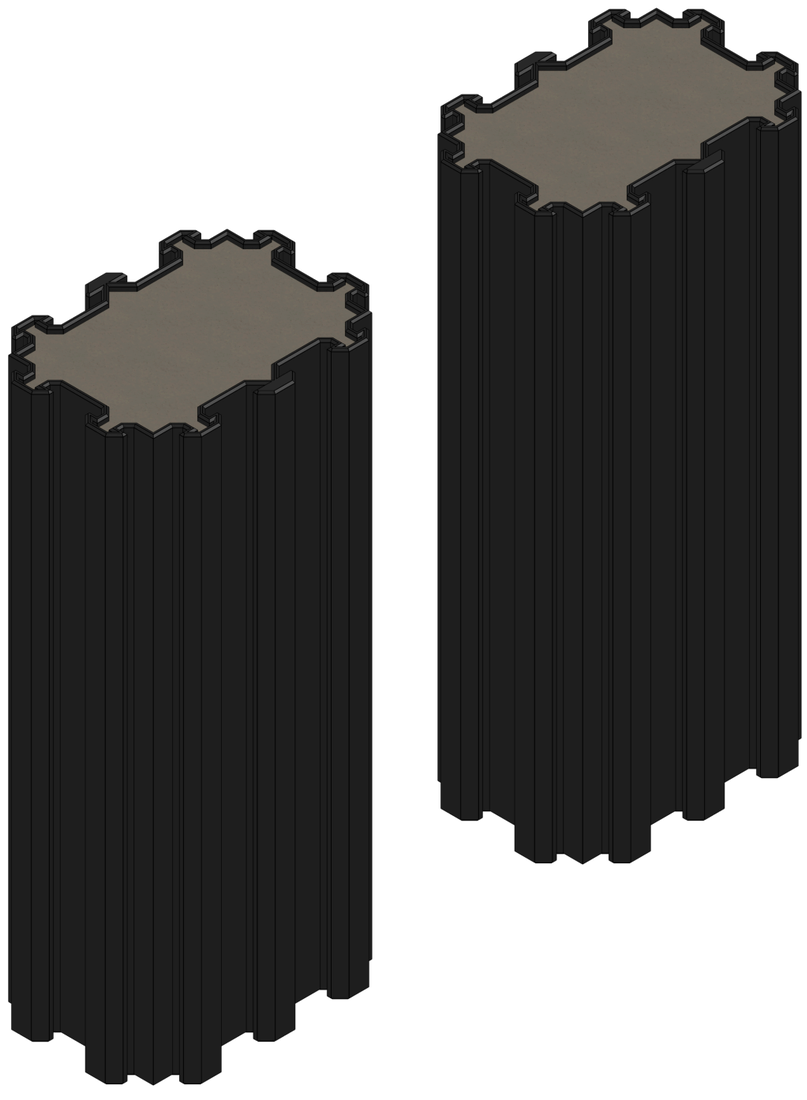
Ensure your material sits just below the top lip of the column ballast



COLUMN BALLAST

The column ballast is designed to house a damping load such as sand or epoxy granite. It is not a requirement but is recommended. Optimal material is granite mixed with just enough epoxy to make it wet. It should *not* flow.

If you decide to use just sand, you should take extra caution. Sand will wear your leadscrews/ballscrews and linear rails.

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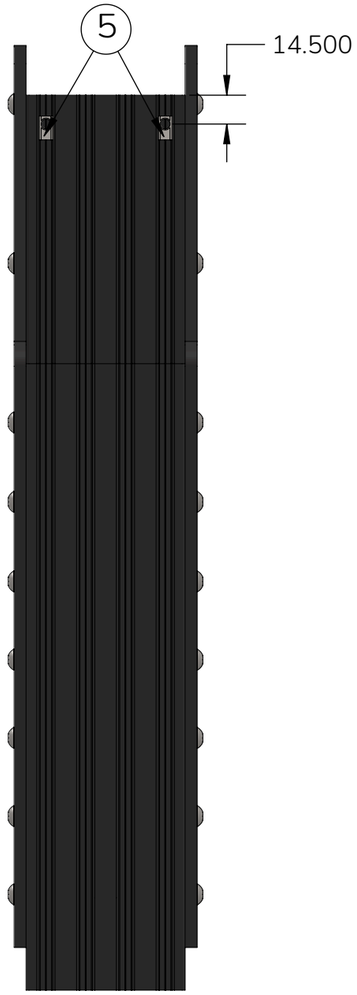
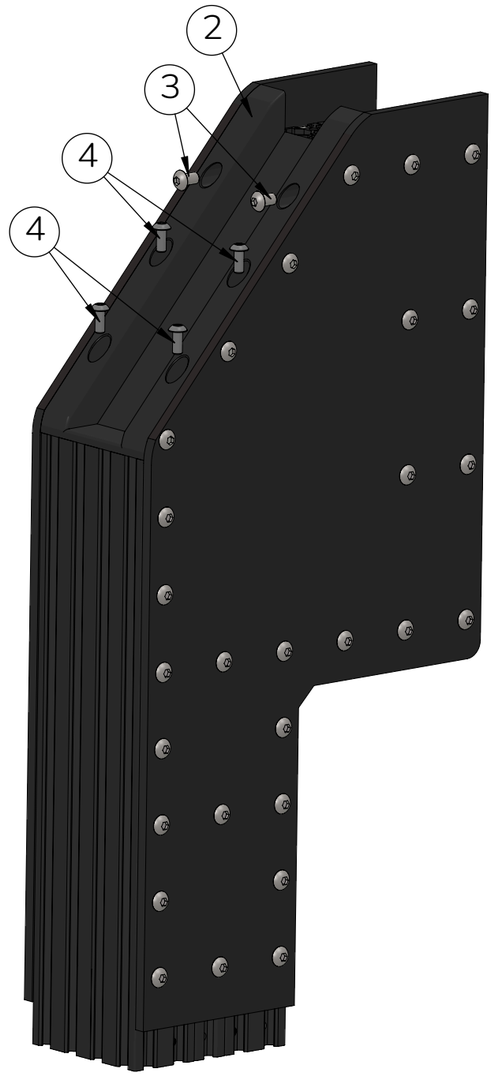
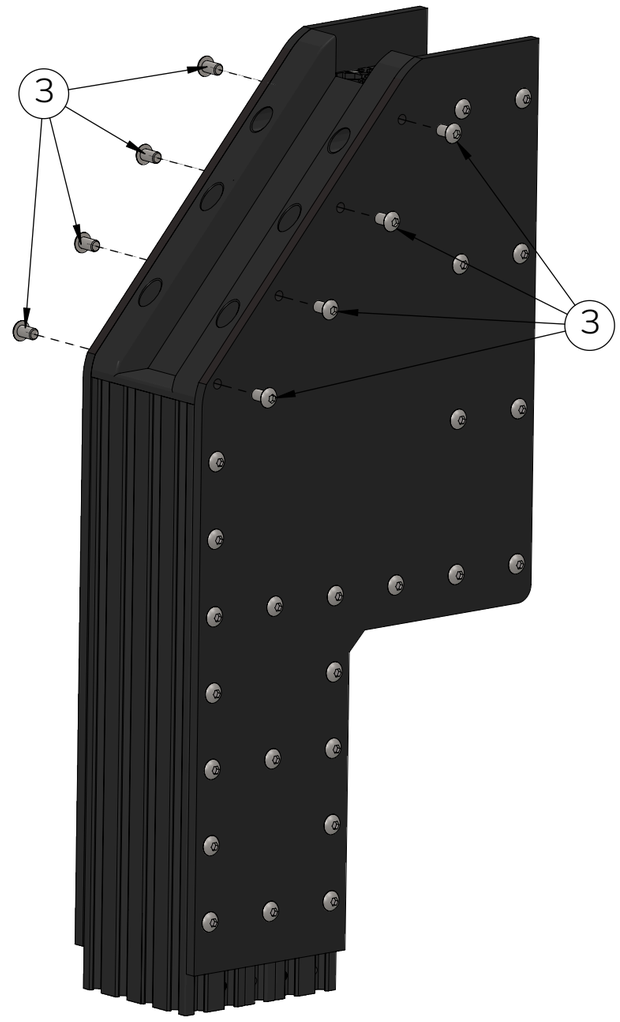
	PART NUMBER	QTY.
2	REAR-CABLE-RUN-ASSY	1
3	ISO-7380-M5X10 BHCS	10
4	ISO-7380-M5X12 BHCS	4
5	M5 DROP IN NUT	2







Z AXIS FRAME

ASSEMBLY STEP 1A

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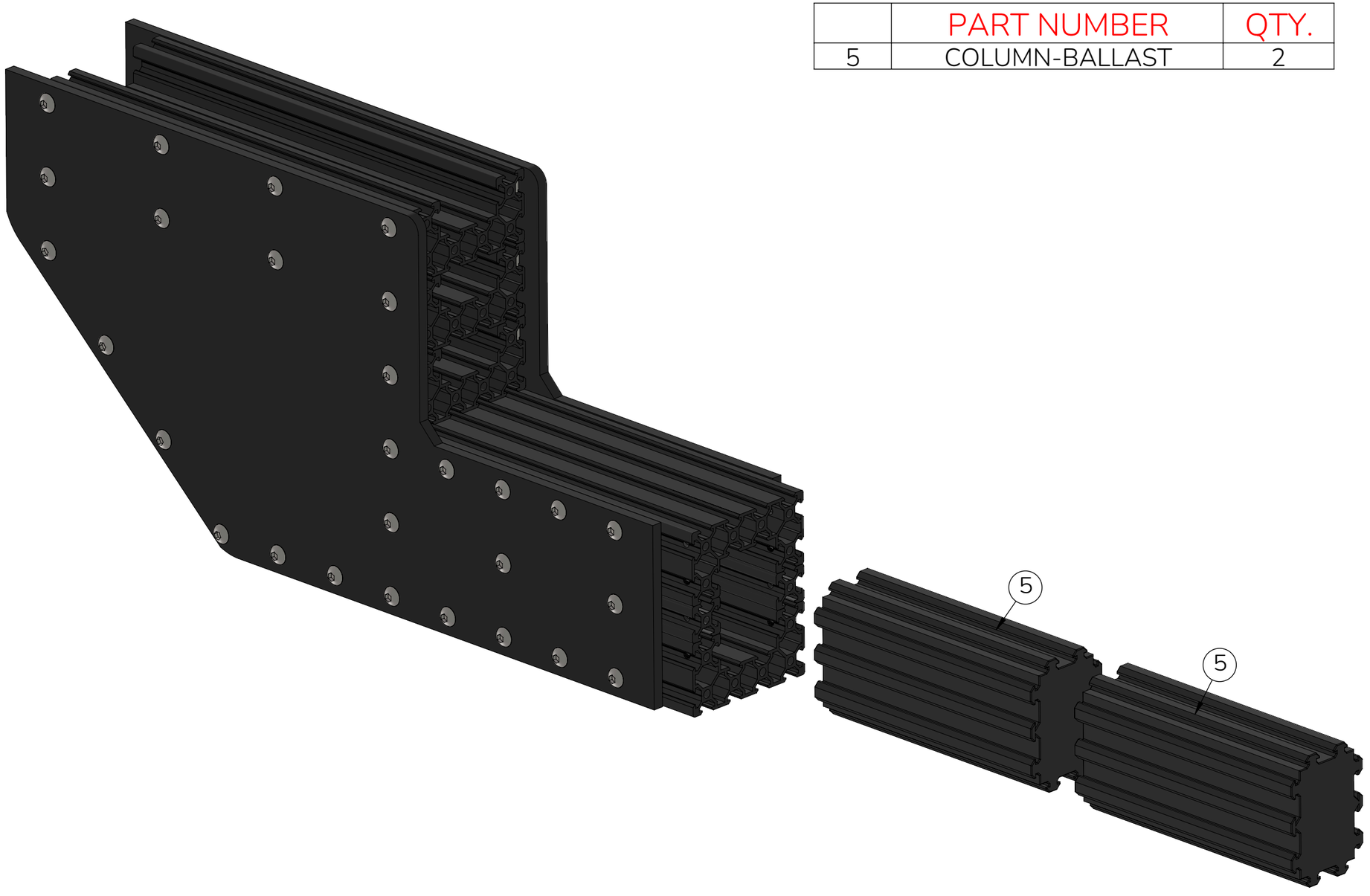
	PART NUMBER	QTY.
5	COLUMN-BALLAST	2



Z AXIS FRAME

ASSEMBLY STEP 1B

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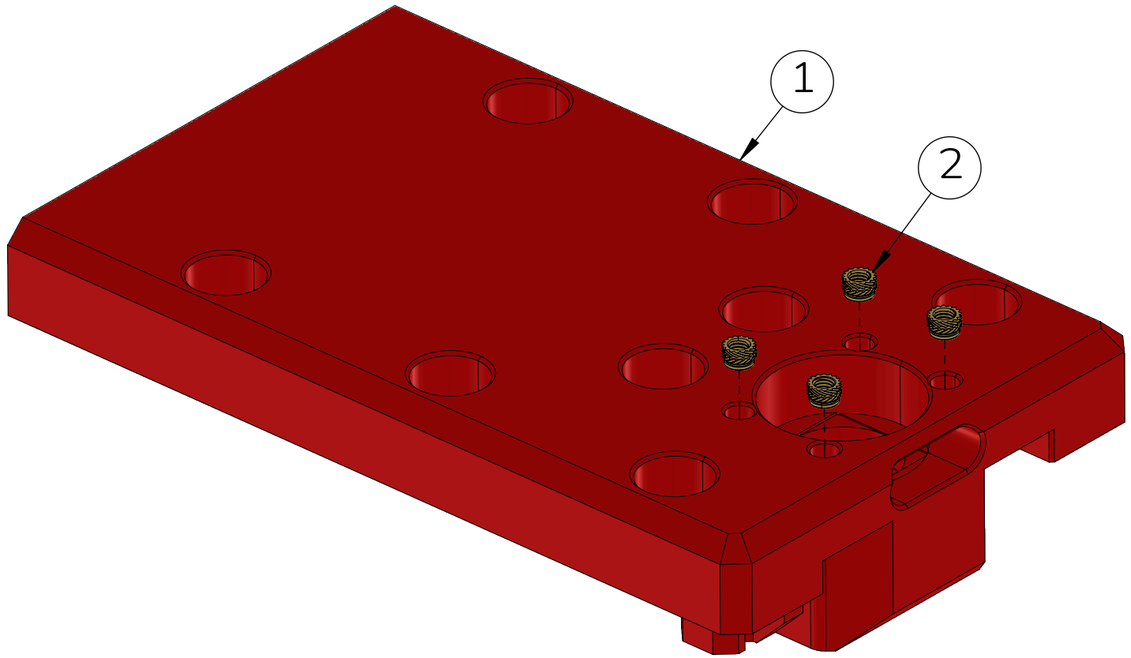
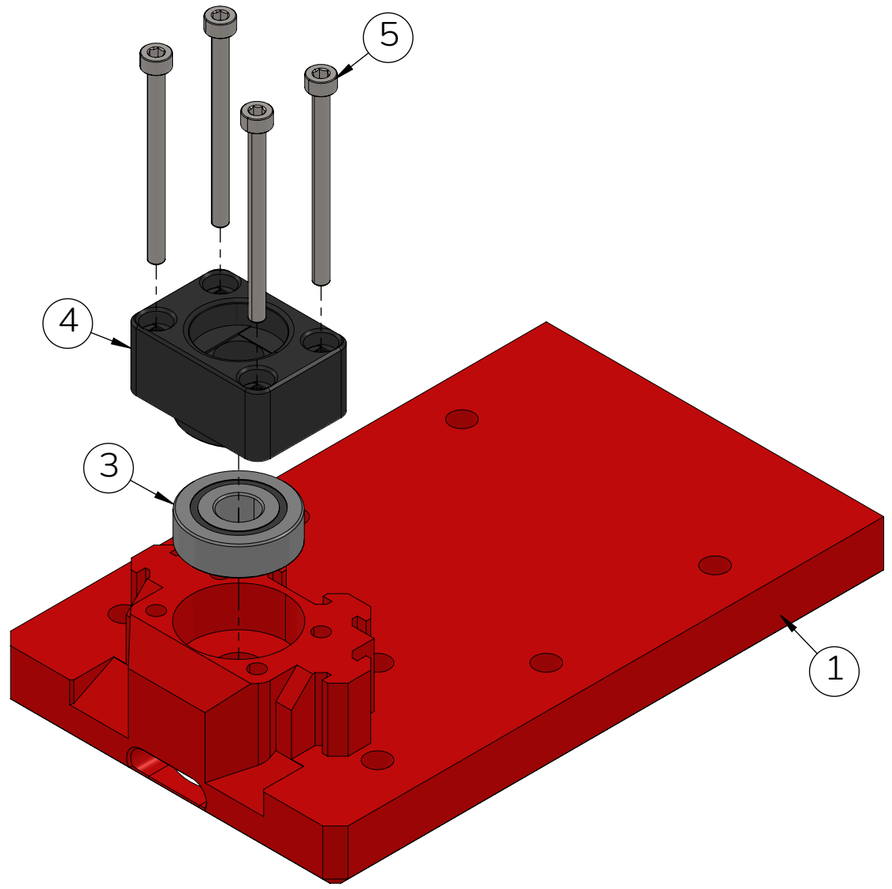
	PART NUMBER	QTY.
1	Z-AXIS BEARING BLOCK	1
2	M3X3MM HEATSET INSERT	4



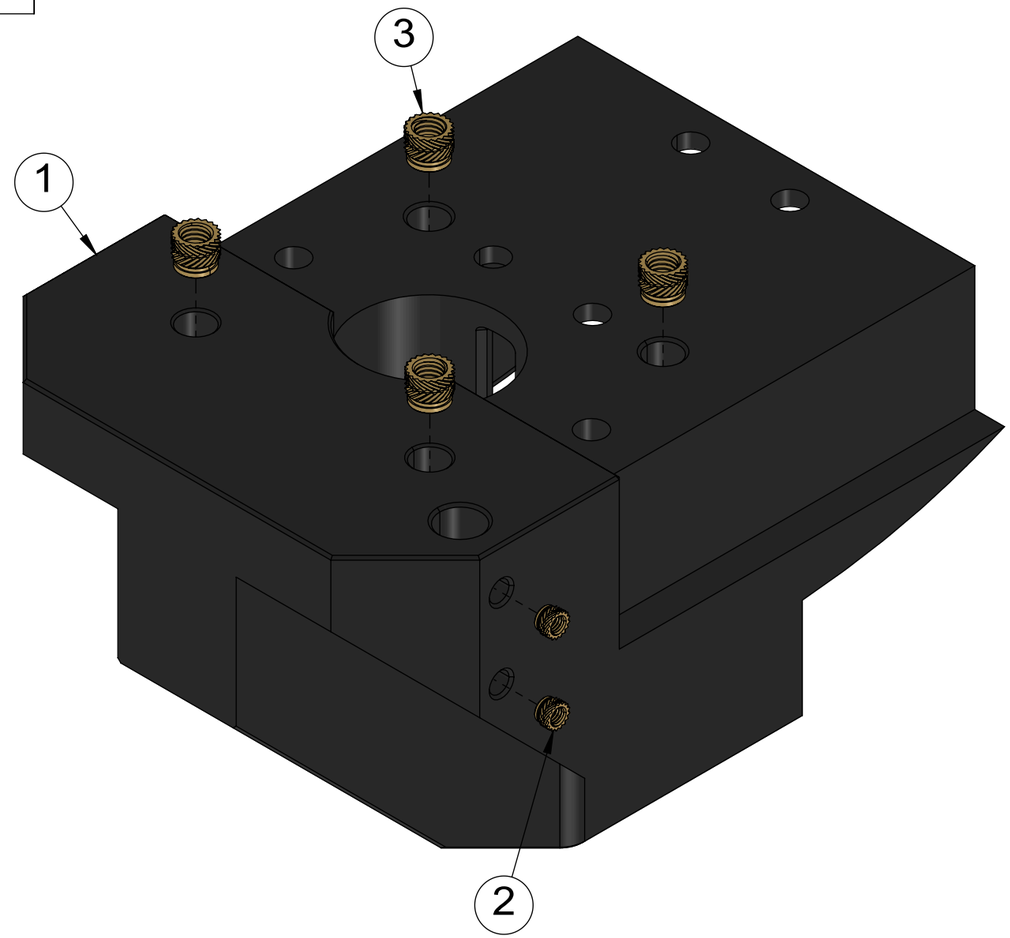
	PART NUMBER	QTY.
1	Z-AXIS BEARING BLOCK	1
3	608ZZ BALL BEARING	1
4	Z-BEARING-BLOCK-INSERT	1
5	ISO-4762-M3X40-SHCS	4



Z AXIS FRAME

BEARING BLOCK ASSEMBLY

	PART NUMBER	QTY.
1	Z-AXIS-MOTOR-MOUNT	1
2	M3x3mm HEATSET INSERT	2
3	M5x5.8mm HEATSET INSERT	4



Z AXIS FRAME

MOTOR MOUNT PRE-ASSEMBLY

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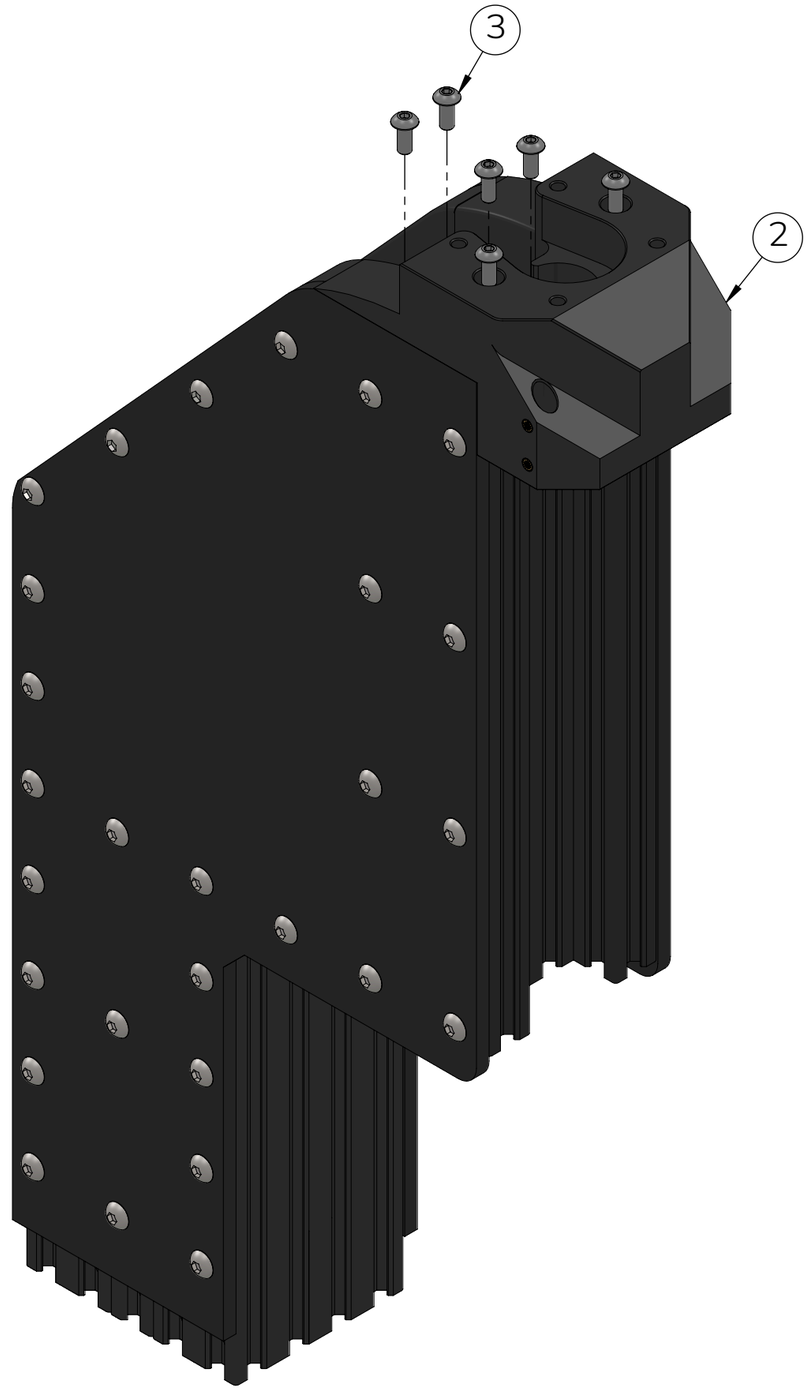
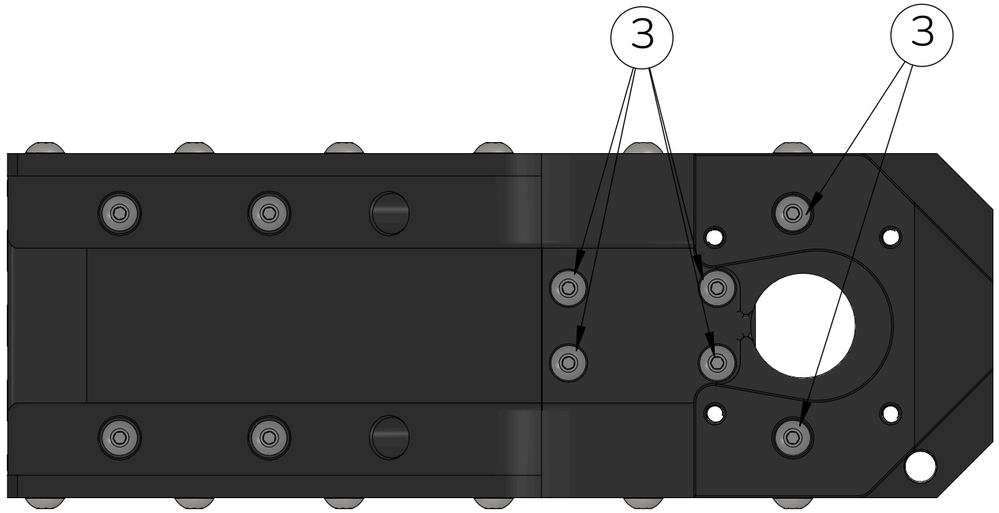
	PART NUMBER	QTY.
2	Z-AXIS-MOTOR-MOUNT-ASSY	1
3	ISO-7380-M5X12 BHCS	6





Z AXIS FRAME

STEP 2A MOTOR MOUNT

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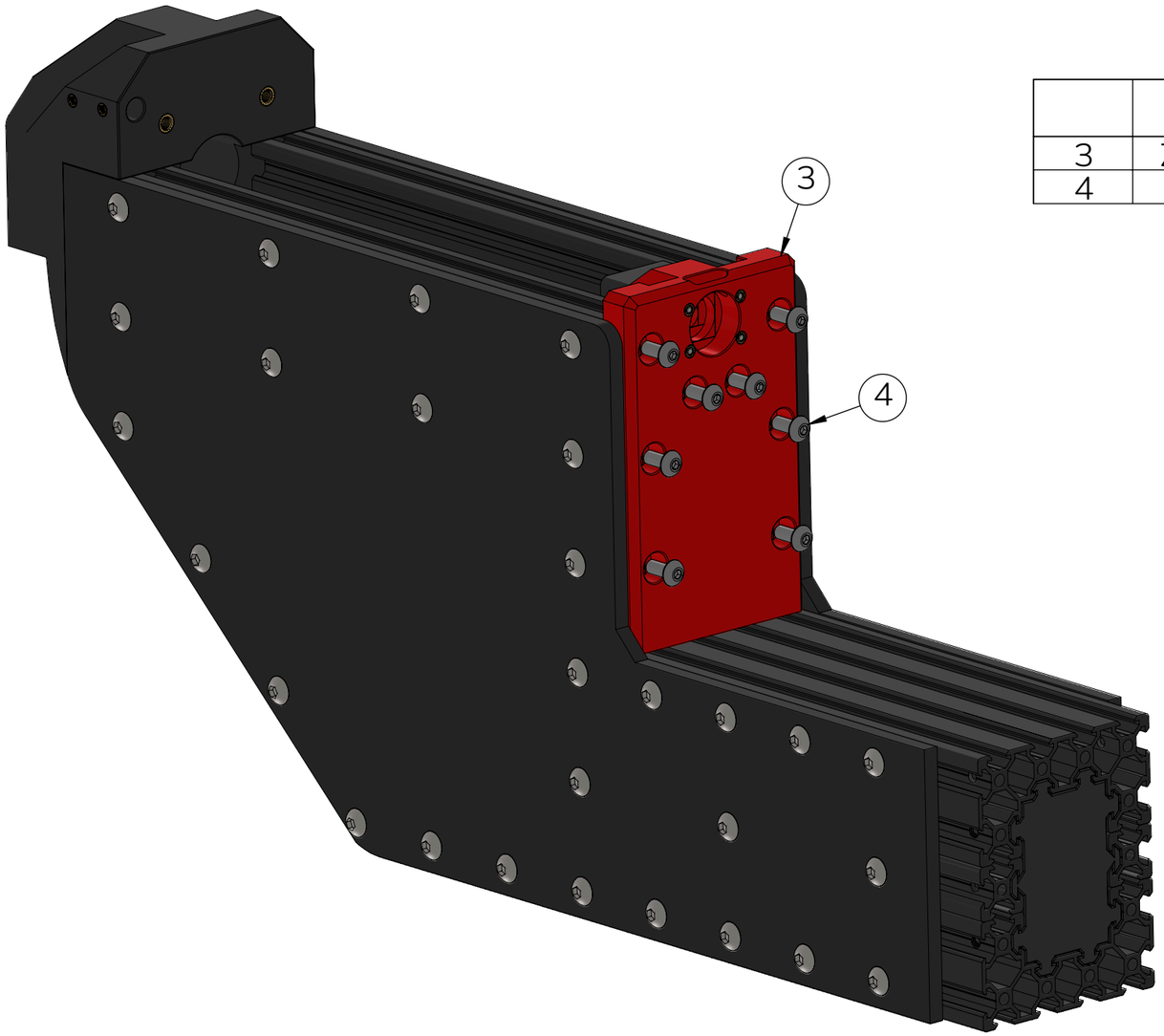

	PART NUMBER	QTY.
3	Z-AXIS-BEARING-BLOCK-ASSY	1
4	ISO-7380-M5X12 BHCS	8

Z AXIS FRAME

STEP 2B BEARING BLOCK

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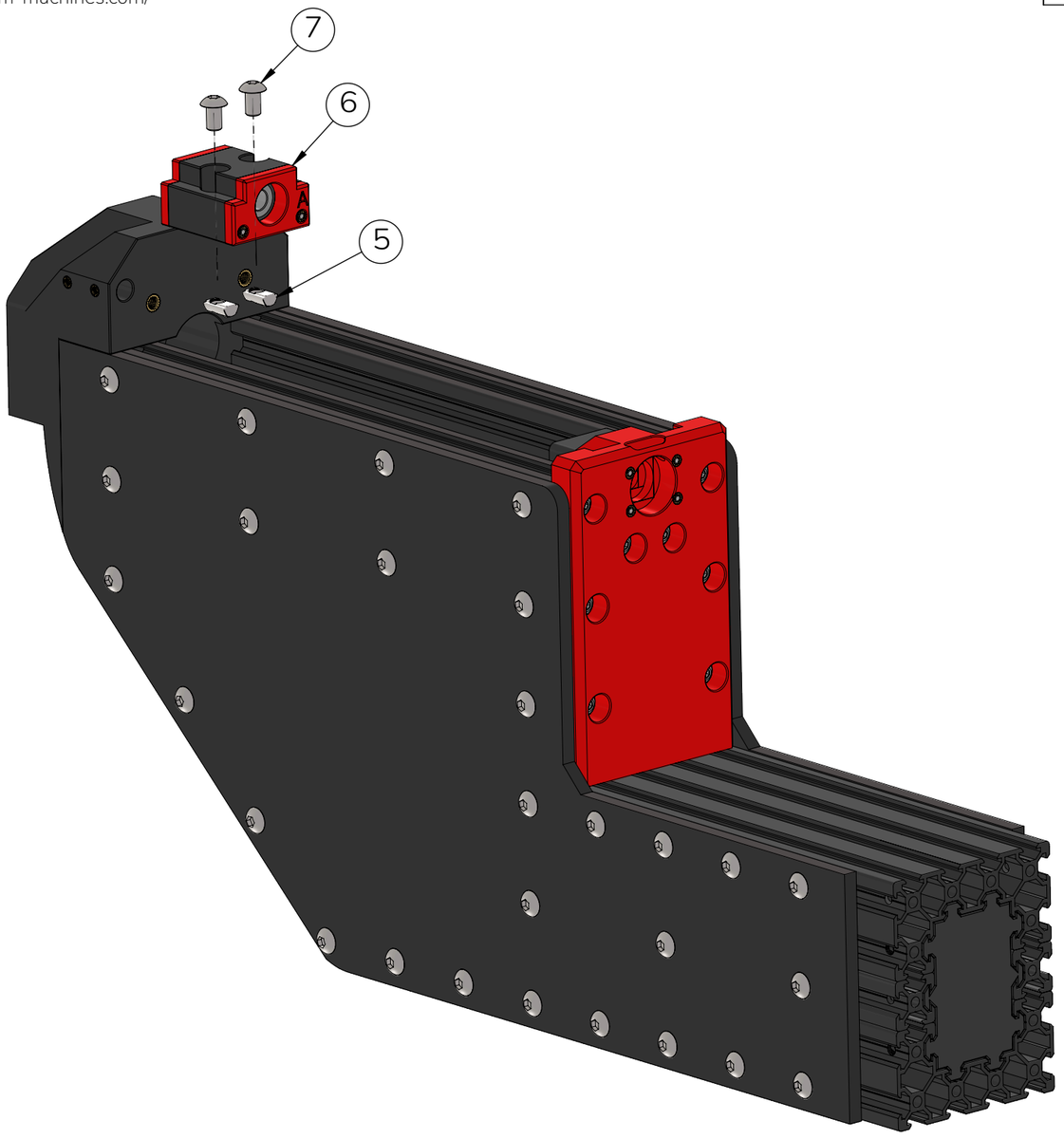
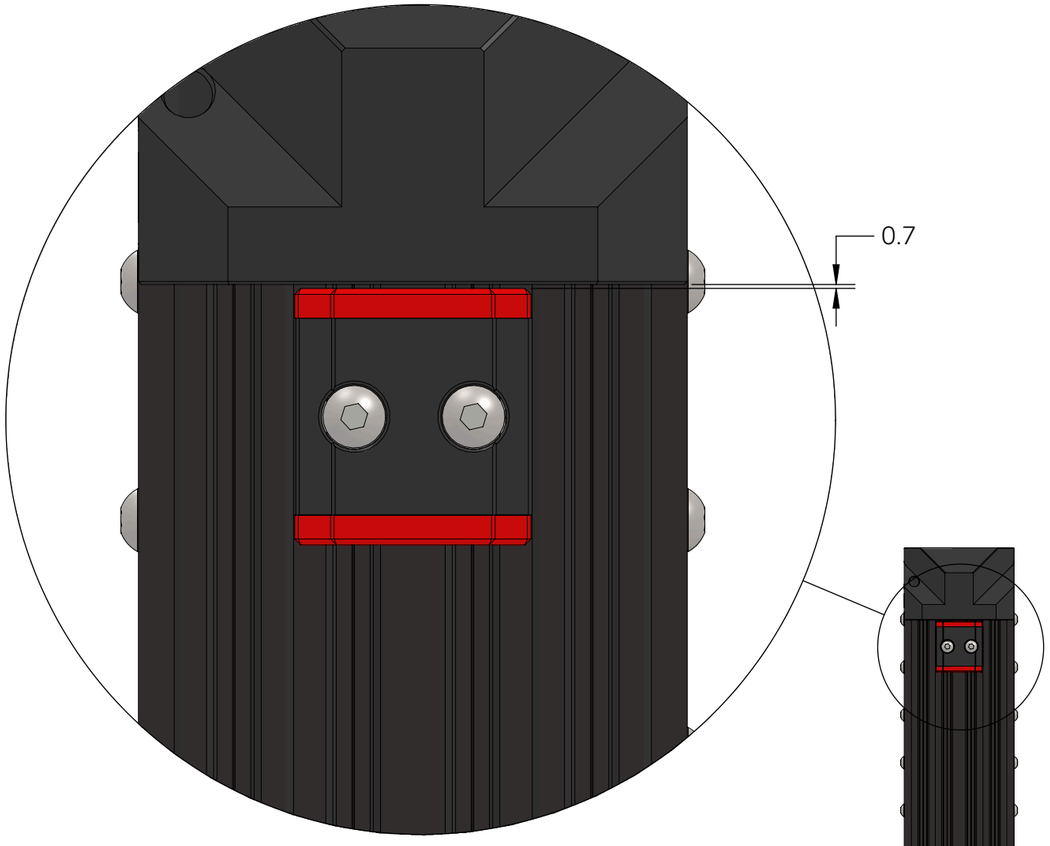
	PART NUMBER	QTY.
5	M5 DROP IN NUT	2
6	FLOATING-BEARING-BLOCK-ASSEMBLY	1
7	ISO-7380-M5X10 BHCS	2





Don’t worry about making this perfect, the floating bearing will be removed in a later step before being installed in it’s final position

Z AXIS FRAME

STEP 2C FLOATING BEARING

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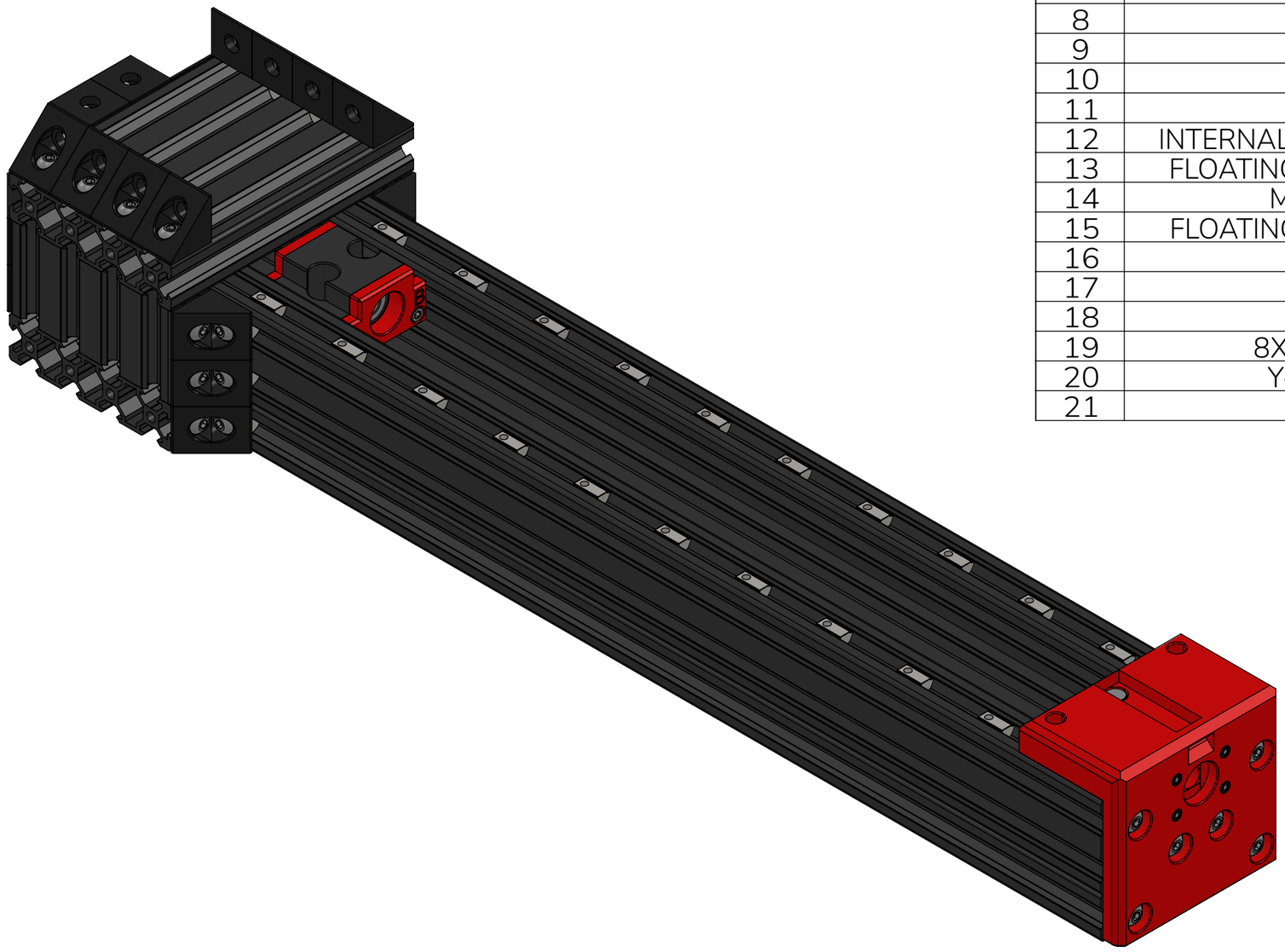

	PART NUMBER	QTY.
1	ISO-7380-M5X8-BHCS	36
2	M5 DROP IN NUT	40
3	ANGLE CORNER CONNECTOR	26
4	C-BEAM-540mm	1
5	V-SLOT-2080X440MM	1
6	V-SLOT 2080X120MM	2
7	ISO-7380-M5X30-BHCS	4
8	V-SLOT 2080X40MM	2
9	ISO-7380-M5X12 BHCS	18
10	ISO-7380-M3x10-BHCS	2
11	M3 DROP IN NUT	22
12	INTERNAL AXIS FLOATING BEARING BLOCK	1
13	FLOATING BEARING BLOCK FACE-PLATE A	1
14	M3X3MM HEATSET INSERT	6
15	FLOATING BEARING BLOCK FACE-PLATE B	1
16	608ZZ BALL BEARING	3
17	ISO-4762-M3X40-SHCS	6
18	Y-BEARING-BLOCK	1
19	8X3MM NEODYMIUM MAGNET	1
20	Y-BEARING-BLOCK-INSERT	1
21	ISO-7380-M5X10 BHCS	2

Y-AXIS FRAME

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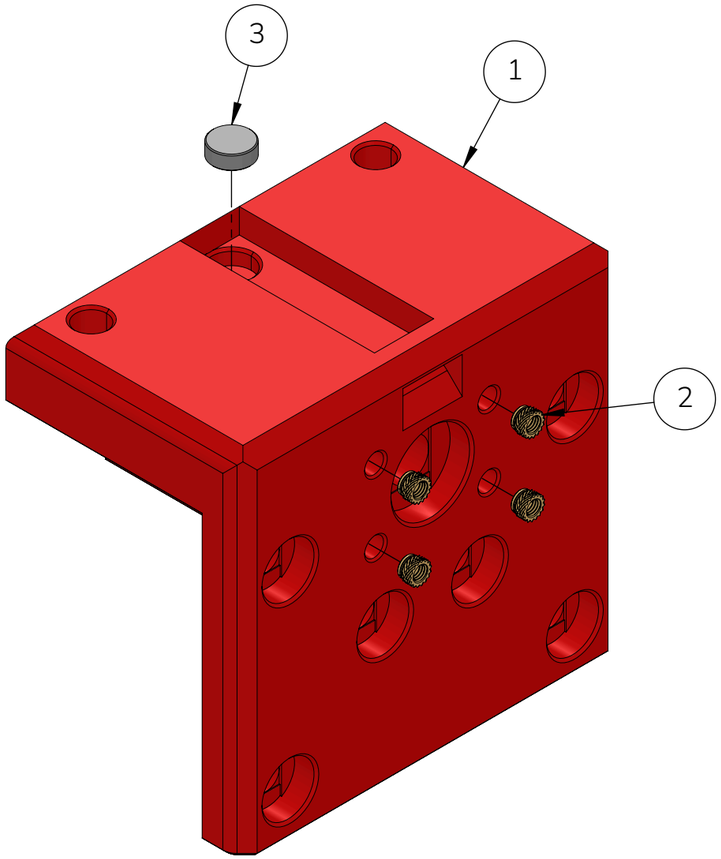
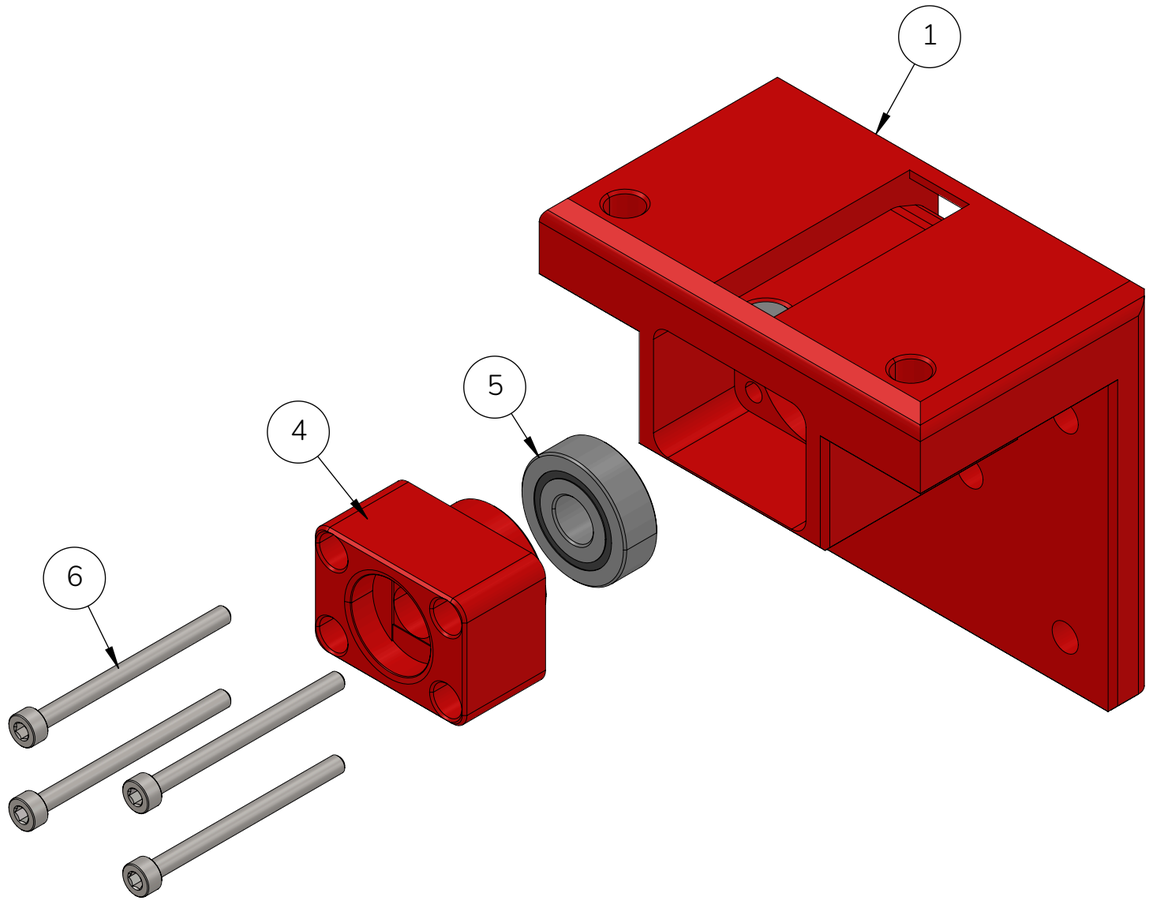
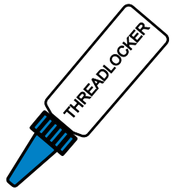
	PART NUMBER	QTY.
1	Y-BEARING-BLOCK	1
2	M3X3MM HEATSET INSERT	4
3	8X3MM NEODYMIUM MAGNET	1
4	Y-BEARING-BLOCK-INSERT	1
5	608ZZ BALL BEARING	1
6	ISO-4762-M3X40-SHCS	4

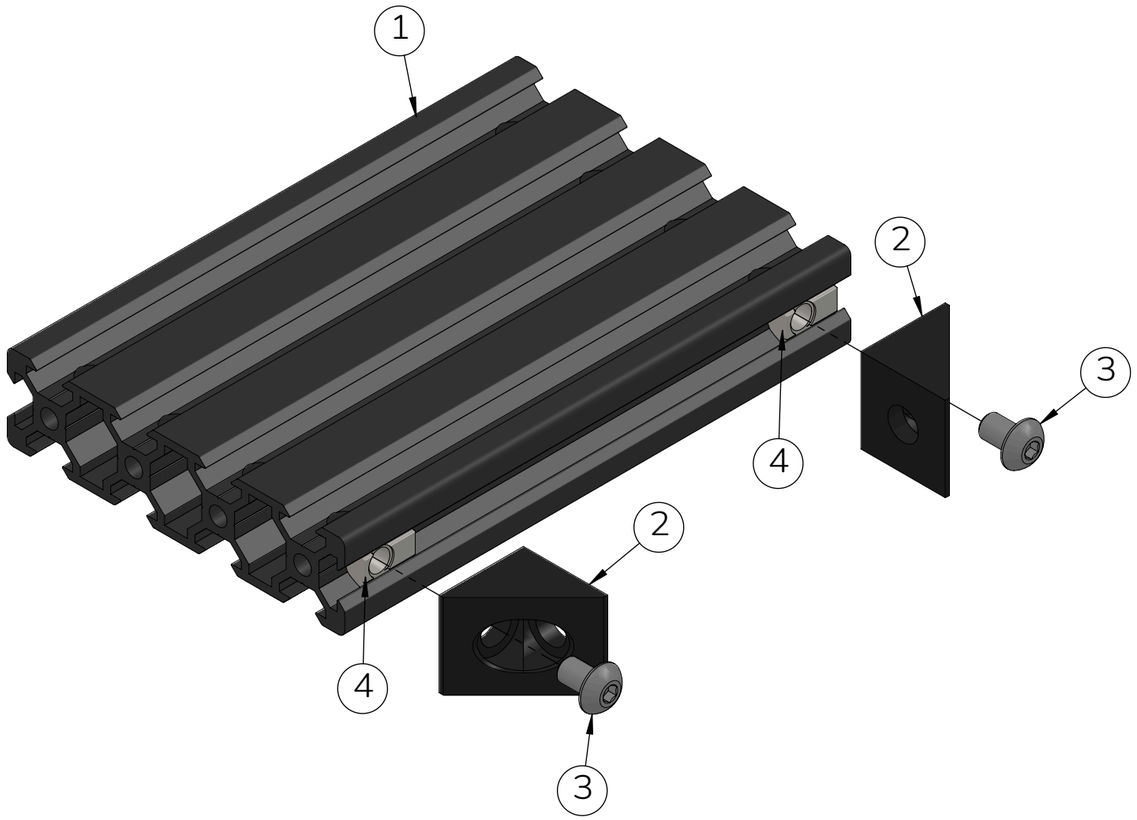




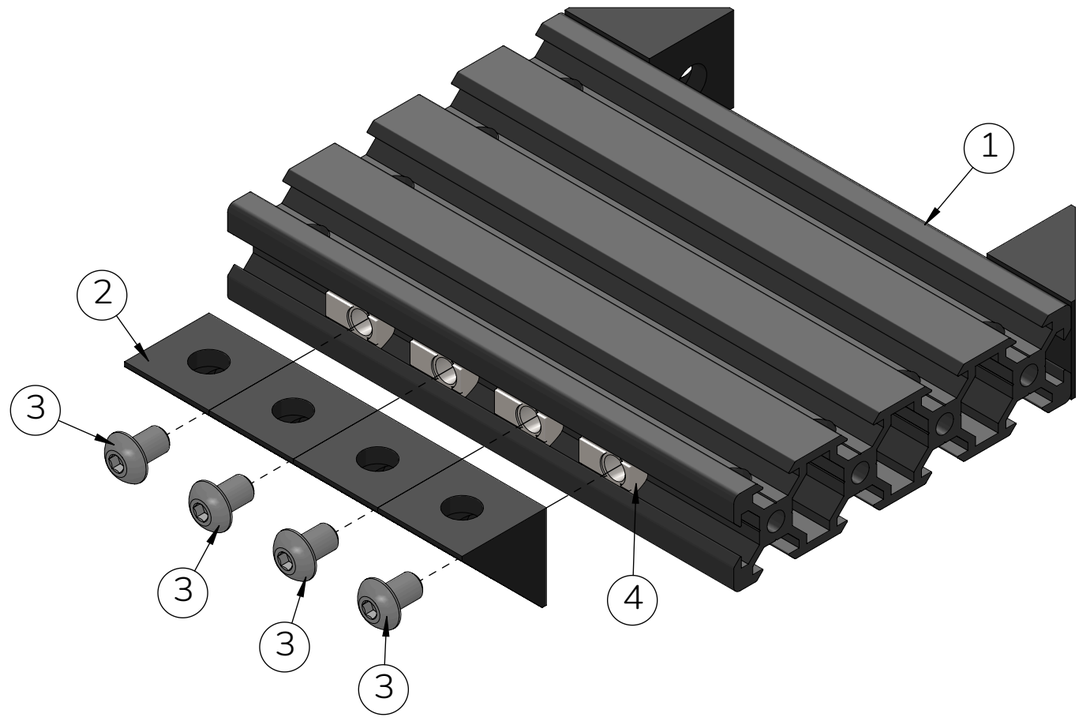
Y AXIS FRAME

BEARING BLOCK PRE-ASSEMBLY





	PART NUMBER	QTY.
1	V-SLOT 2080X120MM	1
2	ANGLE CORNER CONNECTOR	2
3	ISO-7380-M5X8-BHCS	2
4	M5 DROP IN NUT	2



	PART NUMBER	QTY.
1	V-SLOT 2080X120MM	1
2	ANGLE CORNER CONNECTOR	4
3	ISO-7380-M5X8-BHCS	4
4	M5 DROP IN NUT	4

Y-AXIS FRAME

120mm EXTRUSION A SUB-ASSEMBLY

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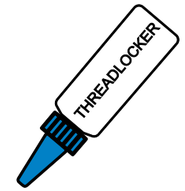
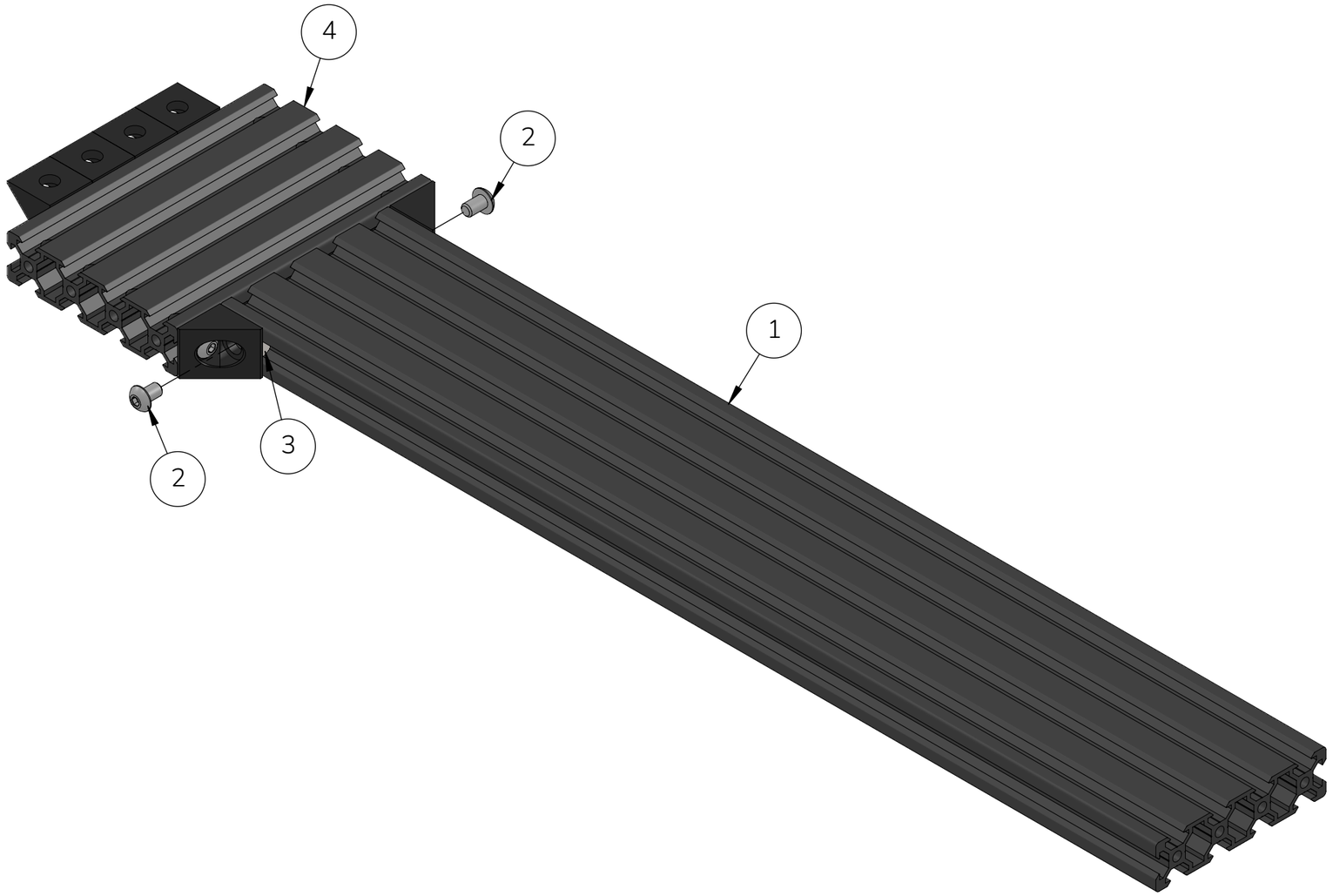
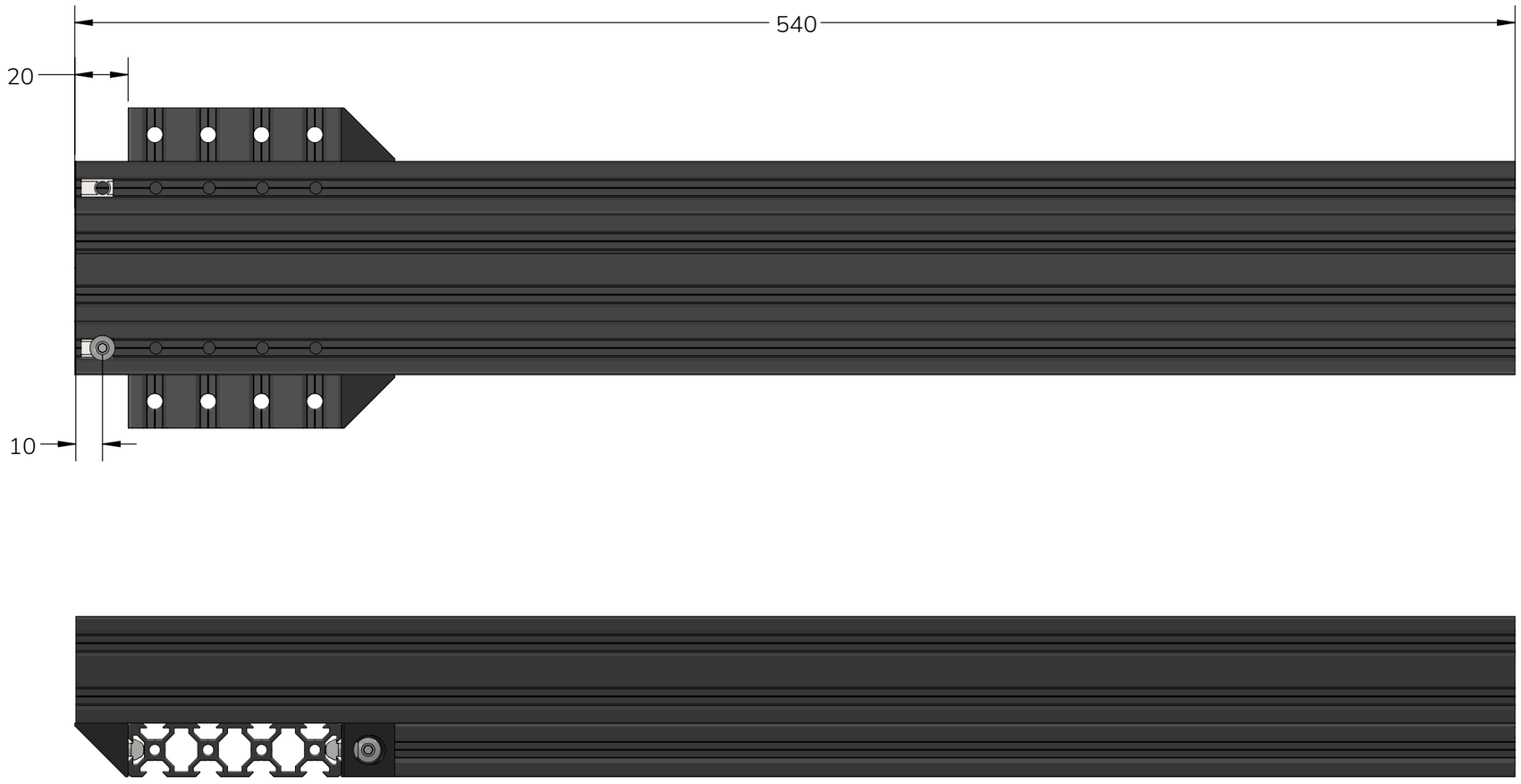
	PART NUMBER	QTY.
1	V-SLOT-2080X440MM	1
2	ISO-7380-M5X8-BHCS	2
3	M5 DROP IN NUT	2
4	V-SLOT-2080X120MM-ASSEMBLY	1





Y-AXIS FRAME

2080 SUB-ASSEMBLY



Y-AXIS FRAME

2080-SUB-ASSEMBLY

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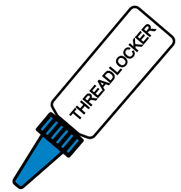
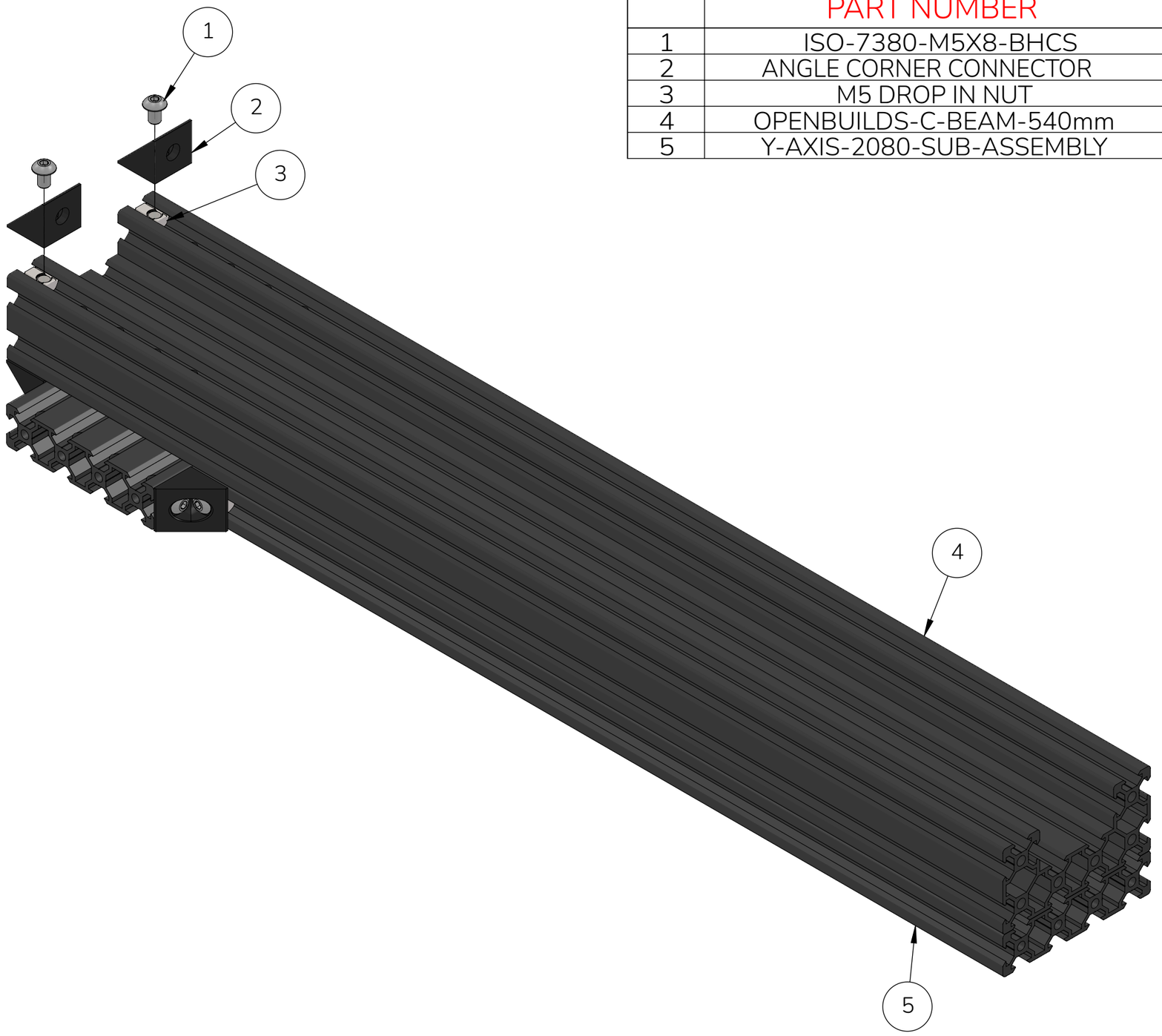
	PART NUMBER	QTY.
1	ISO-7380-M5X8-BHCS	2
2	ANGLE CORNER CONNECTOR	2
3	M5 DROP IN NUT	2
4	OPENBUILDS-C-BEAM-540mm	1
5	Y-AXIS-2080-SUB-ASSEMBLY	1





Y-AXIS FRAME

C-BEAM SUB-ASSEMBLY

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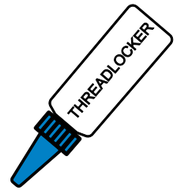
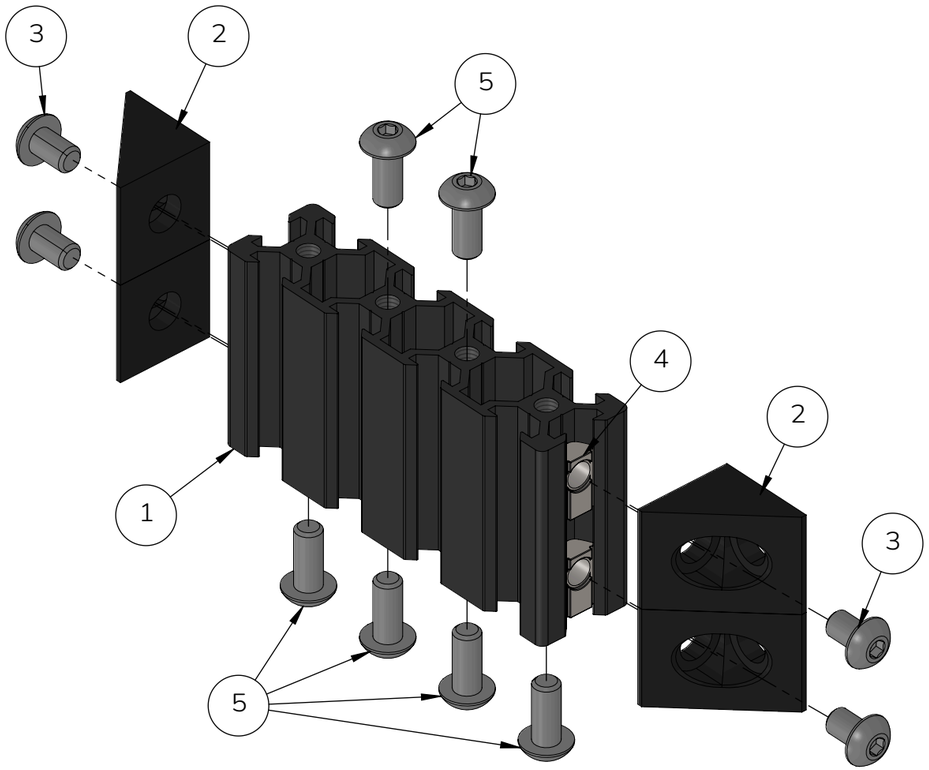
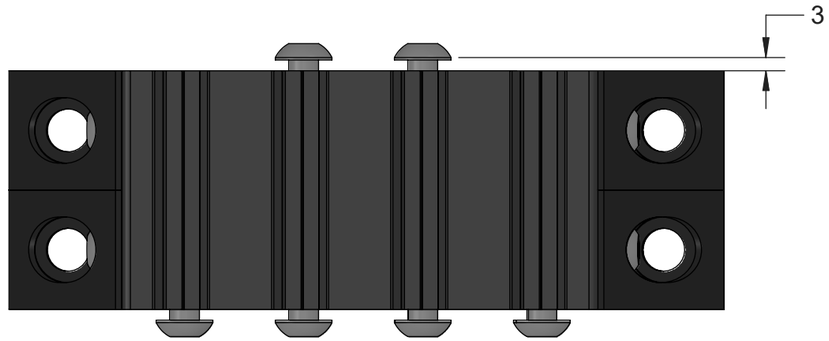


QTY 2

	PART NUMBER	QTY.
1	V-SLOT 2080X40MM	1
2	ANGLE CORNER CONNECTOR	4
3	ISO-7380-M5X8-BHCS	4
4	M5 DROP IN NUT	4
5	ISO-7380-M5X12 BHCS	6





Y-AXIS FRAME

40mm EXTRUSION SUB-ASSEMBLY

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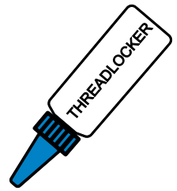
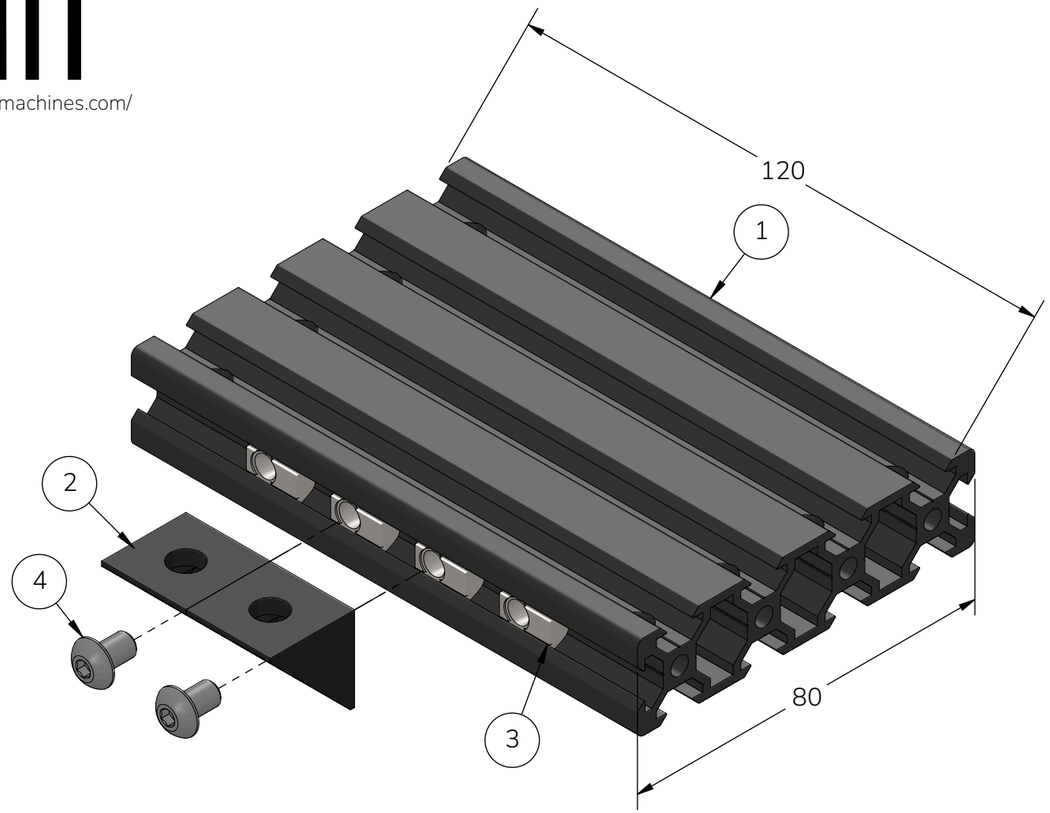
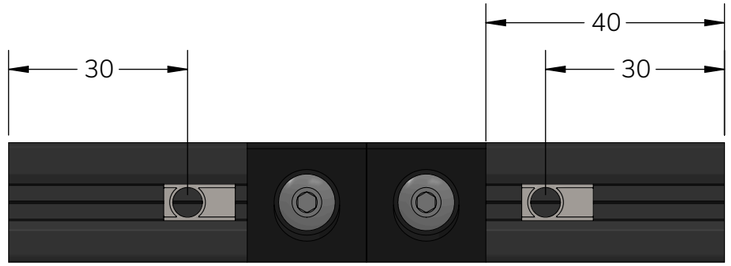
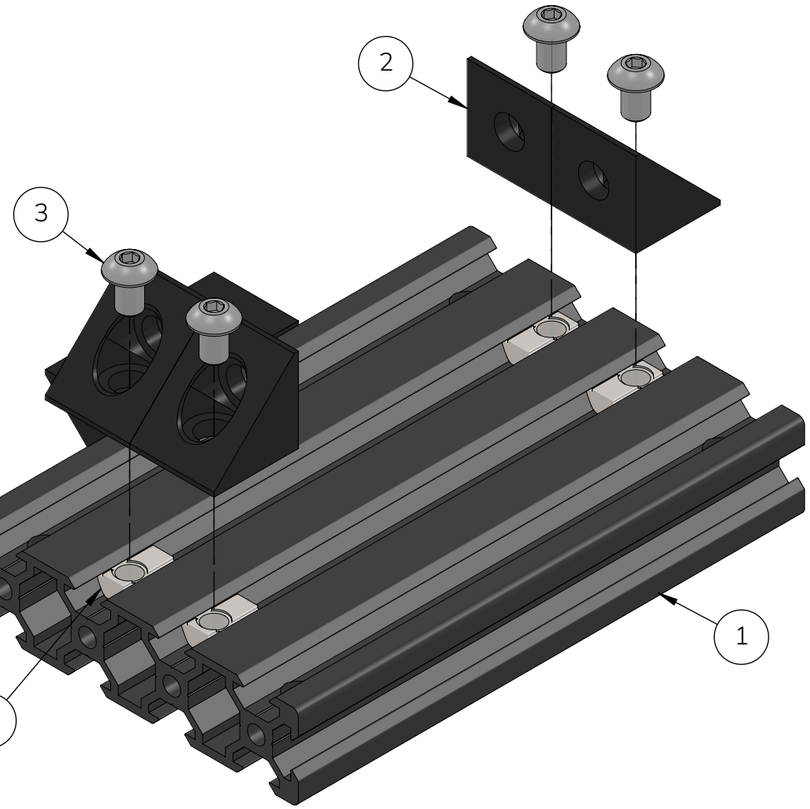
	PART NUMBER	QTY.
1	V-SLOT 2080X120MM	1
2	ANGLE CORNER CONNECTOR	2
3	M5 DROP IN NUT	4
4	ISO-7380-M5X8-BHCS	2









	PART NUMBER	QTY.
1	V-SLOT 2080X120MM	1
2	ANGLE CORNER CONNECTOR	4
3	ISO-7380-M5X8-BHCS	4
4	M5 DROP IN NUT	4

Y-AXIS FRAME

120mm EXTRUSION B SUB-ASSEMBLY

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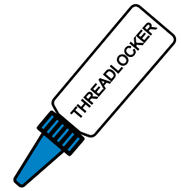
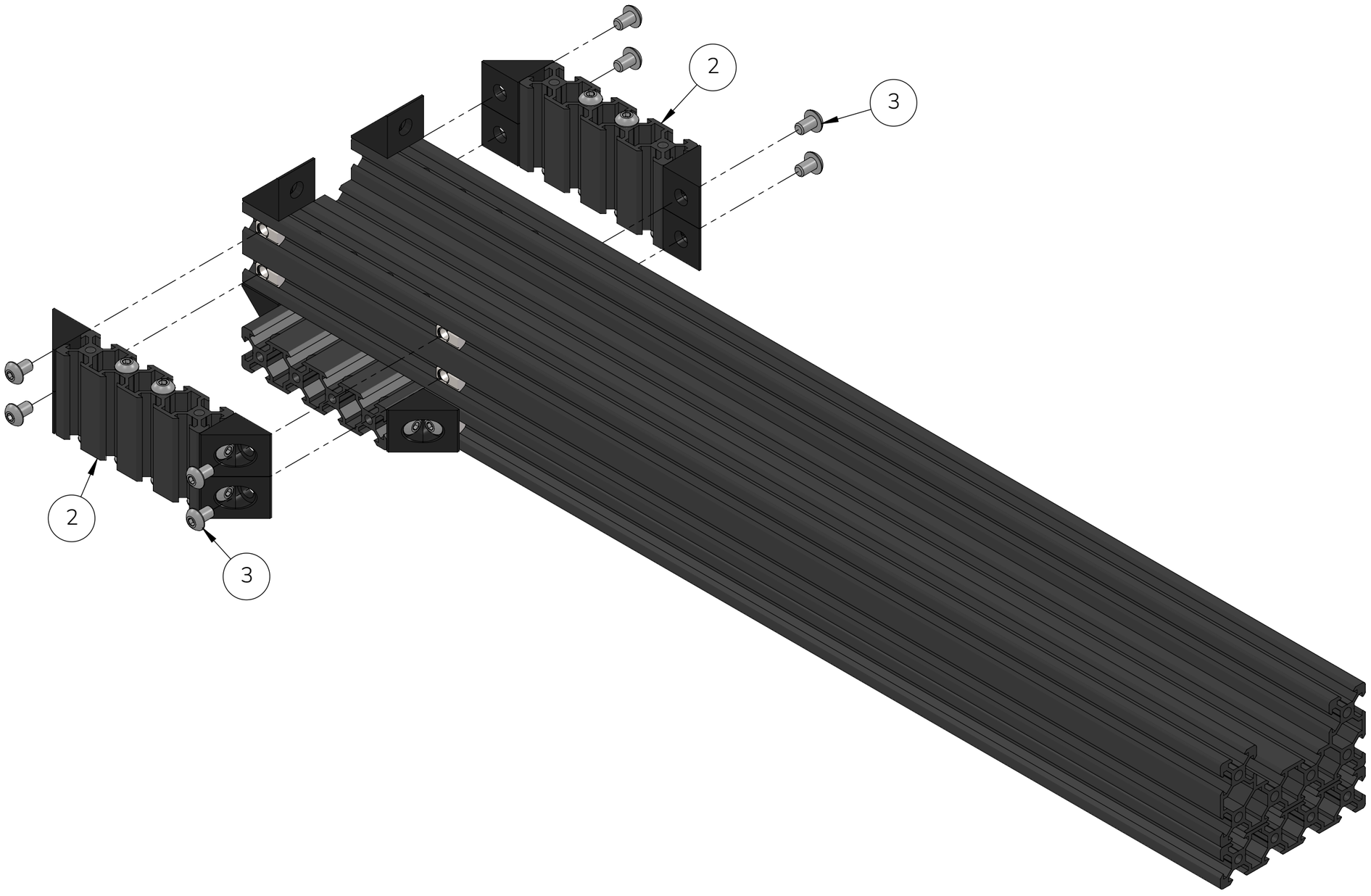
	PART NUMBER	QTY.
1	Y-AXIS-540mm-CBEAM-ASSEMBLY	1
2	Y-AXIS-40mm-EXTRUSION-ASSEMBLY	2
3	ISO-7380-M5X8-BHCS	8





Y AXIS FRAME

ASSEMBLY STEP 1A

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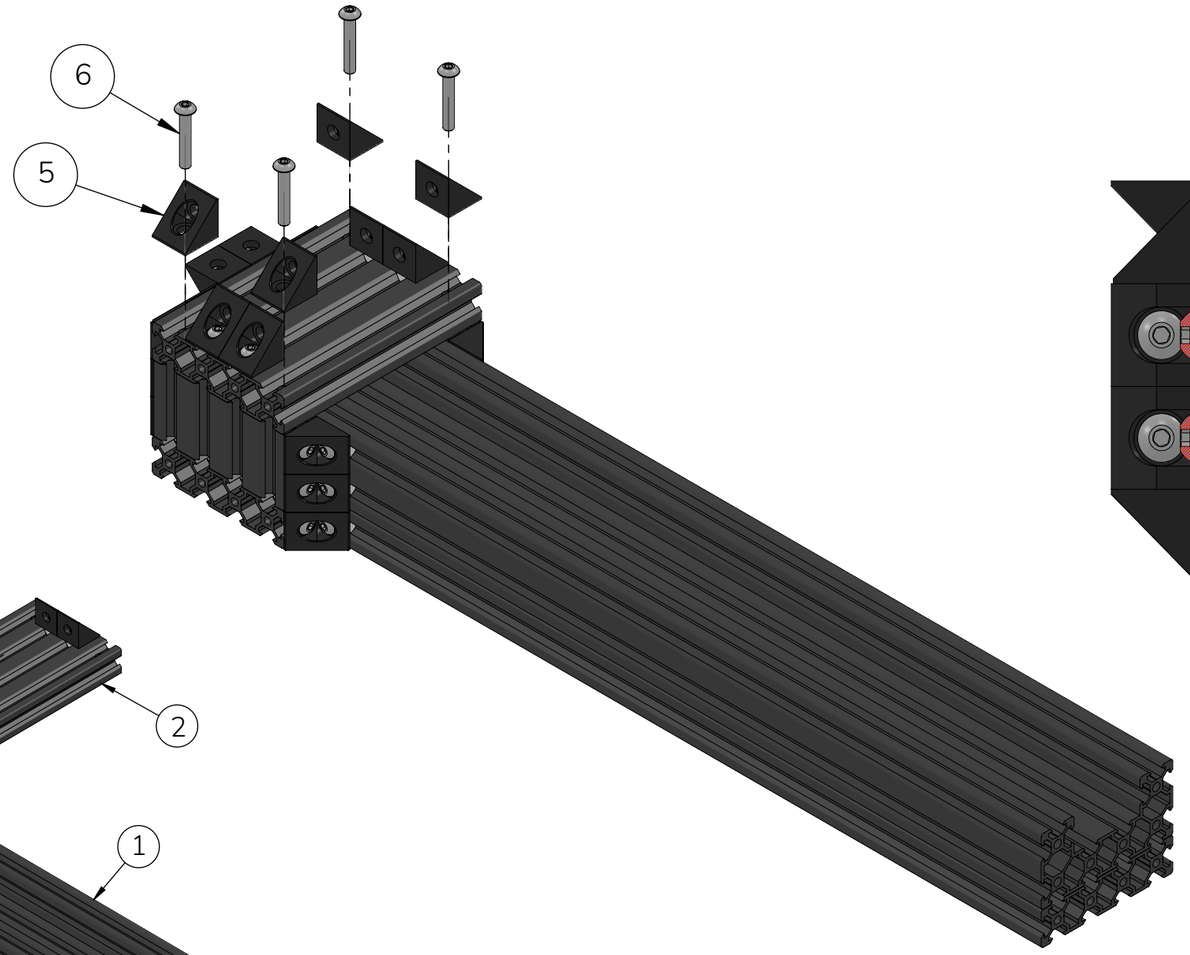
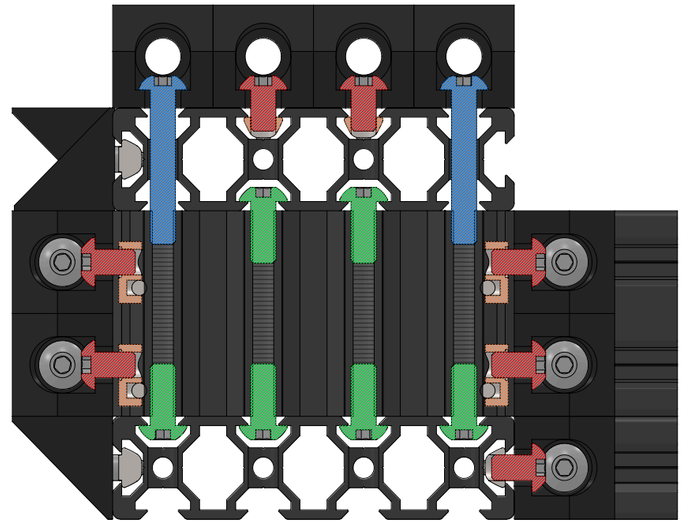
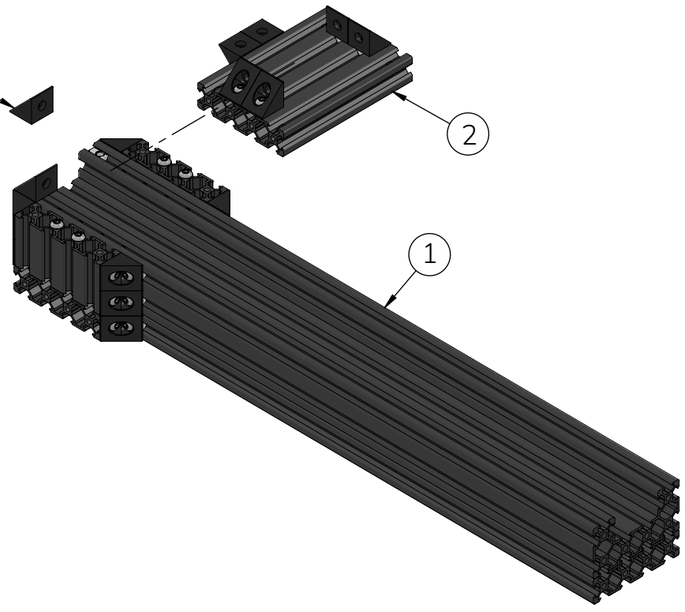
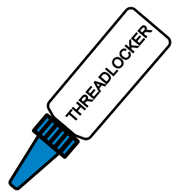






	PART NUMBER	QTY.
1	Y-AXIS-540mm-CBEAM-ASSEMBLY	1
2	Y-AXIS-120mm-EXTRUSION-A-ASSEMBLY	1

	PART NUMBER	QTY.
1	Y-AXIS-540mm-CBEAM-ASSEMBLY	1
5	ANGLE CORNER CONNECTOR	4
6	ISO-7380-M5X30-BHCS	4



Y AXIS FRAME

ASSEMBLY STEP 1B

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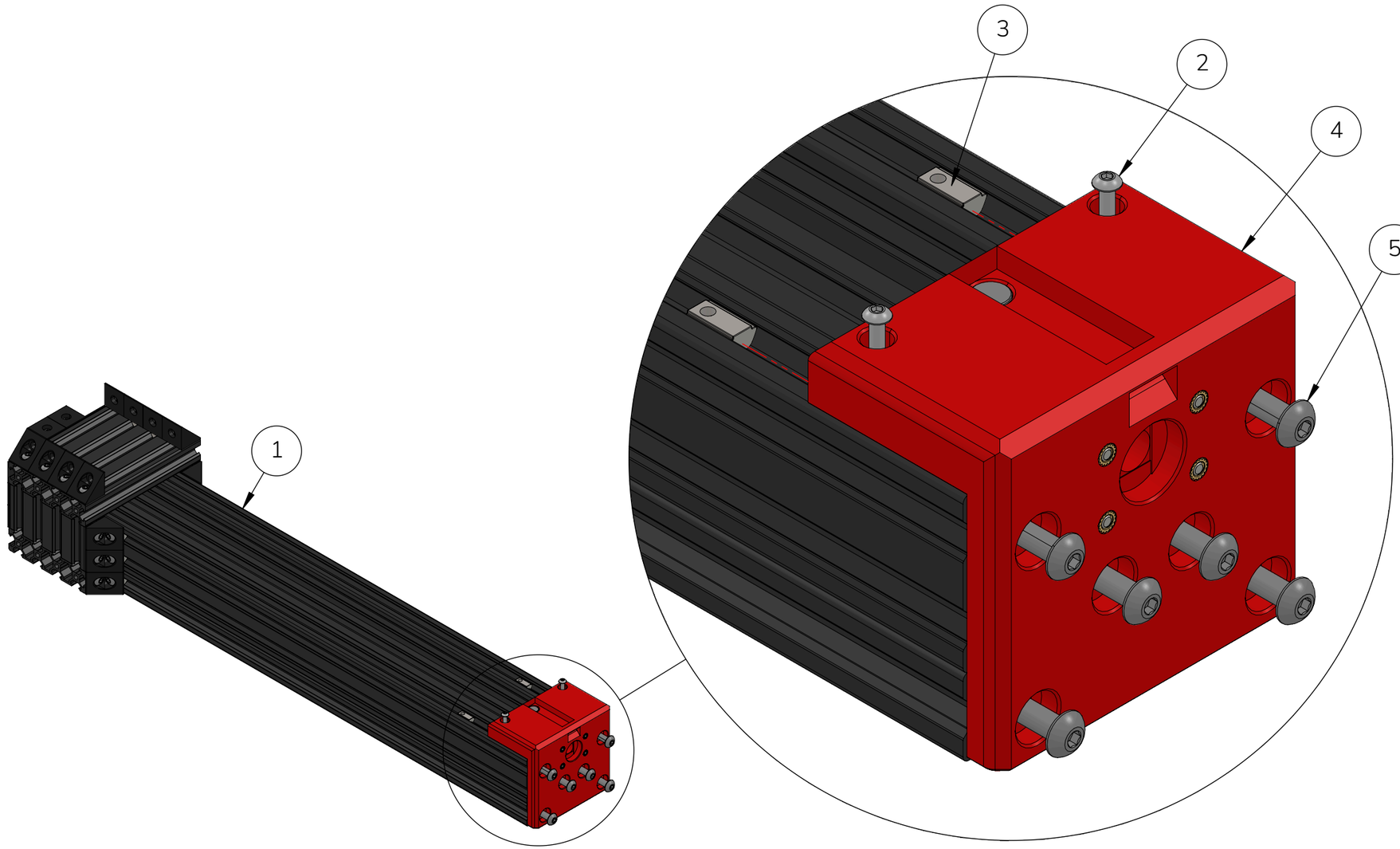
	PART NUMBER	QTY.
1	Y-AXIS-FRAME - STEP1	1
2	ISO-7380-M3X10-BHCS	2
3	M3 DROP IN NUT	2
4	Y-AXIS-BEARING-BLOCK-ASSY	1
5	ISO-7380-M5X12 BHCS	6



Y AXIS FRAME

ASSEMBLY STEP 2A

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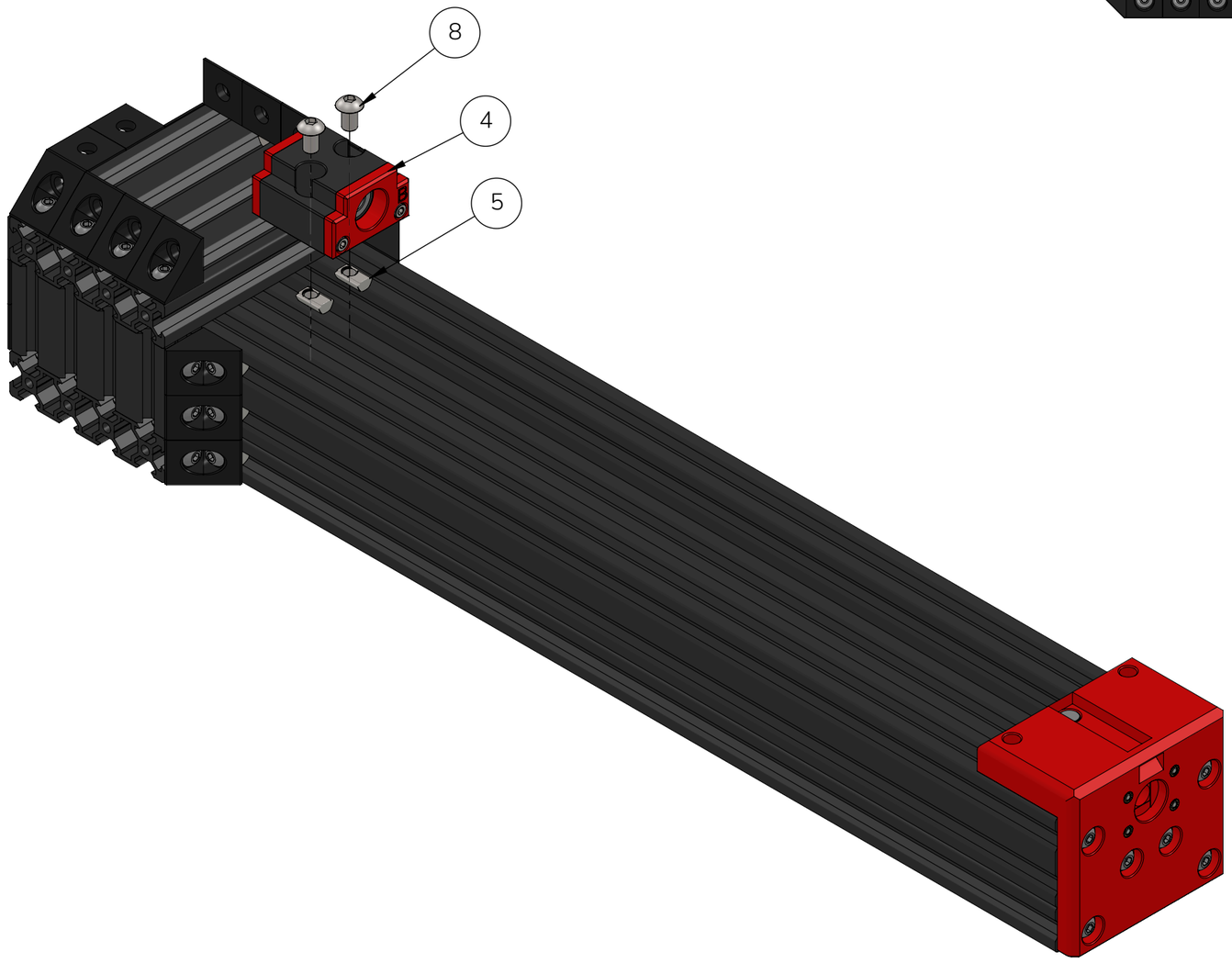

	PART NUMBER	QTY.
1	Y-AXIS-FRAME - STEP1	1
4	FLOATING-BEARING-BLOCK-ASSEMBLY	1
5	M5 DROP IN NUT	2
8	ISO-7380-M5X10 BHCS	2

Y AXIS FRAME

ASSEMBLY STEP 2B

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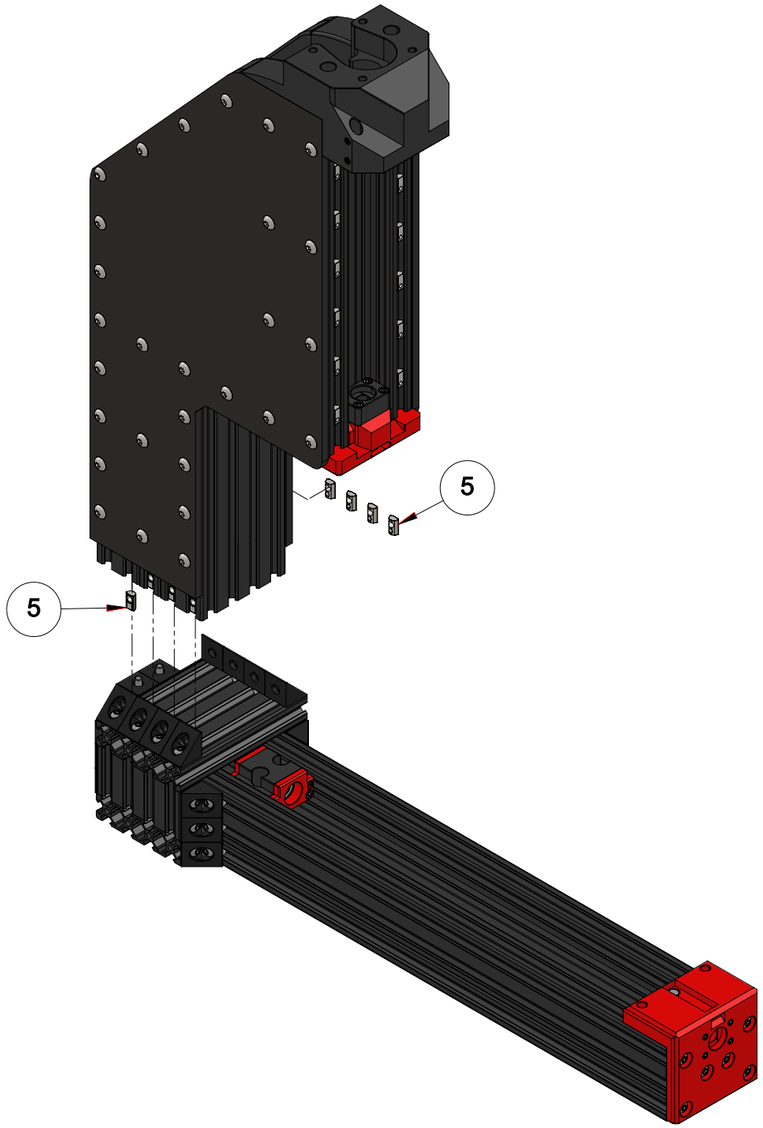
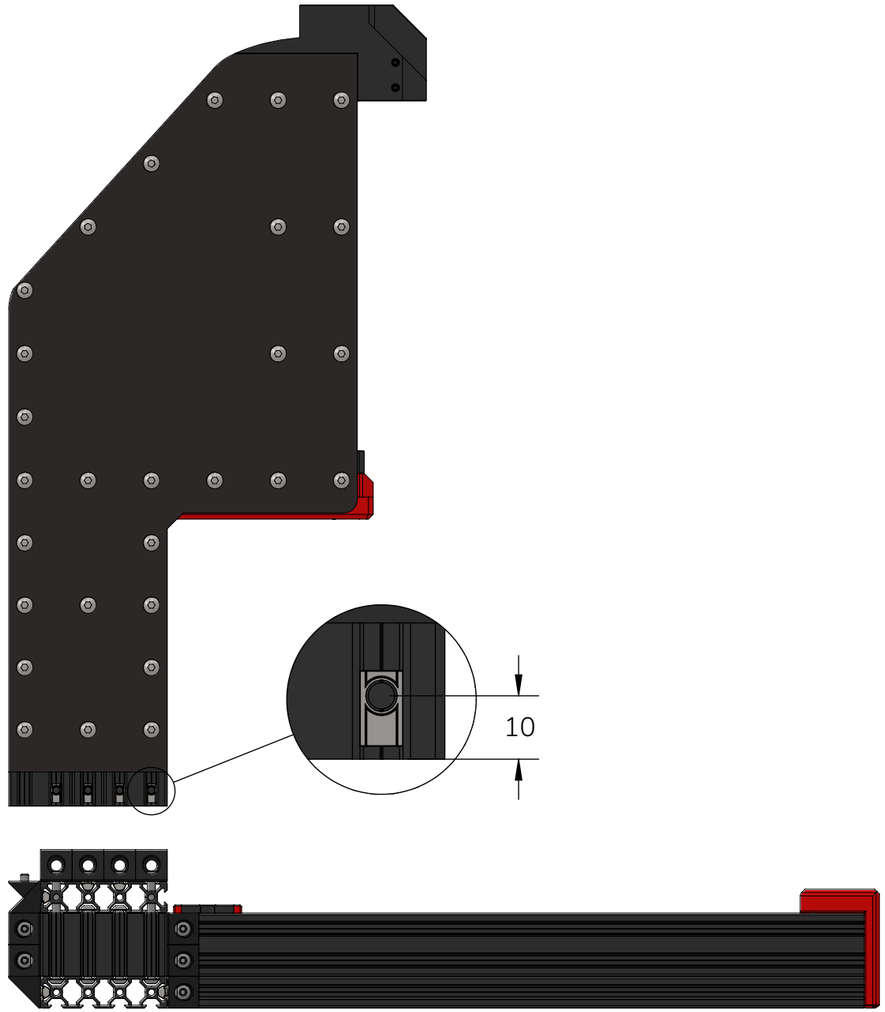
	PART NUMBER	QTY.
1	Y-AXIS-FRAME	1
2	Z-AXIS-FRAME	1
5	M5 DROP IN NUT	8





ZY ASSEMBLY

ASSEMBLY STEP 1A

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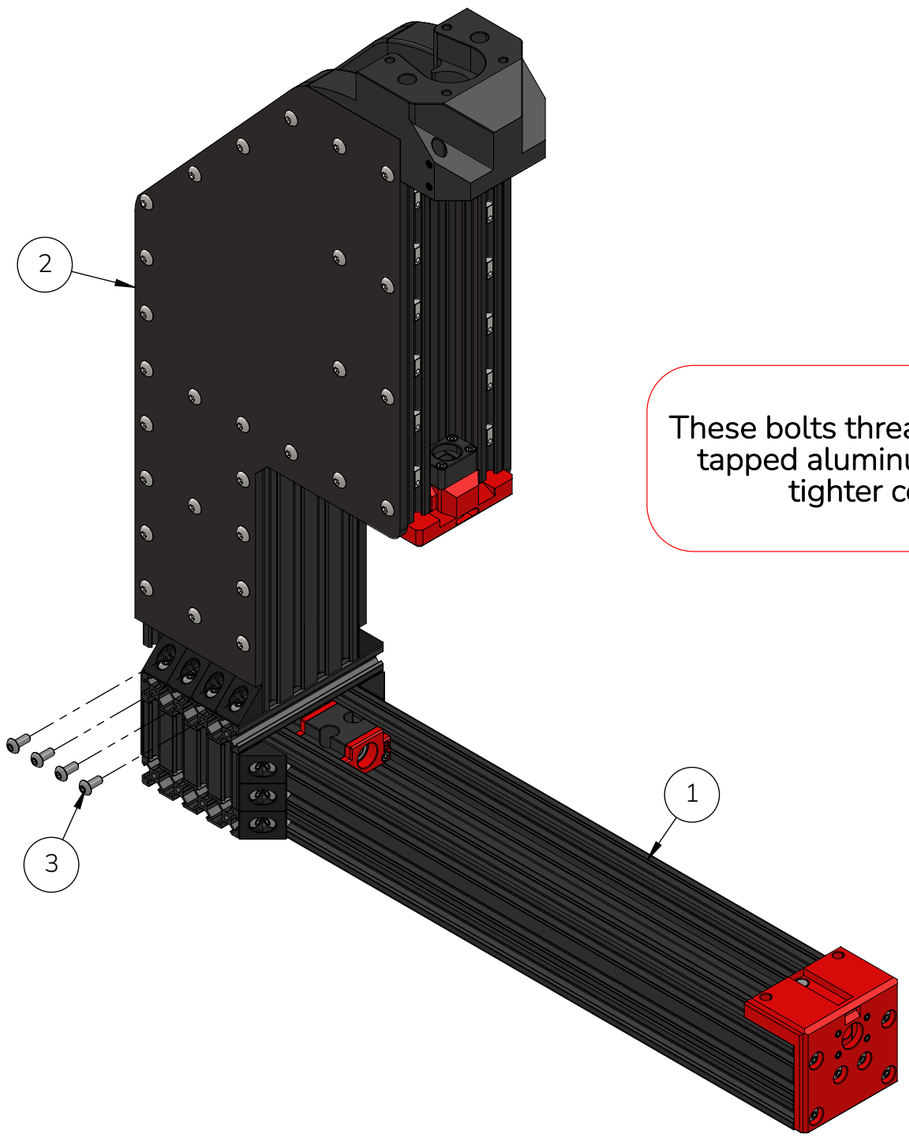
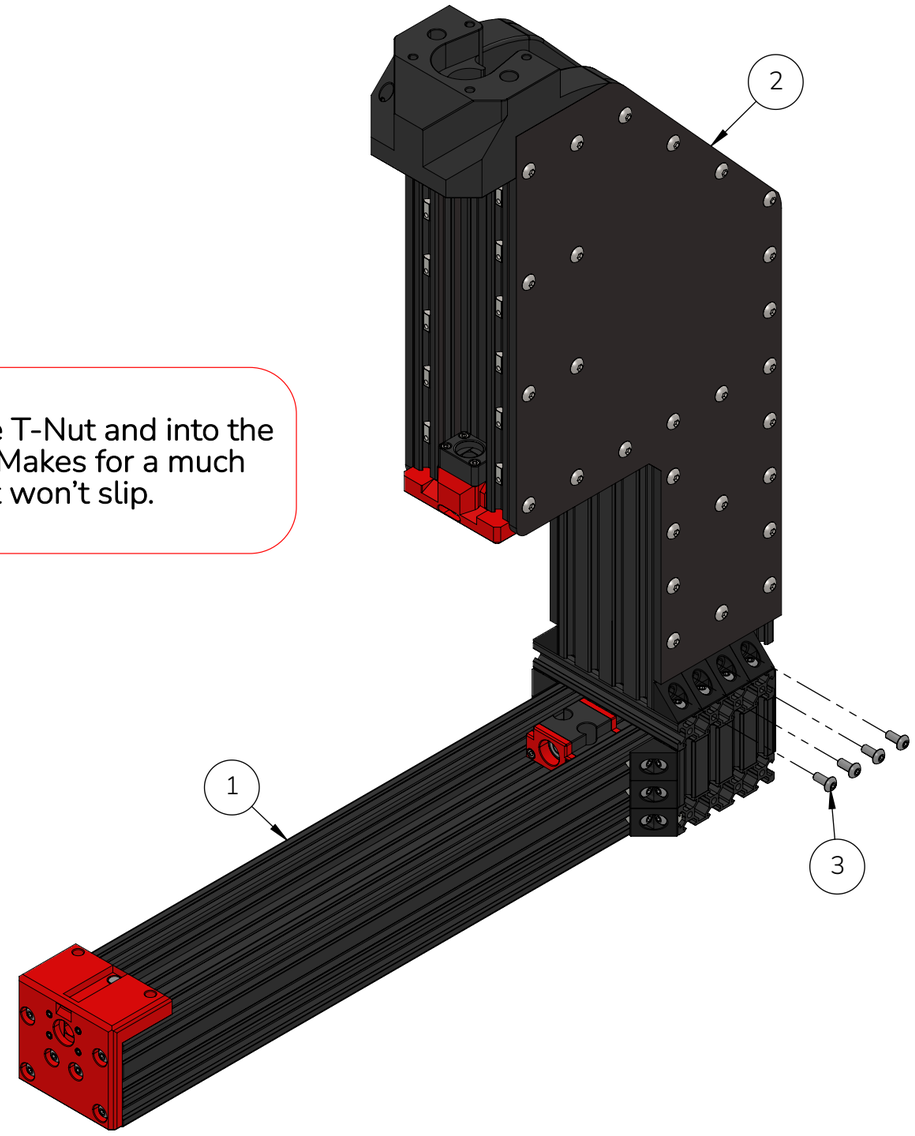
	PART NUMBER	QTY.
1	Y-AXIS-FRAME	1
2	Z-AXIS-FRAME	1
3	ISO-7380-M5X12 BHCS	8



These bolts thread through the T-Nut and into the tapped aluminum extrusion. Makes for a much tighter connection that won't slip.



ZY ASSEMBLY

ASSEMBLY STEP 1B

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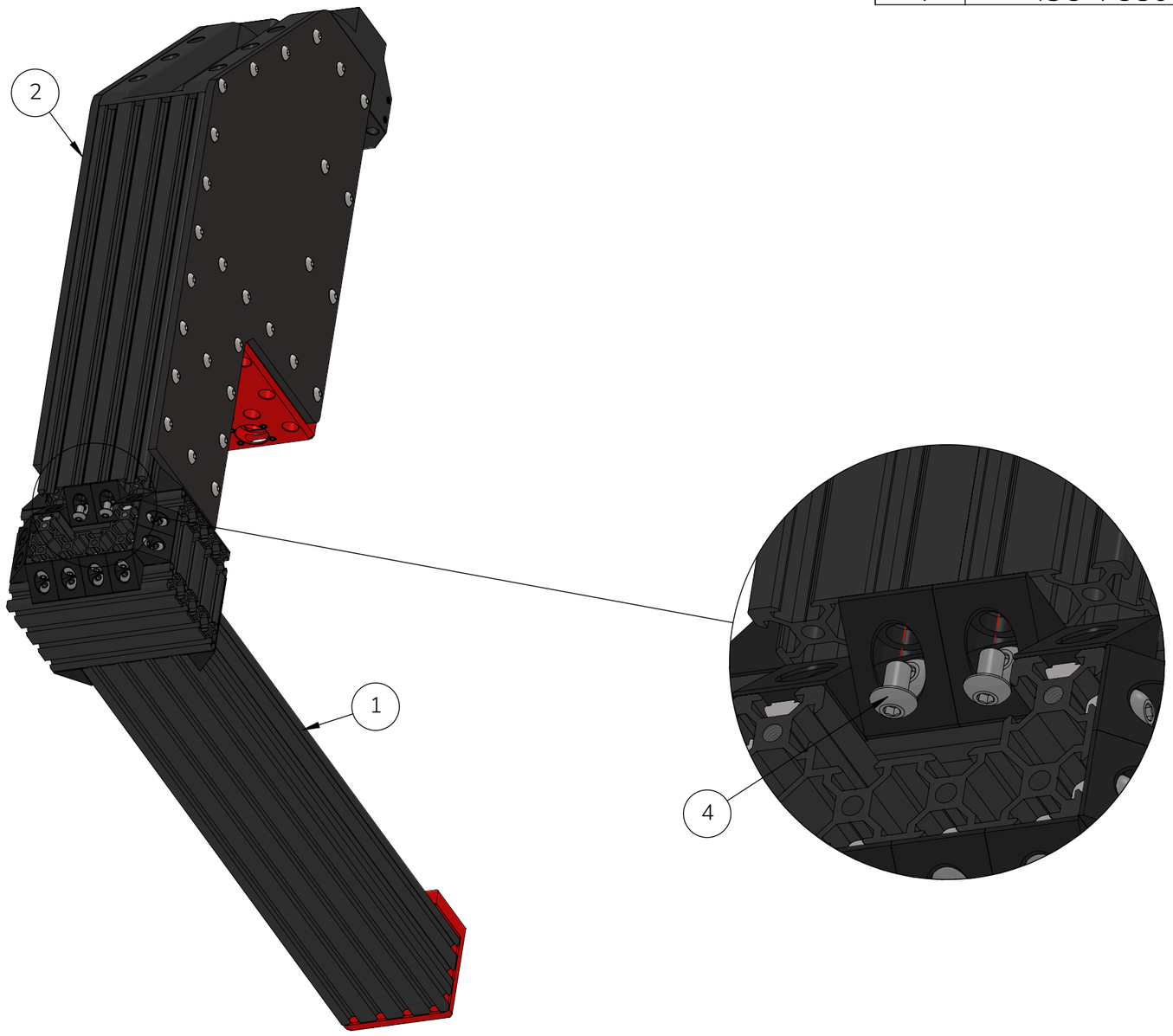
	PART NUMBER	QTY.
1	Y-AXIS-FRAME	1
2	Z-AXIS-FRAME	1
4	ISO-7380-M5X8-BHCS	2



ZY ASSEMBLY

ASSEMBLY STEP 1C

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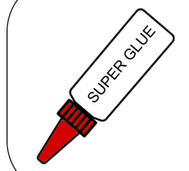
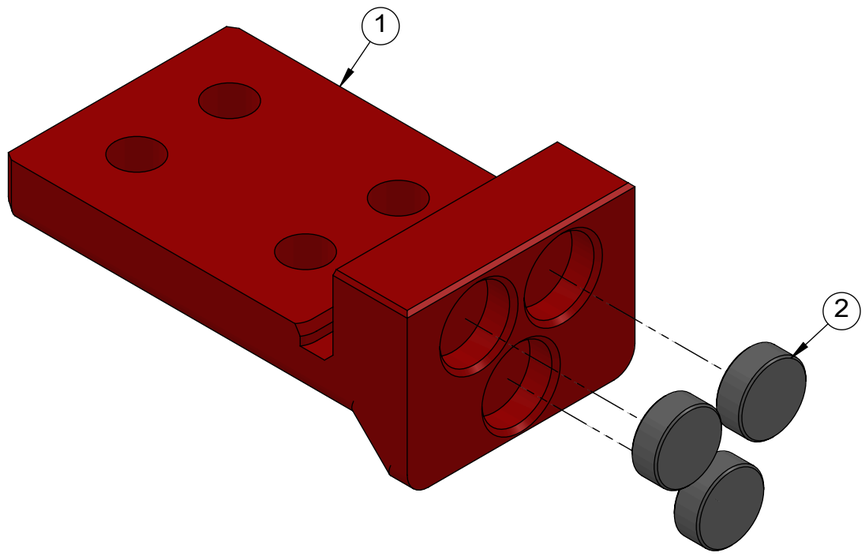
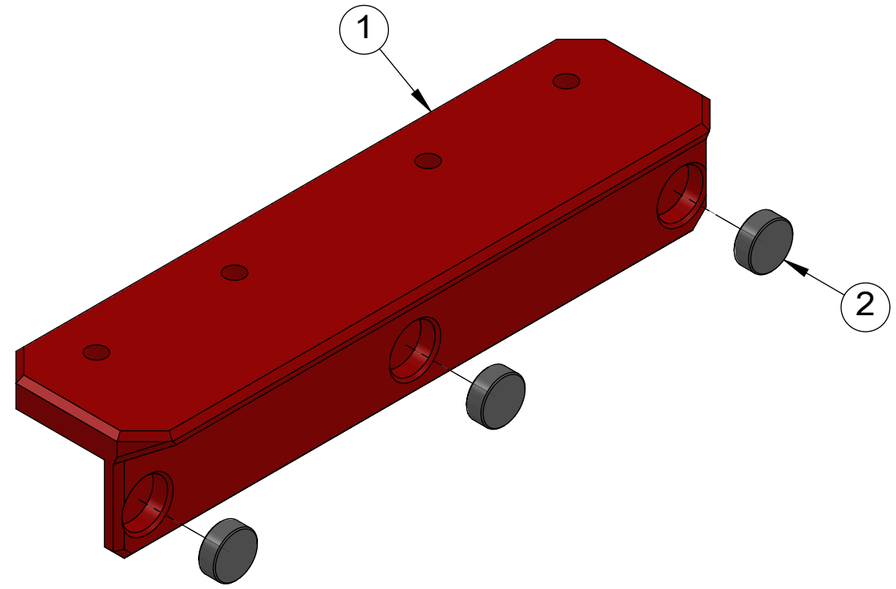


If the magnets aren't tight, you can apply a dab of super glue.

	PART NUMBER	QTY.
1	Y-AXIS-BELLOW-PLATE-MAG-MOUNT	1
2	8X3MM NEODYMIUM MAGNET	3





	PART NUMBER	QTY.
1	WAY-COVER-MAGNET-LOCK	1
2	8X3MM NEODYMIUM MAGNET	3

XY JOINER PLATE PRE-ASSEMBLY

WAY COVER / BELLOWS

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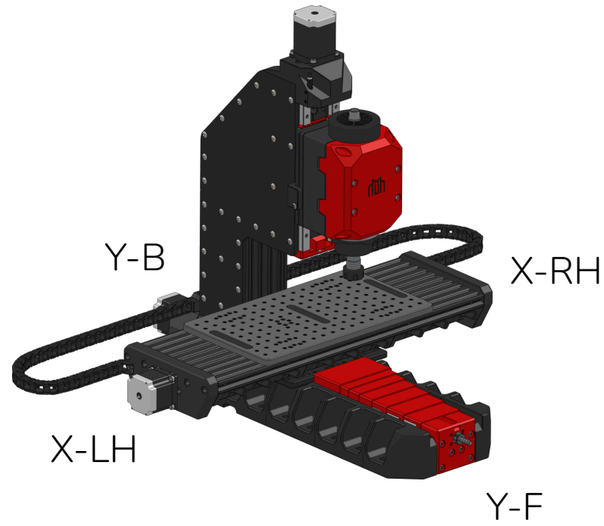
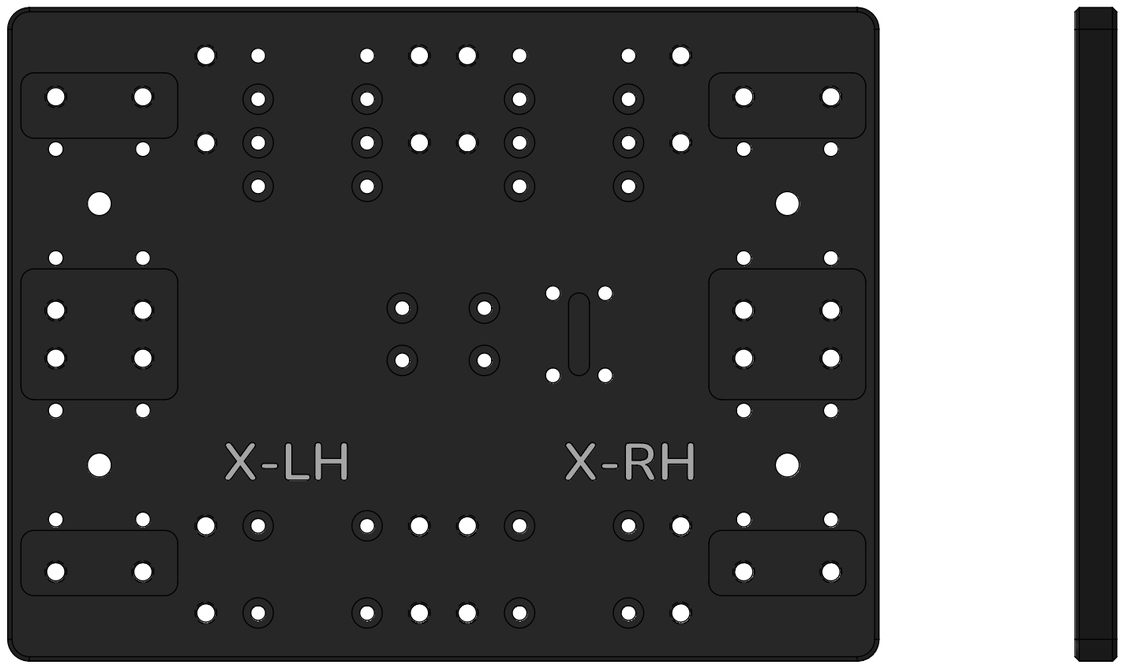
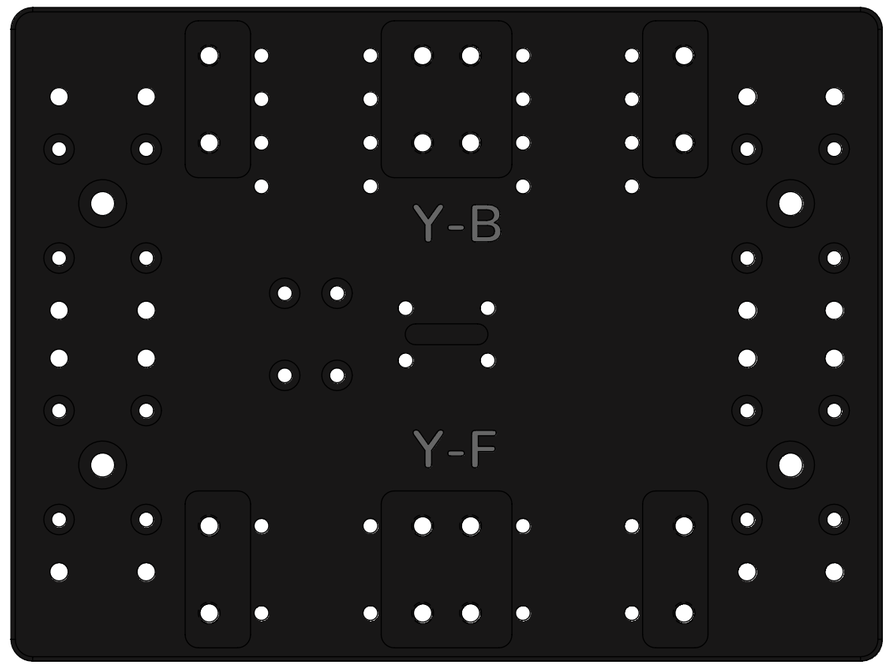


Take your mechanical marking tool and mark your XY Joiner Plate like shown below.





XY JOINER PLATE PRE-ASSEMBLY

REFERENCE

www.millennium-machines.com/

If you are building Milo, follow the instructions on pages 46-68 and then proceed to page 95.

If you are building Miley, follow the instructions on pages 69-94.

	PART NUMBER	QTY.
1	MILO-X-AXIS-EXTRUSION-ASSEMBLY	1
	MILO X AXIS C-BEAM PRE-ASSEMBLY	1
	C BEAM 550mm	1
	M5 DROP IN NUT	26
	V-SLOT-2080X120MM	2
	M5 DROP IN NUT	18
	IS0-7380-M5X10 BHCS	8
2	MILO-X-AXIS-MOTOR-MOUNT-SUBASSEMBLY	1
	X AXIS MOTOR MOUNT	1
	M3X3MM HEATSET INSERT	4
	M5X5.8mm HEATSET INSERT	4
3	MILO-X-AXIS-BEARING-MOUNT-SUBASSEMBLY	1
	X AXIS BEARING BLOCK	1
	M3X3MM HEATSET INSERT	7
	X-BEARING-BLOCK-ASSEMBLY	1
	X-BEARING-BLOCK-INSERT	1
	608ZZ BALL BEARING	1
	ISO-4762-M3X40-SHCS	4
4	MILO TABLE SUPPORT B	2
5	MILO TABLE SUPPORT A	4
6	ISO-7380-M5X12 BHCS	16
7	IS0-7380-M5X10 BHCS	36
8	M3 DROP IN NUT	28

MILO - X AXIS FRAME

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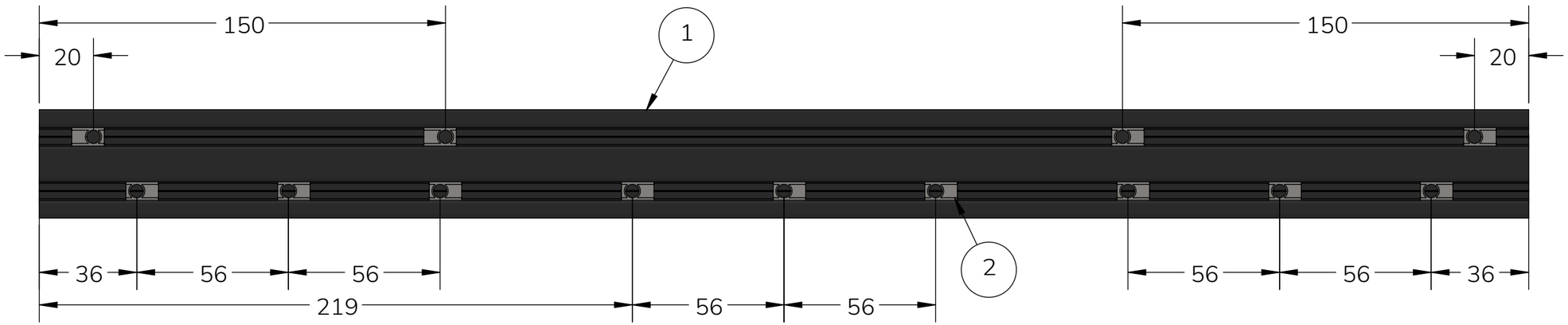
See Manufacturing Drawings for Hole Locations



	PART NUMBER	QTY.
1	C BEAM 550mm	1
2	M5 DROP IN NUT	26

MILO X AXIS FRAME

C BEAM PRE-ASSEMBLY

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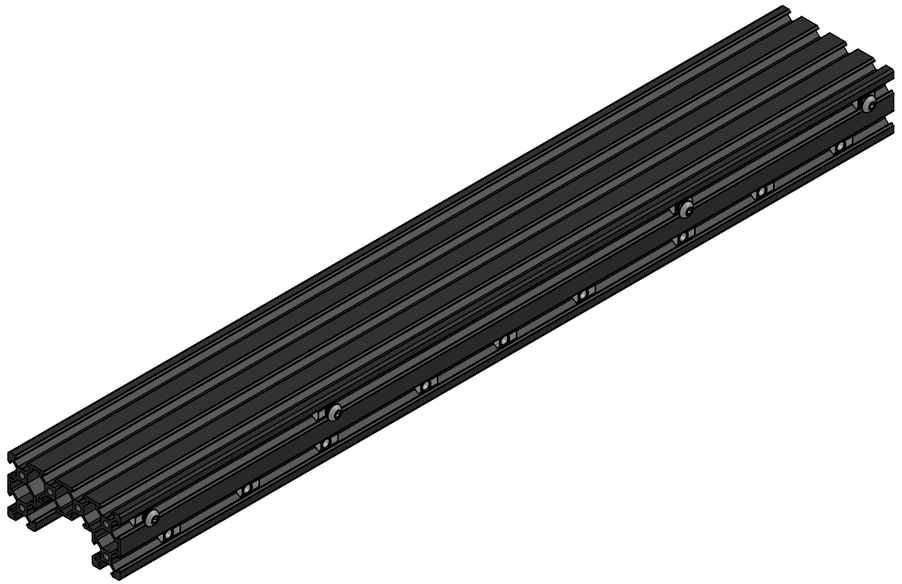
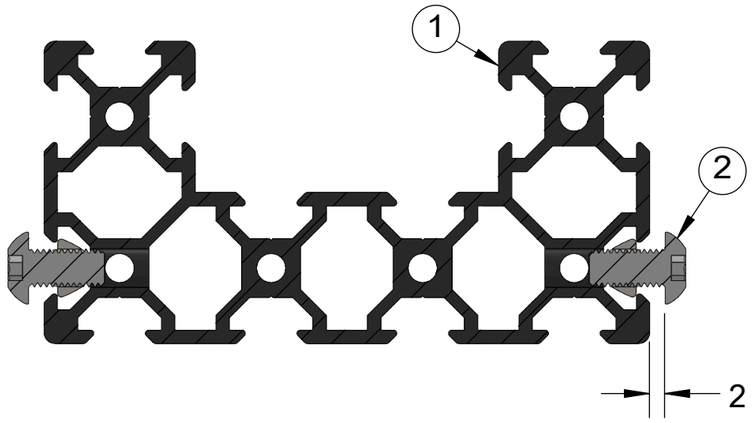


	PART NUMBER	QTY.
1	MILO X AXIS C-BEAM PRE-ASSEMBLY	1
2	IS0-7380-M5X10 BHCS	8



Thread the bolt through the nut and into the hole slightly to hold the bolt in place while sliding the 2080 extrusions into place

MILO X AXIS FRAME

ASSEMBLY STEP 1

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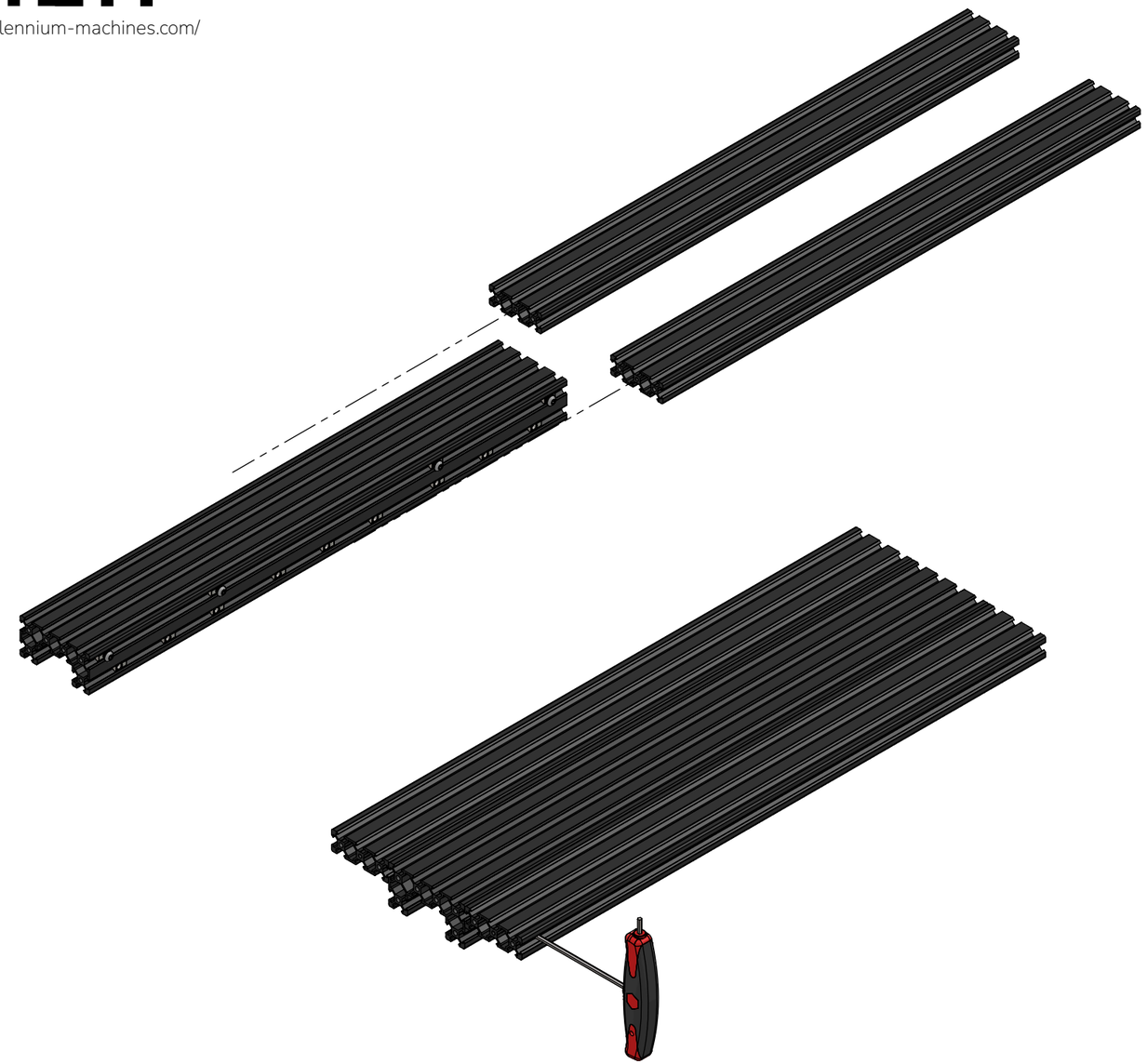
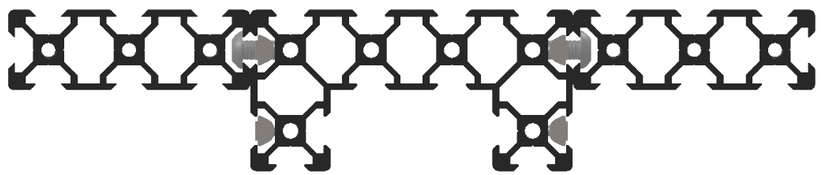


	PART NUMBER	QTY.
1	MILO X AXIS C-BEAM PRE-ASSEMBLY	1
2	V-SLOT-2080X120MM	2
3	IS0-7380-M5X10 BHCS	8



MILO X AXIS FRAME

ASSEMBLY STEP 2

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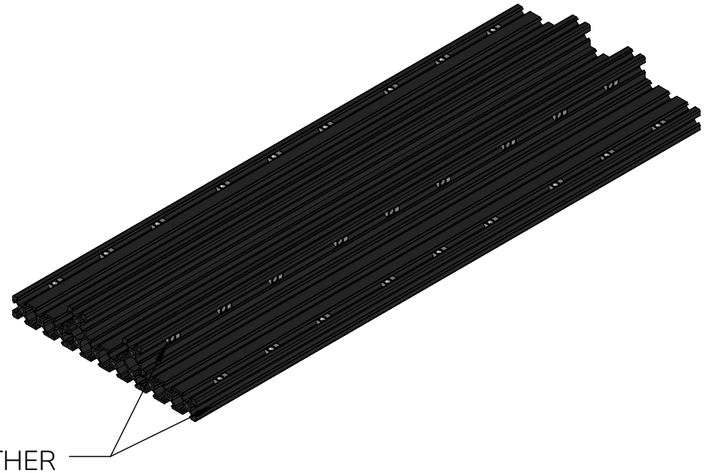
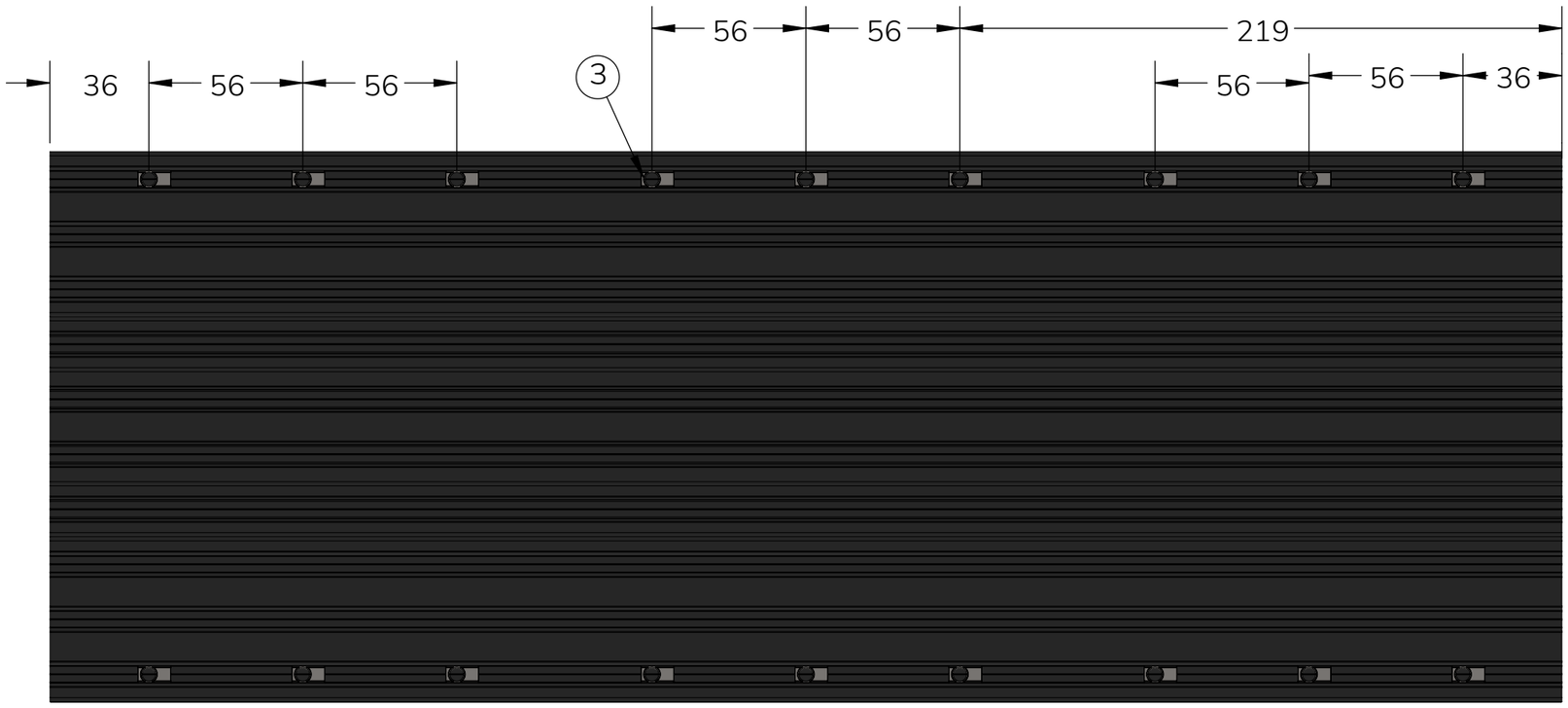
	PART NUMBER	QTY.
3	M5 DROP IN NUT	18





MILO X AXIS FRAME

ASSEMBLY STEP 3

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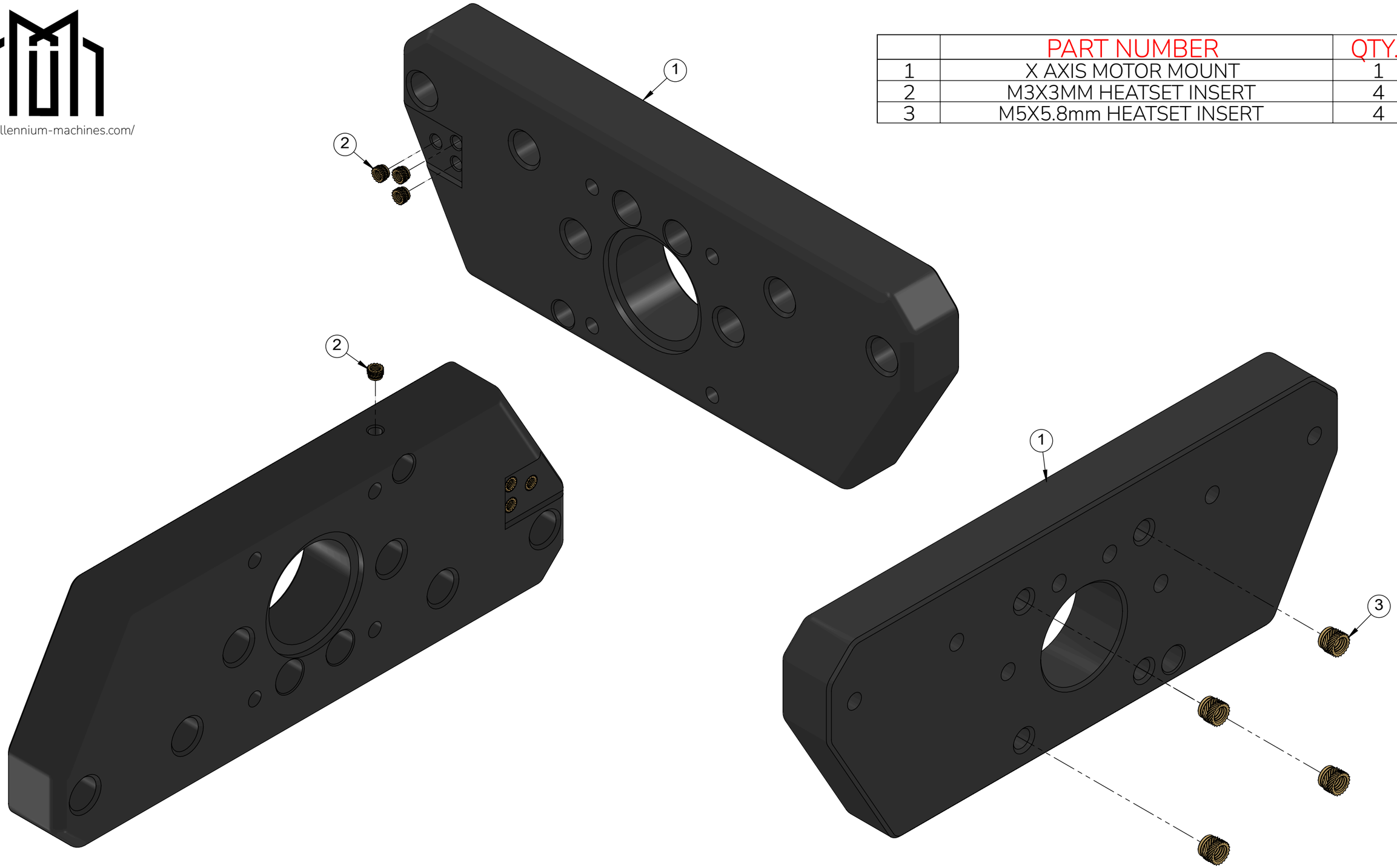
	PART NUMBER	QTY.
1	X AXIS MOTOR MOUNT	1
2	M3X3MM HEATSET INSERT	4
3	M5X5.8mm HEATSET INSERT	4



MILO X AXIS FRAME

MOTOR MOUNT PRE-ASSEMBLY

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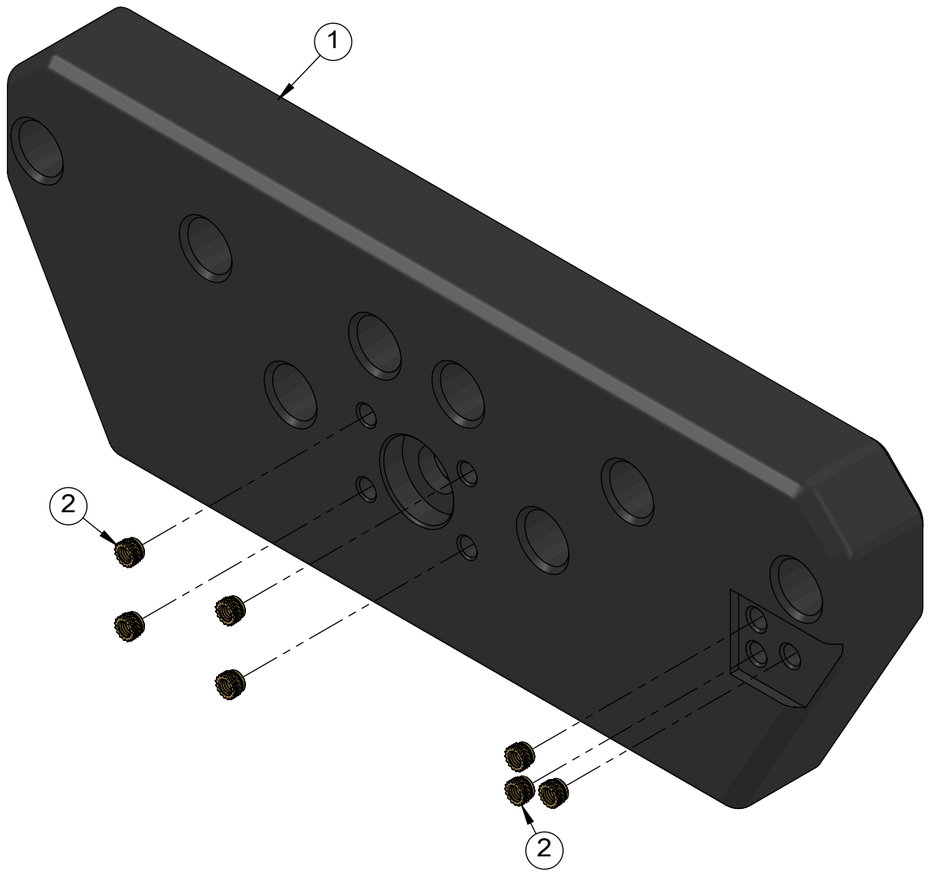
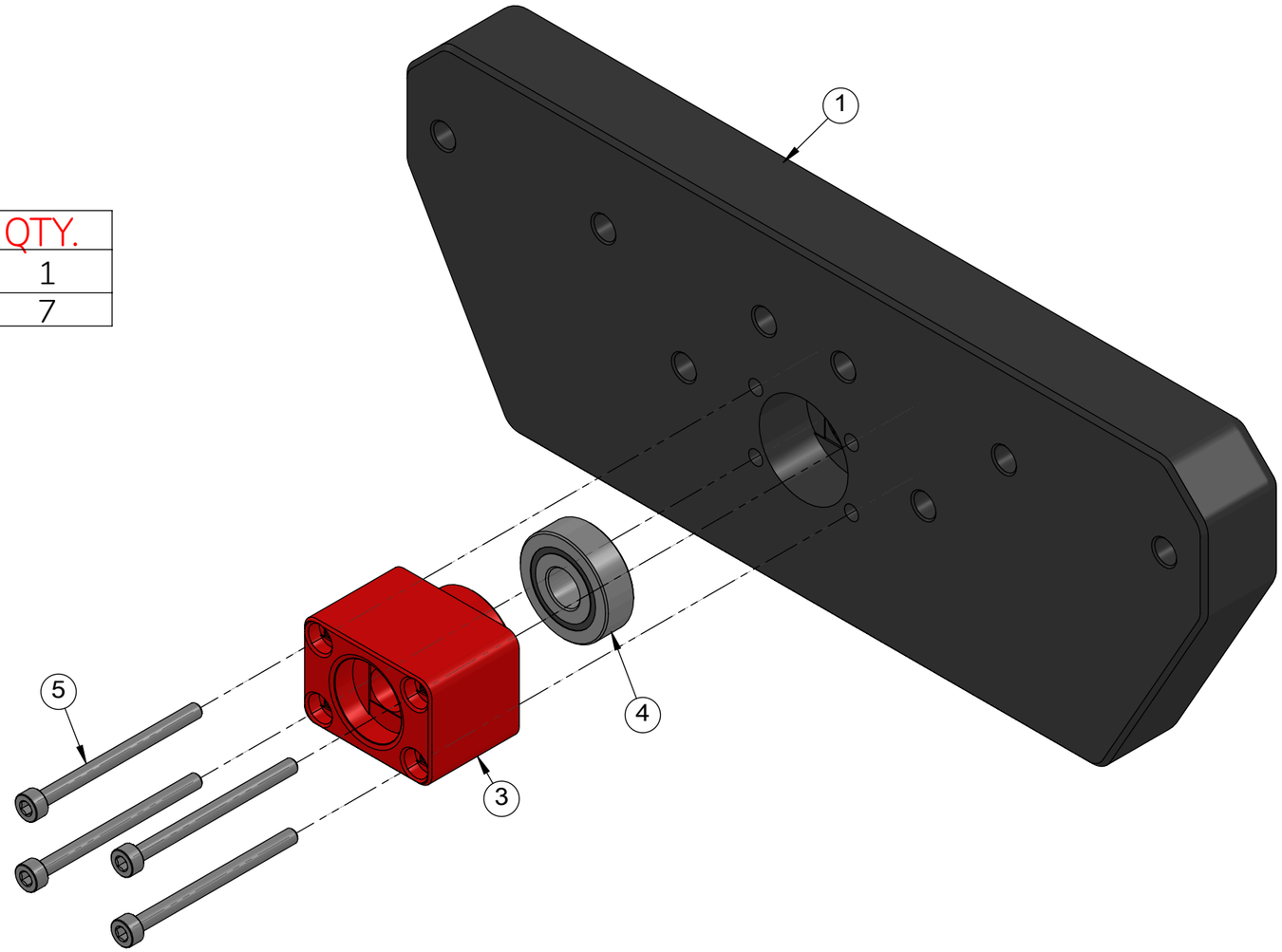
	PART NUMBER	QTY.
1	X AXIS BEARING BLOCK	1
2	M3X3MM HEATSET INSERT	7

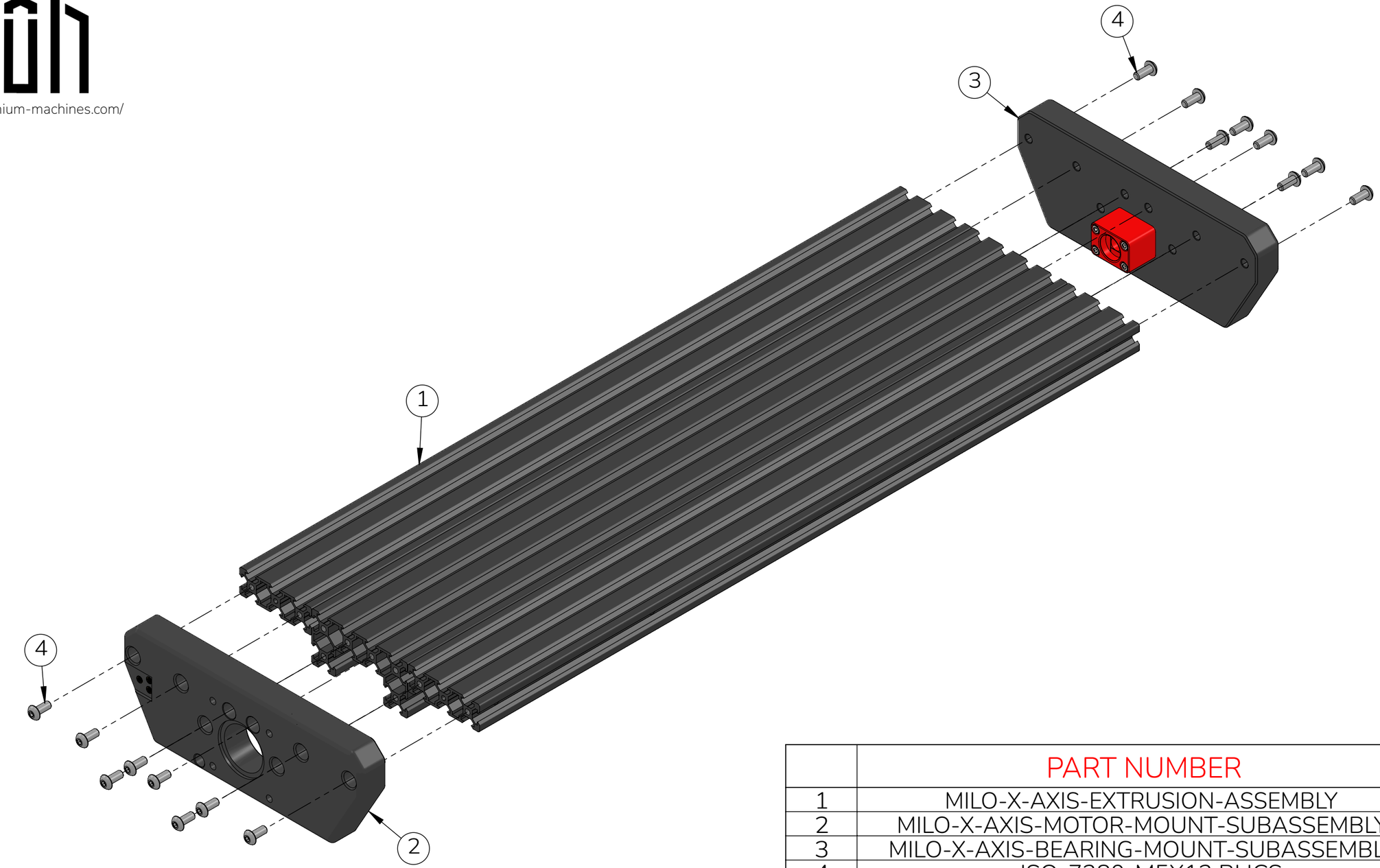




	PART NUMBER	QTY.
1	X AXIS BEARING BLOCK	1
3	X-BEARING-BLOCK-INSERT	1
4	608ZZ BALL BEARING	1
5	ISO-4762-M3X40-SHCS	4

MILO X AXIS FRAME

BEARING BLOCK PRE-ASSEMBLY



	PART NUMBER	QTY.
1	MILO-X-AXIS-EXTRUSION-ASSEMBLY	1
2	MILO-X-AXIS-MOTOR-MOUNT-SUBASSEMBLY	1
3	MILO-X-AXIS-BEARING-MOUNT-SUBASSEMBLY	1
4	ISO-7380-M5X12 BHCS	16

MILO X AXIS FRAME

ASSEMBLY STEP 1

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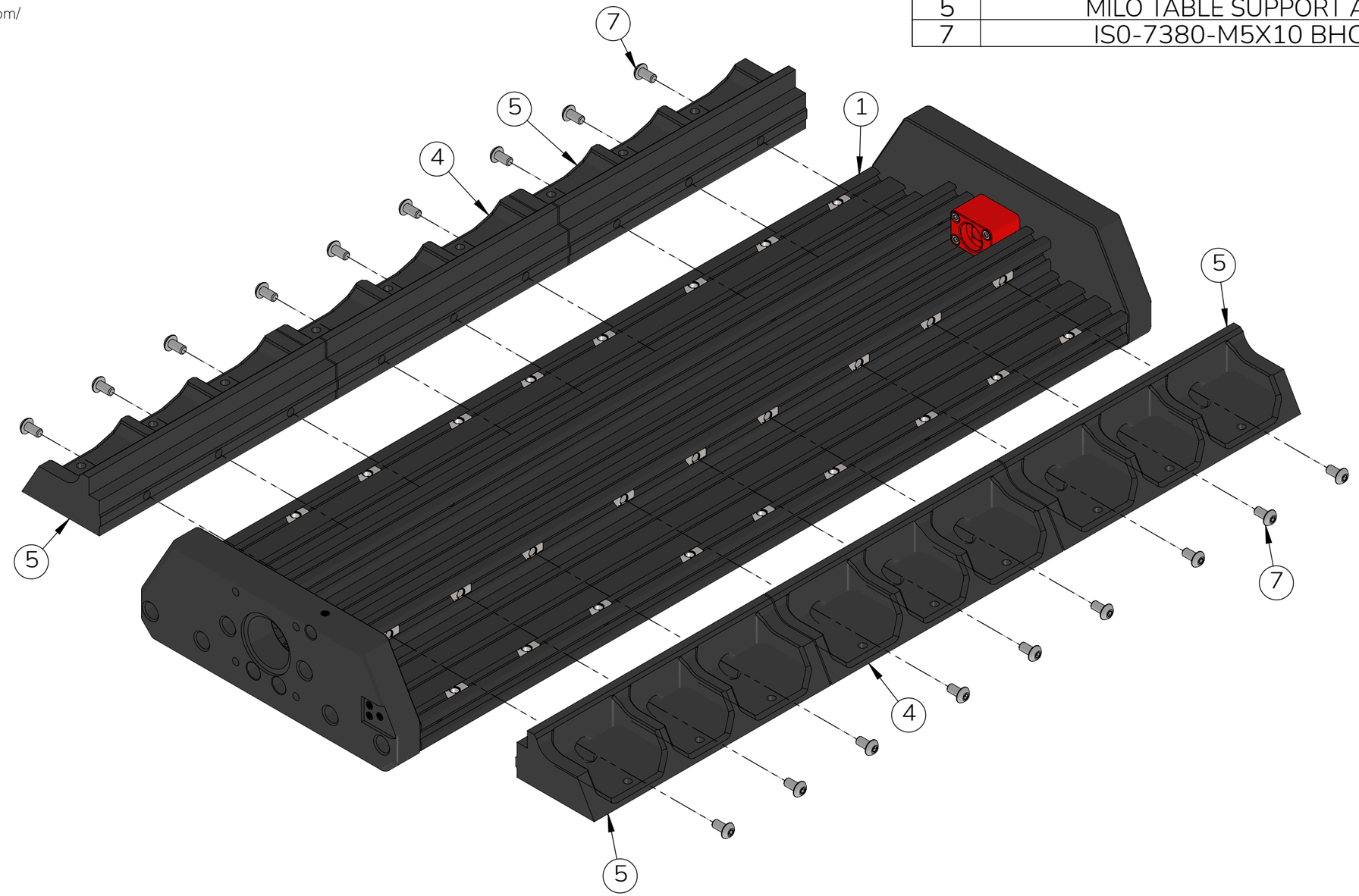
	PART NUMBER	QTY.
1	MILO-X-AXIS-EXTRUSION-ASSEMBLY	1
4	MILO TABLE SUPPORT B	2
5	MILO TABLE SUPPORT A	4
7	IS0-7380-M5X10 BHCS	18



MILO X AXIS FRAME

ASSEMBLY STEP 2

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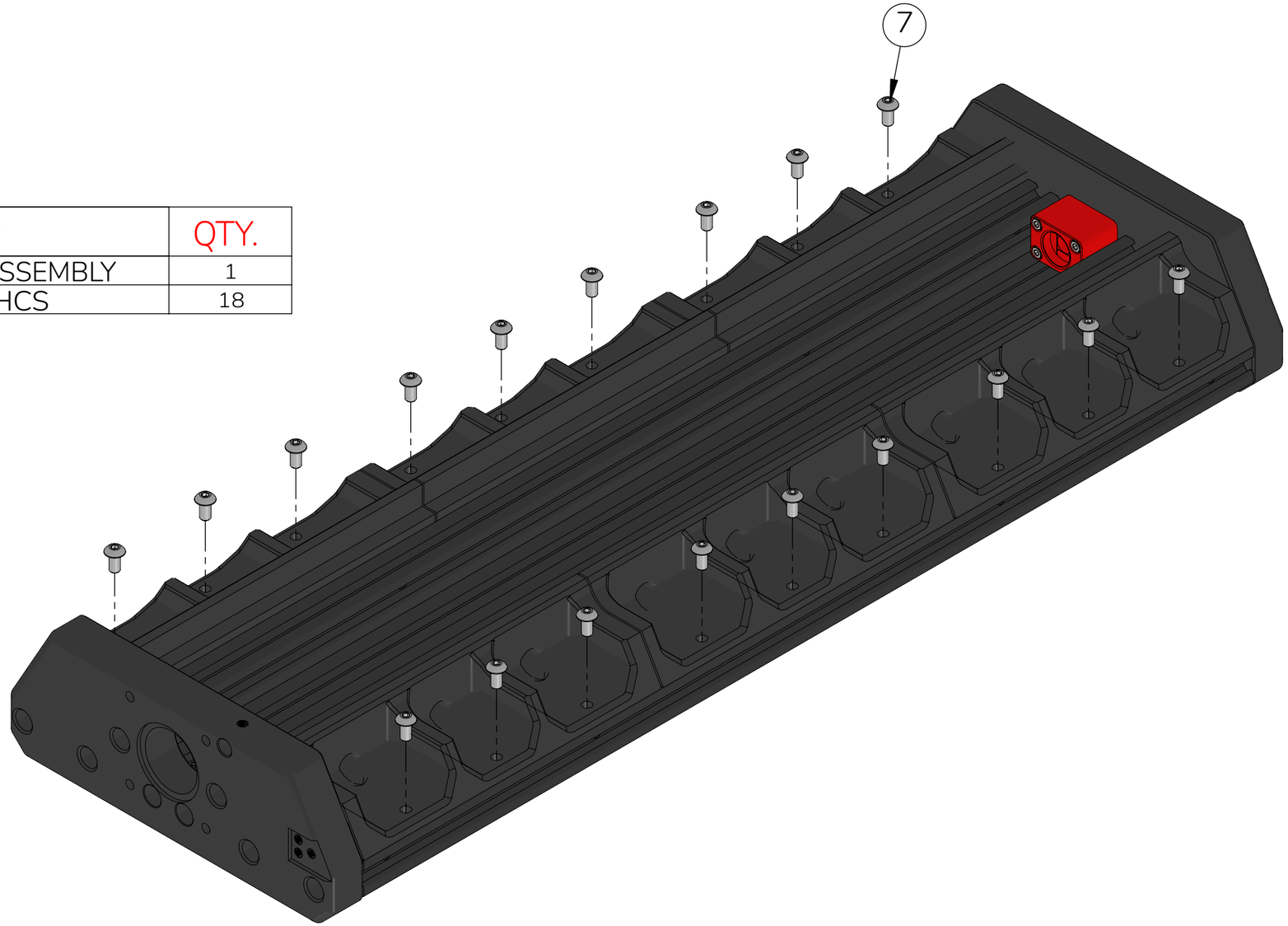
	PART NUMBER	QTY.
1	MILO-X-AXIS-EXTRUSION-ASSEMBLY	1
7	IS0-7380-M5X10 BHCS	18



MILO X AXIS FRAME

ASSEMBLY STEP 3

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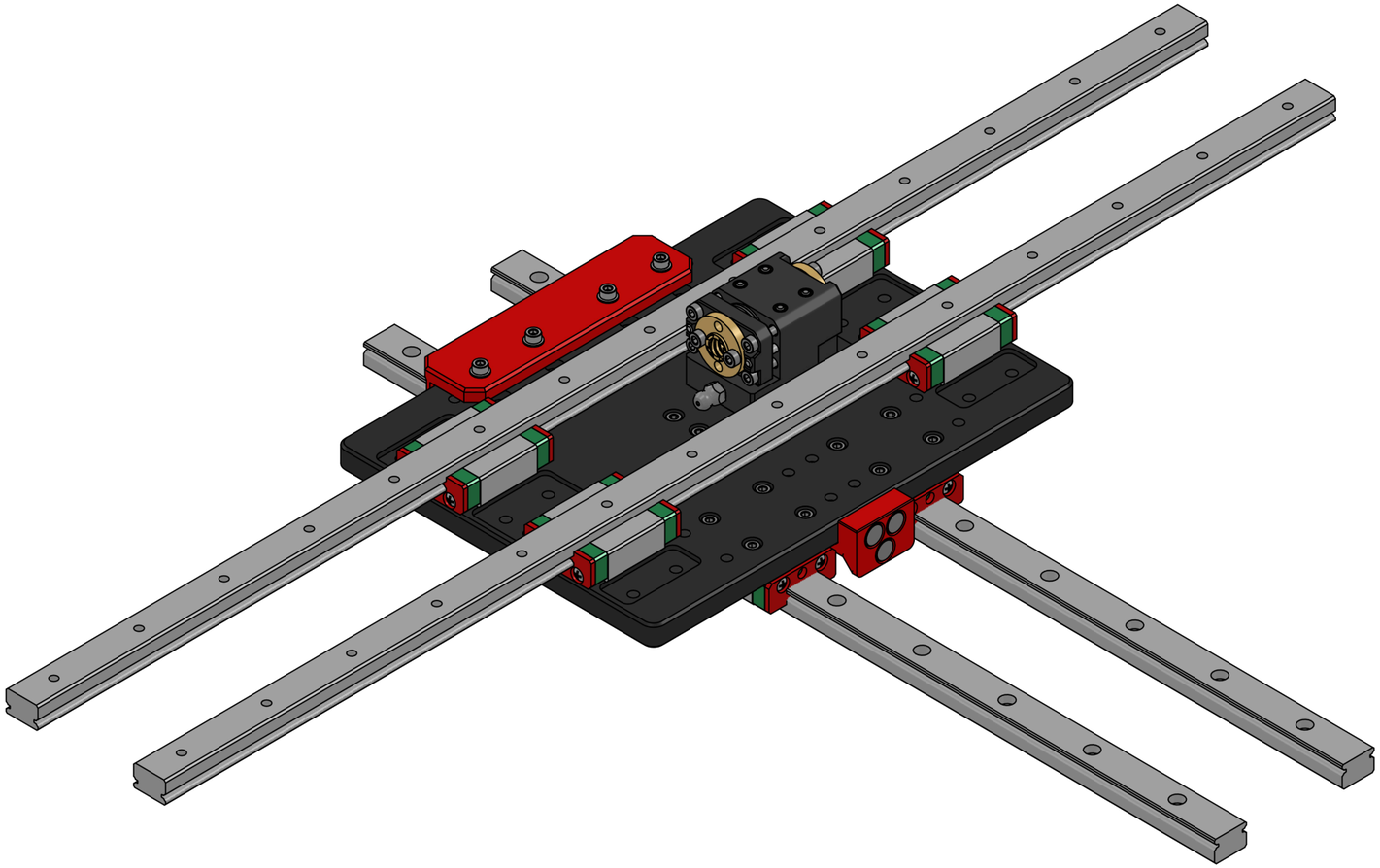

	PART NUMBER	QTY.
1	X-Y JOINER-PLATE	1
2	AB-BLOCK-MAIN-ASSEMBLY	2
	ThrED-AB-BLOCK-ASSY	1
	ThrED-AB-BLOCK	1
	M3X3MM HEATSET INSERT	10
	M5X5.8mm HEATSET INSERT	1
	TR8X8 BRASS LEADSCREW NUT	1
	ISO-4762-M3X30-SHCS	2
	ISO-2338 M3-20mm PIN	2
	DIN-71412 GREASE-NIPPLE	1
	ThrED-COMPRESSION-PLATE-ASSEMBLY	1
	ThrED-COMPRESSION-PLATE	1
	M3X3MM HEATSET INSERT	2
	TR8X8 BRASS LEADSCREW NUT	1
	ISO-4762-M3X10-SHCS	2
	ISO-4762-M3X40-SHCS	4
	DIN-2093 10.2mm ID - 20mm OD BELLEVILLE WASHER	2
3	ISO-2338-M5-15mm-PIN	2
4	ISO-4762-M3X40-SHCS	8
5	X-AXIS-LINEAR-RAIL-ASSEMBLY	2
	MGN15 RAIL 550mm	1
	MGN15C RAIL CARRIAGE	2
6	Y-AXIS-LINEAR-RAIL-ASSEMBLY	2
	MGN15-RAIL-400mm	1
	MGN15C RAIL CARRIAGE	2
7	WAY-COVER-MAGNET-LOCK-ASSEMBLY	1
	WAY-COVER-MAGNET-LOCK	1
	8X3MM NEODYMIUM MAGNET	3
8	BELLOW PLATE MAG MOUNT ASSEMBLY	1
	Y-AXIS-BELLOW-PLATE-MAG-MOUNT	1
	8X3MM NEODYMIUM MAGNET	3
9	ISO-10642-M5X12MM-FHCS	4
10	ISO-4762-M3X10-SHCS	28
11	ISO-4762-M3X20-SHCS	4
12	ISO-7089 M3 WASHER	4

MILO XY JOINER PLATE

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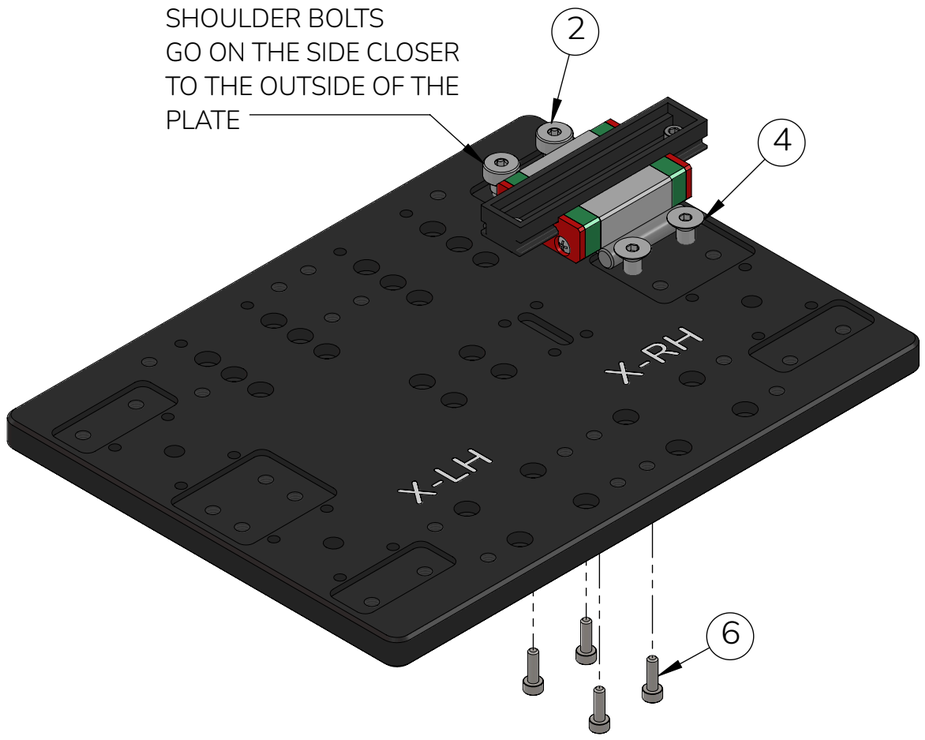
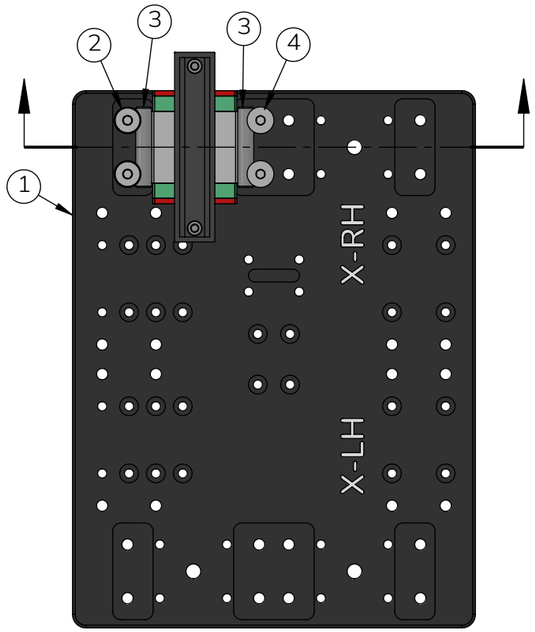
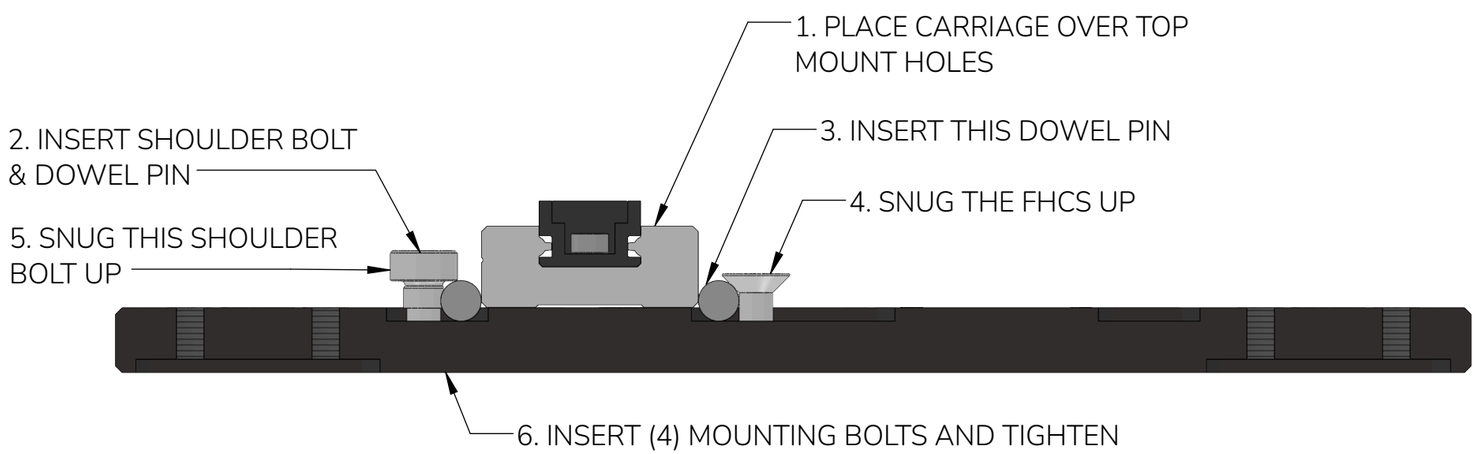
	PART NUMBER	QTY.
1	X-Y JOINER-PLATE	1
2	ISO-7379-M5X6MM-THREAD-M6X4MM-SHOULDER	2
3	ISO-2338-M6-30mm-PIN	2
4	ISO-10642-M5X12MM-FHCS	2
5	CARRIAGE RETAINER & BEARING ASSEMBLY	1
6	ISO-4762-M3X10-SHCS	4



Reuse the FHCS bolts & shoulder bolts for each carriage. Be sure to remove the bolts and dowel pins completely after installing all of the carriages.





MILO XY JOINER PLATE

ASSEMBLY STEP 1

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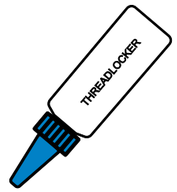
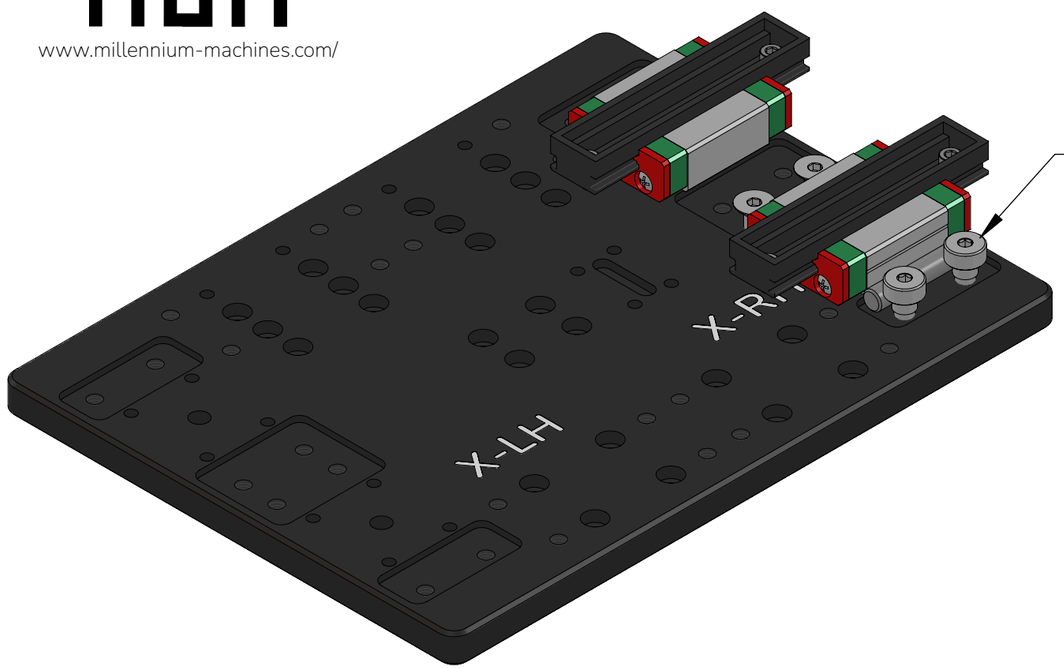
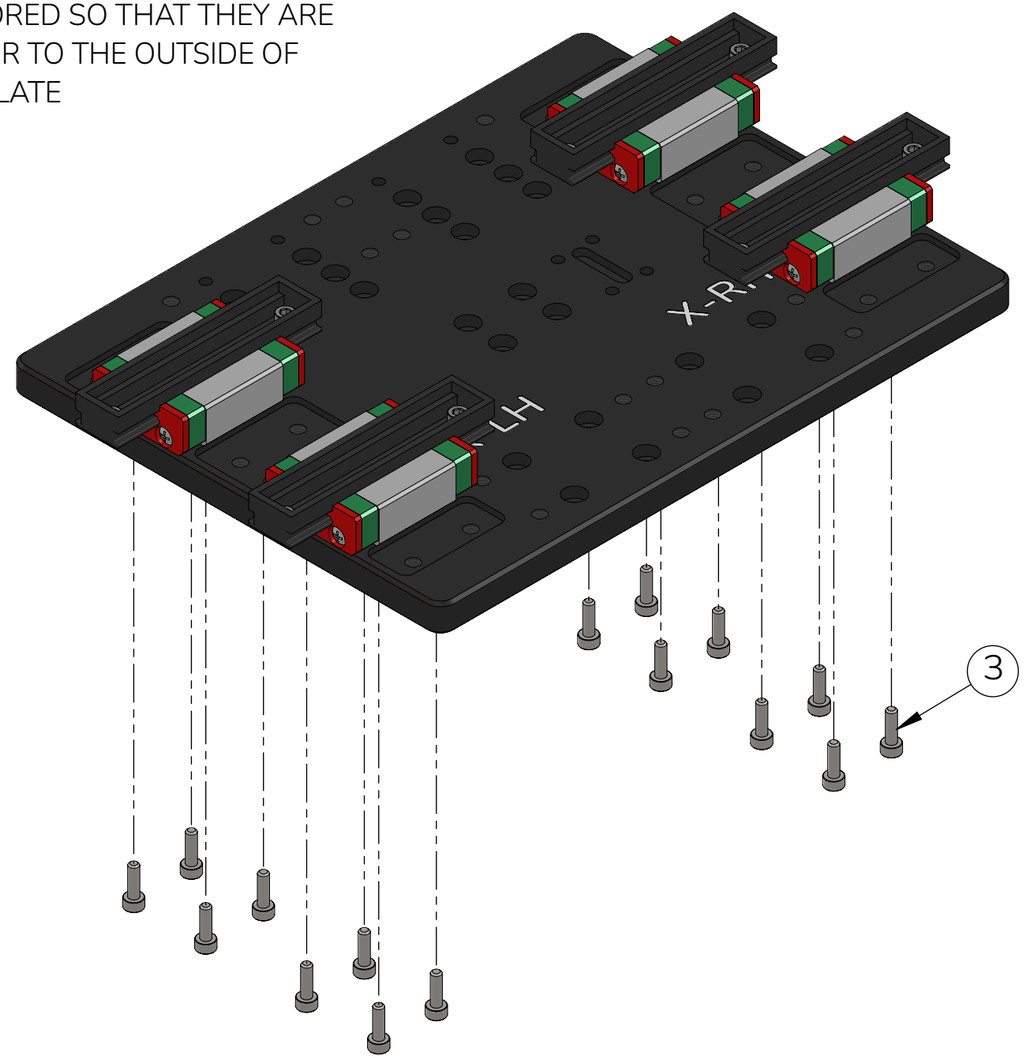


	PART NUMBER	QTY.
1	X-Y JOINER-PLATE	1
2	CARRIAGE RETAINER & BEARING ASSEMBLY	4
3	ISO-4762-M3X10-SHCS	16



NOTICE THE SHOULDER BOLTS MIRRORED SO THAT THEY ARE CLOSER TO THE OUTSIDE OF THE PLATE



MILO XY JOINER PLATE

ASSEMBLY STEP 2

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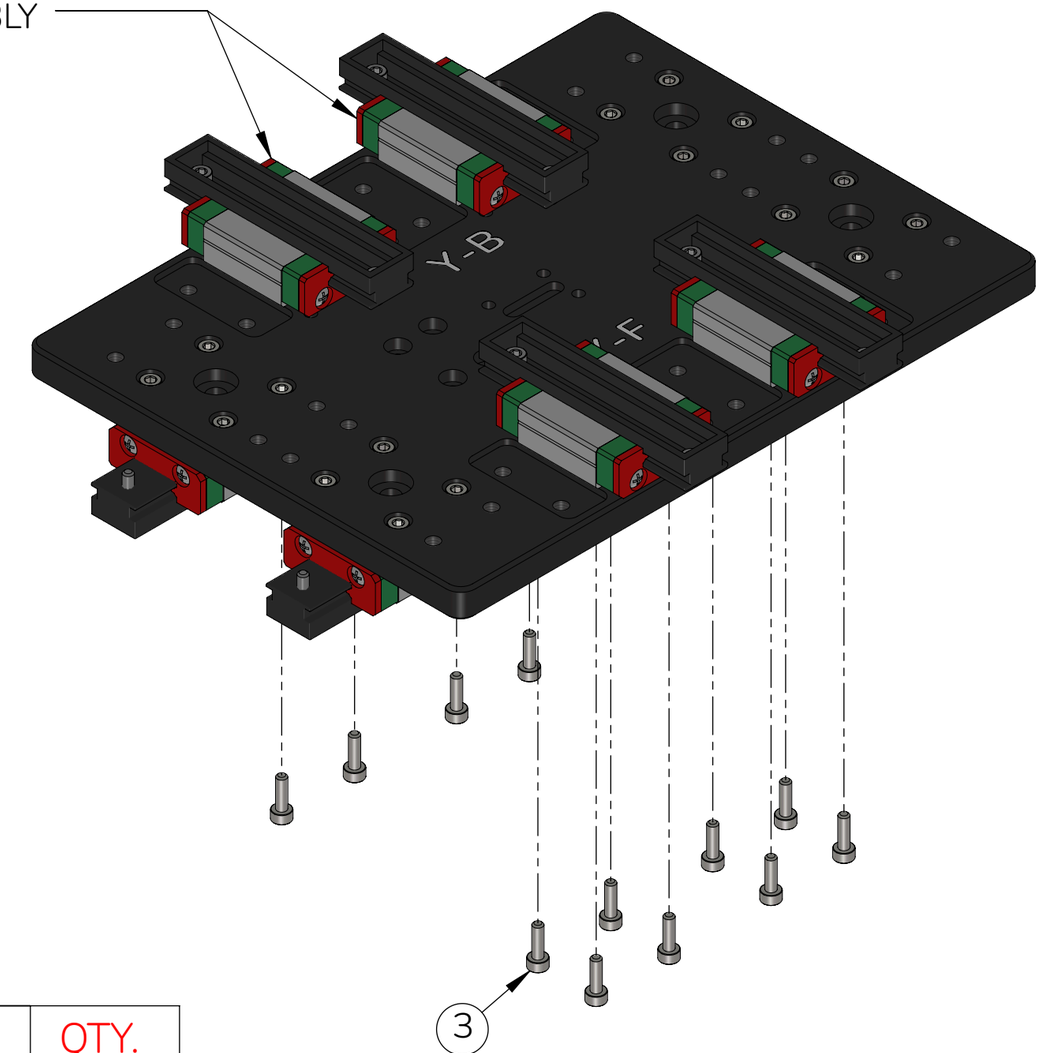
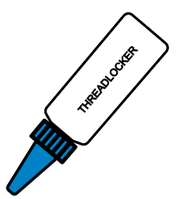
THESE CARRIAGES WILL ONLY GET THE 2
BACK SIDE BOLTS FOR NOW. WE WILL ADD
THE OTHER 2 WHEN WE MOUNT THE BELLOW
MAG MOUNT ASSEMBLY





	PART NUMBER	QTY.
1	X-Y JOINER-PLATE	1
2	CARRIAGE RETAINER & BEARING ASSEMBLY	4
3	ISO-4762-M3X10-SHCS	12

MILO XY JOINER PLATE

ASSEMBLY STEP 3

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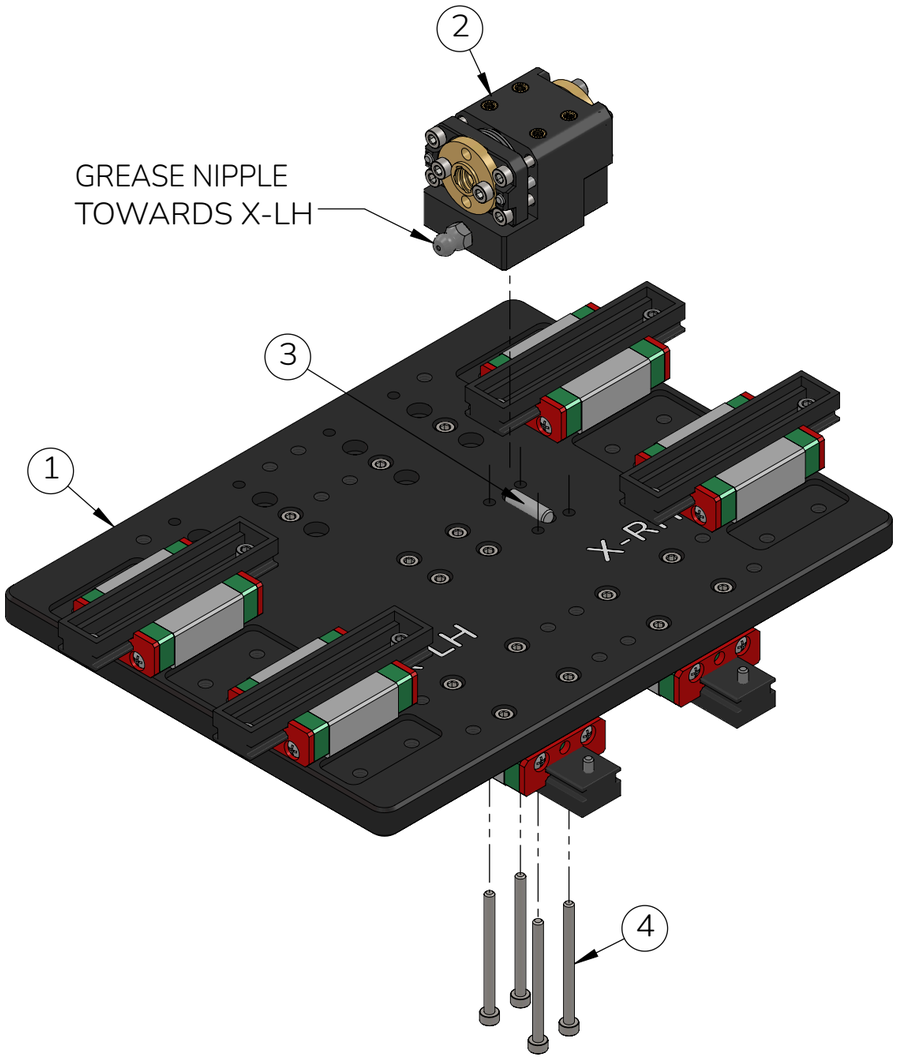
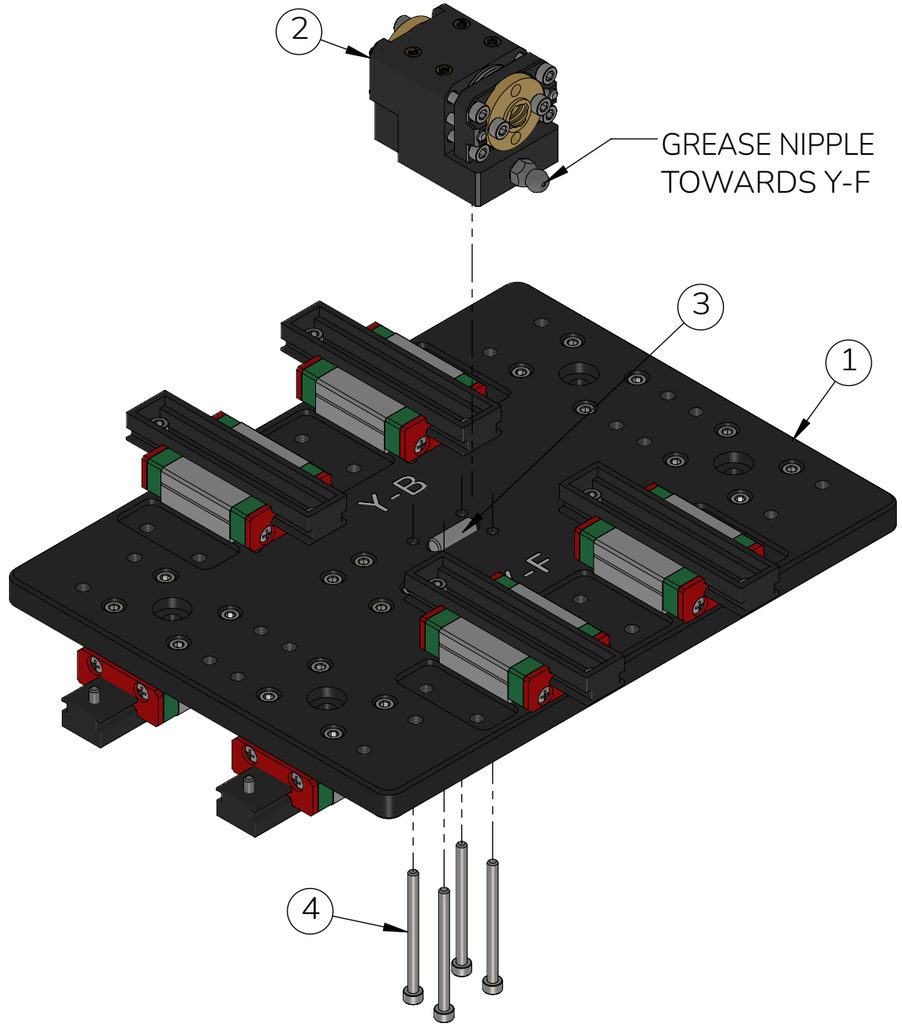
	PART NUMBER	QTY.
1	X-Y JOINER-PLATE	1
2	AB-BLOCK-MAIN-ASSEMBLY	2
3	ISO-2338-M5-15mm-PIN	2
4	ISO-4762-M3X40-SHCS	8





MILO XY JOINER PLATE

ASSEMBLY STEP 4

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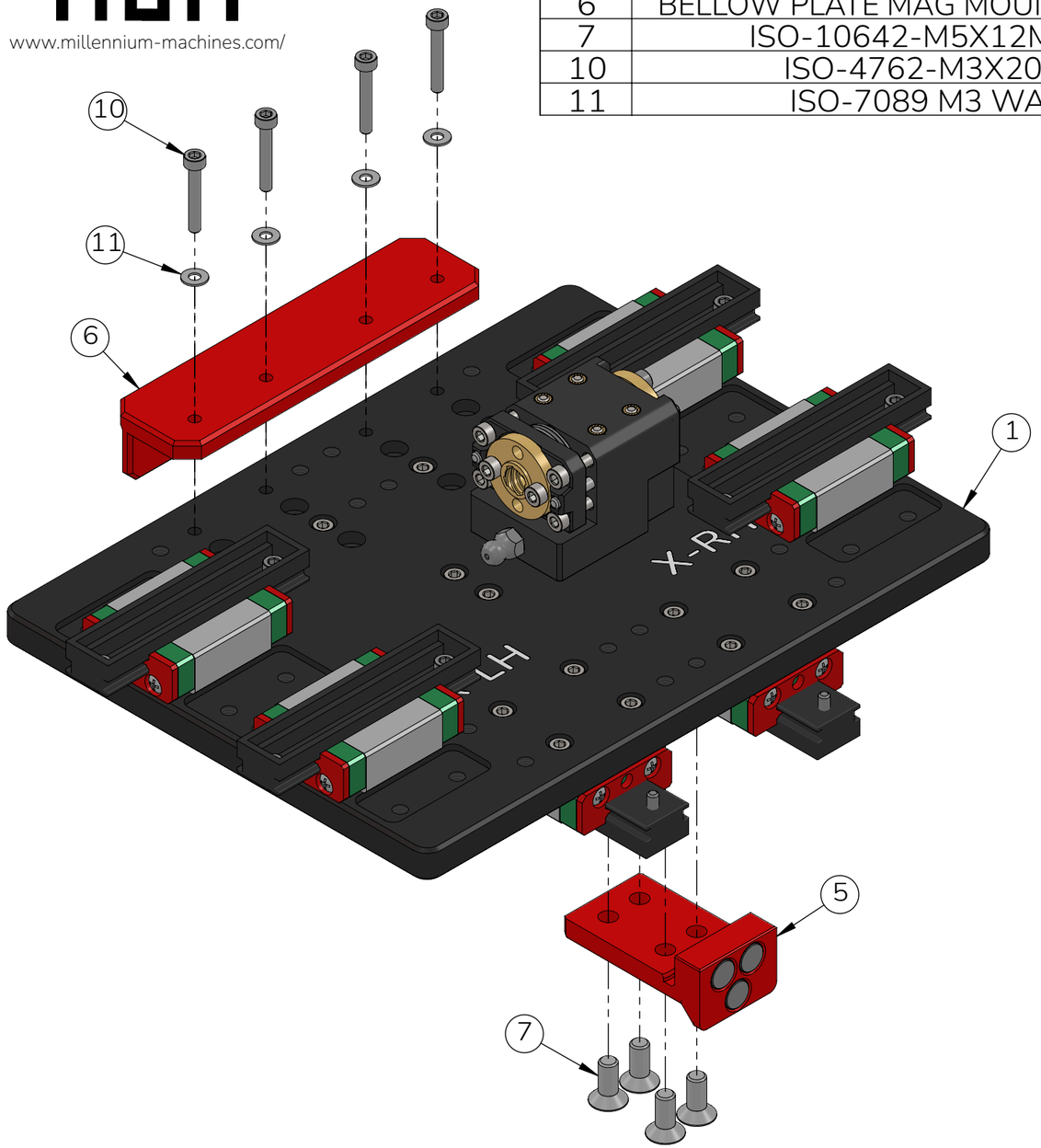
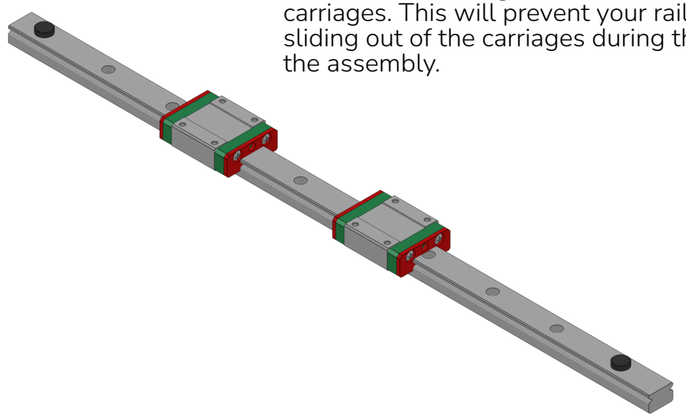
	PART NUMBER	QTY.
1	X-Y JOINER-PLATE	1
5	WAY-COVER-MAGNET-LOCK-ASSEMBLY	1
6	BELLOW PLATE MAG MOUNT ASSEMBLY	1
7	ISO-10642-M5X12MM-FHCS	4
10	ISO-4762-M3X20-SHCS	4
11	ISO-7089 M3 WASHER	4



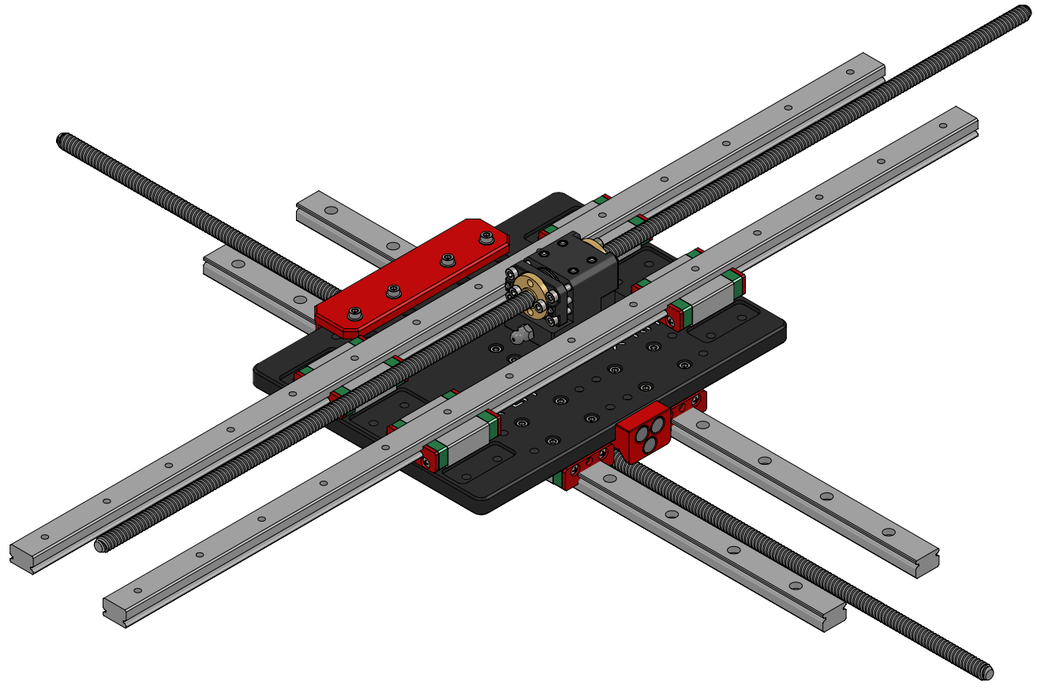
Linear Rail Carriage Safety

If your rails came with rubber stoppers, stick them in after sliding the rails back onto the carriages. This will prevent your rails from sliding out of the carriages during the rest of the assembly.



1. Loosen the bolts in the carriage retainer and slide the rails back onto the carriages

2. Follow the instructions from earlier and thread in both the X & Y axis leadscrew



MILO XY JOINER PLATE

ASSEMBLY STEP 5

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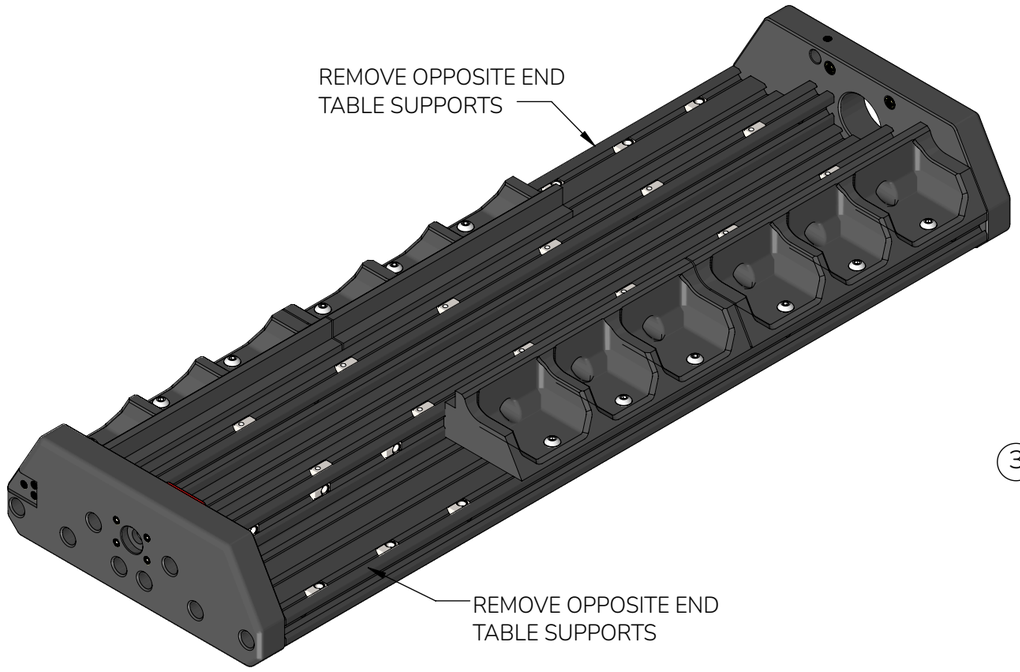
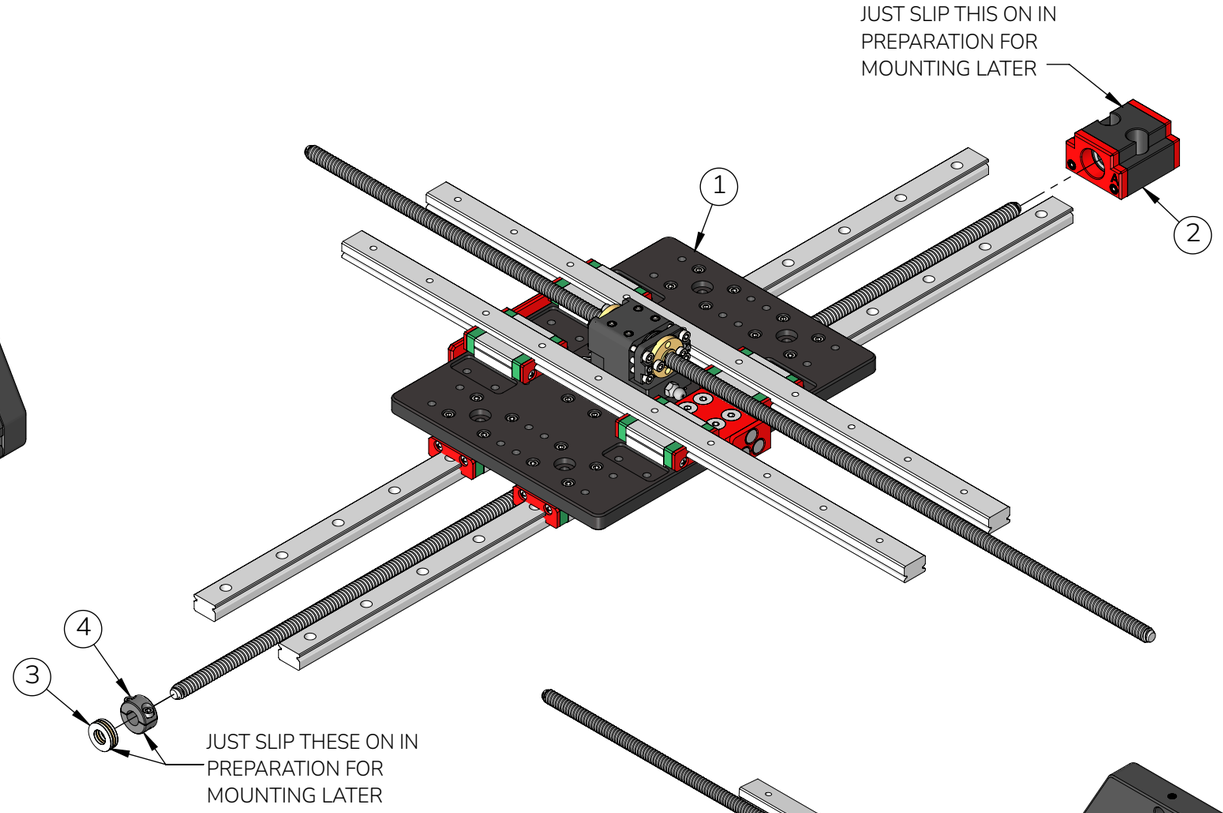
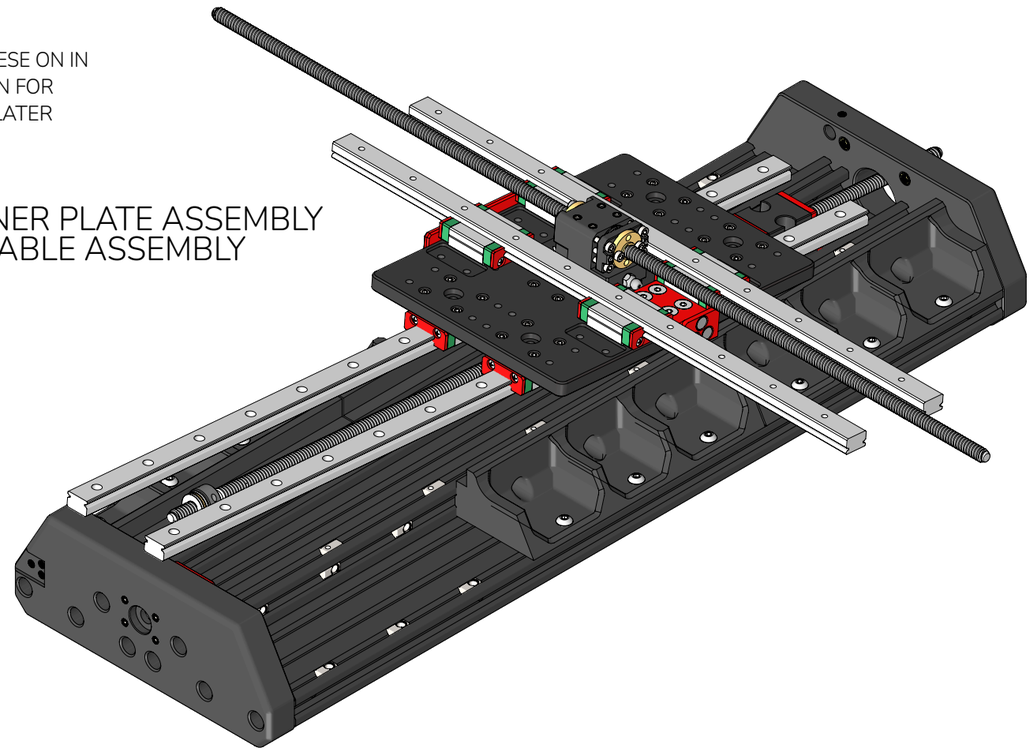




TILT XY-JOINER PLATE ASSEMBLY
INTO THE TABLE ASSEMBLY



	PART NUMBER	QTY.
1	MILO-XY-JOINER-PLATE-ASSEMBLY	1
2	FLOATING-BEARING-BLOCK-ASSEMBLY	1
3	F8-16M THRUST BEARING	1
4	DOUBLE SPLIT SHAFT COLLAR 8X18X9MM	1

MILO X AXIS FRAME

ASSEMBLY STEP 4

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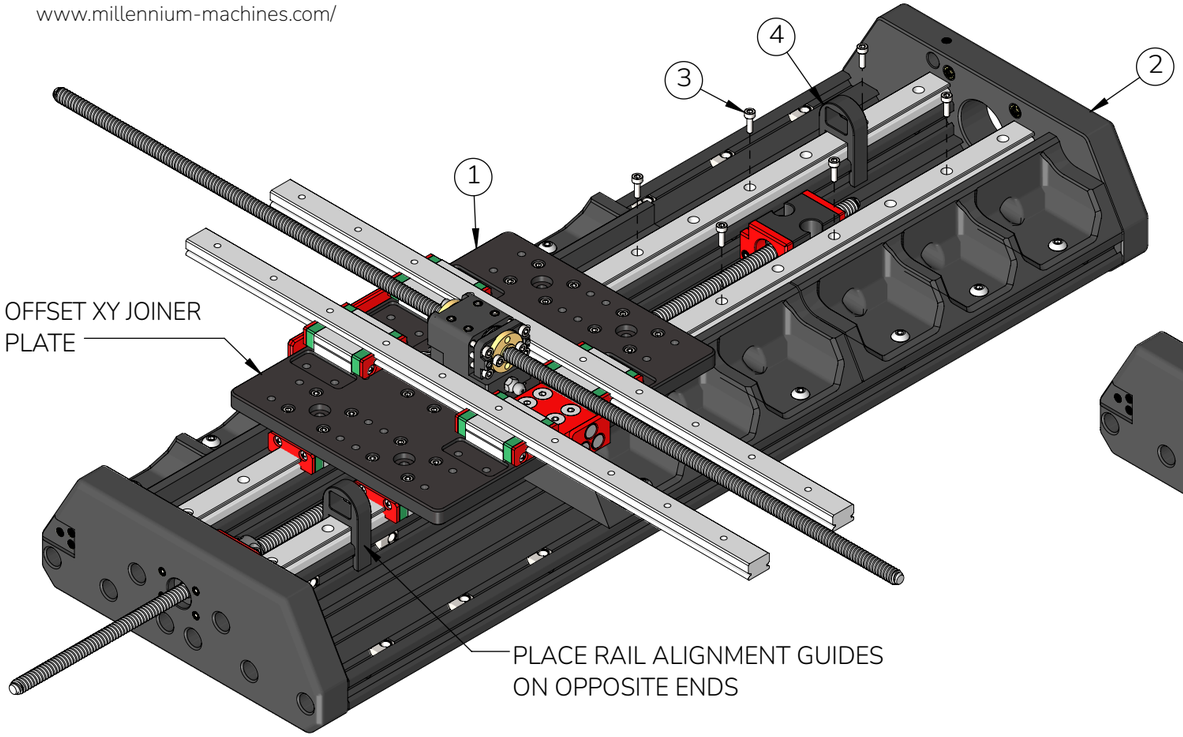

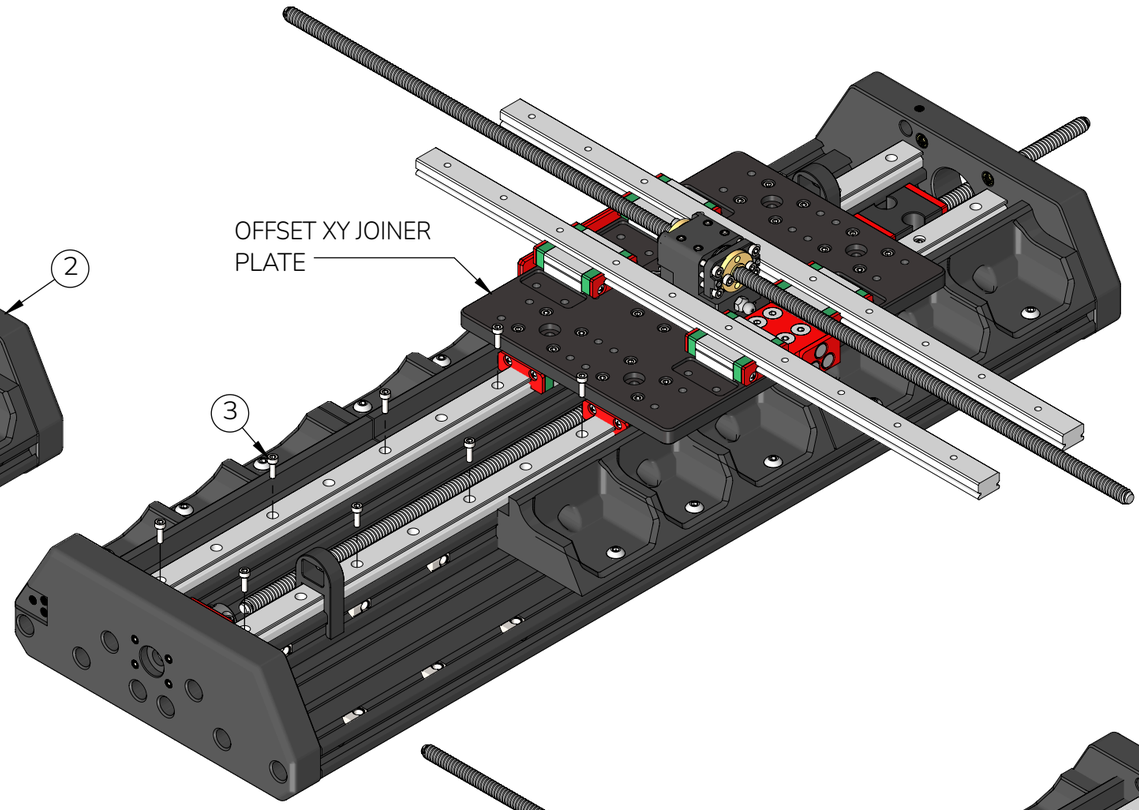

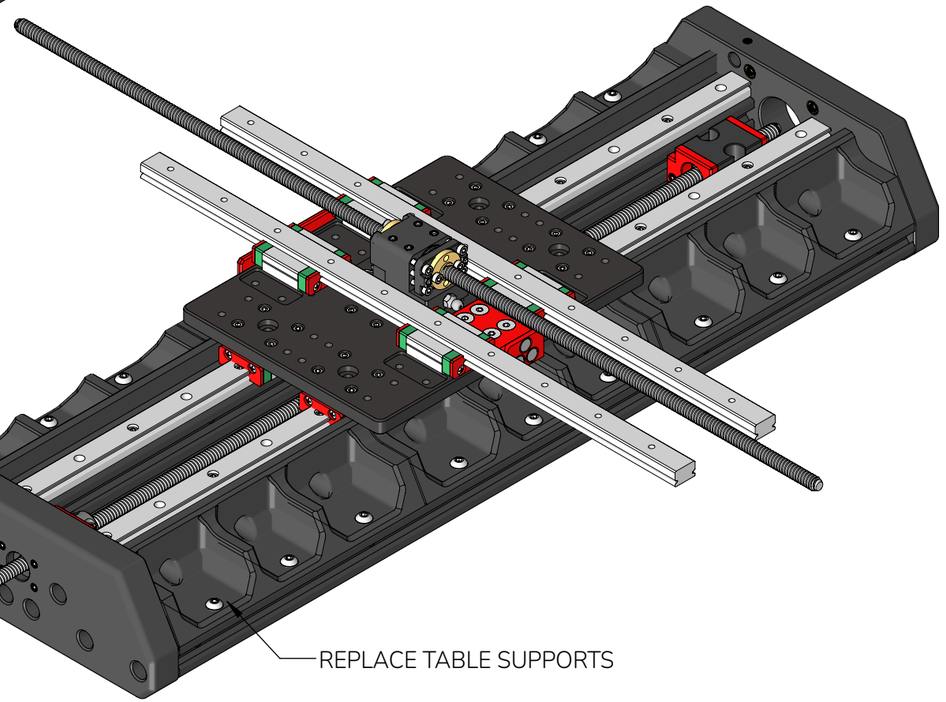

	PART NUMBER	QTY.
1	MILO-XY-JOINER-PLATE-ASSEMBLY	1
2	MILO-X-AXIS-ASSEMBLY	1
3	ISO-4762-M3X10-SHCS	14
4	RAIL ALIGNMENT GUIDE	2

MILO X AXIS FRAME

ASSEMBLY STEP 5

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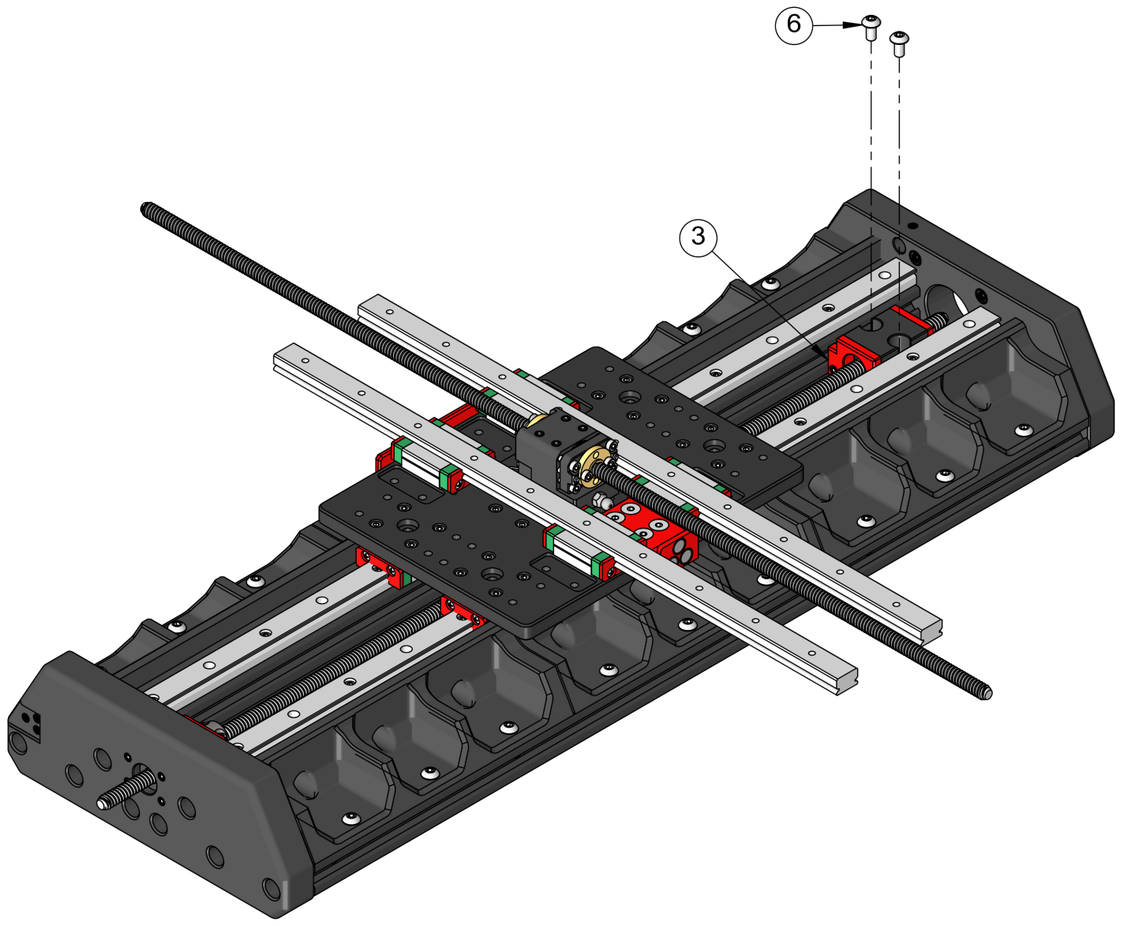
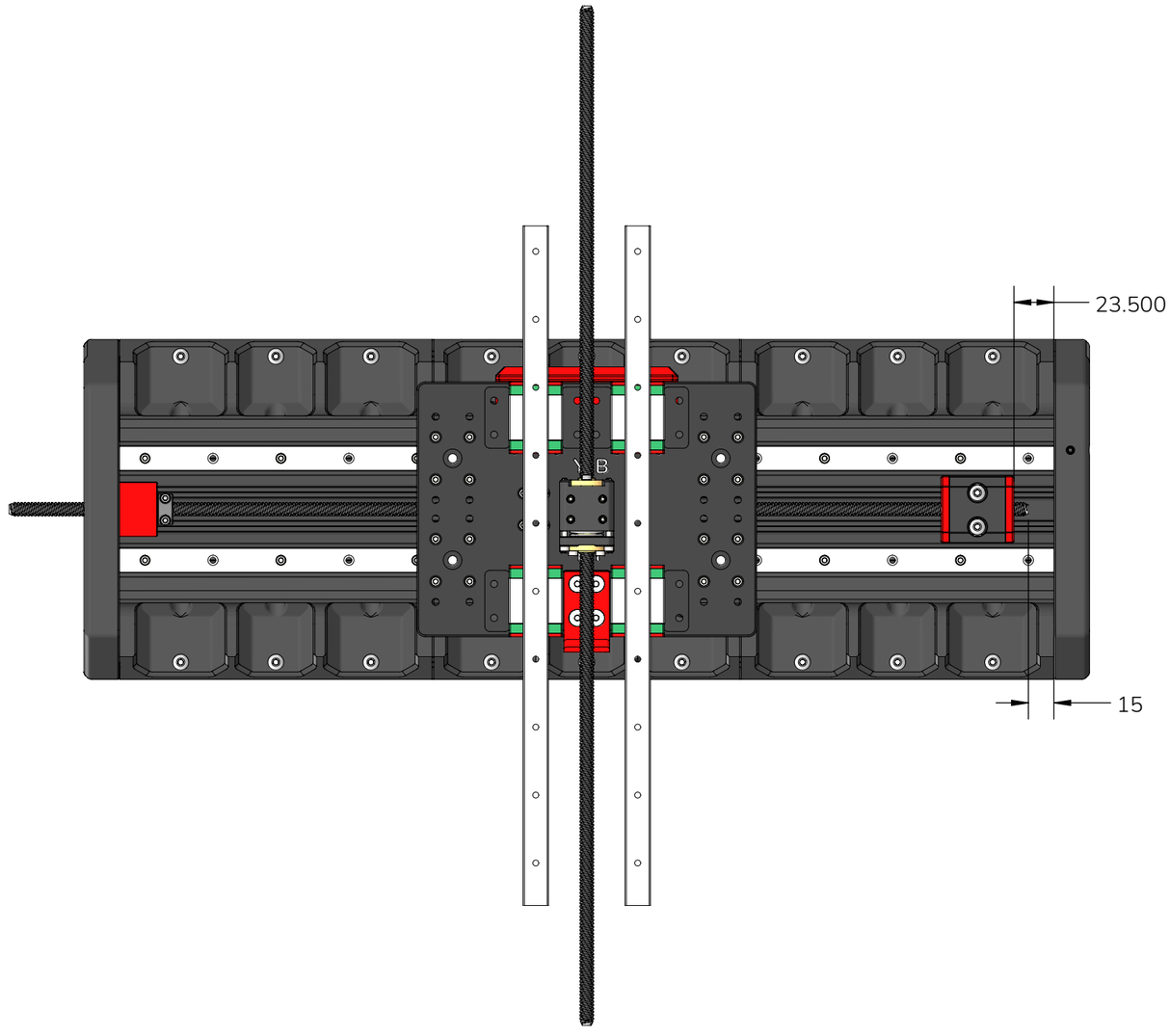
	PART NUMBER	QTY.
1	MILO-XY-JOINER-PLATE-ASSEMBLY	1
2	MILO-X-AXIS-ASSEMBLY	1
3	FLOATING-BEARING-BLOCK-ASSEMBLY	1
4	M5 DROP IN NUT	2
6	IS0-7380-M5X10 BHCS	2





MILO X AXIS FRAME

ASSEMBLY STEP 6

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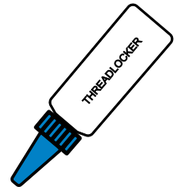
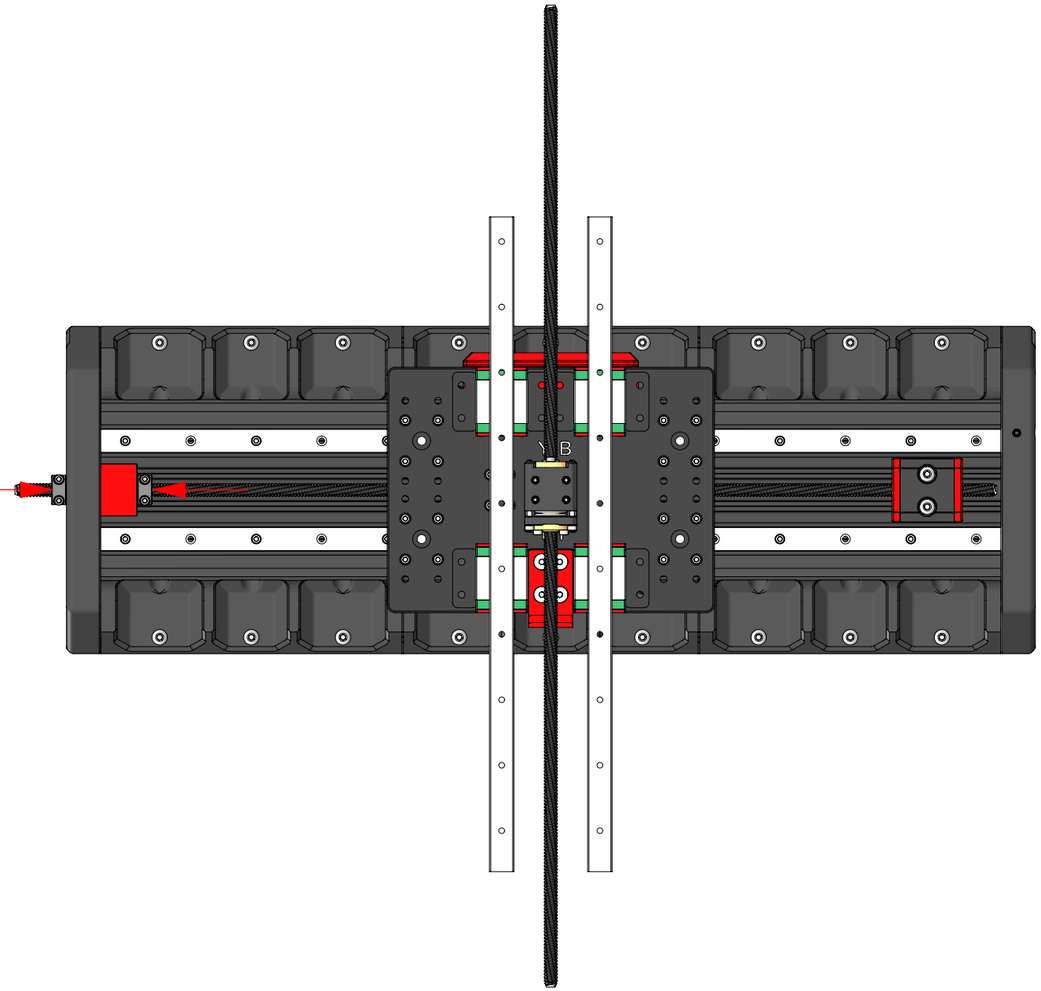
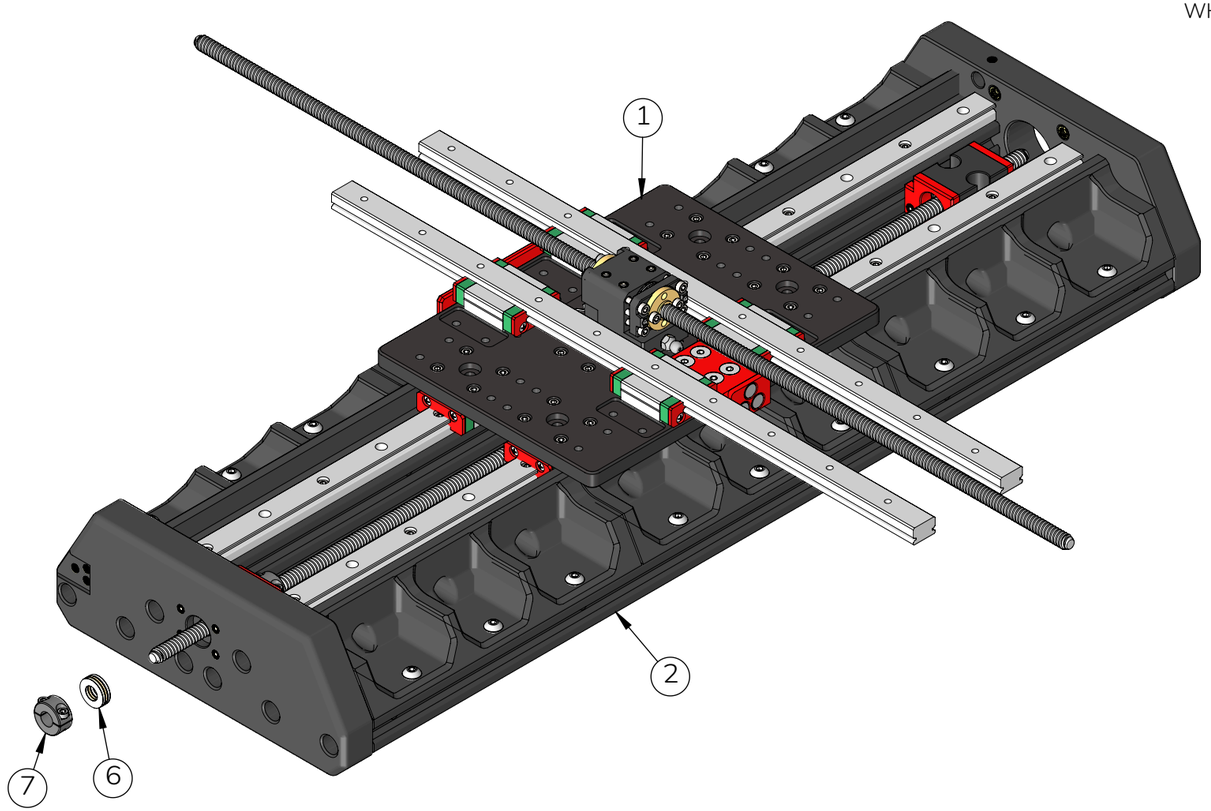


Shaft Collar Bolts

TIGHTEN BOTH
SHAFT COLLARS

APPLY PRESSURE
WHEN TIGHTENING



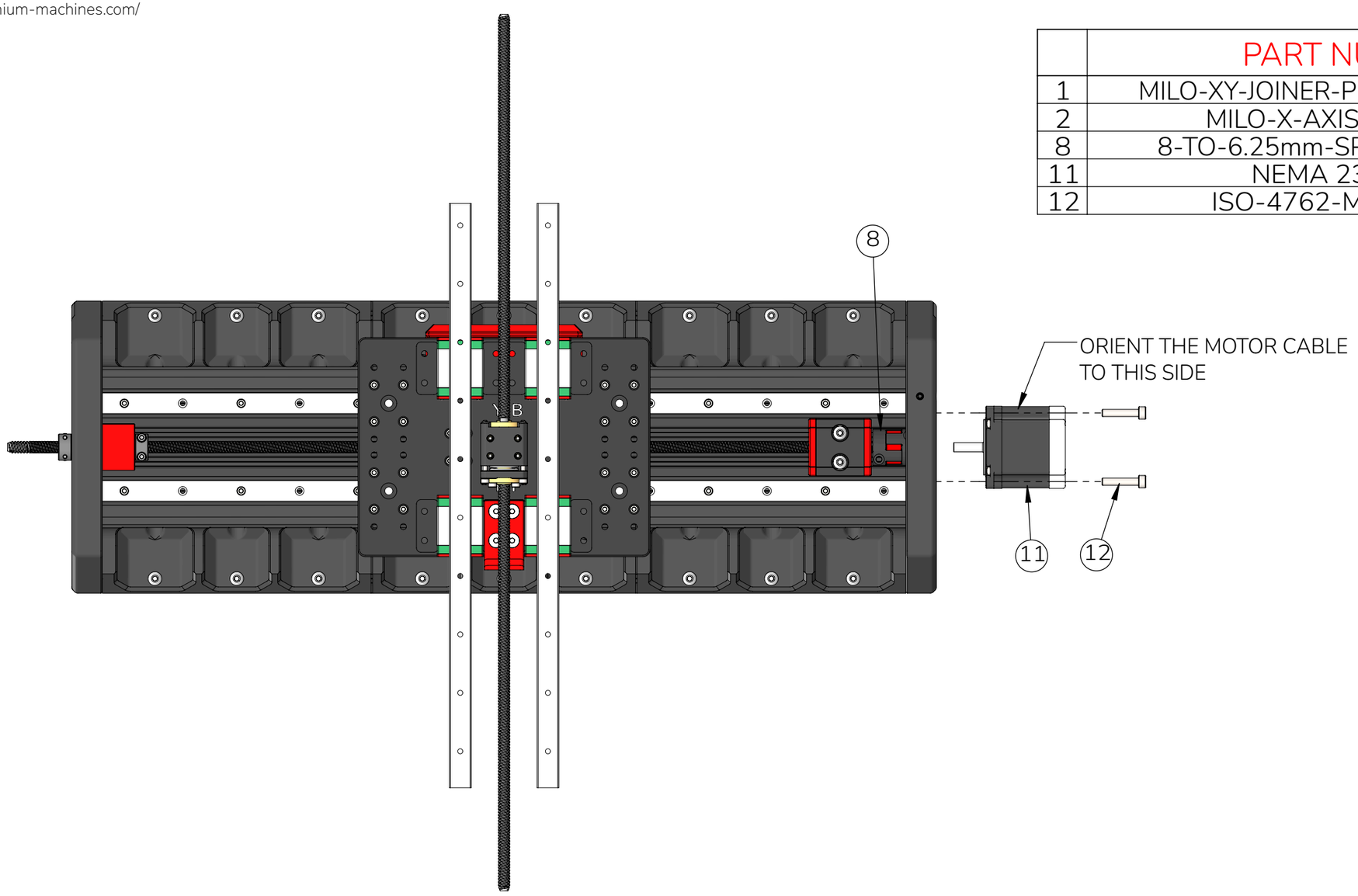


	PART NUMBER	QTY.
1	MILO-XY-JOINER-PLATE-ASSEMBLY	1
2	MILO-X-AXIS-ASSEMBLY	1
6	F8-16M THRUST BEARING	1
7	DOUBLE SPLIT SHAFT COLLAR 8X18X9MM	1

MILO X AXIS FRAME

ASSEMBLY STEP 7

	PART NUMBER	QTY.
1	MILO-XY-JOINER-PLATE-ASSEMBLY	1
2	MILO-X-AXIS-ASSEMBLY	1
8	8-TO-6.25mm-SPIDER-COUPLER	1
11	NEMA 23 - 56mm	1
12	ISO-4762-M5X25-SHCS	4



MILO X AXIS FRAME

ASSEMBLY STEP 8

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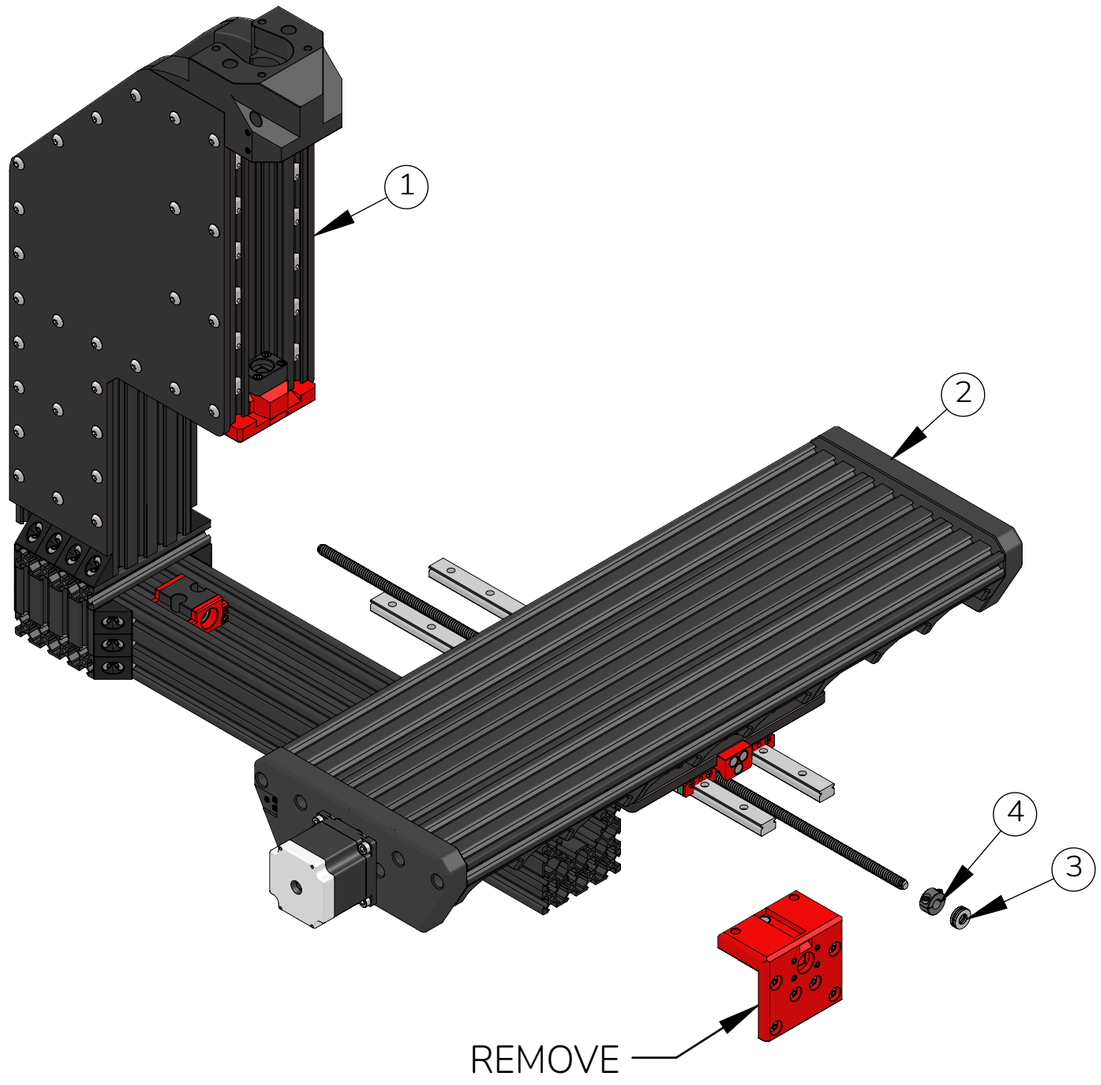
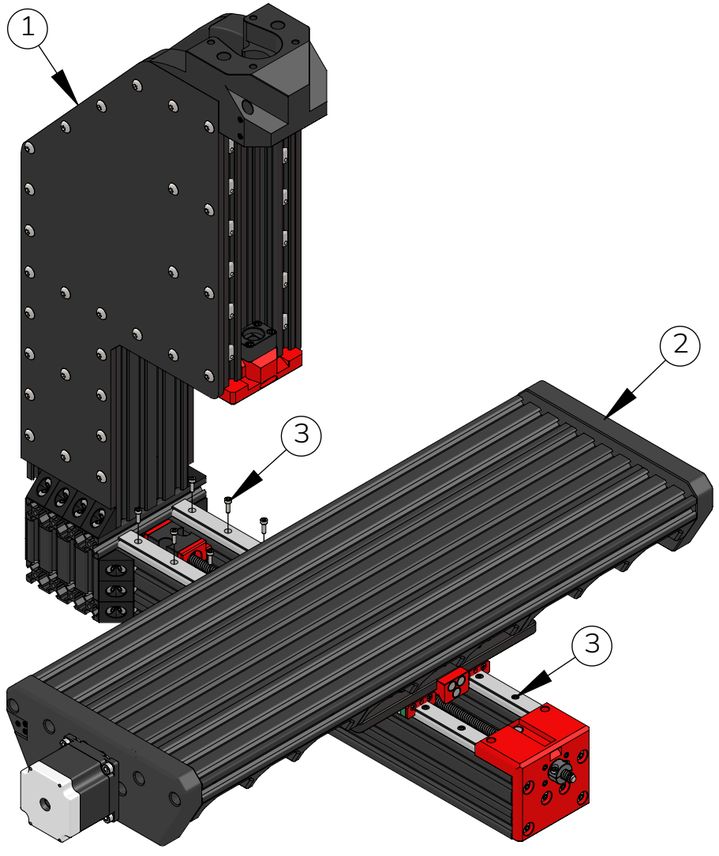
	PART NUMBER	QTY.
1	ZY-ASSEMBLY	1
2	MILO-XY-SUBASSEMBLY	1
3	F8-16M THRUST BEARING	1
4	DOUBLE SPLIT SHAFT COLLAR 8X18X9MM	1





	PART NUMBER	QTY.
1	ZY-ASSEMBLY	1
2	MILO-XY-SUBASSEMBLY	1
3	ISO-4762-M3X10-SHCS	20

MILO SUB-ASSEMBLY-A

STEP 1

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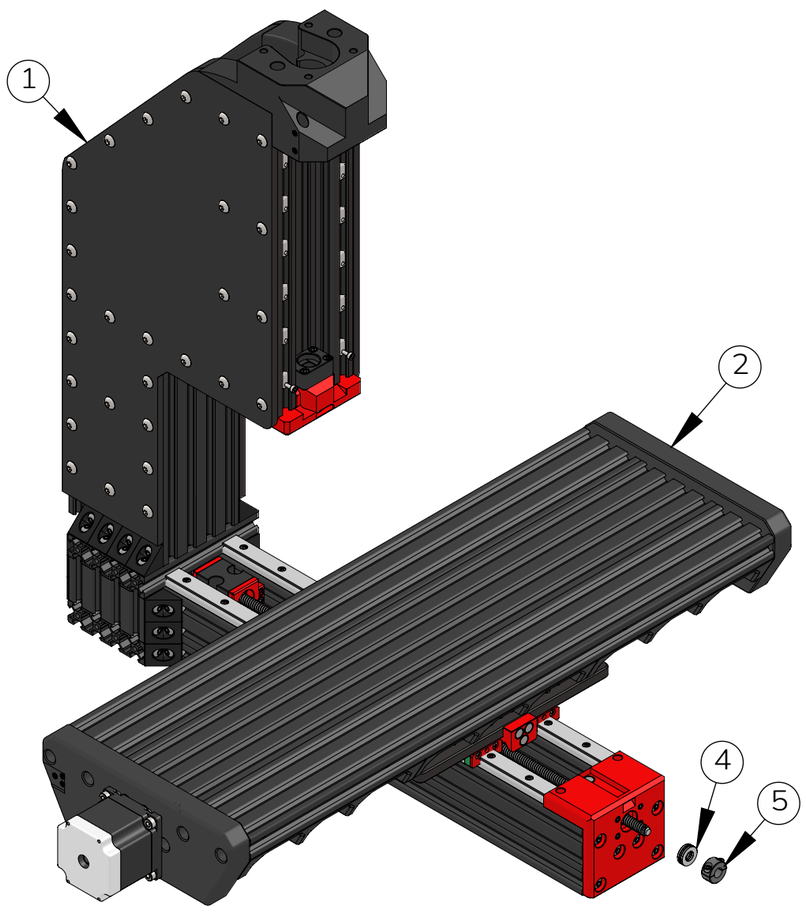
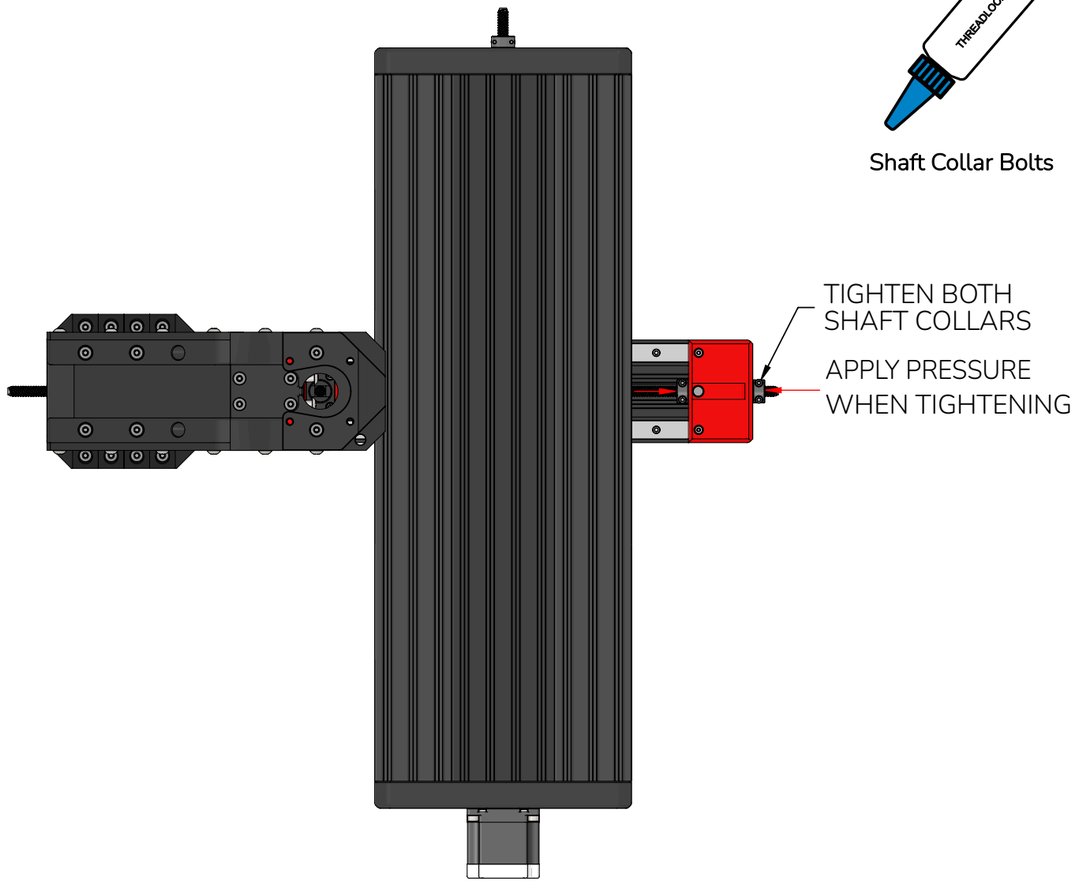
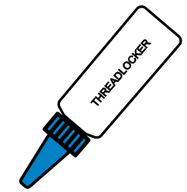
	PART NUMBER	QTY.
1	ZY-ASSEMBLY	1
2	MILO-XY-SUBASSEMBLY	1
4	F8-16M THRUST BEARING	2
5	DOUBLE SPLIT SHAFT COLLAR 8X18X9MM	2







Shaft Collar Bolts

MILO SUB-ASSEMBLY-A

STEP 2

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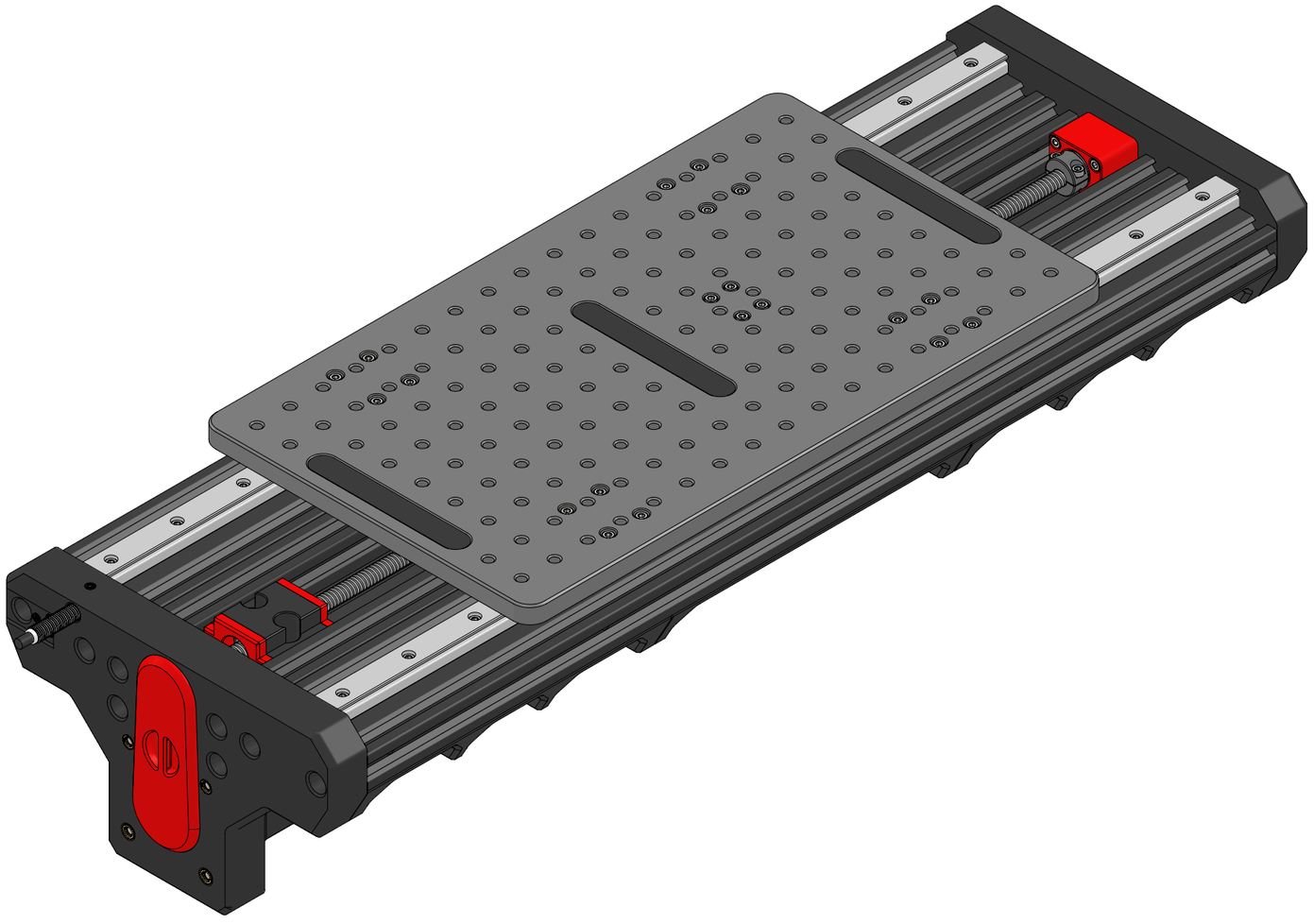
If you are building Milo, skip to page 96.

If you are building Miley, follow the instructions on pages 69-94.



	PART NUMBER	QTY.
1	MILEY-X-AXIS-ASSEMBLY-A	1
2	X-AXIS-LINEAR-RAIL-ASSEMBLY	2
3	ISO-4762-M3X10-SHCS	44
4	M3 DROP IN NUT	28
5	F8-16M THRUST BEARING	1
6	DOUBLE SPLIT SHAFT COLLAR 8X18X9MM	1
7	NEMA 23 - 56mm	1
8	GT2-40T-ID-6.35mm	1
9	120mm-GATE-GT2-CLOSED-LOOP-BELT	1
10	ISO-4026-M3X10-SET SCREW	3
11	ISO-4762-M5X25-SHCS	4
12	MILEY-X-AXIS-BELT-COVER-ASSEMBLY	1
13	LJ8A3-2-Z/AY	1
14	MILEY-X-AXIS-FIXATED-ASSEMBLY	1
15	MILEY TABLE SUPPORT B	2
16	MILEY TABLE SUPPORT A	4
17	ISO-7380-M5X10 BHCS	38
18	M5 DROP IN NUT	2
19	GT2-40T-ID-8mm	1

MILEY X AXIS

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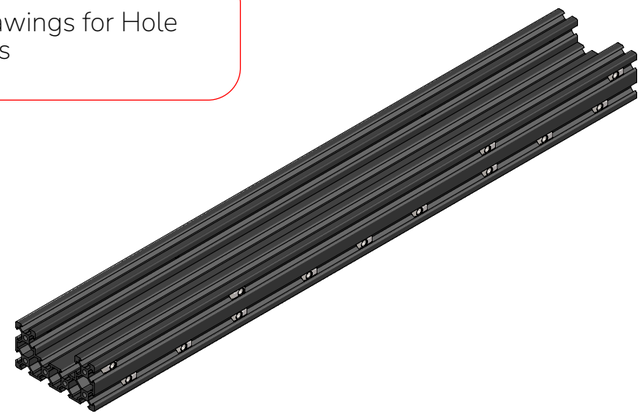
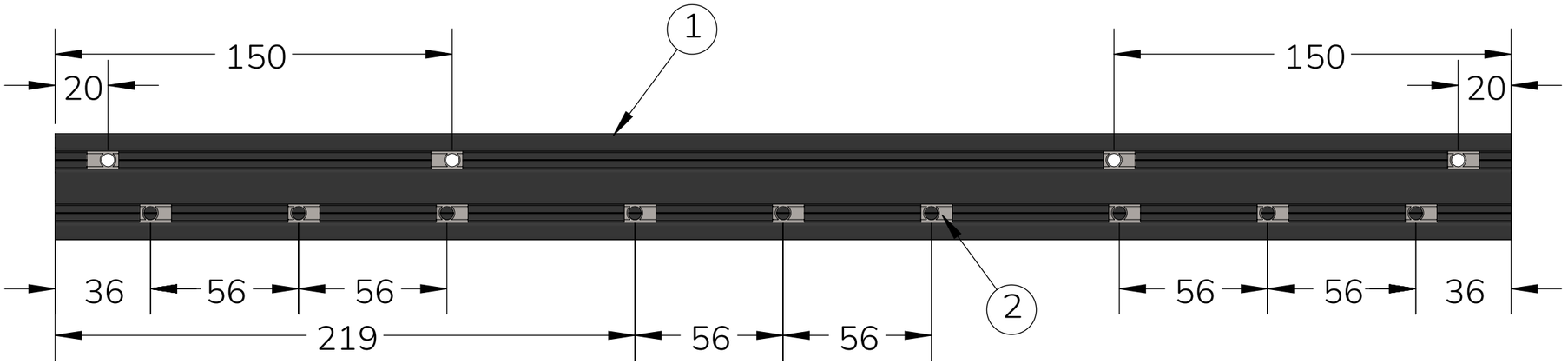
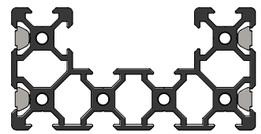


See Manufacturing Drawings for Hole Locations





	PART NUMBER	QTY.
1	C BEAM 550mm	1
2	M5 DROP IN NUT	26

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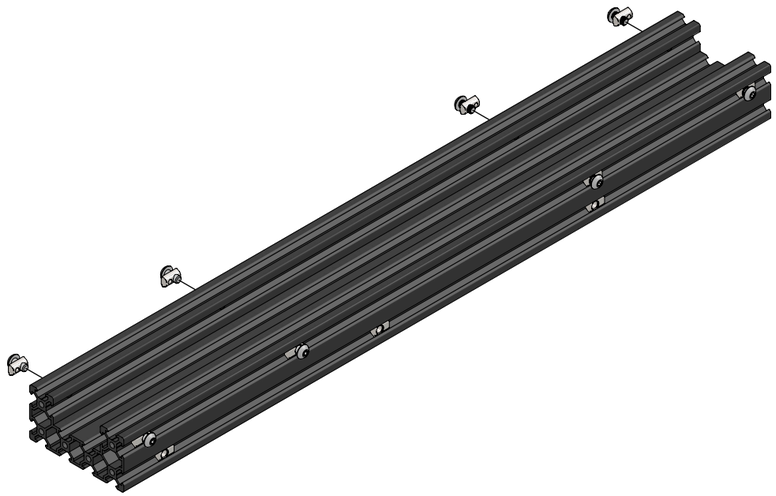
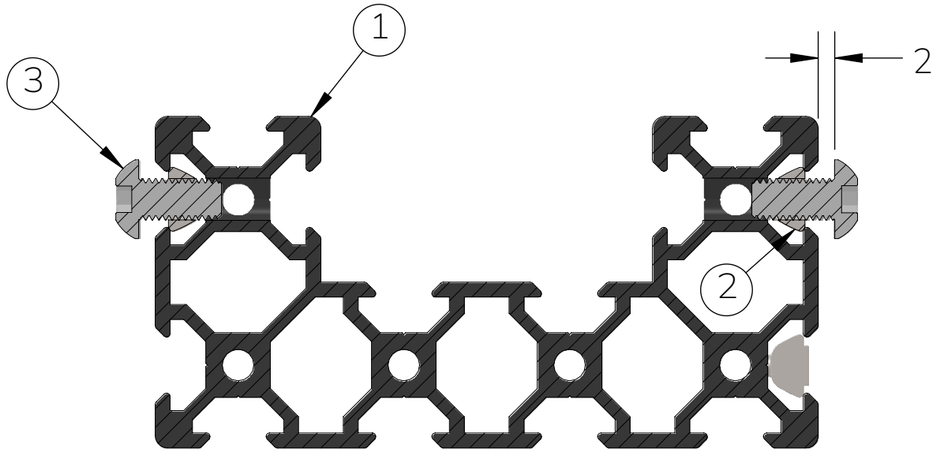
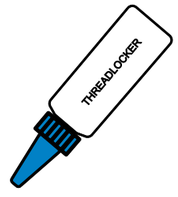
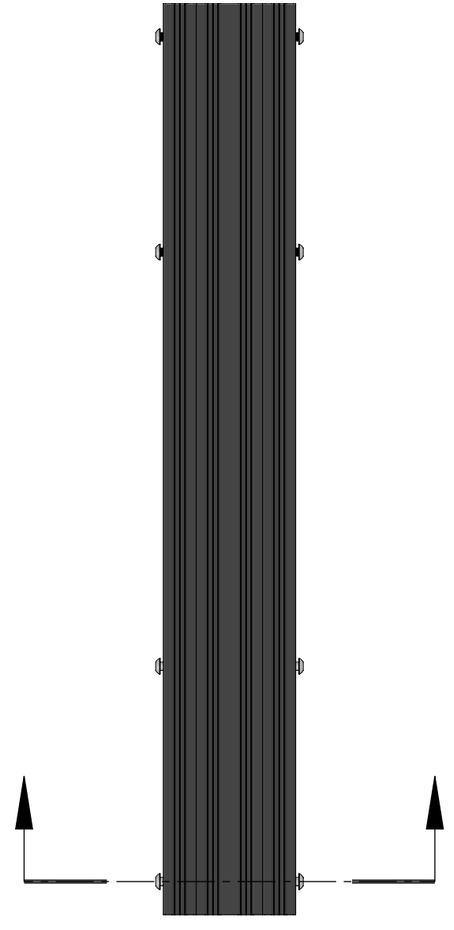






Thread the bolt through the nut and into the hole slightly to hold the bolt in place while sliding the 2080 extrusions into place



MILEY X AXIS

ASSEMBLY STEP 1

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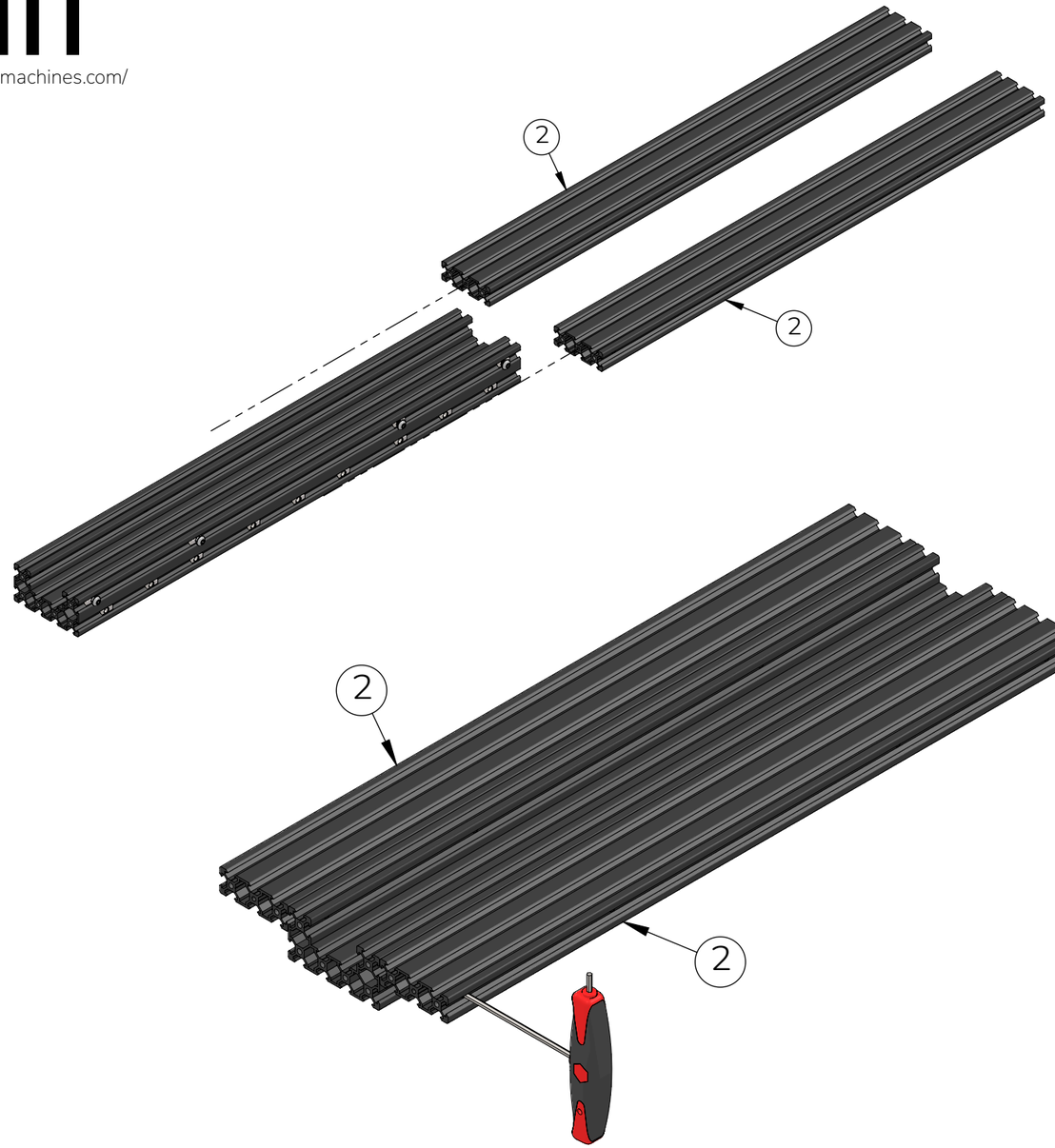
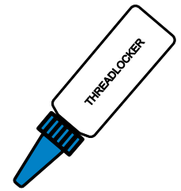
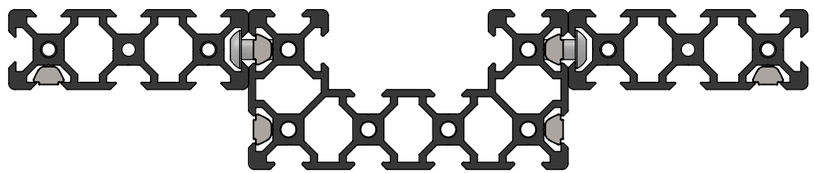




	PART NUMBER	QTY.
1	C BEAM 550mm	1
2	V-SLOT-2080X120MM	2



MILEY X AXIS

ASSEMBLY STEP 2

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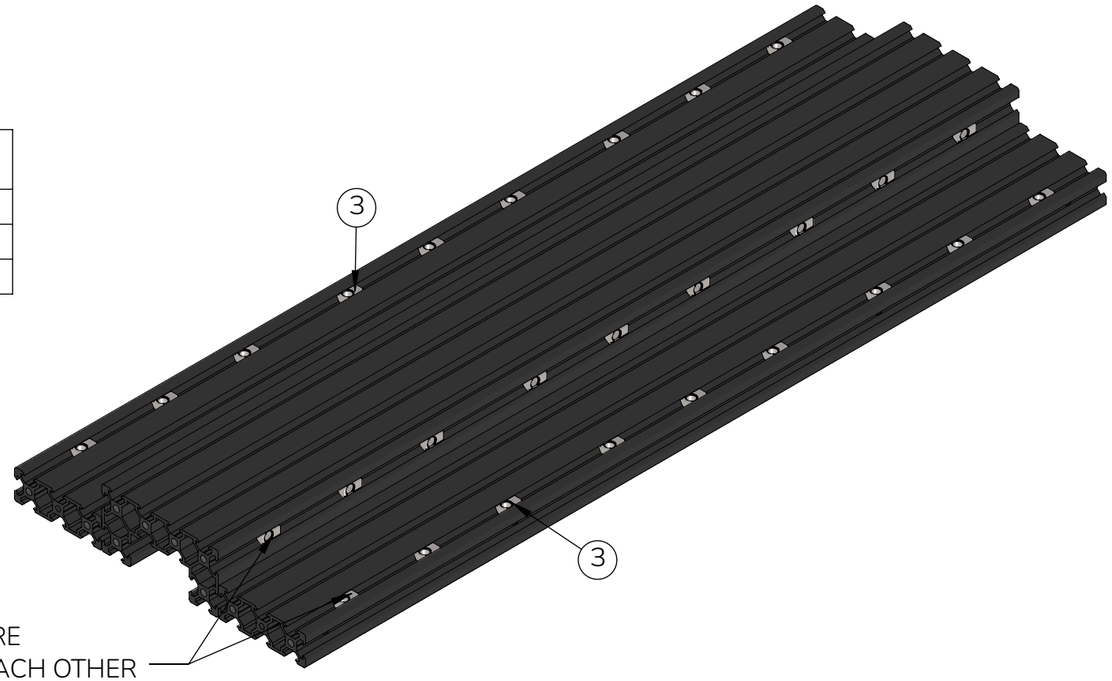
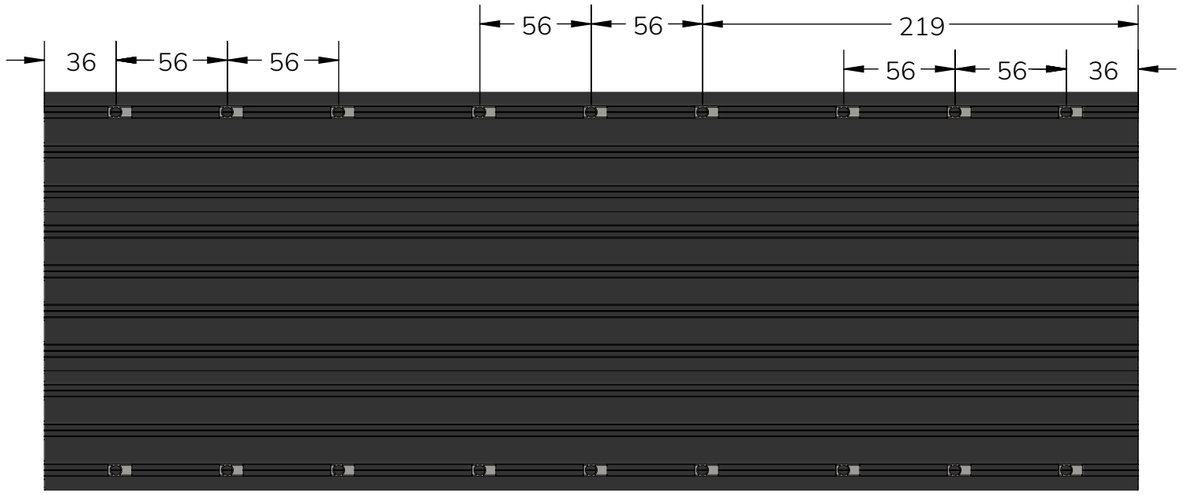
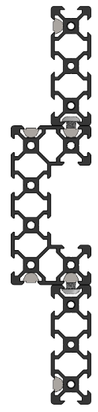
	PART NUMBER	QTY.
1	C BEAM 550mm	1
2	V-SLOT-2080X120MM	2
3	M5 DROP IN NUT	36







MILEY X AXIS

ASSEMBLY STEP 3

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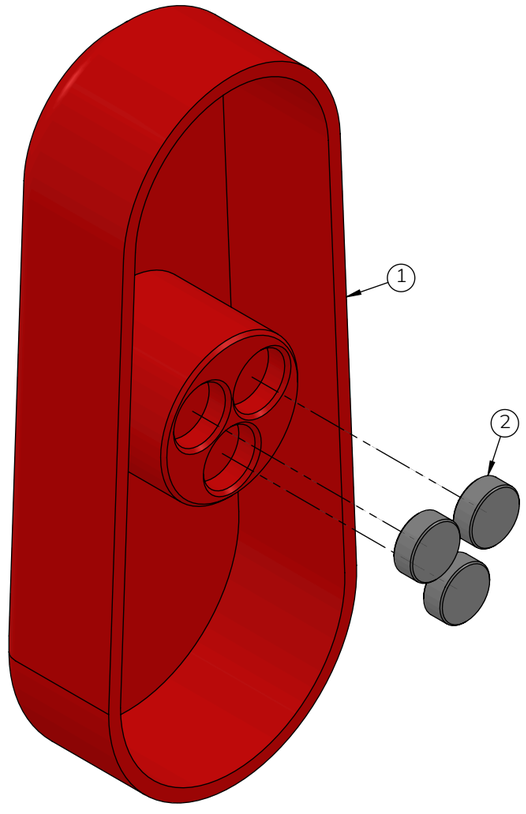
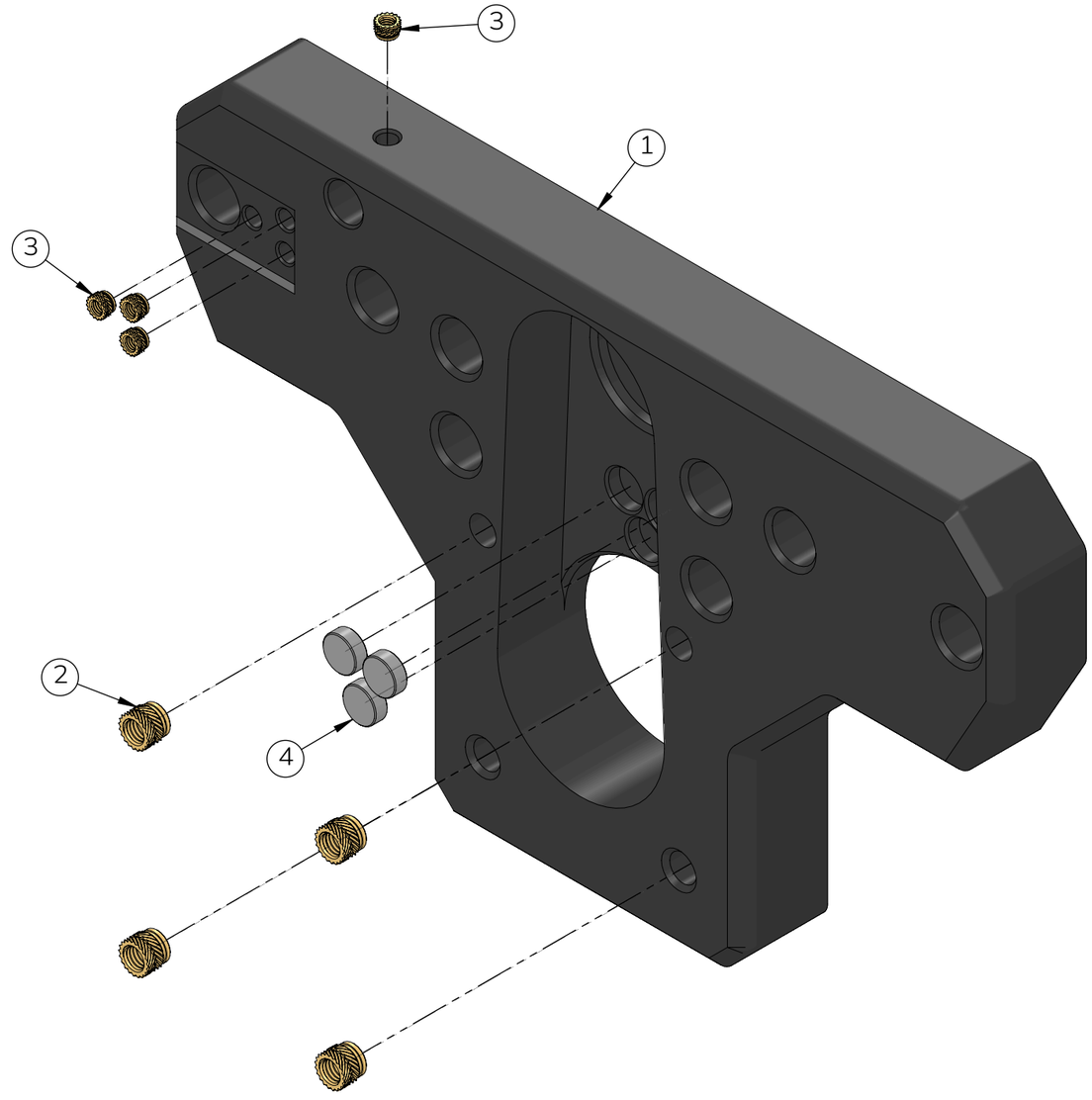


	PART NUMBER	QTY.
1	X-AXIS-BELT-COVER	1
2	8X3MM NEODYMIUM MAGNET	3



	PART NUMBER	QTY.
1	MILEY-X-AXIS-MOTOR-MOUNT	1
2	M5X5.8mm HEATSET INSERT	4
3	M3X3MM HEATSET INSERT	4
4	8X3MM NEODYMIUM MAGNET	3

MILEY X AXIS

MOTOR MOUNT PRE-ASSEMBLY

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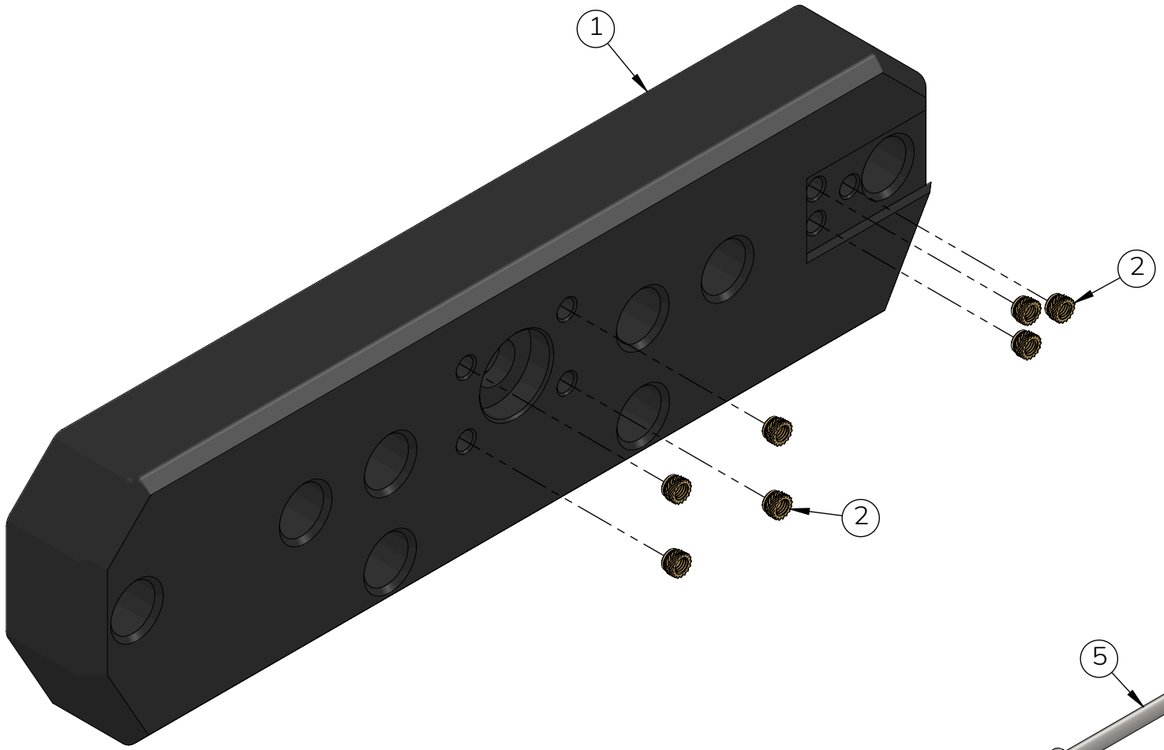
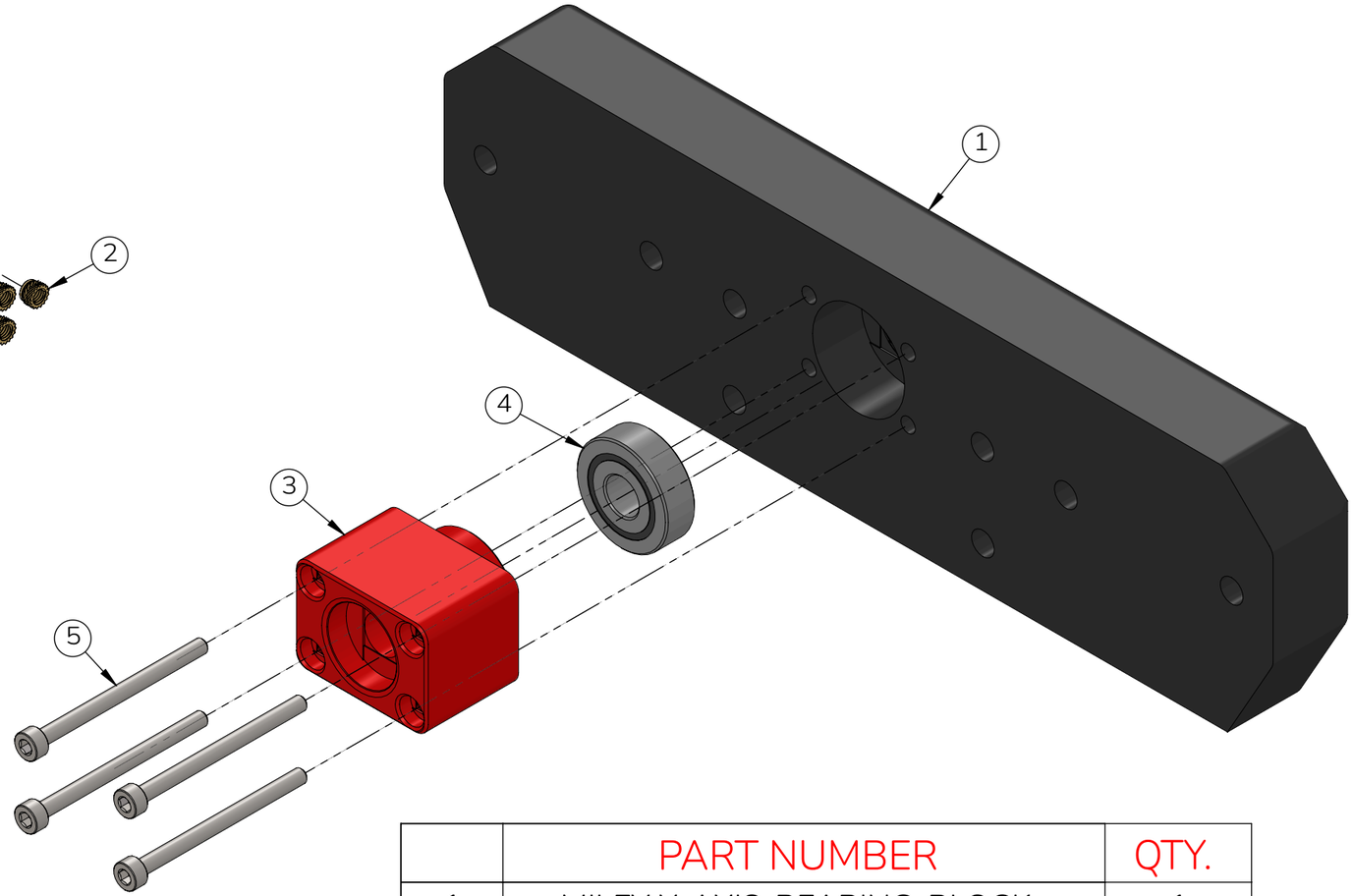


	PART NUMBER	QTY.
1	MILEY X-AXIS-BEARING-BLOCK	1
2	M3X3MM HEATSET INSERT	7



	PART NUMBER	QTY.
1	MILEY X-AXIS-BEARING-BLOCK	1
3	X-BEARING-BLOCK-INSERT	1
4	608ZZ BALL BEARING	1
5	ISO-4762-M3X40-SHCS	4

MILEY X AXIS

BEARING BLOCK PRE-ASSEMBLY

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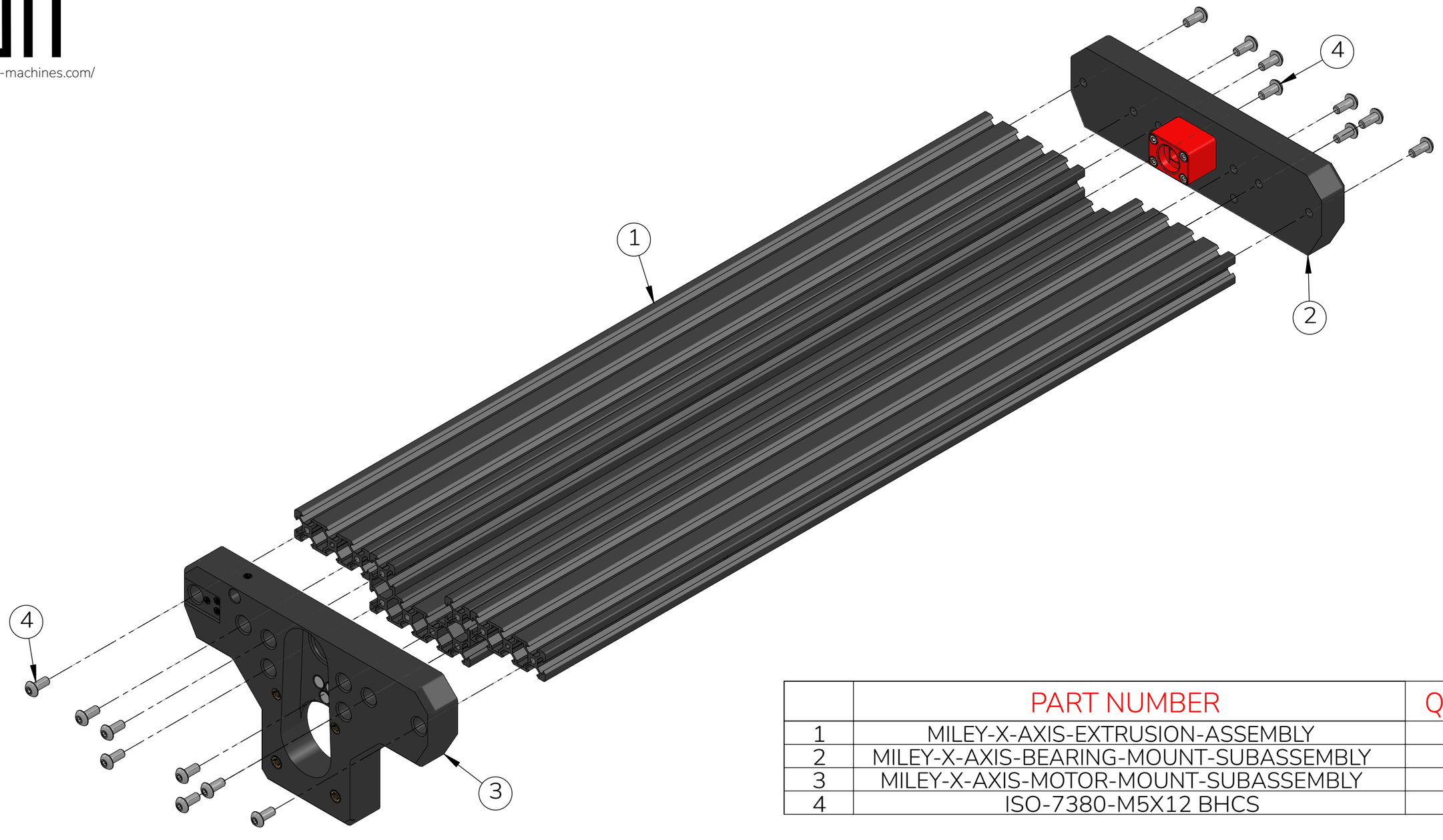

	PART NUMBER	QTY.
1	MILEY-X-AXIS-EXTRUSION-ASSEMBLY	1
2	MILEY-X-AXIS-BEARING-MOUNT-SUBASSEMBLY	1
3	MILEY-X-AXIS-MOTOR-MOUNT-SUBASSEMBLY	1
4	ISO-7380-M5X12 BHCS	16

MILEY X AXIS

ASSEMBLY STEP 4

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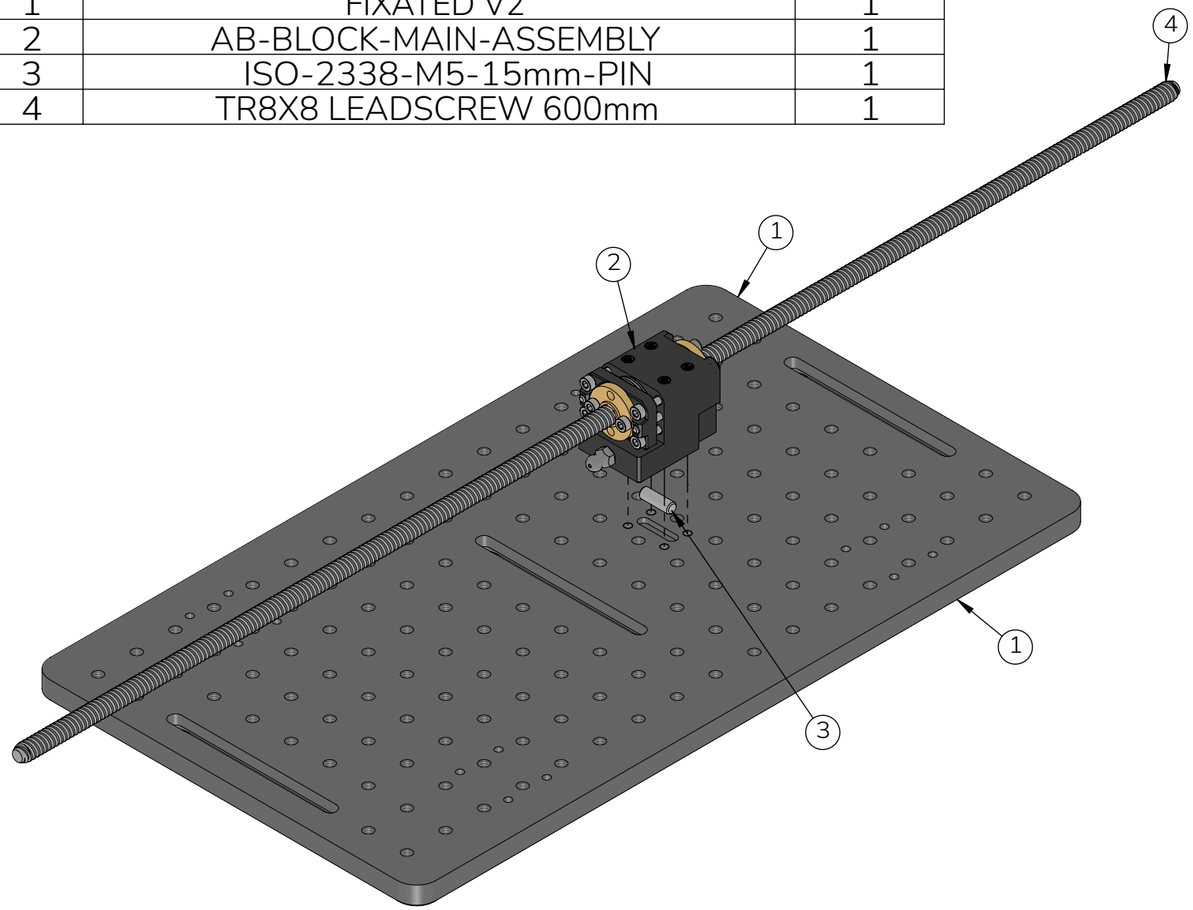
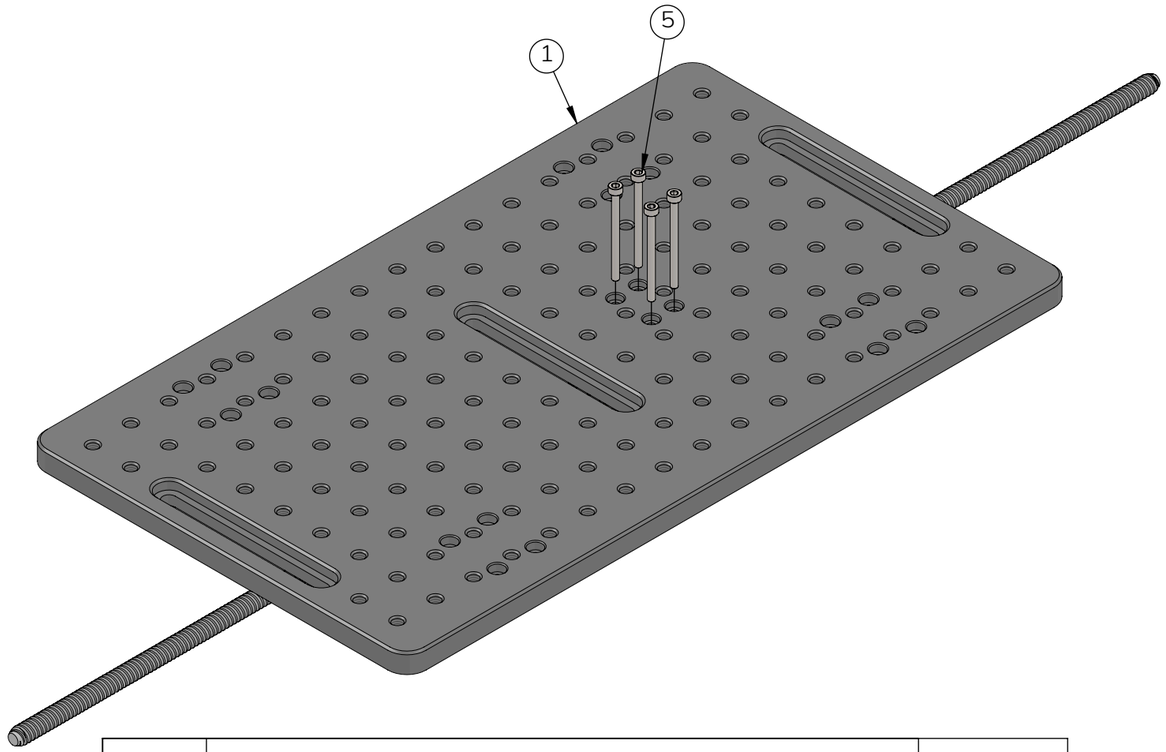
	PART NUMBER	QTY.
1	FIXATED V2	1
2	AB-BLOCK-MAIN-ASSEMBLY	1
3	ISO-2338-M5-15mm-PIN	1
4	TR8X8 LEADSCREW 600mm	1





	PART NUMBER	QTY.
1	FIXATED V2	1
5	ISO-4762-M3X40-SHCS	4

MILEY X AXIS

FIXATED ASSEMBLY

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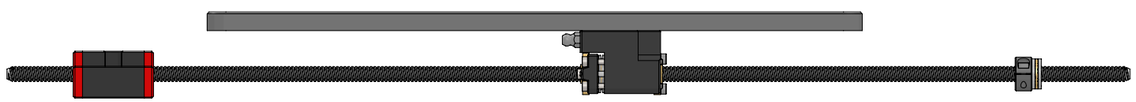
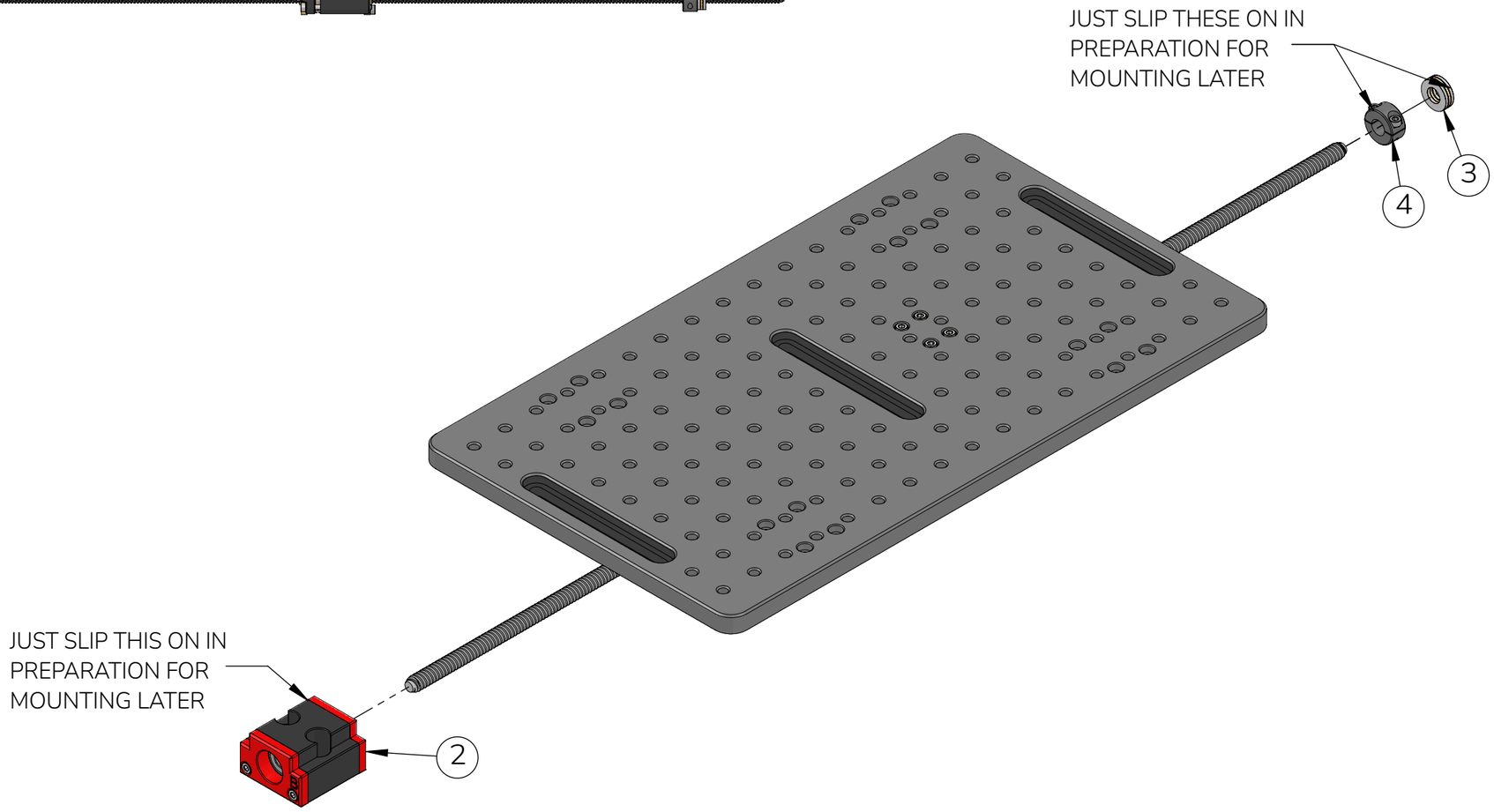
	PART NUMBER	QTY.
1	FIXATED V2	1
2	FLOATING-BEARING-BLOCK-ASSEMBLY	1
3	F8-16M THRUST BEARING	1
4	DOUBLE SPLIT SHAFT COLLAR 8X18X9MM	1





MILEY X AXIS
FIXATED ASSEMBLY

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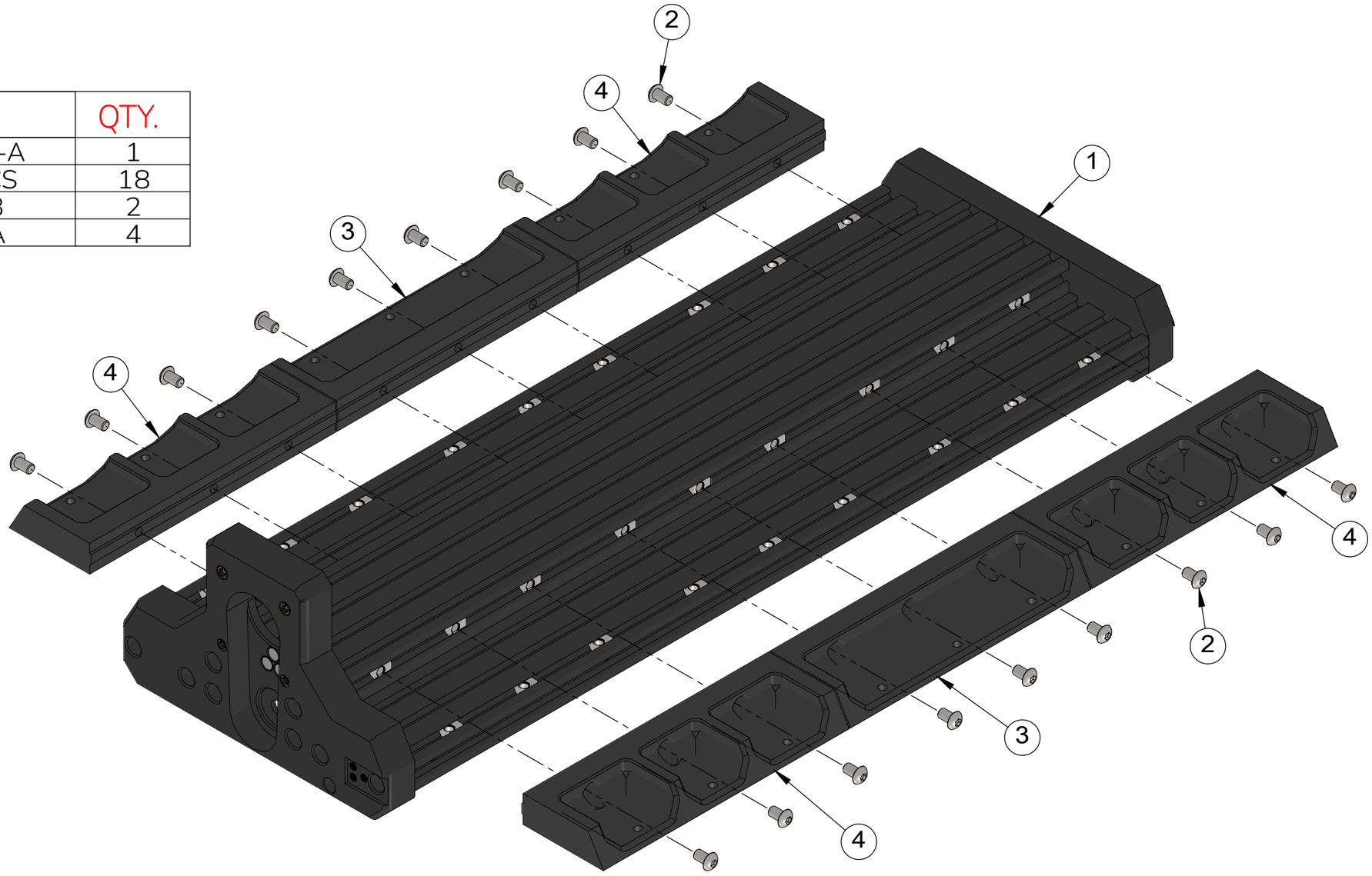
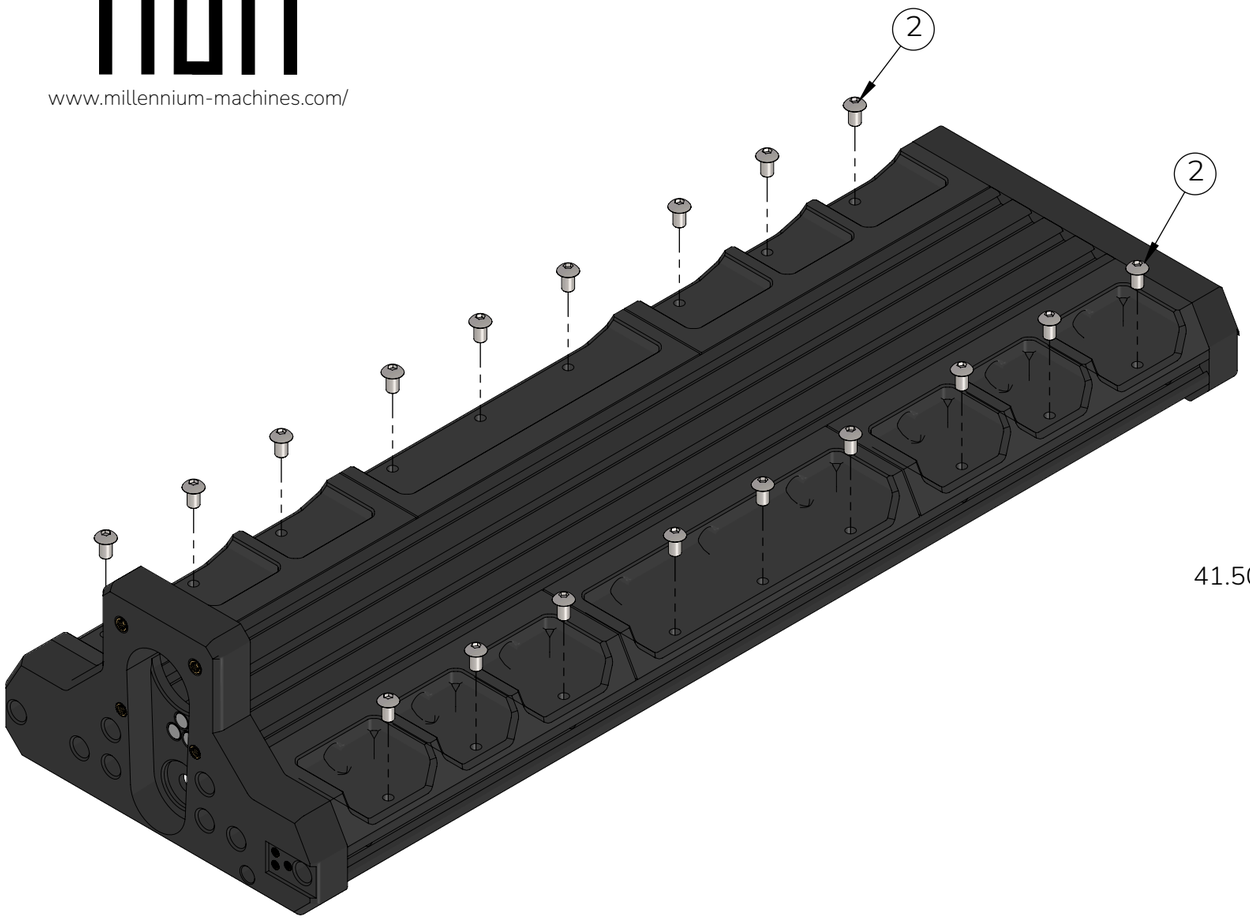
	PART NUMBER	QTY.
1	MILEY-X-AXIS-ASSEMBLY-A	1
2	ISO-7380-M5X10 BHCS	18
3	MILEY TABLE SUPPORT B	2
4	MILEY TABLE SUPPORT A	4



MILEY X AXIS

ASSEMBLY STEP 5

www.millennium-machines/



	PART NUMBER	QTY.
1	MILEY-X-AXIS-ASSEMBLY-A	1
2	ISO-7380-M5X10 BHCS	18

	PART NUMBER	QTY.
1	MILEY-X-AXIS-ASSEMBLY-A	1
5	M5 DROP IN NUT	2

MILEY X AXIS

ASSEMBLY STEP 6

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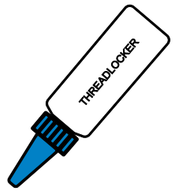
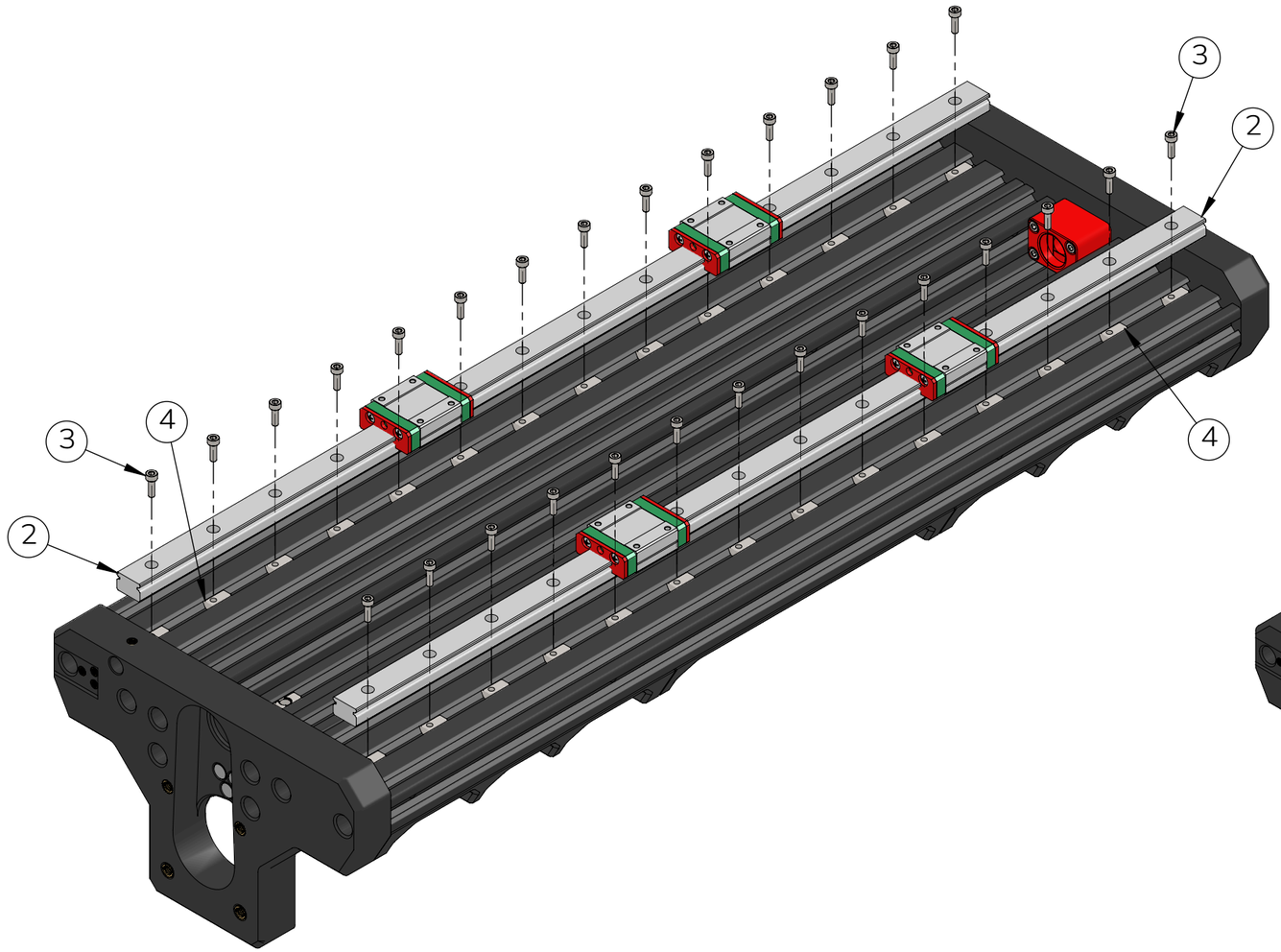
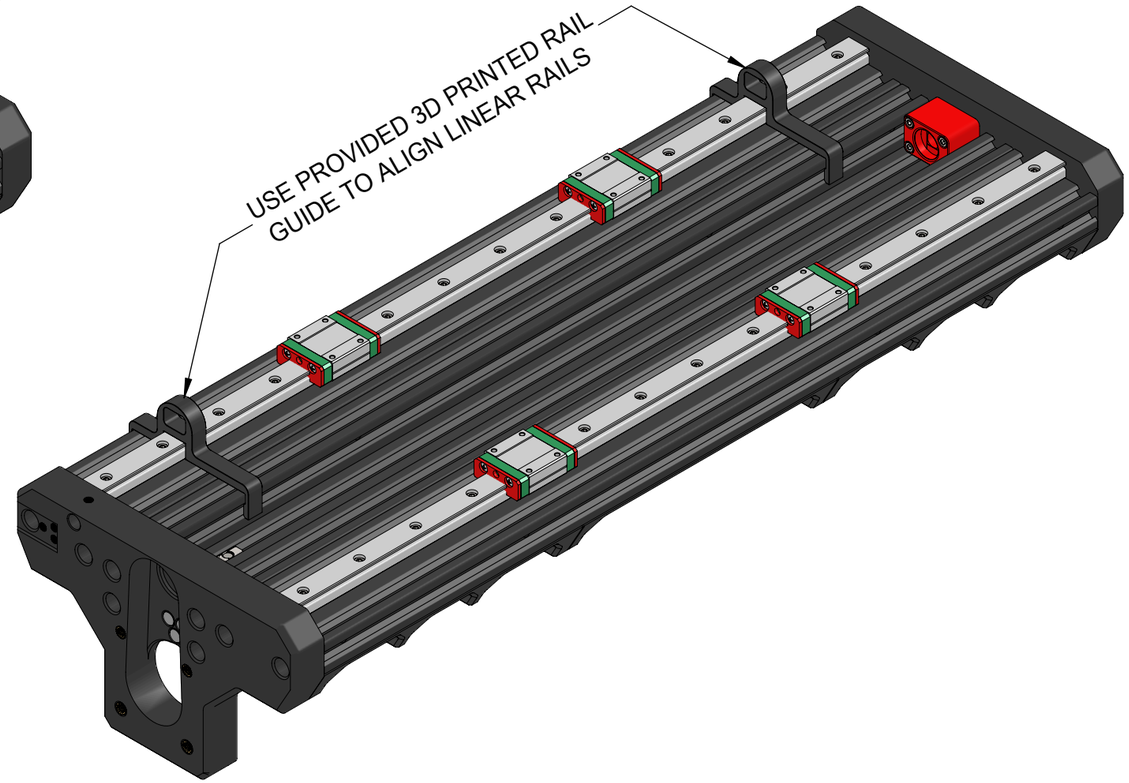
	PART NUMBER	QTY.
1	MILEY-X-AXIS-ASSEMBLY-A	1
2	X-AXIS-LINEAR-RAIL-ASSEMBLY	2
3	ISO-4762-M3X10-SHCS	28
4	M3 DROP IN NUT	28







MILEY X AXIS

ASSEMBLY STEP 7

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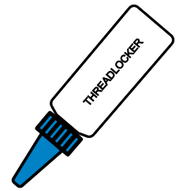
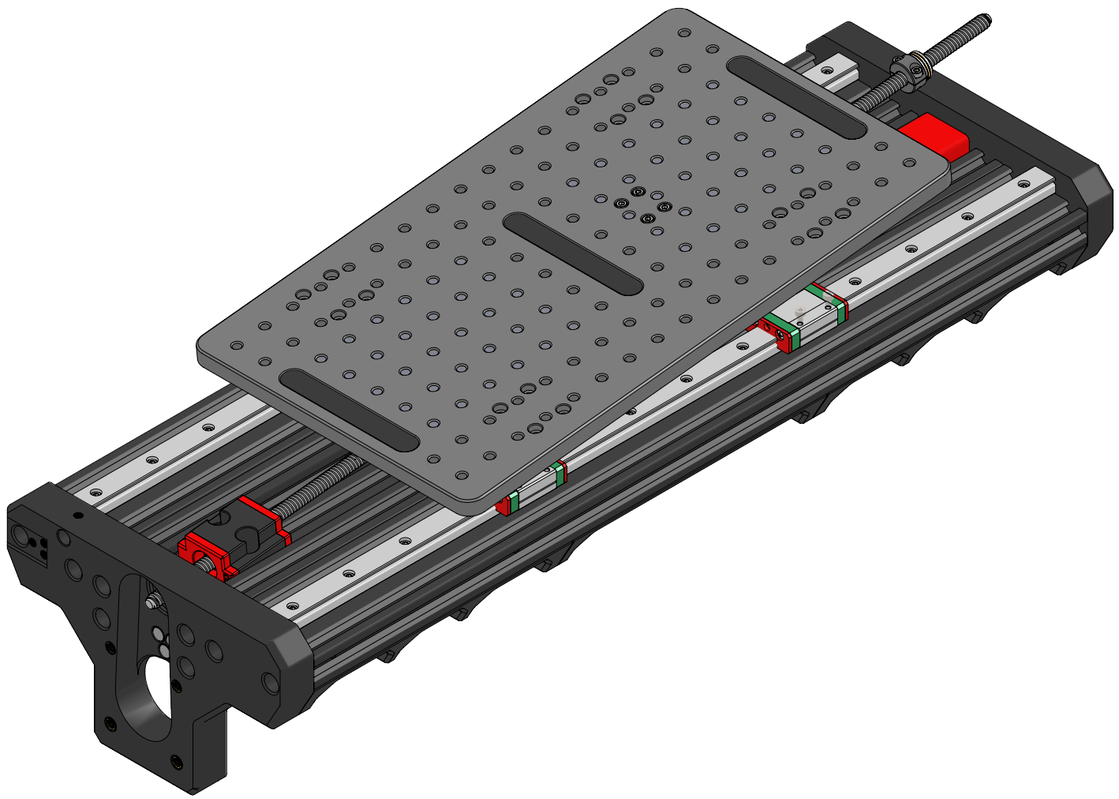
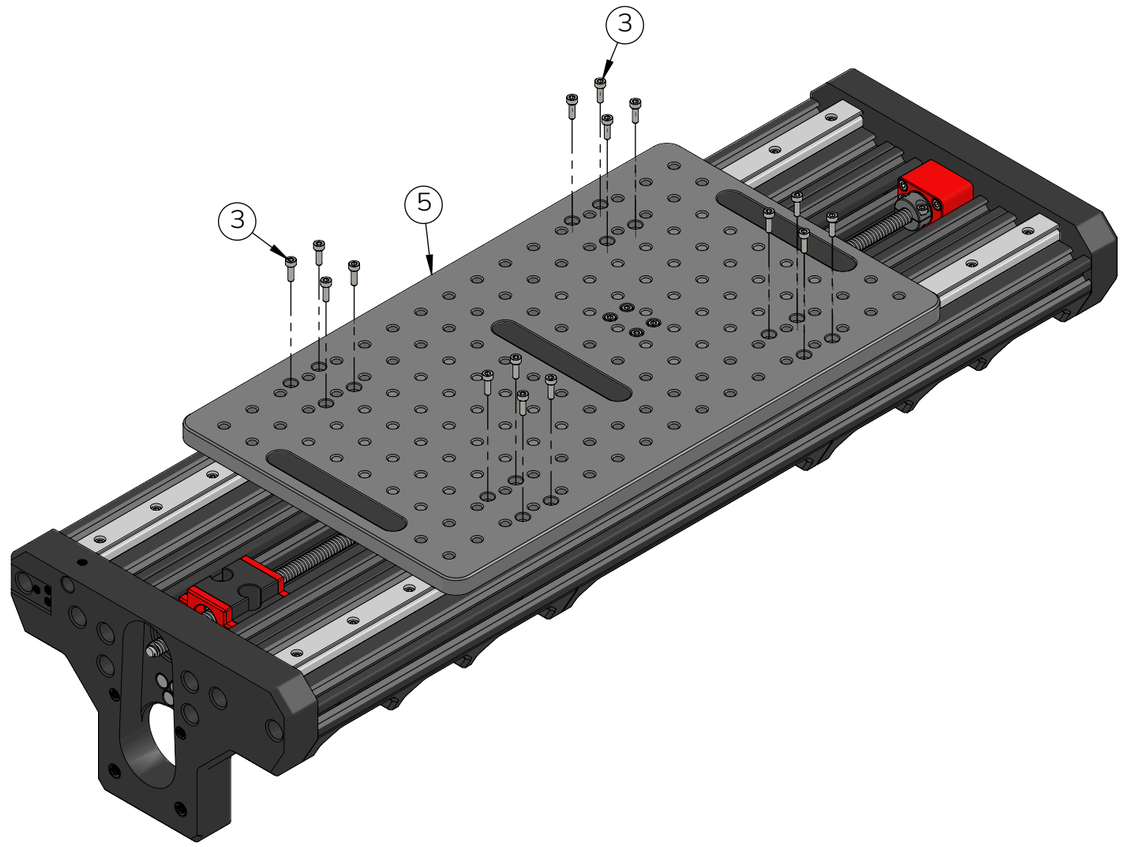
	PART NUMBER	QTY.
1	MILEY-X-AXIS-ASSEMBLY-A	1
3	ISO-4762-M3X10-SHCS	16
5	MILEY-X-AXIS-FIXATED-ASSEMBLY	1







MILEY X AXIS

ASSEMBLY STEP 8

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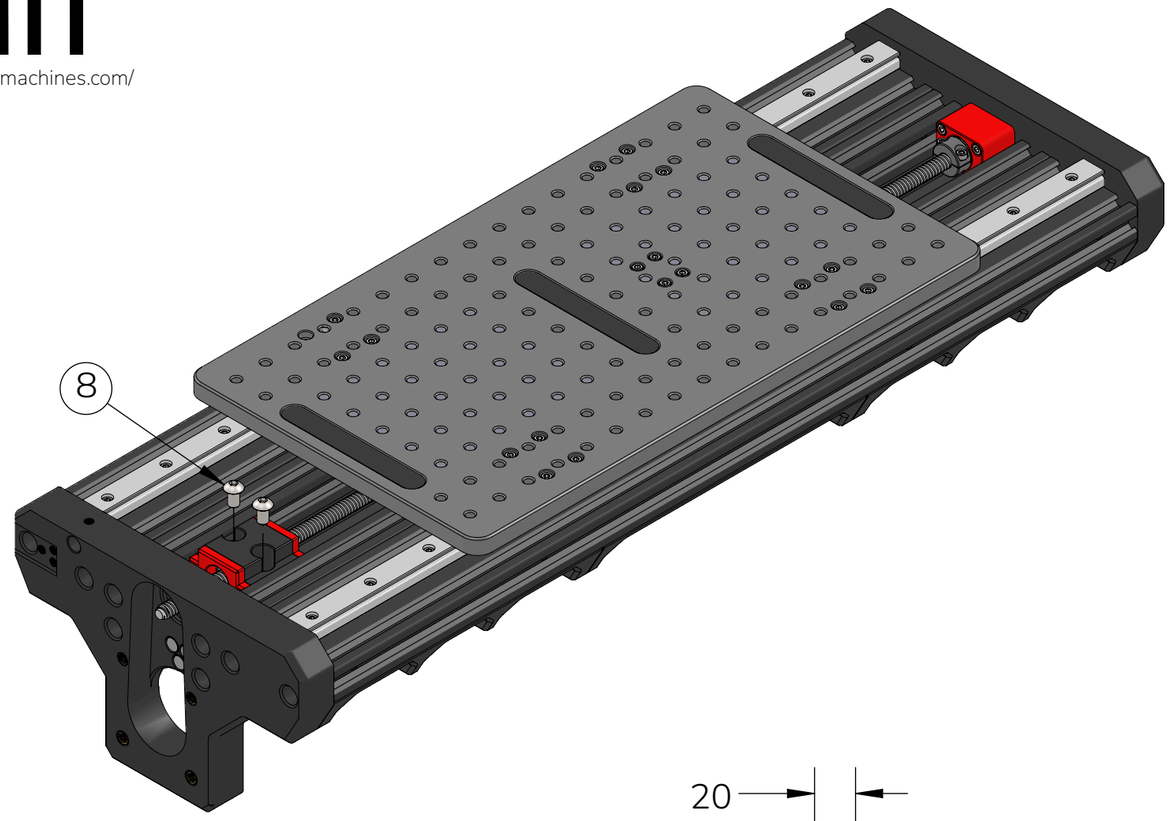
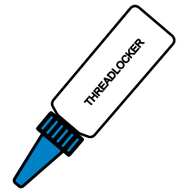
	PART NUMBER	QTY.
1	MILEY-X-AXIS-ASSEMBLY-A	1
5	MILEY-X-AXIS-FIXATED-ASSEMBLY	1
8	ISO-7380-M5X10 BHCS	2
10	F8-16M THRUST BEARING	1
11	DOUBLE SPLIT SHAFT COLLAR 8X18X9MM	1

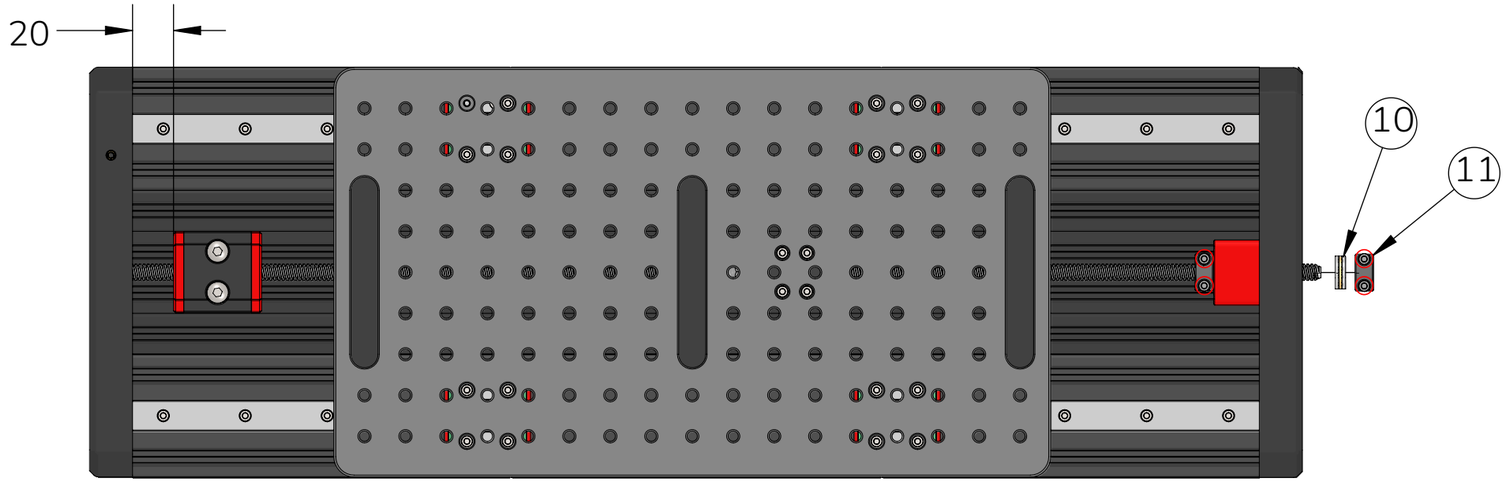




Shaft Collar Bolts

TIGHTEN BOTH
SHAFT COLLARS

APPLY PRESSURE
WHEN TIGHTENING



MILEY X AXIS

ASSEMBLY STEP 9

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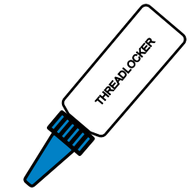
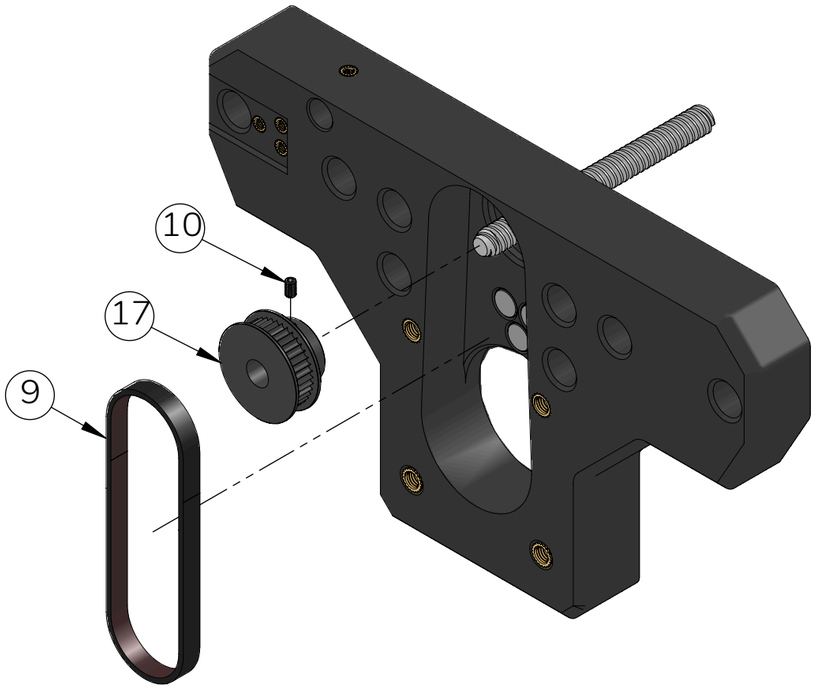
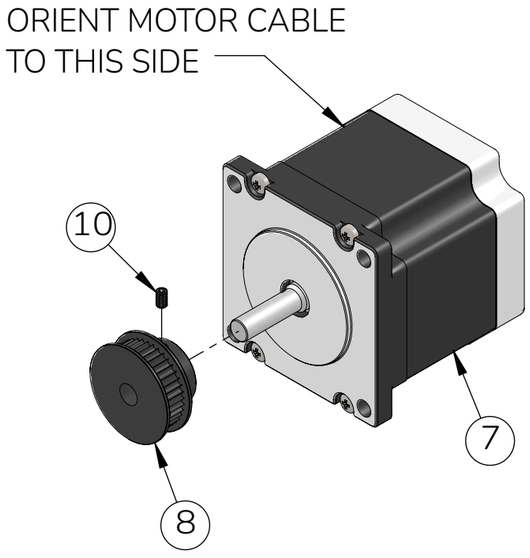
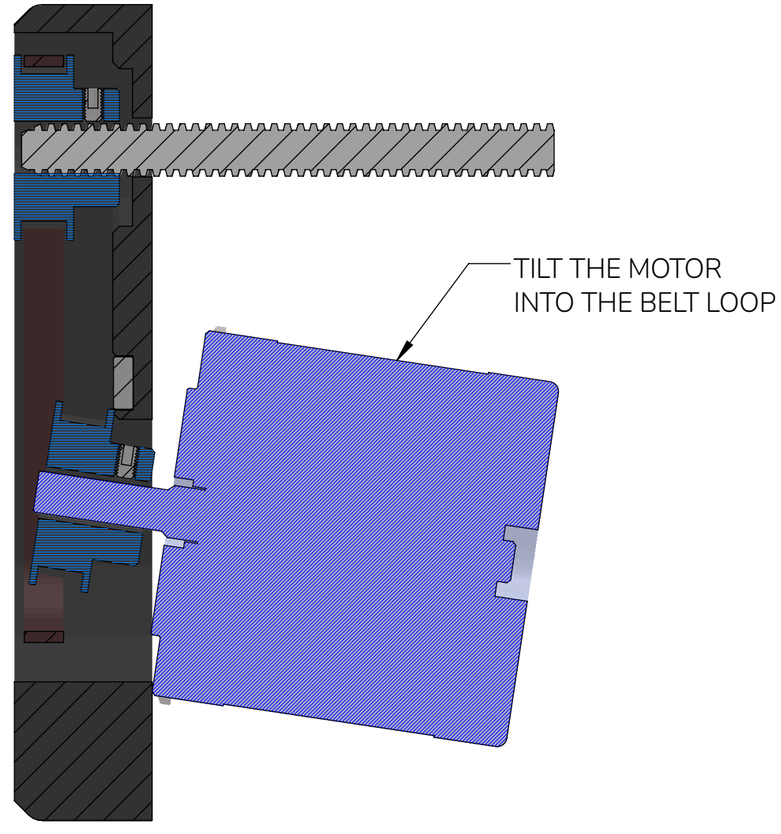
	PART NUMBER	QTY.
1	MILEY-X-AXIS-ASSEMBLY-A	1
7	NEMA 23 - 56mm	1
8	GT2-40T-ID-6.35mm	1
9	120mm-GATE-GT2-CLOSED-LOOP-BELT	1
10	ISO-4026-M3X10-SET SCREW	2
17	GT2-40T-ID-8mm	1



On the Set Screws







MILEY X AXIS

ASSEMBLY STEP 10

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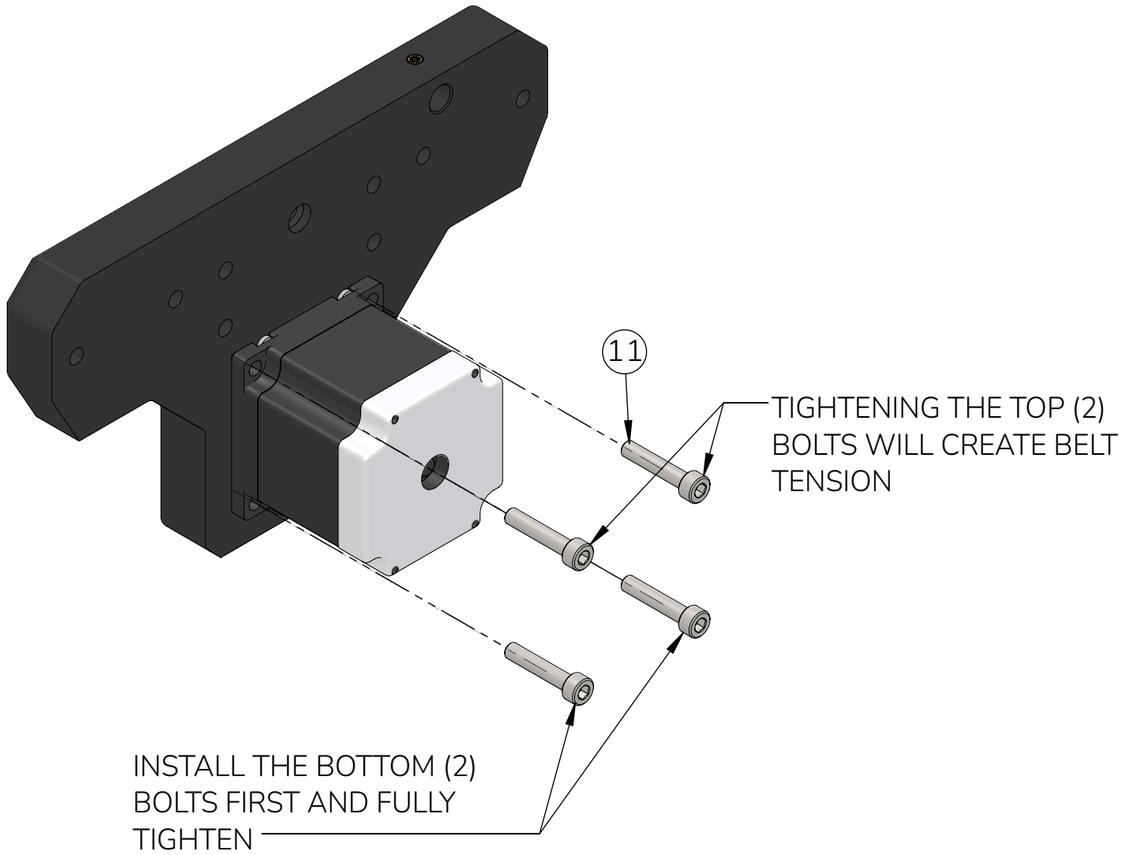
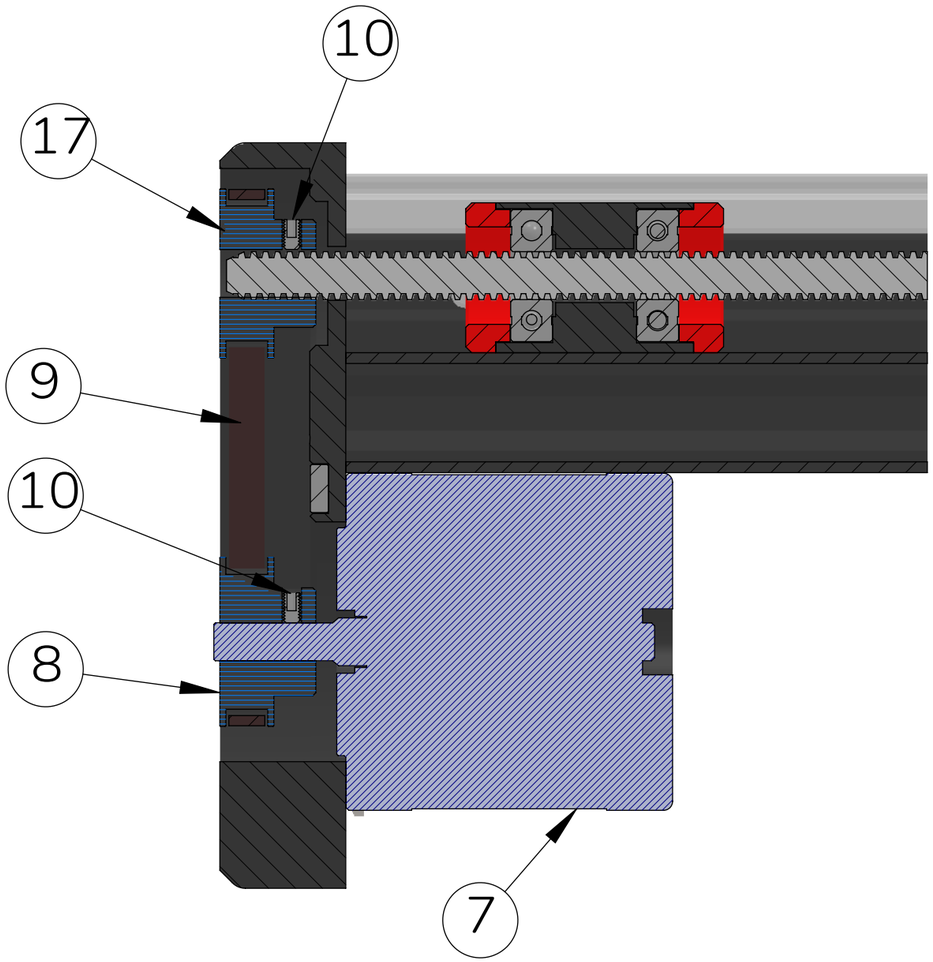
	PART NUMBER	QTY.
1	MILEY-X-AXIS-ASSEMBLY-A	1
7	NEMA 23 - 56mm	1
8	GT2-40T-ID-6.35mm	2
9	120mm-GATE-GT2-CLOSED-LOOP-BELT	1
10	ISO-4026-M3X10-SET SCREW	2
11	ISO-4762-M5X25-SHCS	4





MILEY X AXIS

ASSEMBLY STEP 11

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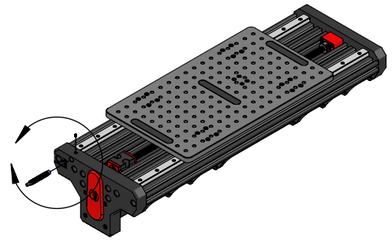

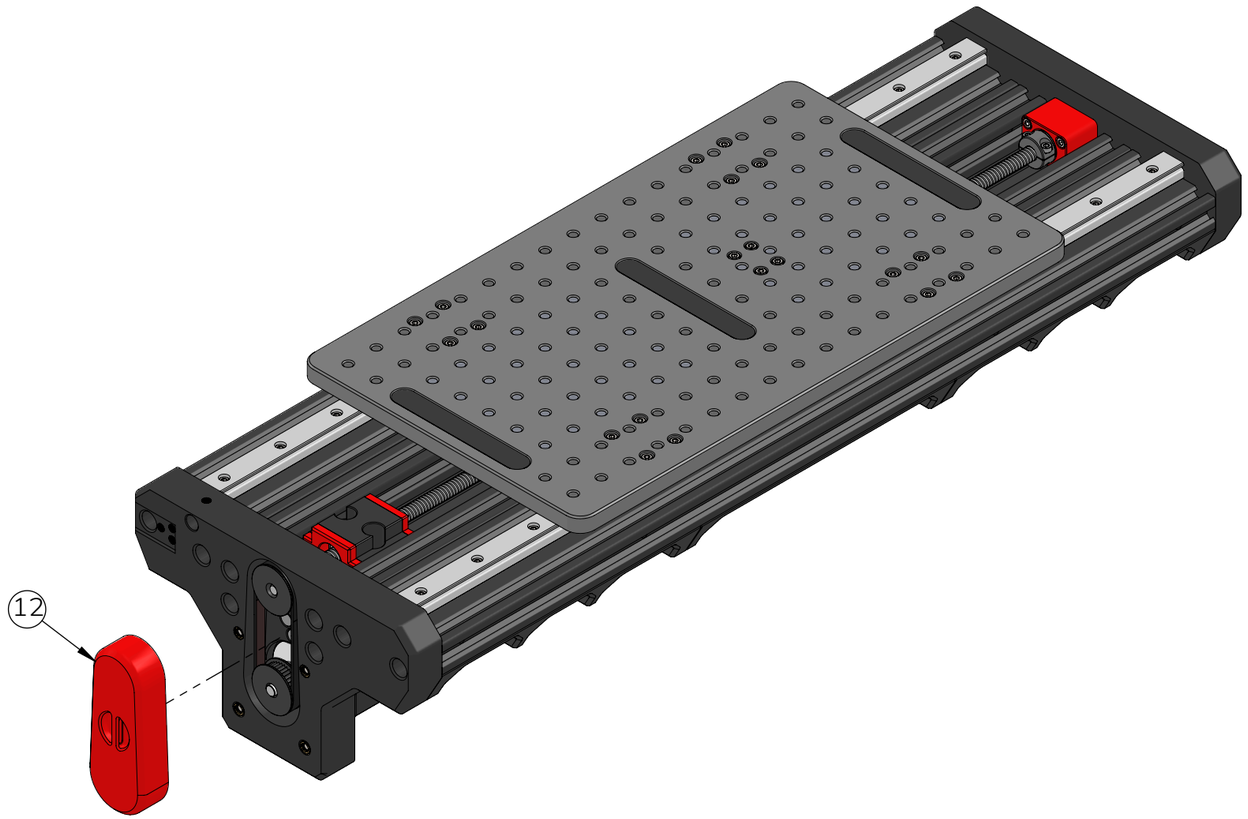

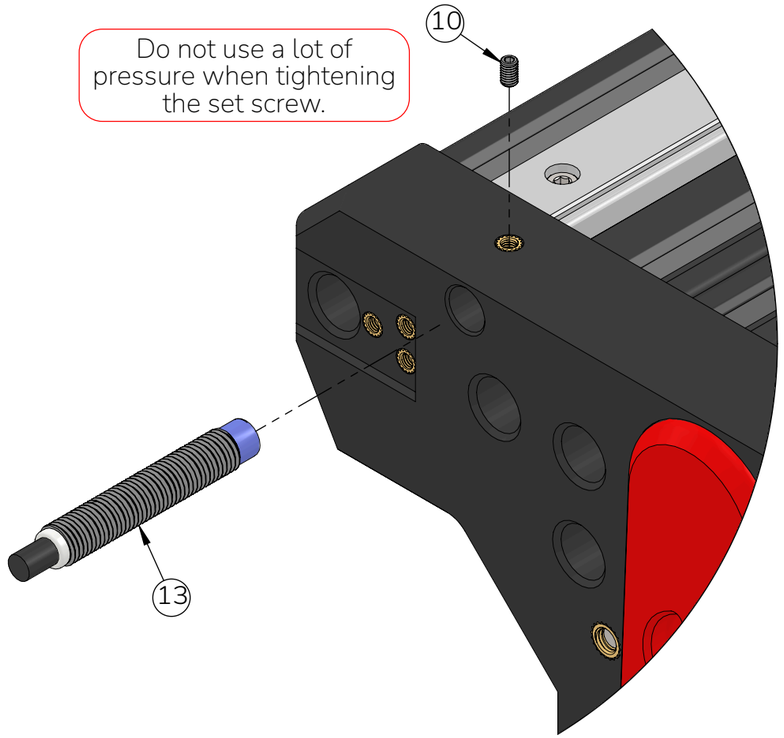

	PART NUMBER	QTY.
1	MILEY-X-AXIS-ASSEMBLY-A	1
12	MILEY-X-AXIS-BELT-COVER-ASSEMBLY	1

	PART NUMBER	QTY.
1	MILEY-X-AXIS-ASSEMBLY-A	1
10	ISO-4026-M3X10-SET SCREW	1
13	LJ8A3-2-Z/AY	1

MILEY X AXIS

ASSEMBLY STEP 12

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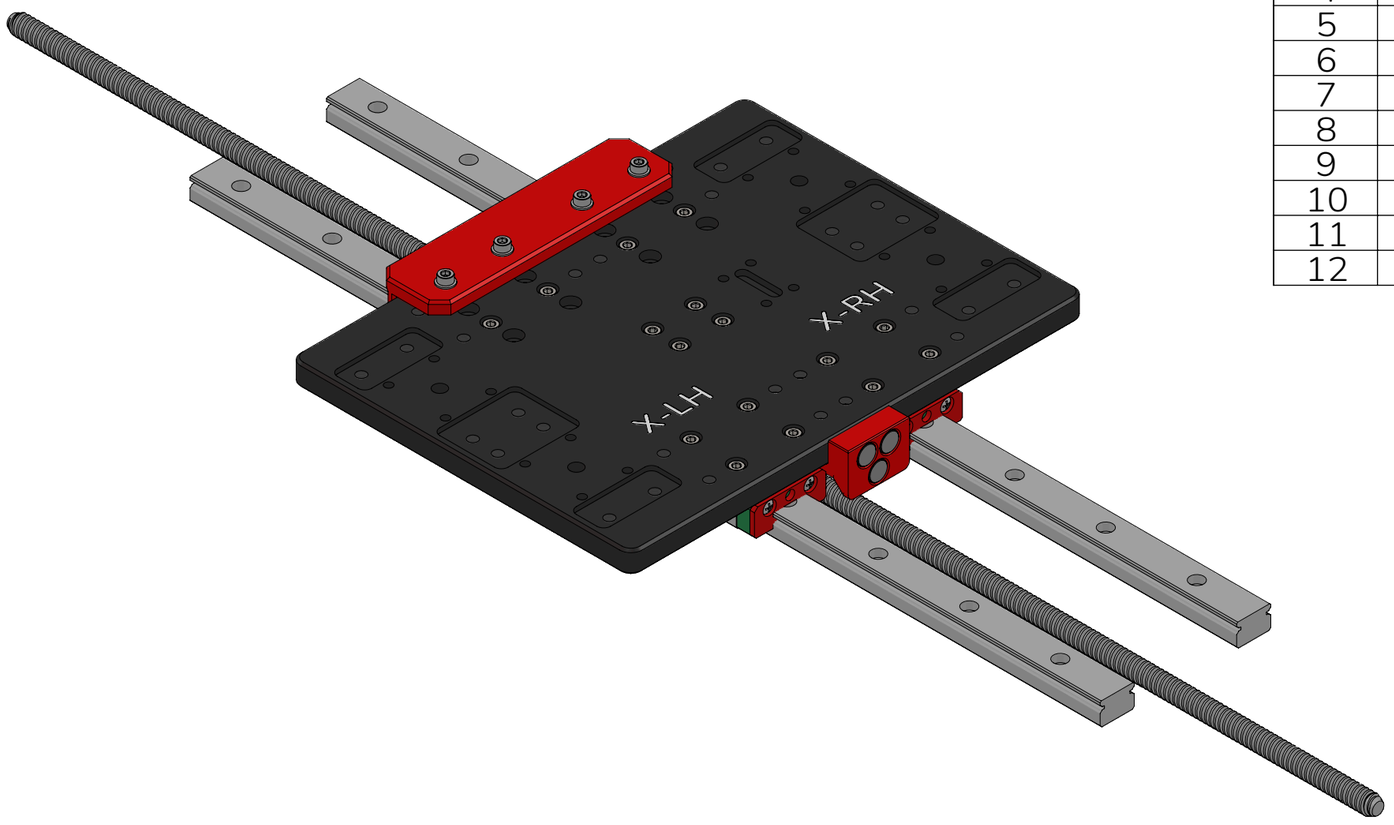

	PART NUMBER	QTY.
1	X-Y JOINER-PLATE	1
2	AB-BLOCK-MAIN-ASSEMBLY	1
3	ISO-2338-M5-15mm-PIN	1
4	ISO-4762-M3X40-SHCS	4
5	Y-AXIS-LINEAR-RAIL-ASSEMBLY	2
6	WAY-COVER-MAGNET-LOCK-ASSEMBLY	1
7	BELLOW PLATE MAG MOUNT ASSEMBLY	1
8	ISO-10642-M5X12MM-FHCS	4
9	ISO-4762-M3X10-SHCS	12
10	ISO-4762-M3X20-SHCS	4
11	ISO-7089 M3 WASHER	4
12	TR8X8 LEADSCREW 600mm	1

MILEY XY JOINER PLATE

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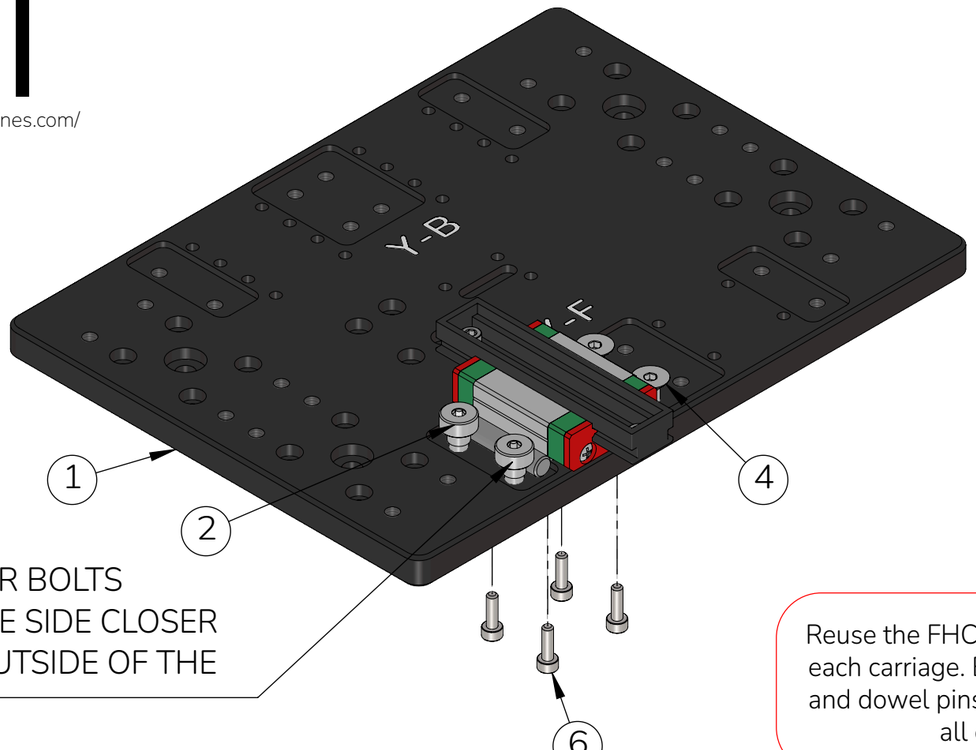
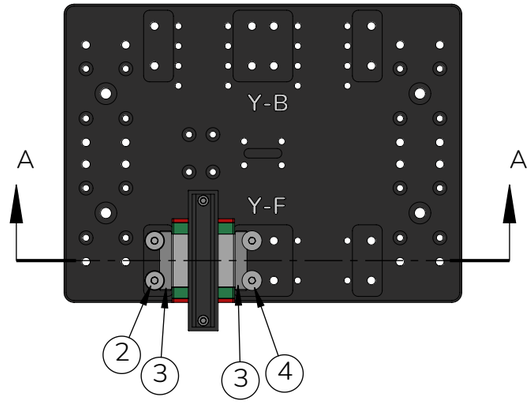
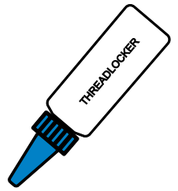
	PART NUMBER	QTY.
1	X-Y JOINER-PLATE	1
2	ISO-7379-M5X6MM-THREAD-M6X4MM-SHOULDER	2
3	ISO-2338-M6-30mm-PIN	2
4	ISO-10642-M5X12MM-FHCS	2
5	CARRIAGE RETAINER & BEARING ASSEMBLY	1
6	ISO-4762-M3X10-SHCS	4

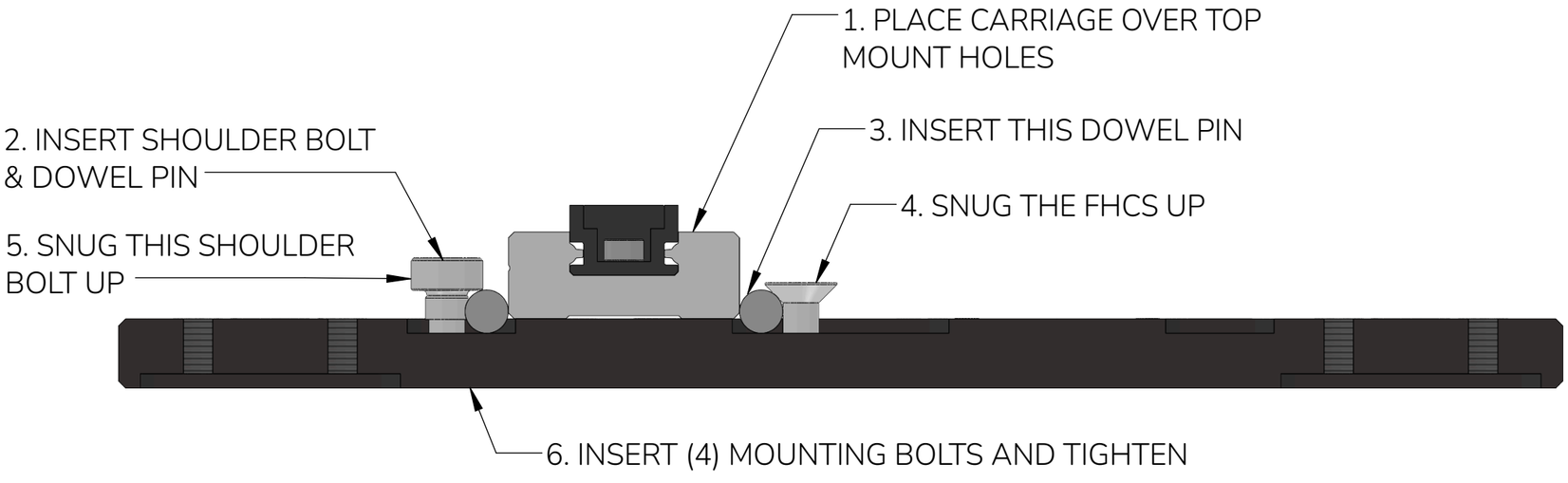


Reuse the FHCS bolts & shoulder bolts for each carriage. Be sure to remove the bolts and dowel pins completely after installing all of the carriages.





On Part #6 Only



MILEY XY JOINER PLATE

ASSEMBLY STEP 1

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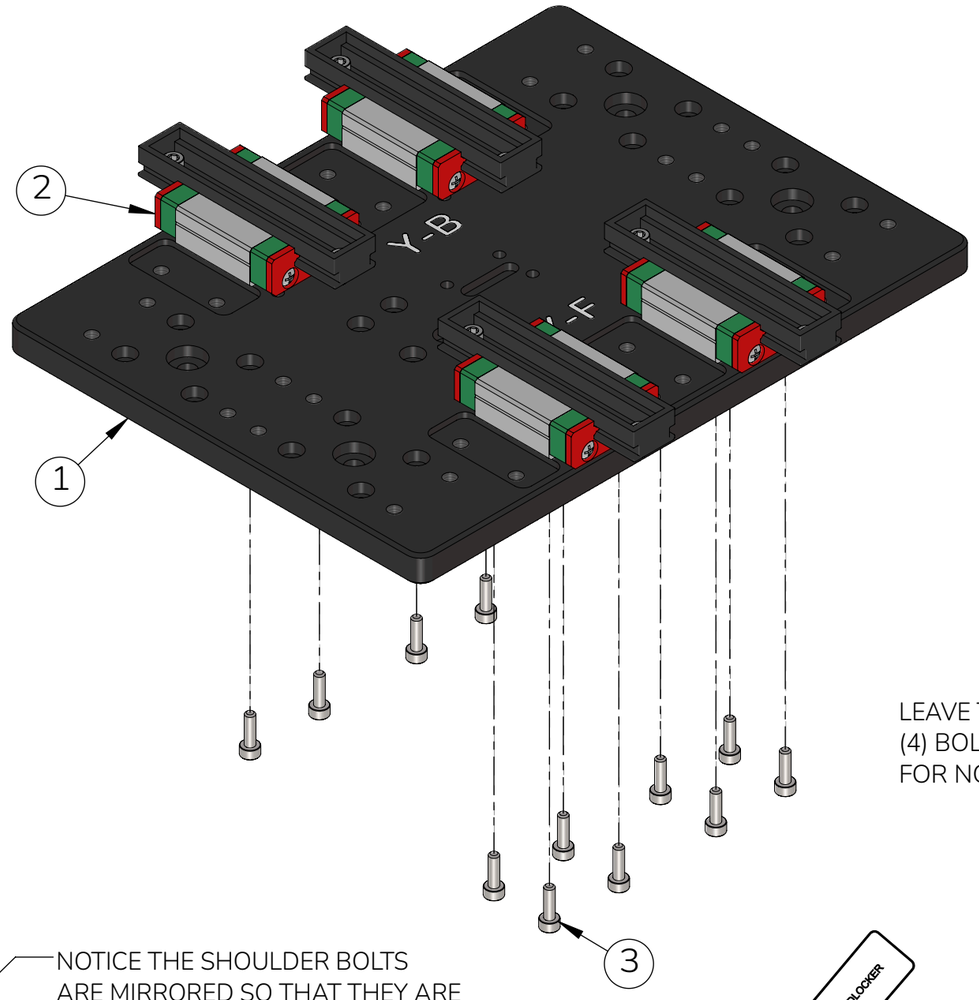
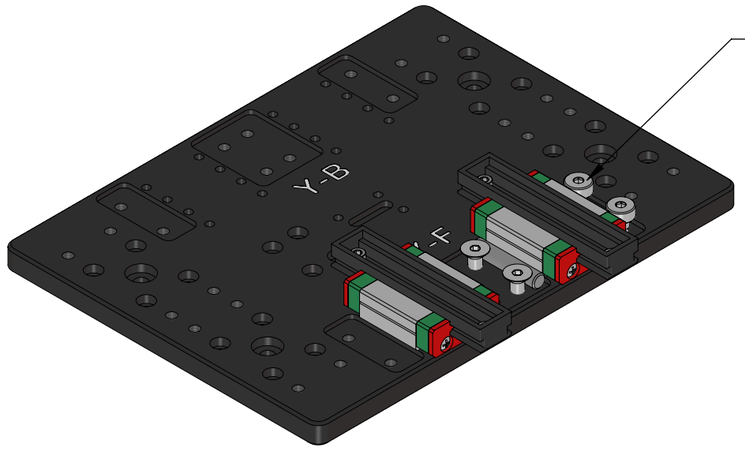
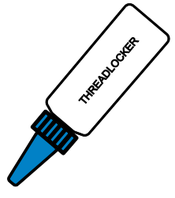
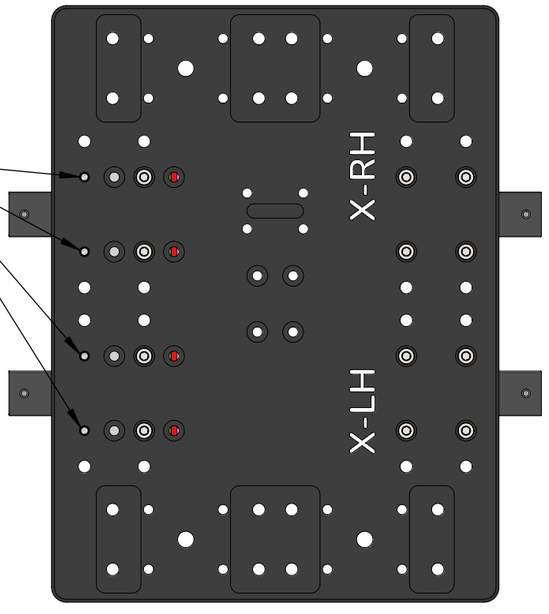




NOTICE THE SHOULDER BOLTS ARE MIRRORED SO THAT THEY ARE CLOSER TO THE OUTSIDE OF THE PLATE



LEAVE THESE (4) BOLTS OUT FOR NOW



	PART NUMBER	QTY.
1	X-Y JOINER-PLATE	1
2	CARRIAGE RETAINER & BEARING ASSEMBLY	4
3	ISO-4762-M3X10-SHCS	12

MILEY XY JOINER PLATE

ASSEMBLY STEP 2

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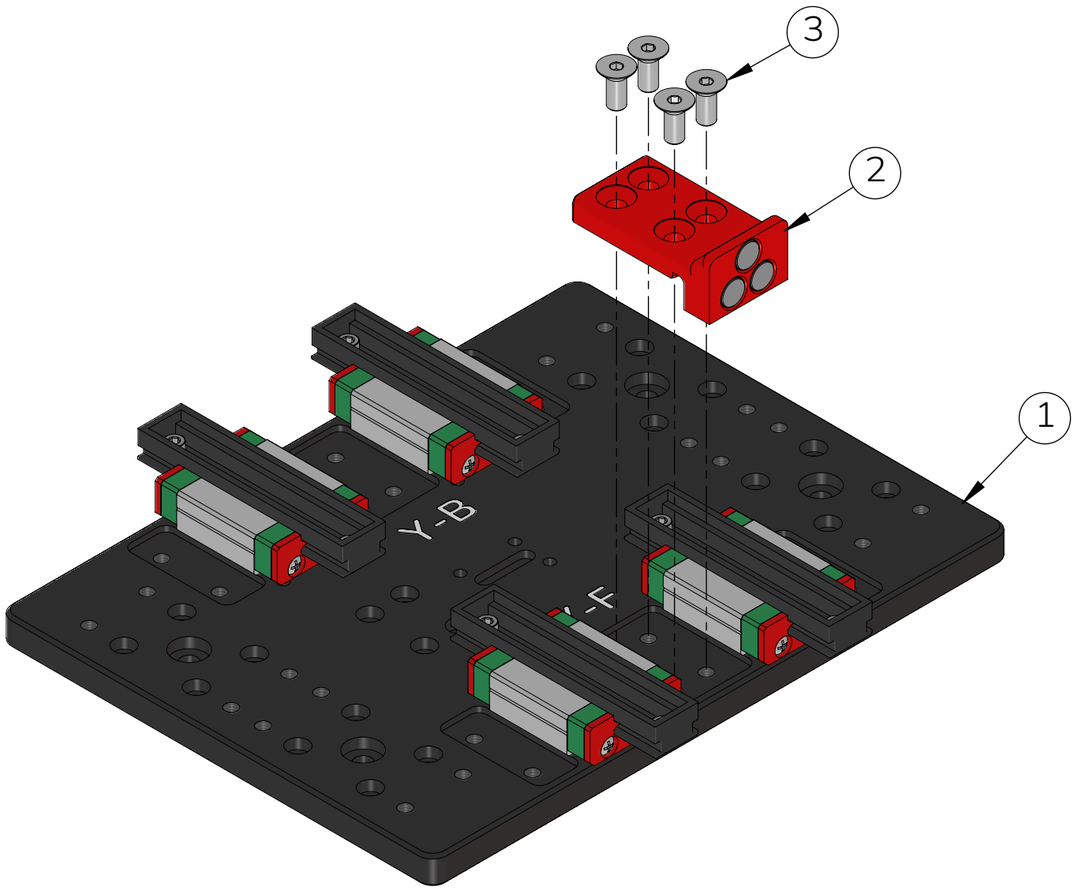
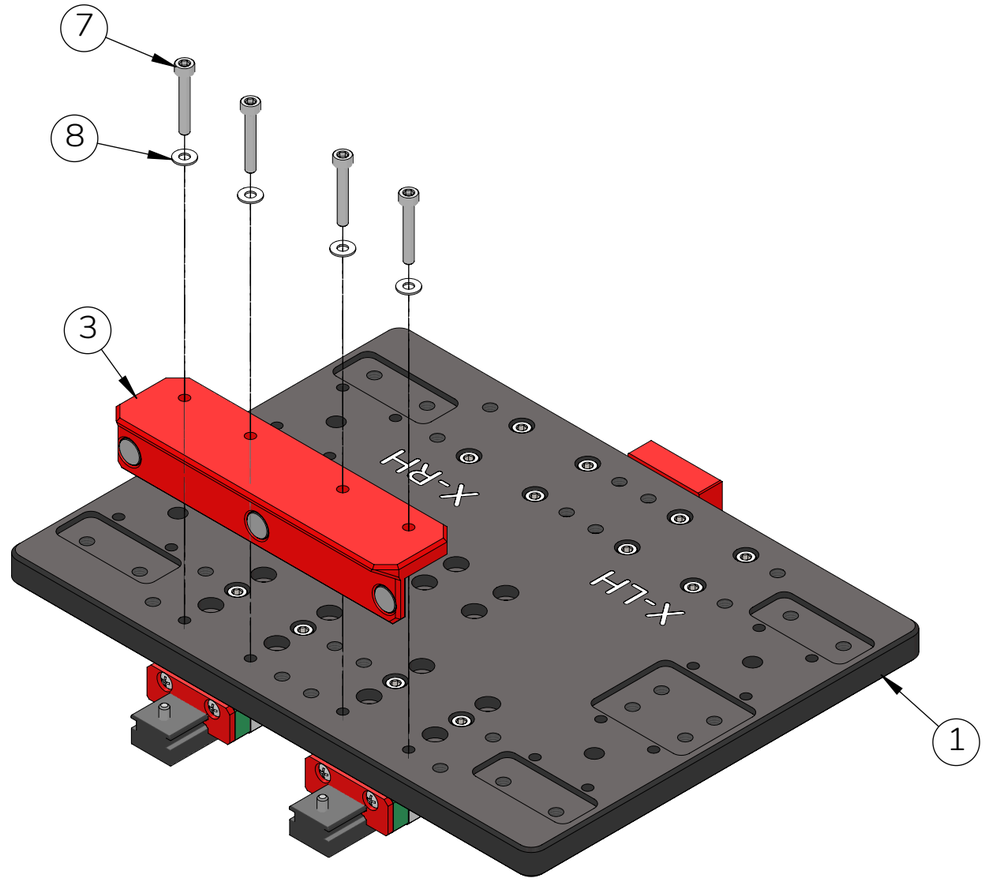


	PART NUMBER	QTY.
1	X-Y JOINER-PLATE	1
2	WAY-COVER-MAGNET-LOCK-ASSEMBLY	1
3	ISO-10642-M5X12MM-FHCS	4



	PART NUMBER	QTY.
1	X-Y JOINER-PLATE	1
3	BELLOW PLATE MAG MOUNT ASSEMBLY	1
7	ISO-4762-M3X20-SHCS	4
8	ISO-7089 M3 WASHER	4

MILEY XY JOINER PLATE

ASSEMBLY STEP 3

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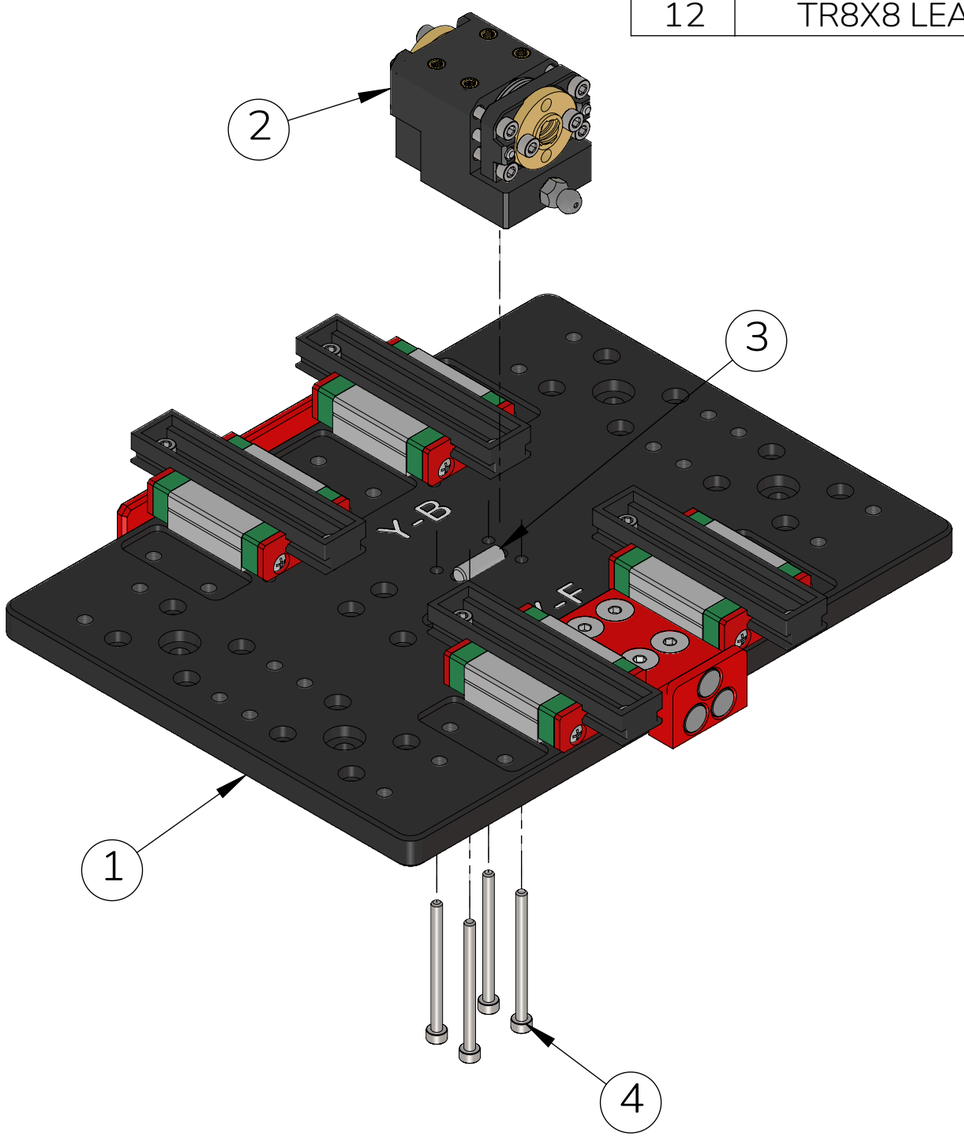
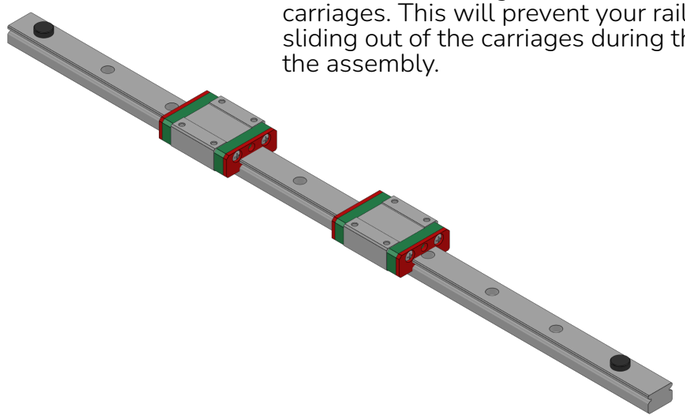
	PART NUMBER	QTY.
1	X-Y JOINER-PLATE	1
2	AB-BLOCK-MAIN-ASSEMBLY	1
3	ISO-2338-M5-15mm-PIN	1
4	ISO-4762-M3X40-SHCS	4
12	TR8X8 LEADSCREW 600mm	1



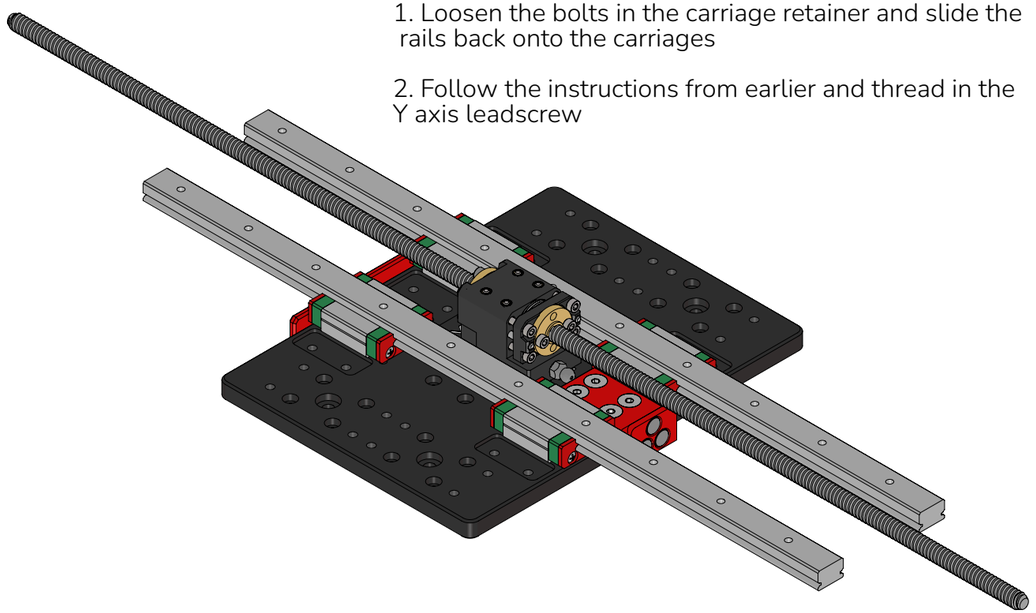
Linear Rail Carriage Safety

If your rails came with rubber stoppers, stick them in after sliding the rails back onto the carriages. This will prevent your rails from sliding out of the carriages during the rest of the assembly.



1. Loosen the bolts in the carriage retainer and slide the rails back onto the carriages

2. Follow the instructions from earlier and thread in the Y axis leadscrew



MILEY XY JOINER PLATE

ASSEMBLY STEP 4

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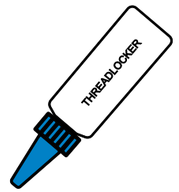
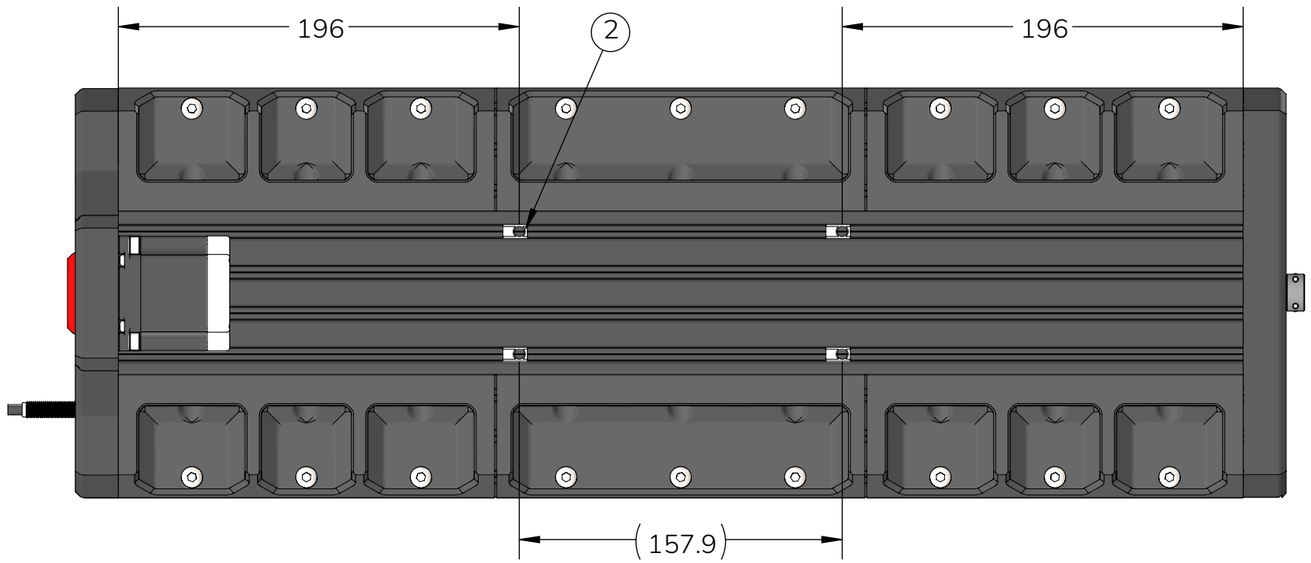
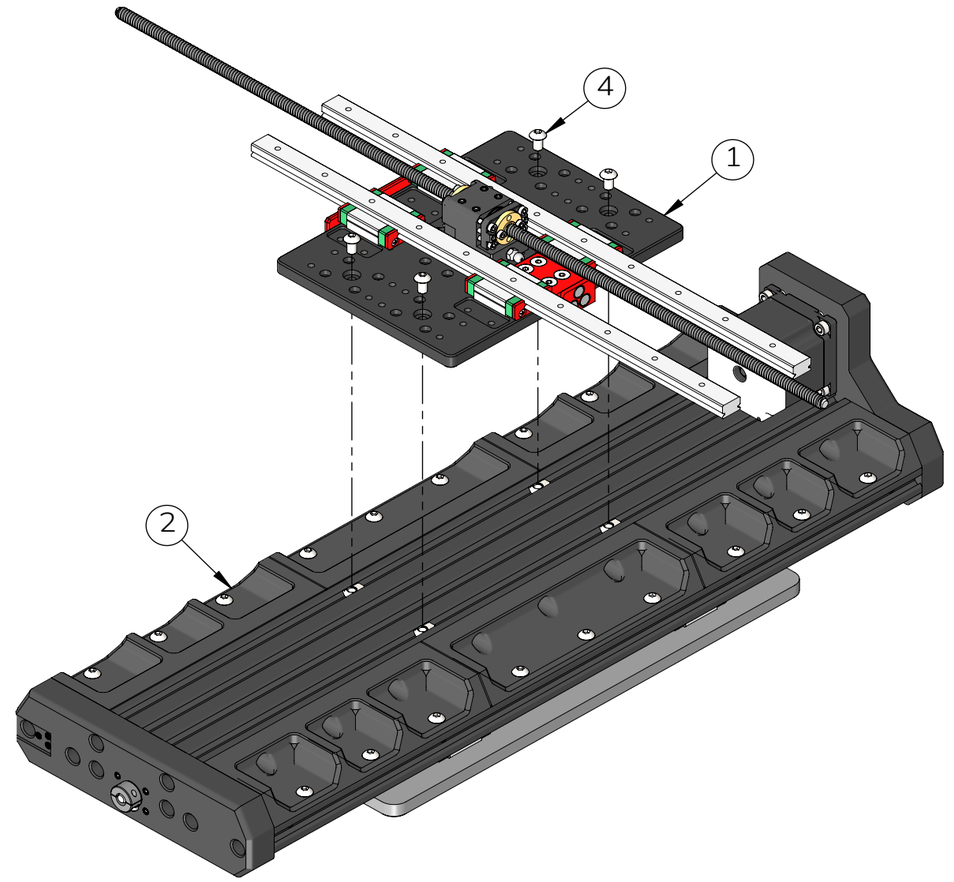
	PART NUMBER	QTY.
1	MILEY-X-AXIS-ASSEMBLY	1
2	M5 DROP IN NUT	4





	PART NUMBER	QTY.
1	MILEY-XY-JOINER-PLATE-ASSEMBLY	1
2	MILEY-X-AXIS-ASSEMBLY	1
4	ISO-7380-M5X10 BHCS	4



MILEY XY SUB-ASSEMBLY

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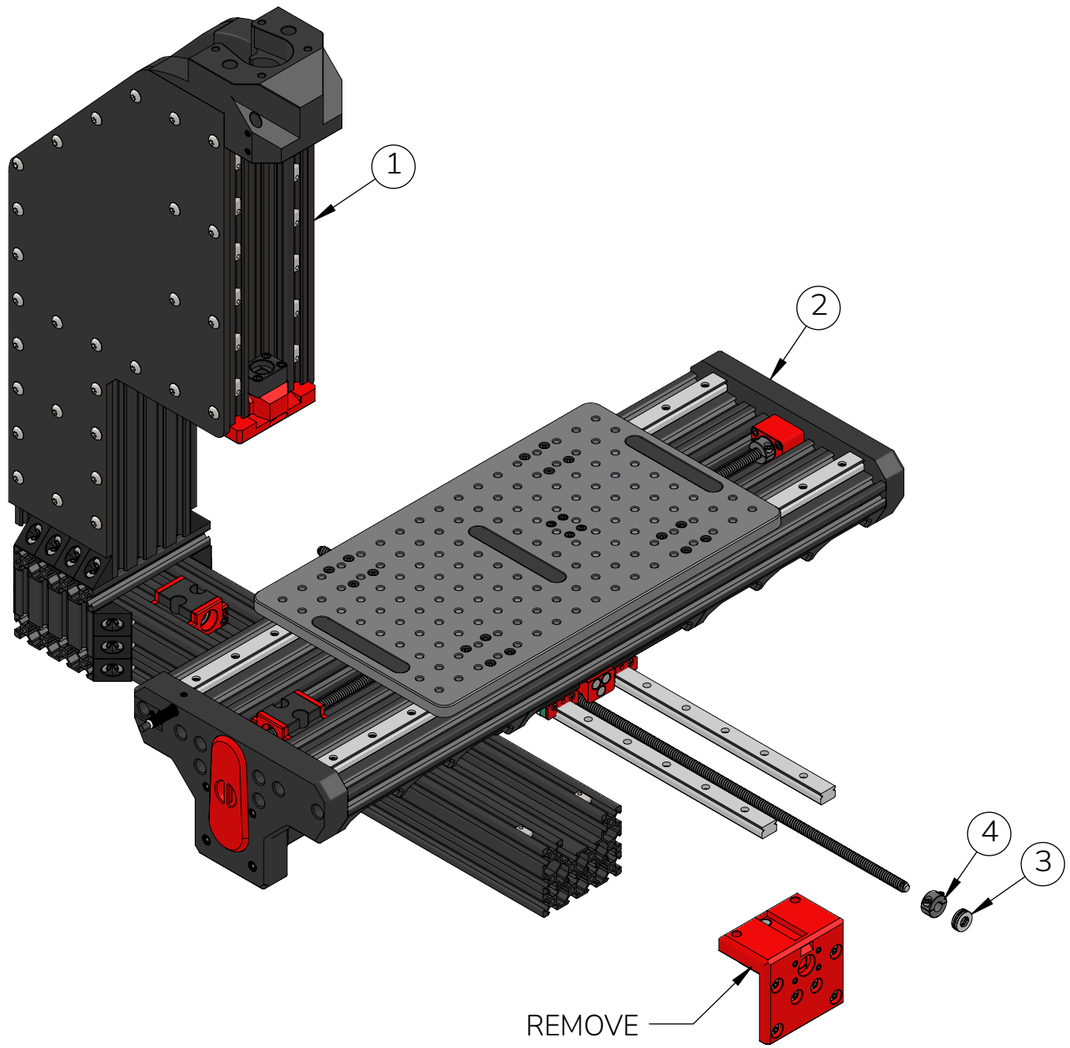
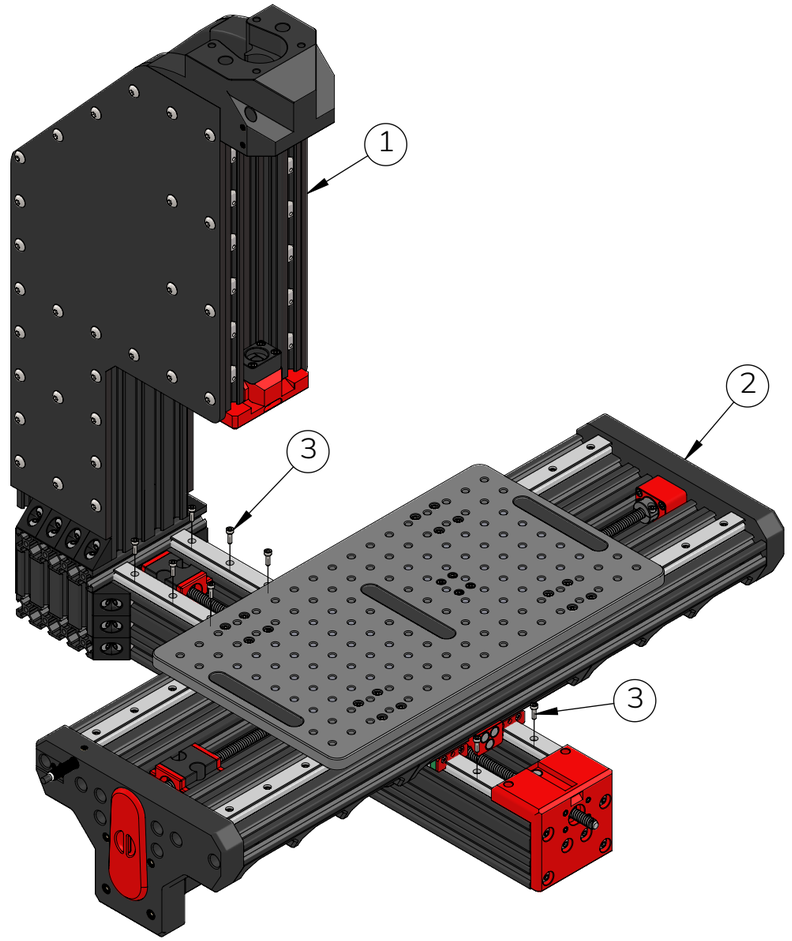
	PART NUMBER	QTY.
1	ZY-ASSEMBLY	1
2	MILEY-XY-SUBASSEMBLY	1
3	F8-16M THRUST BEARING	1
4	DOUBLE SPLIT SHAFT COLLAR 8X18X9MM	1





	PART NUMBER	QTY.
1	ZY-ASSEMBLY	1
2	MILEY-XY-SUBASSEMBLY	1
3	ISO-4762-M3X10-SHCS	20

MILEY SUB-ASSEMBLY A

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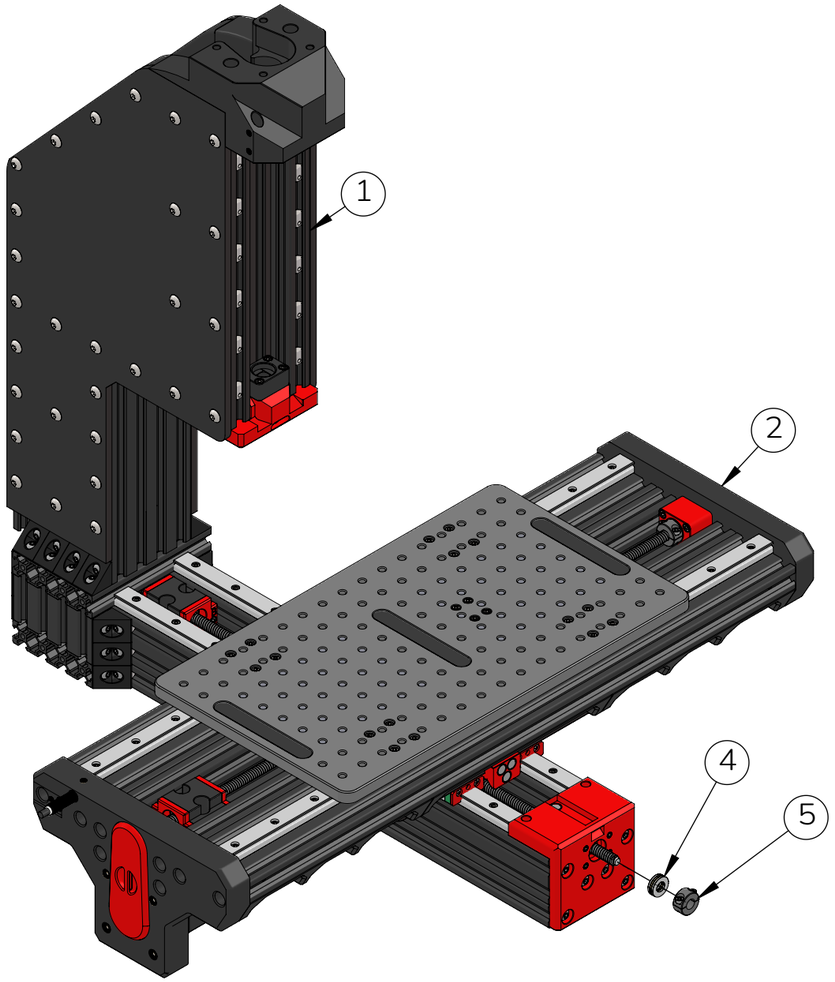
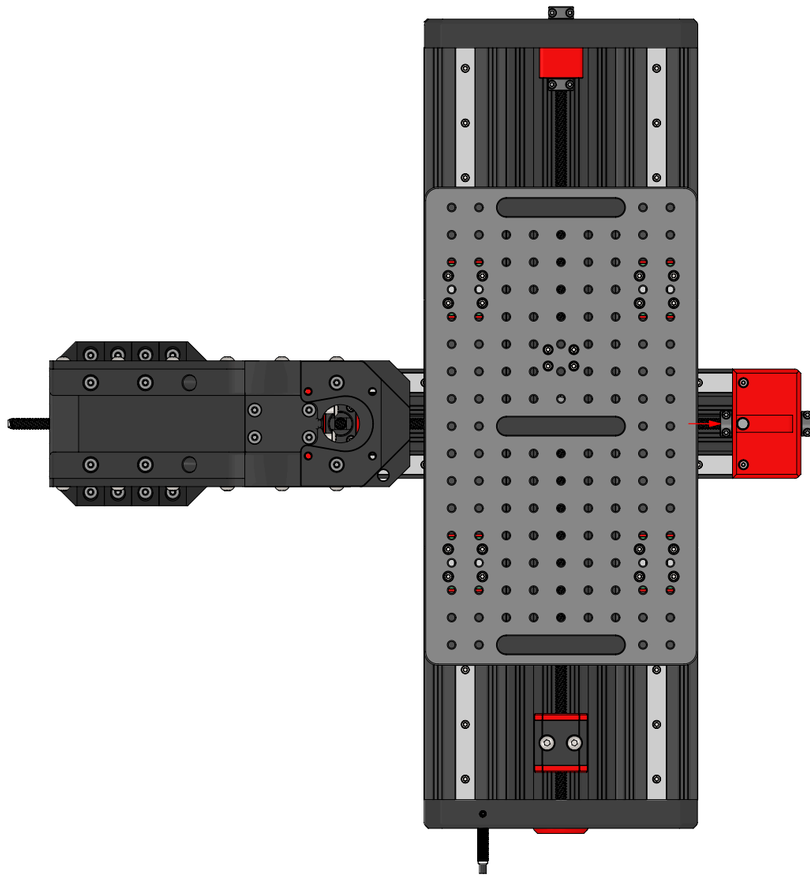
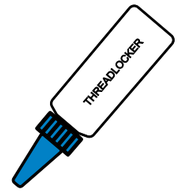
This concludes the Miley specific assembly instructions. The remaining steps will show Milo but will still apply to Miley.

	PART NUMBER	QTY.
1	ZY-ASSEMBLY	1
2	MILEY-XY-SUBASSEMBLY	1
4	F8-16M THRUST BEARING	1
5	DOUBLE SPLIT SHAFT COLLAR 8X18X9MM	1







Shaft Collar Bolts

MILEY SUB-ASSEMBLY A

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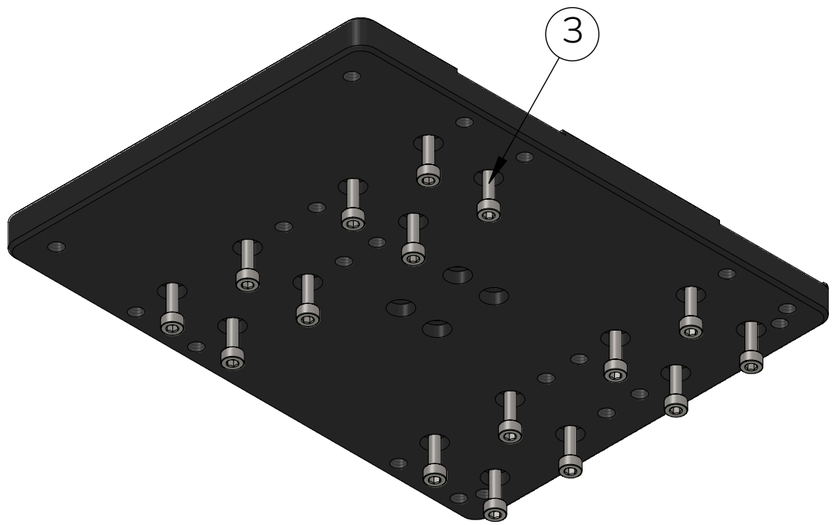
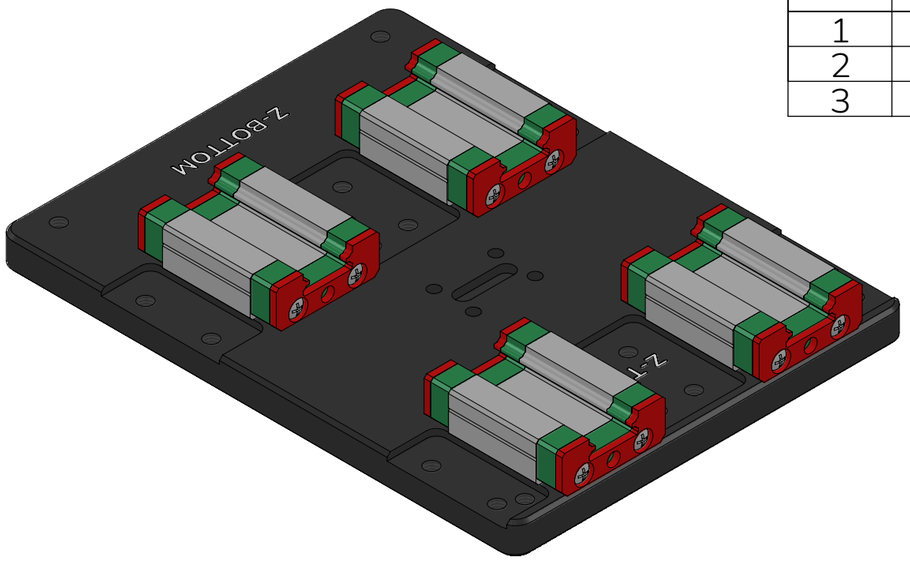
Mount Z Axis Carriages

Follow the instructions from earlier and mount and tram the Z Axis Carriages to the Z Axis Mount Plate





	PART NUMBER	QTY.
1	Z MOUNTING PLATE	1
2	Z-AXIS-LINEAR-RAIL-ASSEMBLY	2
3	ISO-4762-M3X10-SHCS	16

Z MOUNT PLATE ASSEMBLY

ASSEMBLY STEP 1

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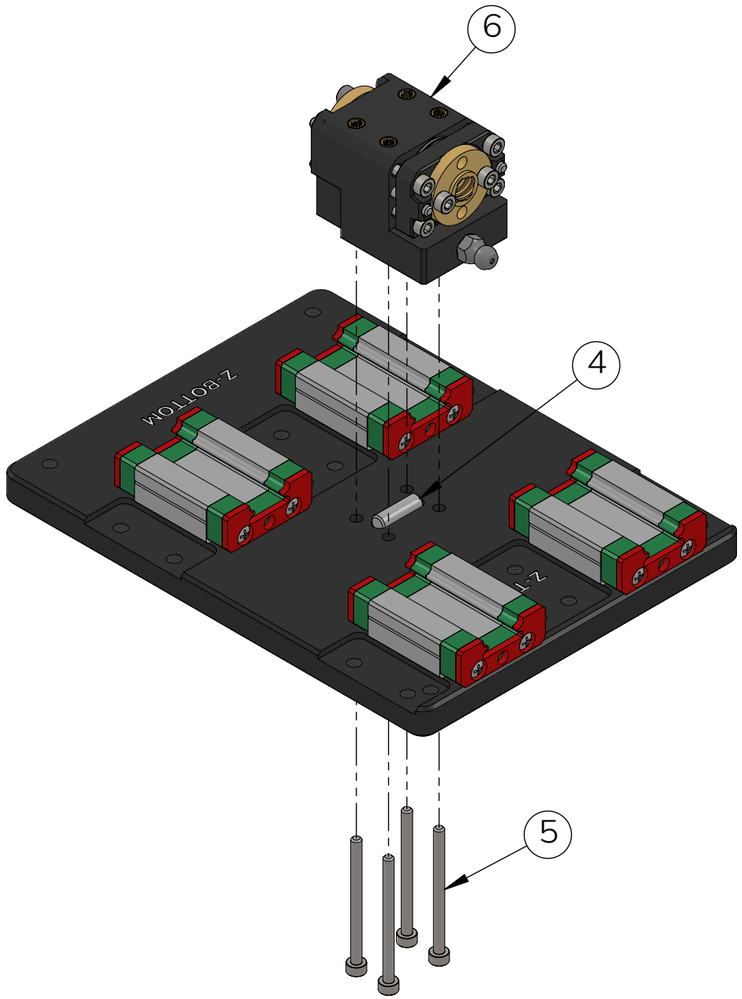
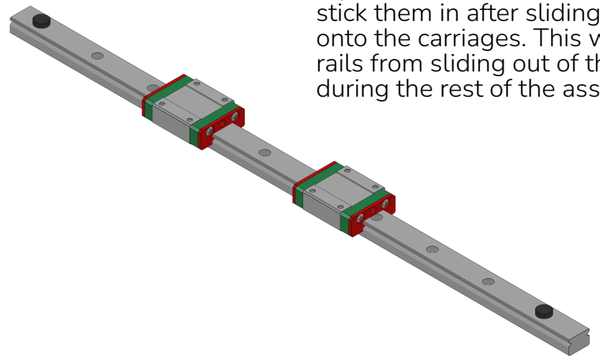
	PART NUMBER	QTY.
1	Z MOUNTING PLATE	1
4	ISO-2338-M5-15mm-PIN	1
5	ISO-4762-M3X40-SHCS	4
6	AB-BLOCK-MAIN-ASSEMBLY	1



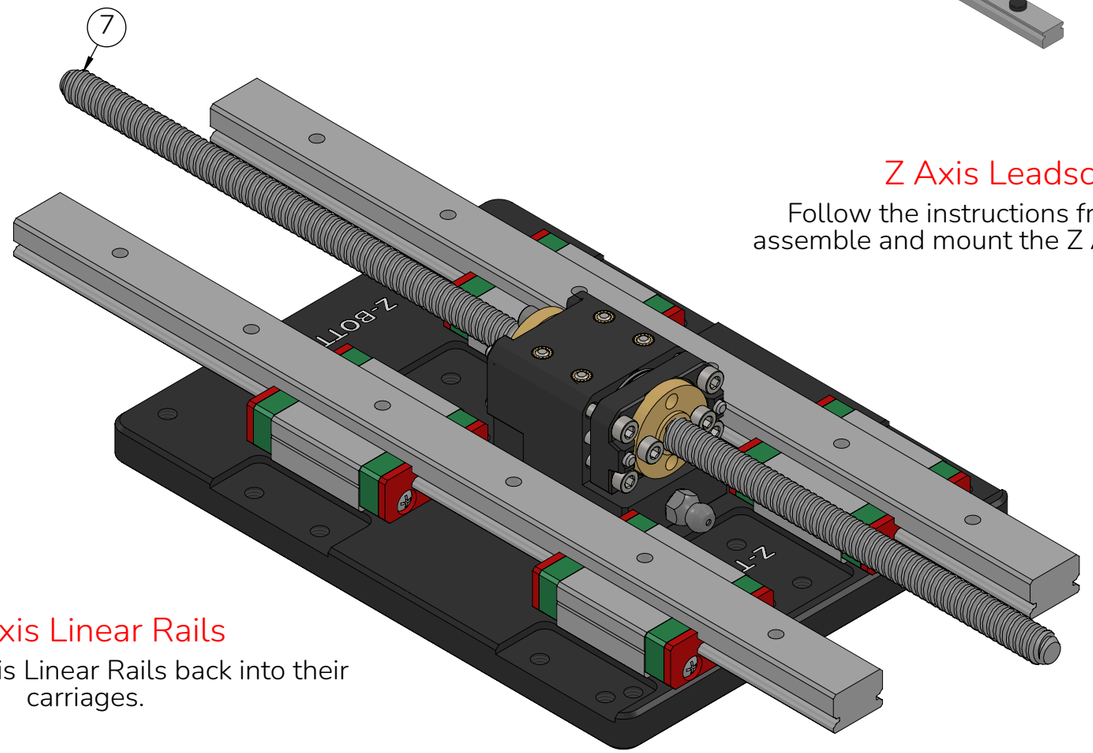
Linear Rail Carriage Safety

If your rails came with rubber stoppers, stick them in after sliding the rails back onto the carriages. This will prevent your rails from sliding out of the carriages during the rest of the assembly.



Z Axis Leadscrew

Follow the instructions from earlier to assemble and mount the Z Axis Leadscrew



Z Axis Linear Rails

Slide the Z Axis Linear Rails back into their carriages.

	PART NUMBER	QTY.
1	Z MOUNTING PLATE	1
7	TR8X8 LEADSCREW 600mm	1

Z MOUNT PLATE ASSEMBLY

ASSEMBLY STEP 2

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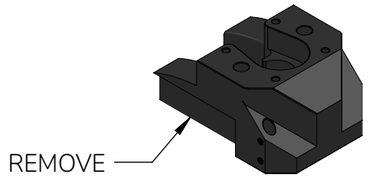
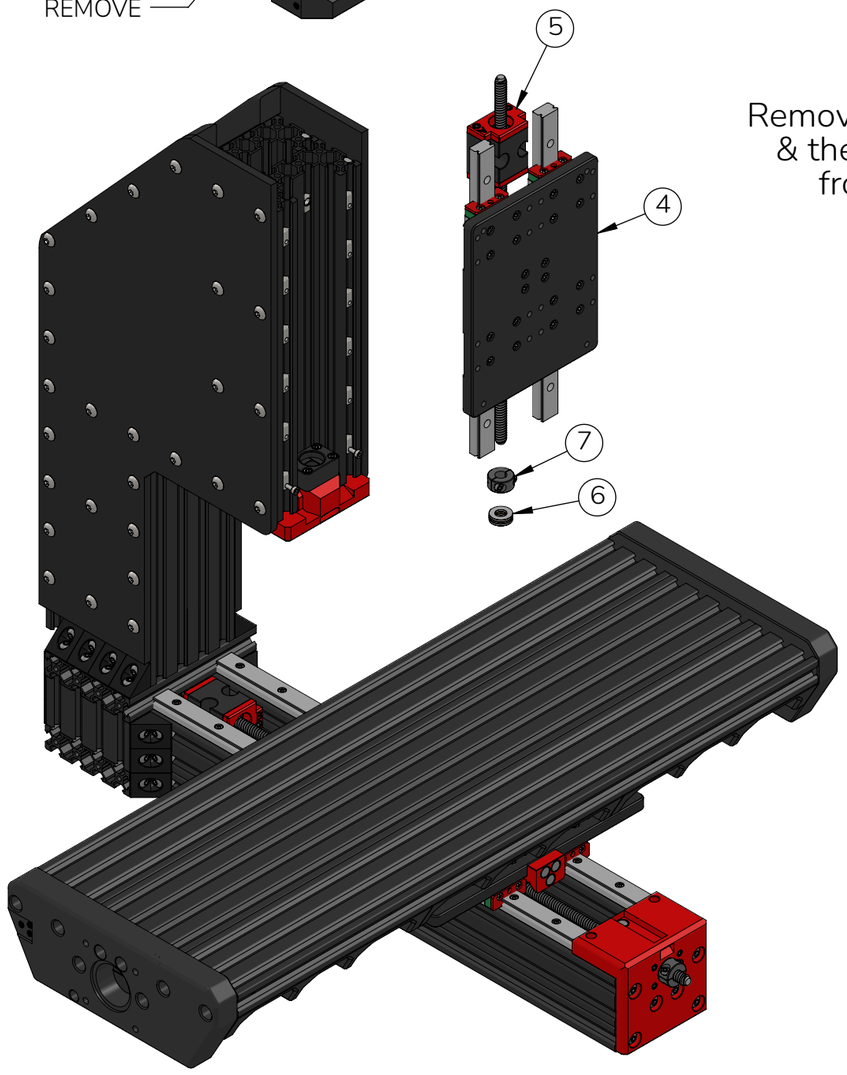
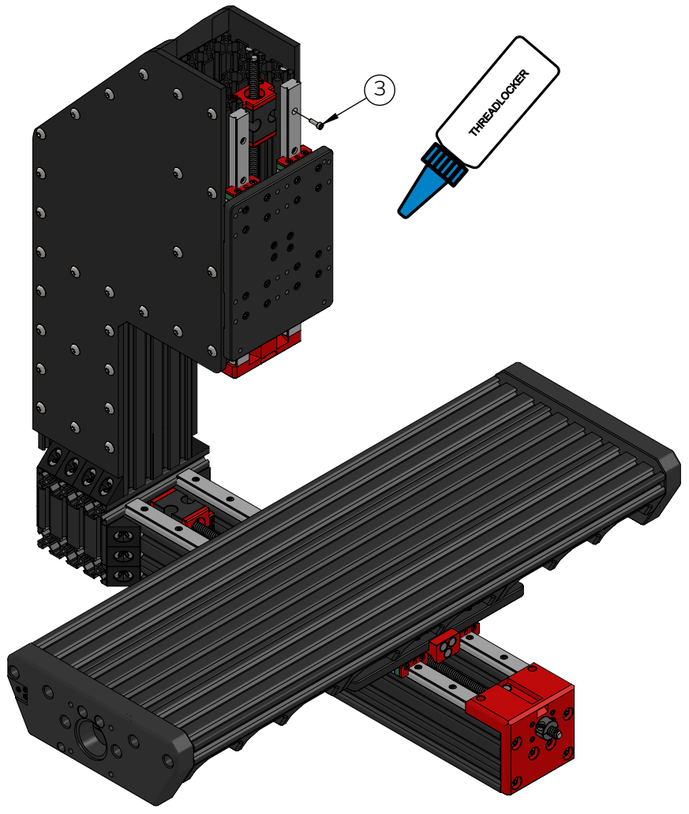
	PART NUMBER	QTY.
1	ZY-ASSEMBLY	1
4	Z-AXIS-SUBASSEMBLY-A	1
5	FLOATING-BEARING-BLOCK-ASSEMBLY	1
6	F8-16M THRUST BEARING	1
7	DOUBLE SPLIT SHAFT COLLAR 8X18X9MM	1





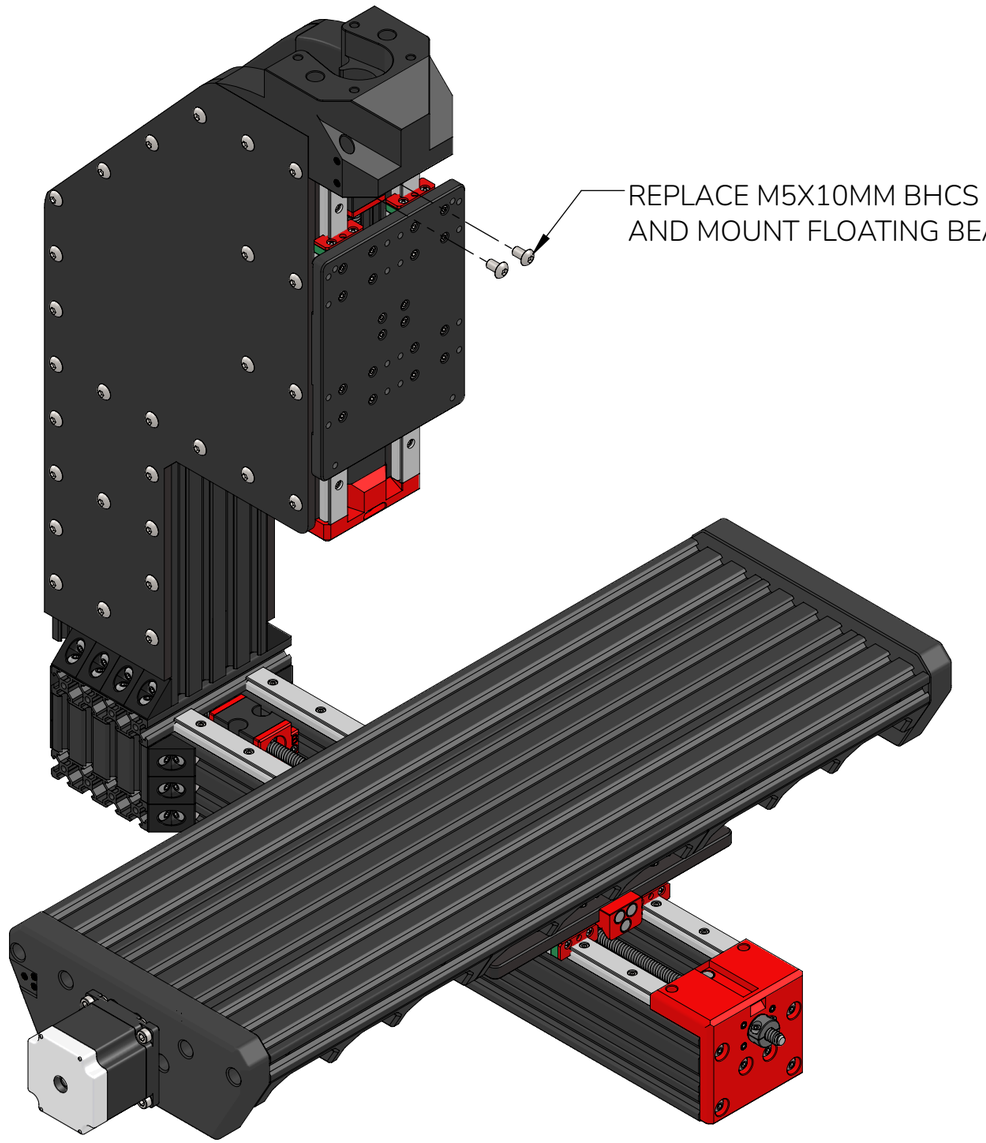
Remove the Z Axis Motor Mount
& the Floating Bearing Block
from the ZY-Assembly.

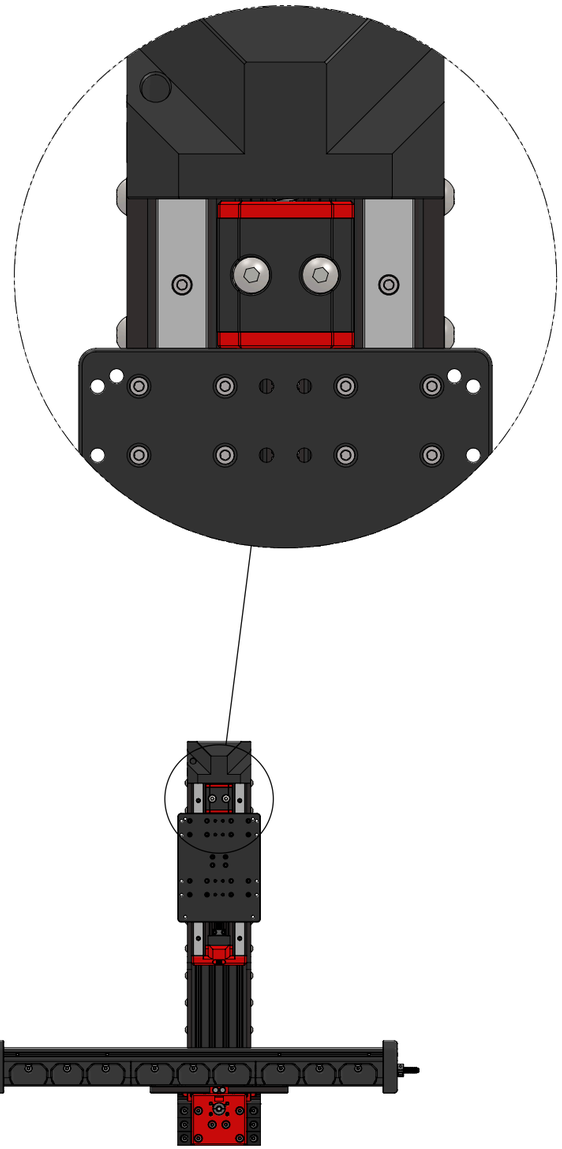


	PART NUMBER	QTY.
1	ZY-ASSEMBLY	1
3	ISO-4762-M3X10-SHCS	12
4	Z-AXIS-SUBASSEMBLY-A	1

MILO SUB-ASSEMBLY A

Z AXIS





MILO SUB-ASSEMBLY A

Z AXIS

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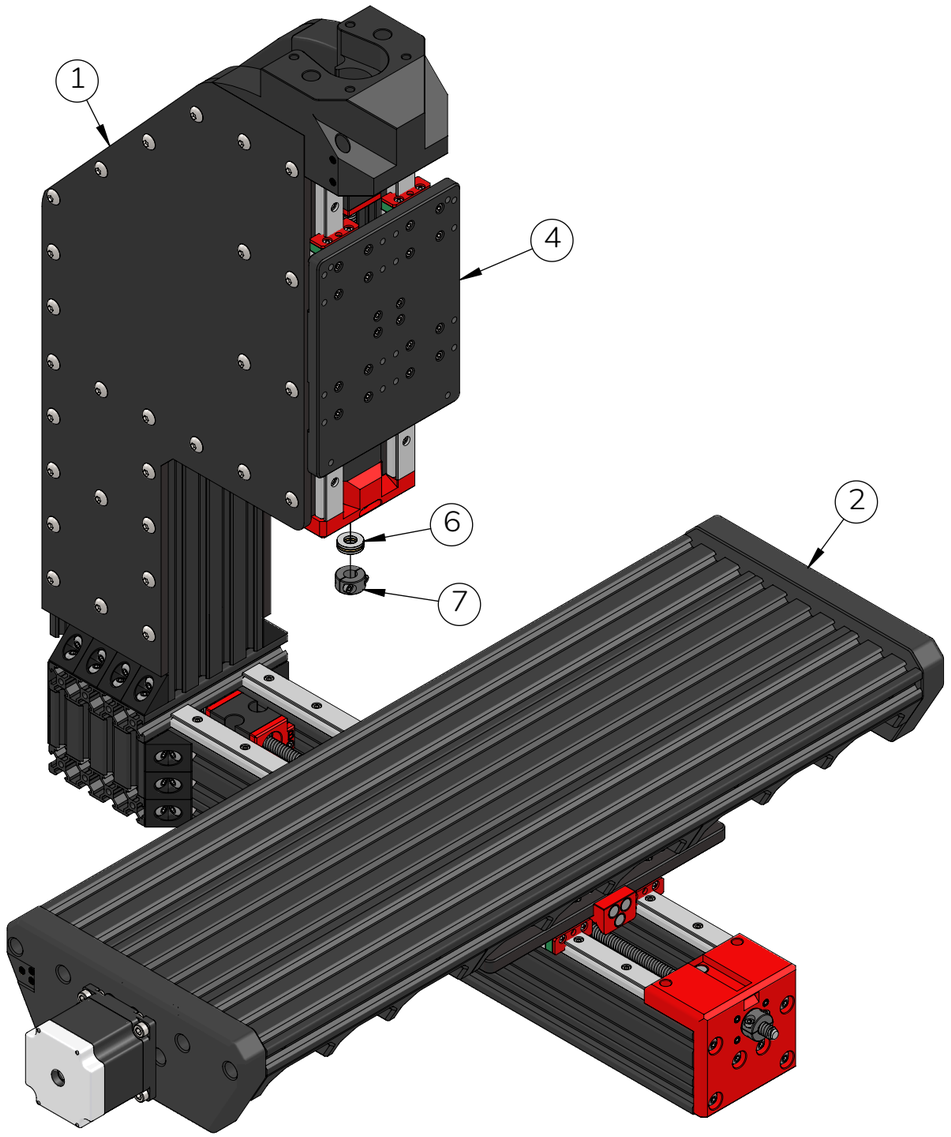
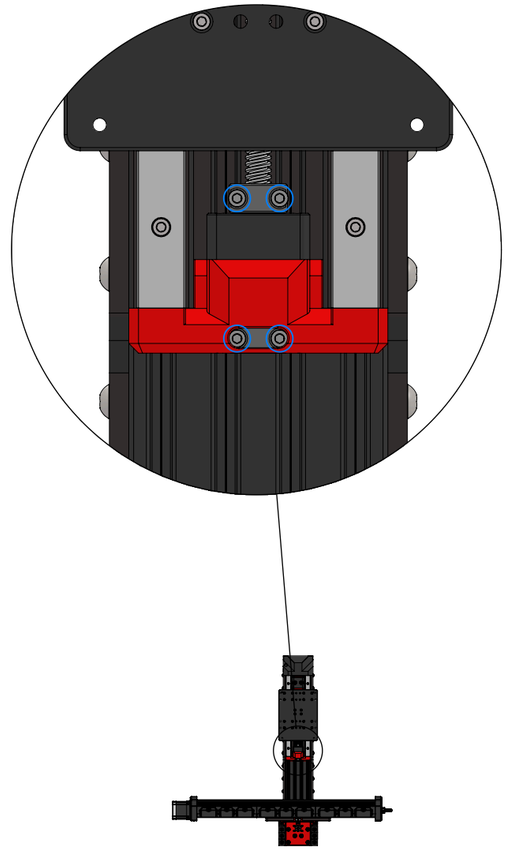


	PART NUMBER	QTY.
1	ZY-ASSEMBLY	1
2	MILO-XY-SUBASSEMBLY	1
4	Z-AXIS-SUBASSEMBLY-A	1
6	F8-16M THRUST BEARING	1
7	DOUBLE SPLIT SHAFT COLLAR 8X18X9MM	1

Apply inward pressure to the shaft collars and tighten.



MILO SUB-ASSEMBLY A

Z AXIS

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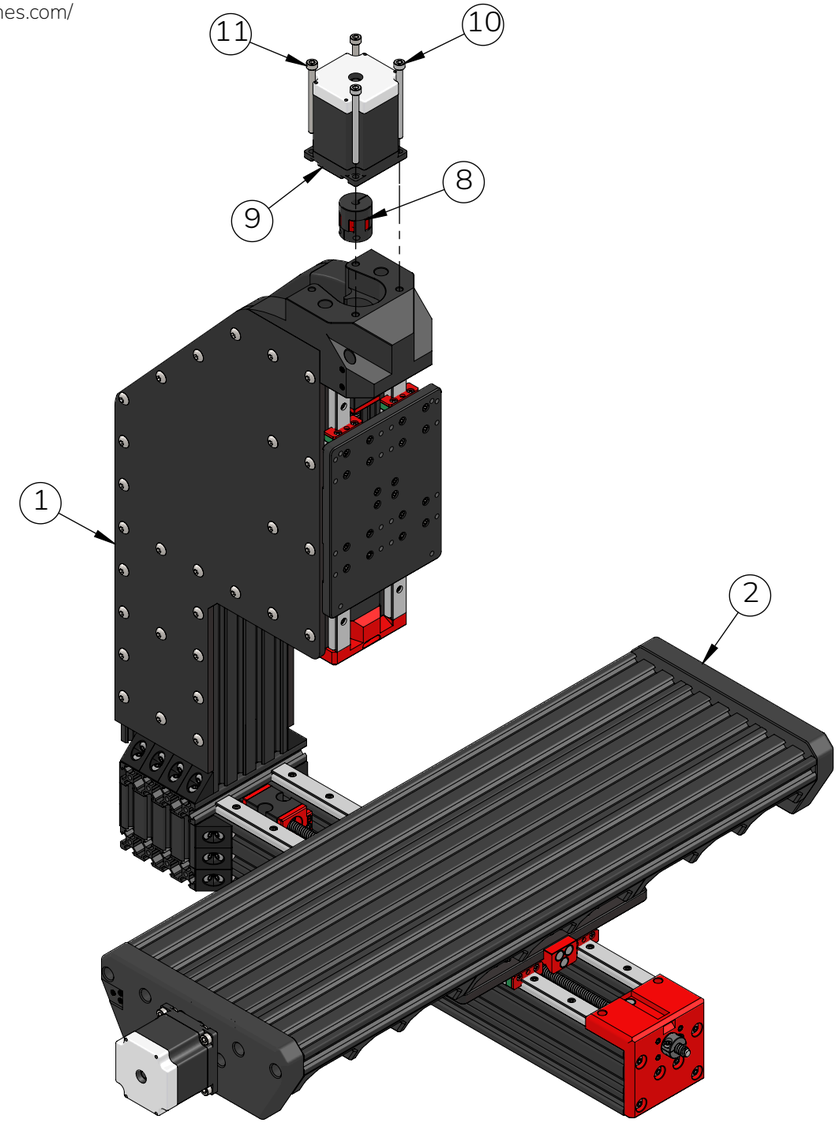
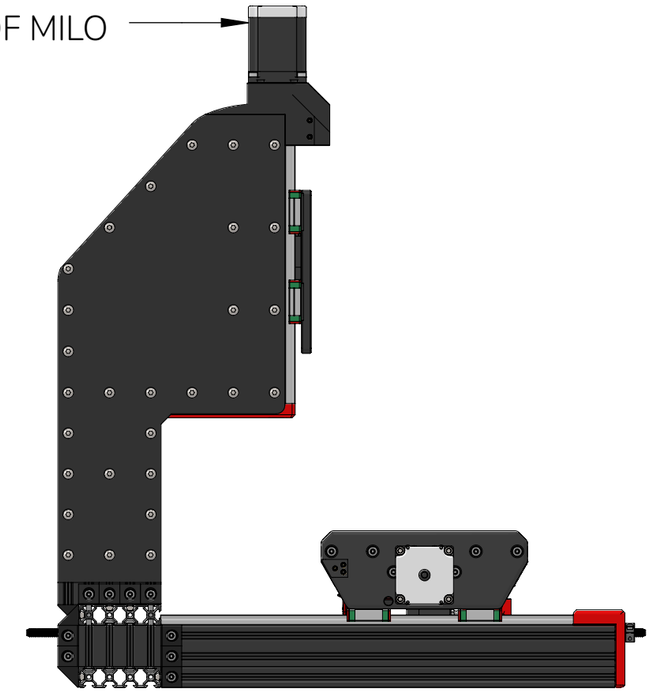
	PART NUMBER	QTY.
1	ZY-ASSEMBLY	1
2	MILO-XY-SUBASSEMBLY	1
8	8-TO-6.25mm-SPIDER-COUPLER	1
9	NEMA23-76mm	1
10	ISO-4762-M5X65-SHCS	2
11	ISO-4762-M5X60-SHCS	2



ORIENT MOTOR CABLE
TO THE BACK OF MILO



MILO SUB-ASSEMBLY A

Z AXIS

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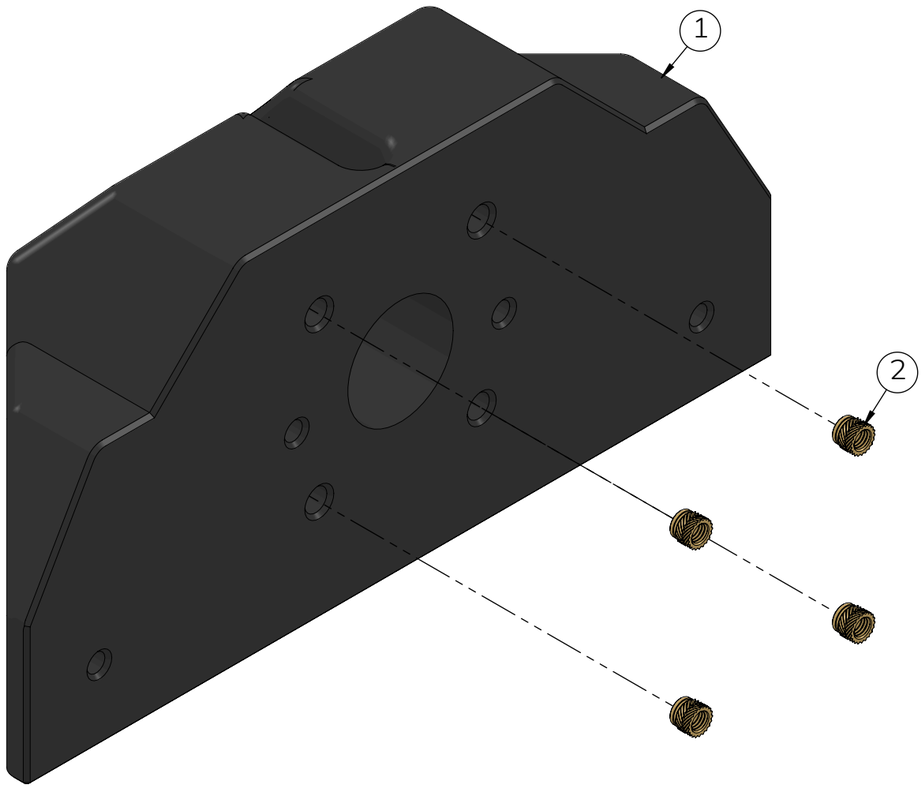
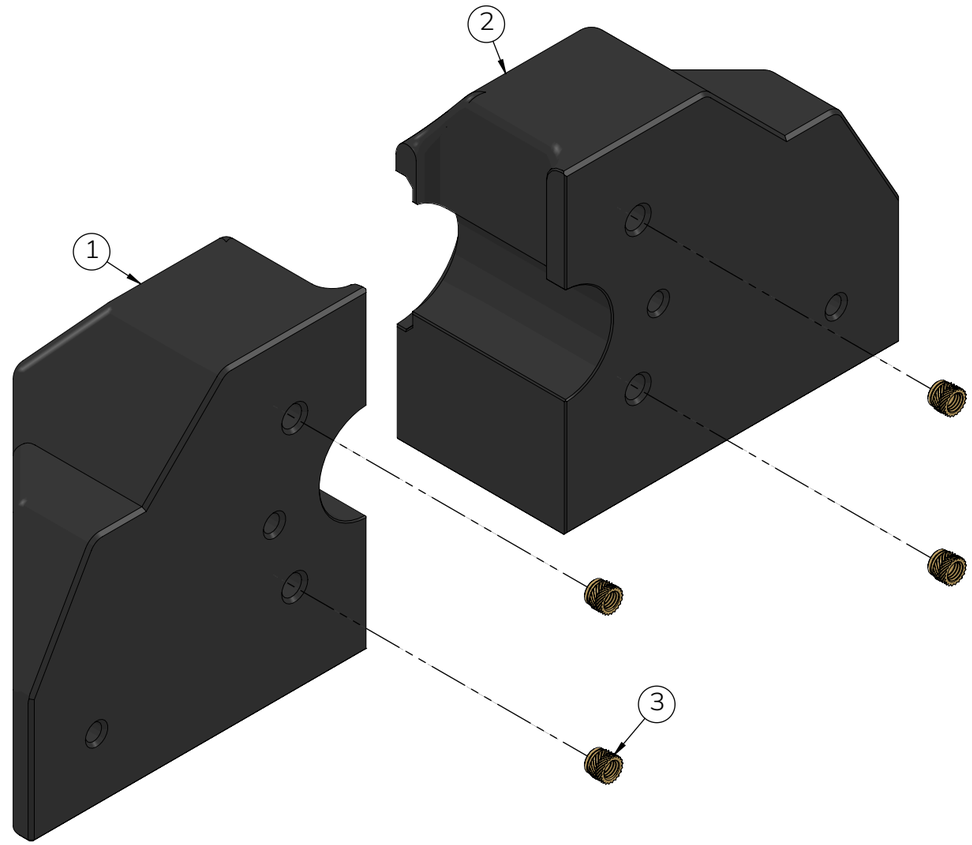
Y Axis Motor Mount

Two different versions are provided. If your 3D printer isn't big enough to print the mono-body version, print the split version



OR



	PART NUMBER	QTY.
1	Y-AXIS-MOTOR-MOUNT	1
2	M5X5.8mm HEATSET INSERT	4

	PART NUMBER	QTY.
1	Y-AXIS-MOTOR-MOUNT-A	1
2	Y-AXIS-MOTOR-MOUNT-B	1
3	M5X5.8mm HEATSET INSERT	4

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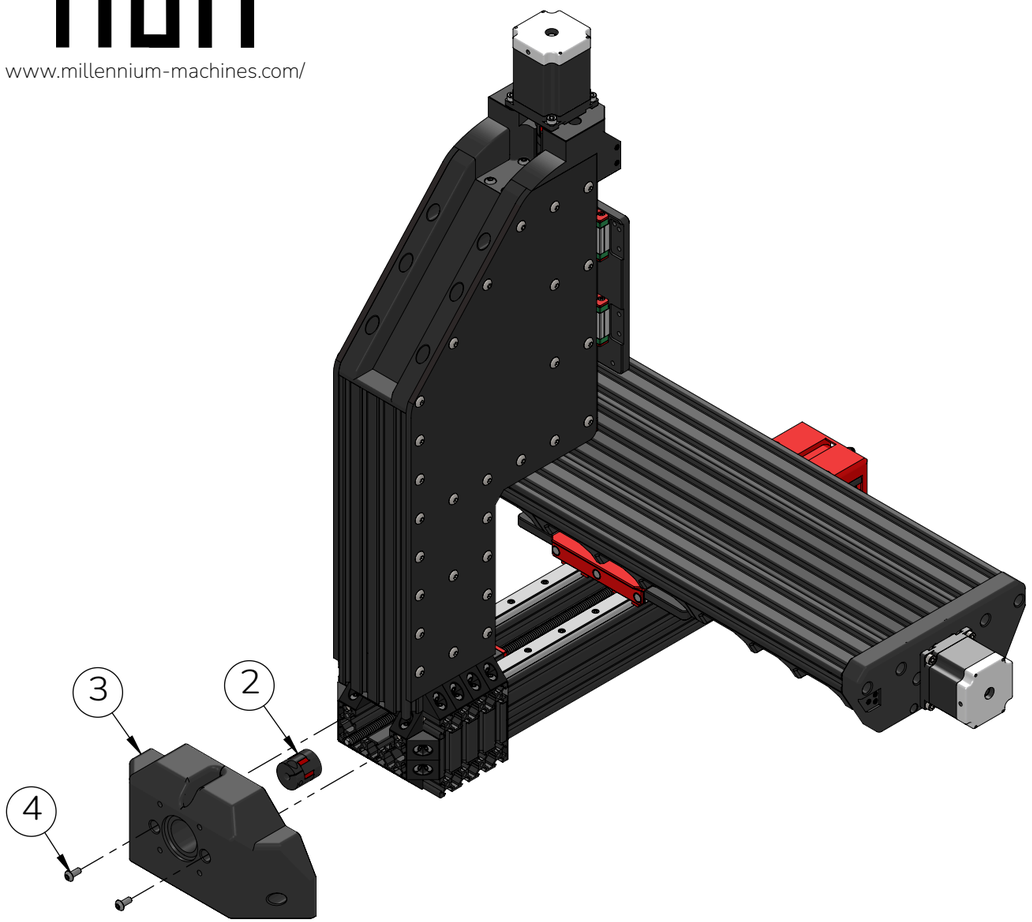
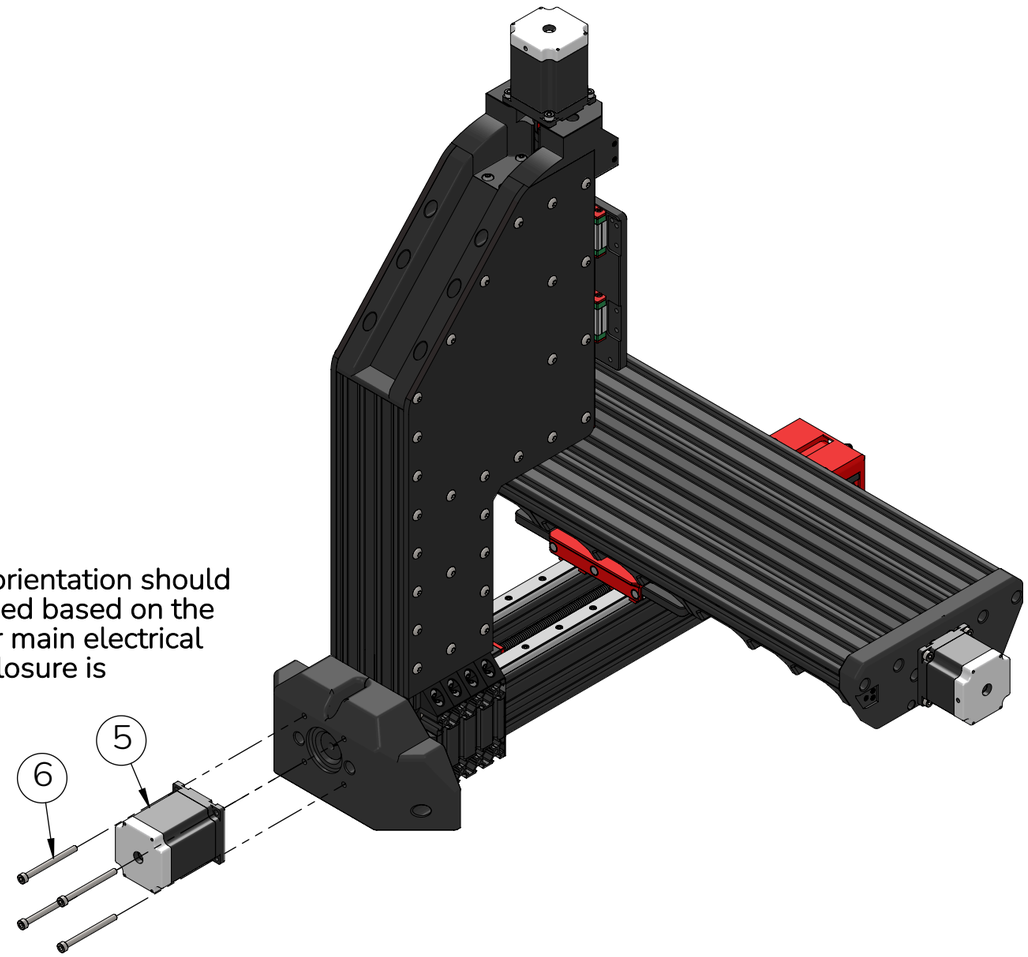


	PART NUMBER	QTY.
1	MILO-SUBASSEMBLY-A	1
2	8-TO-6.25mm-SPIDER-COUPLER	1
3	Y-AXIS-MOTOR-MOUNT-SUBASSEMBLY	1
4	ISO-7380-M5X12 BHCS	2

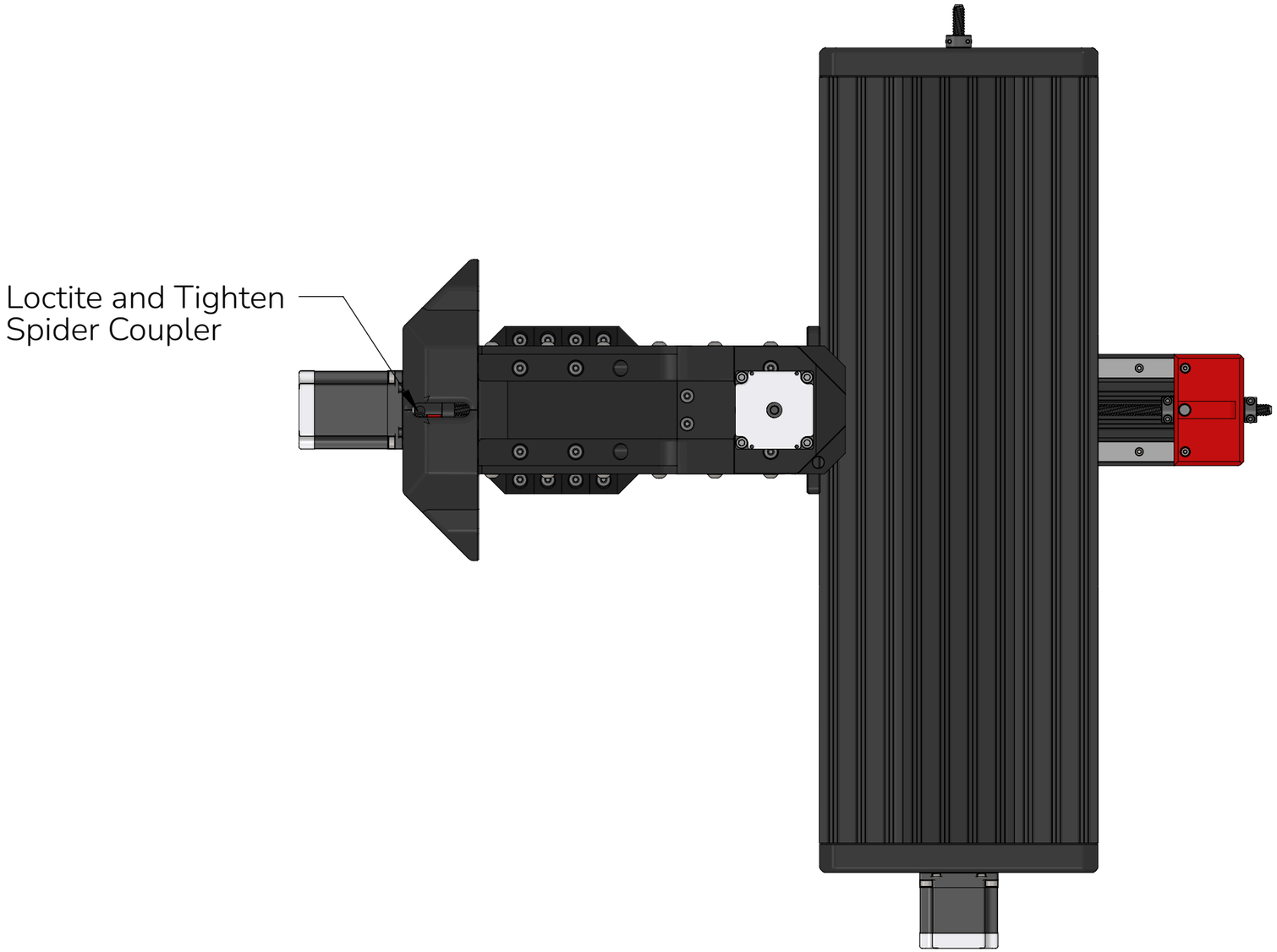
Motor cable orientation should be determined based on the where your main electrical enclosure is



	PART NUMBER	QTY.
1	MILO-SUBASSEMBLY-A	1
5	NEMA23-76mm	1
6	ISO-4762-M5X60-SHCS	4

MILO SUB-ASSEMBLY A

Y AXIS - ASSEMBLY STEP 1



MILO SUB-ASSEMBLY A

Y AXIS - ASSEMBLY STEP 2

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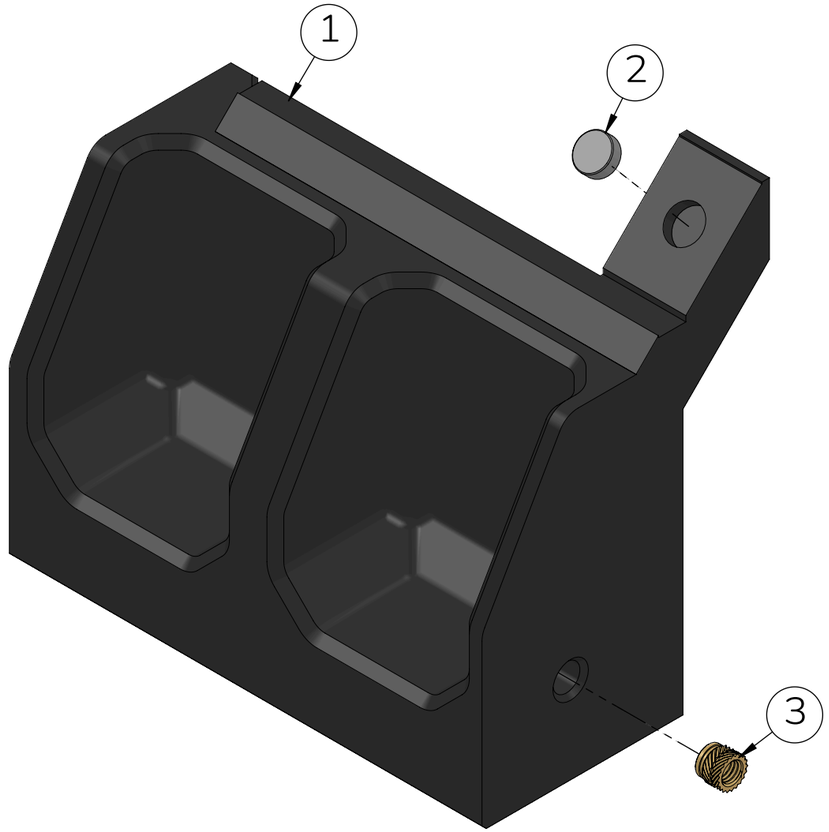
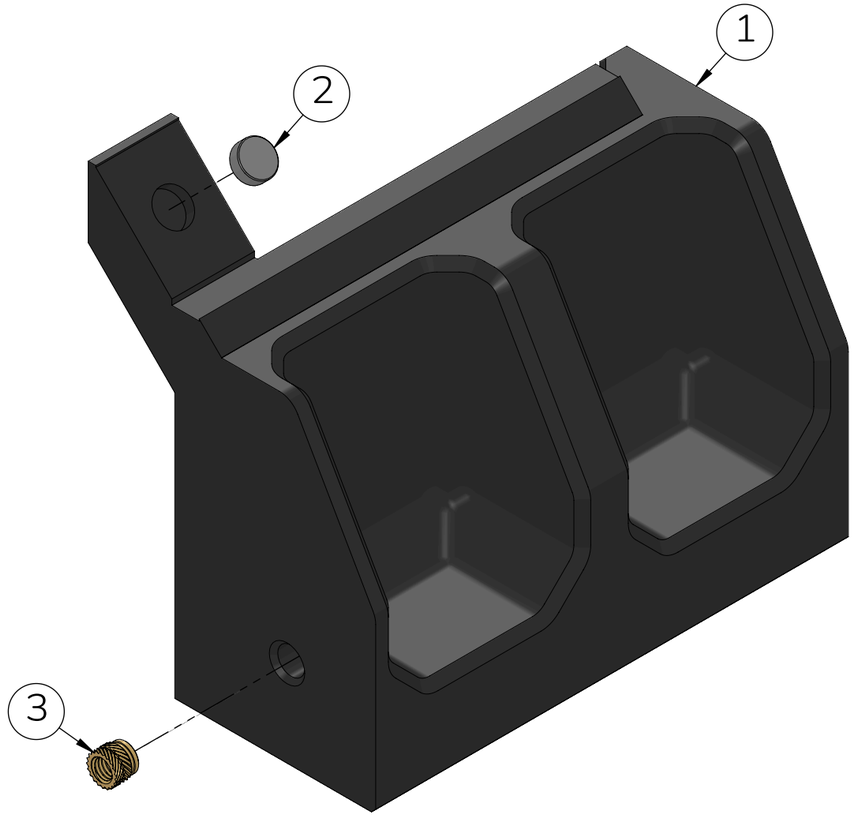


	PART NUMBER	QTY.
1	Y-AXIS-HOLD-DOWN-D1	1
2	8X3MM NEODYMIUM MAGNET	1
3	M5X5.8mm HEATSET INSERT	1



	PART NUMBER	QTY.
1	Y-AXIS-HOLD-DOWN-D2	1
2	8X3MM NEODYMIUM MAGNET	1
3	M5X5.8mm HEATSET INSERT	1

Y AXIS HOLD DOWNS

D1/D2 PRE-ASSEMBLY

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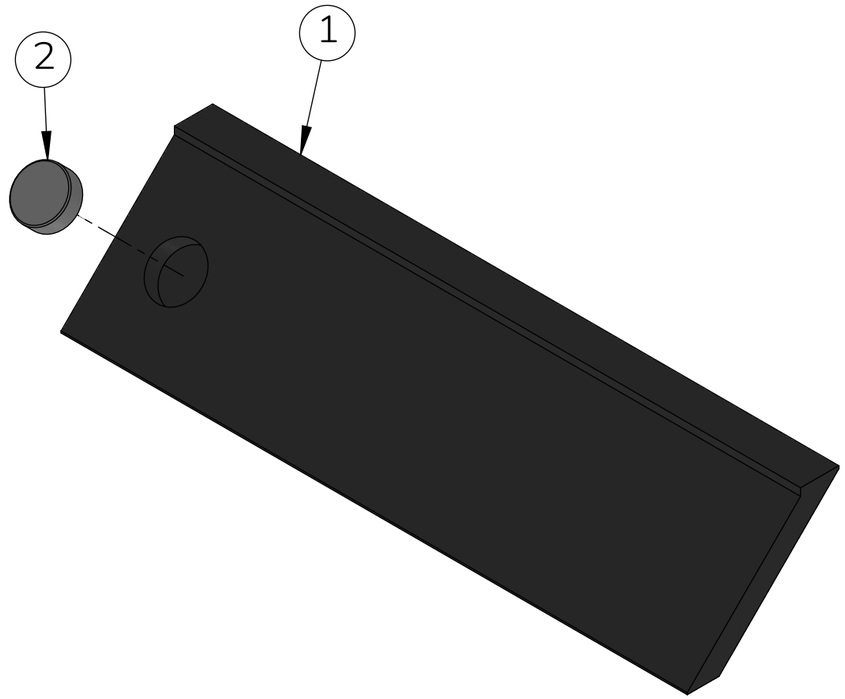
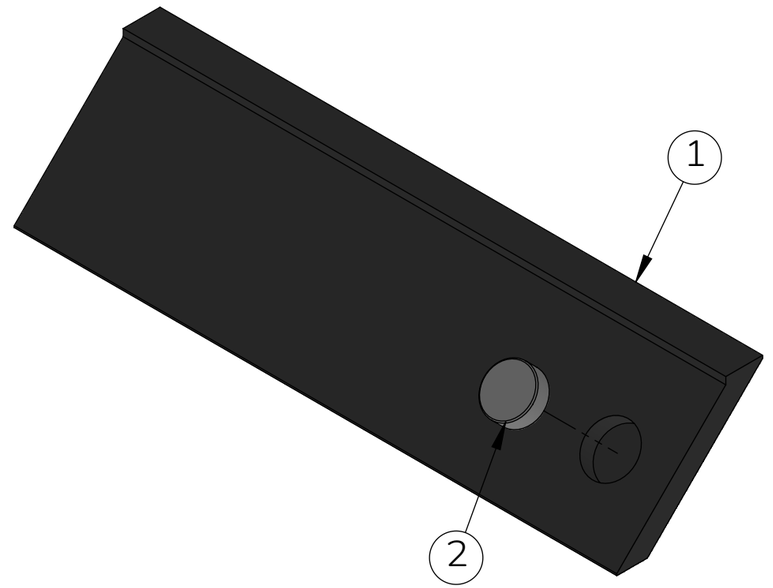


	PART NUMBER	QTY.
1	DUST-COVER-A	1
2	8X3MM NEODYMIUM MAGNET	1



	PART NUMBER	QTY.
1	DUST-COVER B	1
2	8X3MM NEODYMIUM MAGNET	1

Y AXIS HOLD DOWNS

DUST COVER PRE-ASSEMBLY

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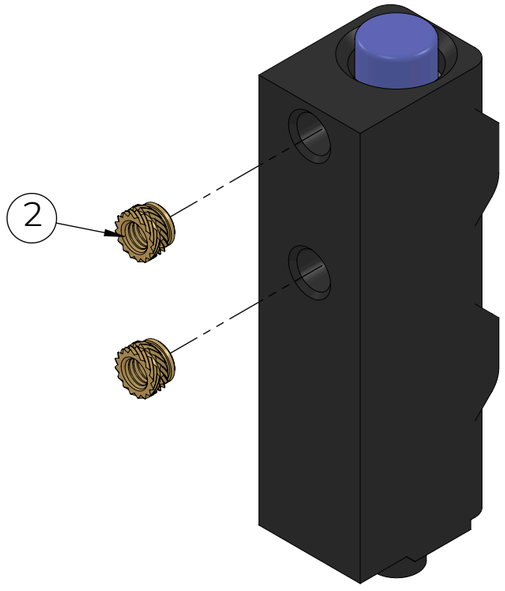
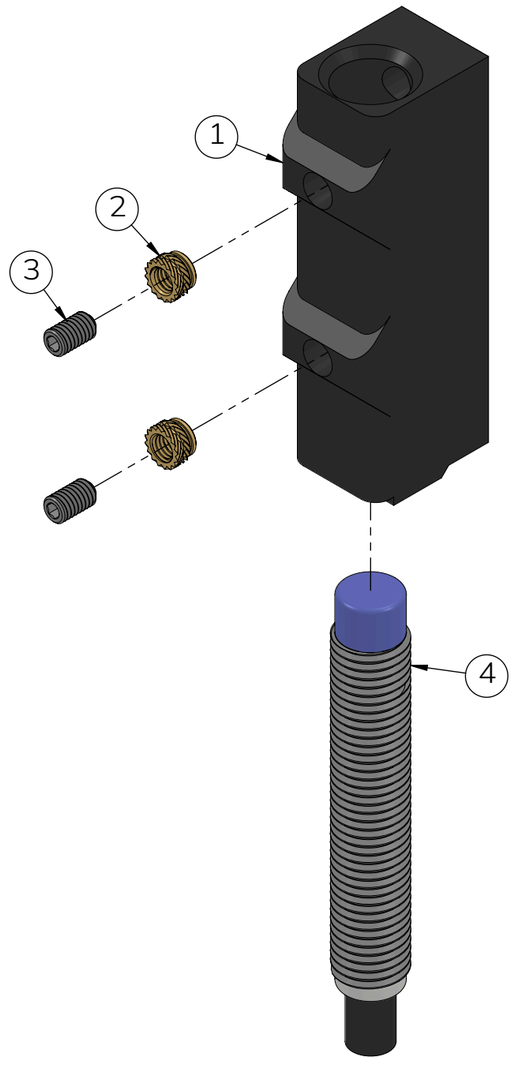
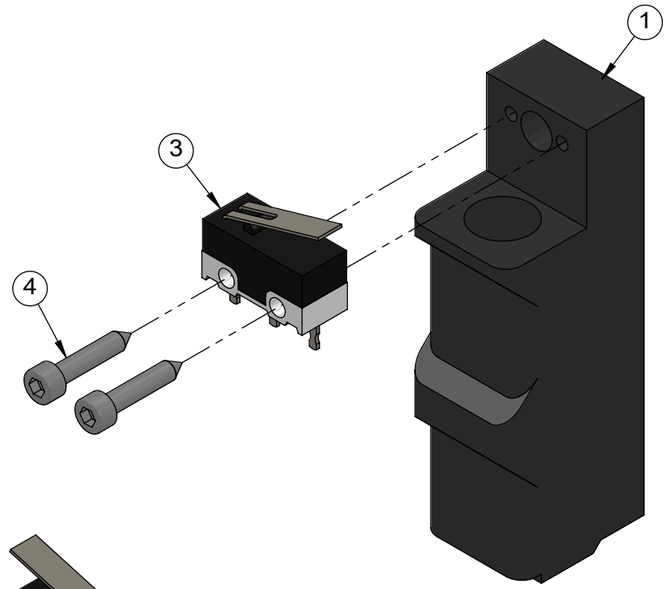
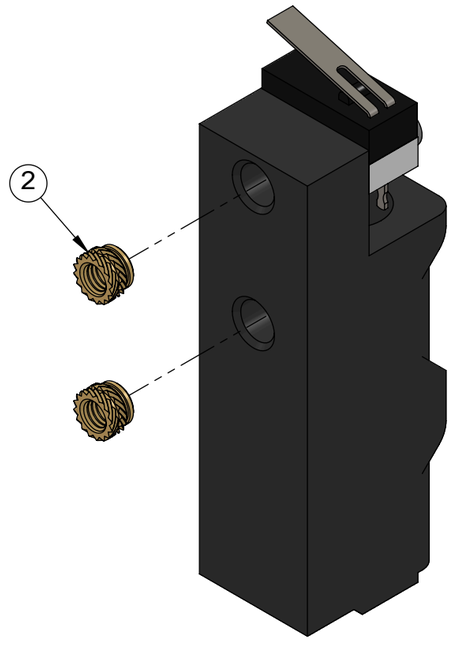




If you want to home above 1000mm/min, use the micro switch for the Y Axis.

OR





	PART NUMBER	QTY.
1	INDUCTIVE-PROBE-MOUNT-Y	1
2	M3X3MM HEATSET INSERT	4
3	ISO-4026-M3X10-SET SCREW	2
4	LJ8A3-2-Z/AY	1

	PART NUMBER	QTY.
1	MICRO-SWITCH-MOUNT-Y	1
2	M3X3MM HEATSET INSERT	2
3	D2F-01L	1
4	M2X10-SELF-TAPPING-SCREW-SHCS	2

Y AXIS HOMING SWITCH

PRE-ASSEMBLY

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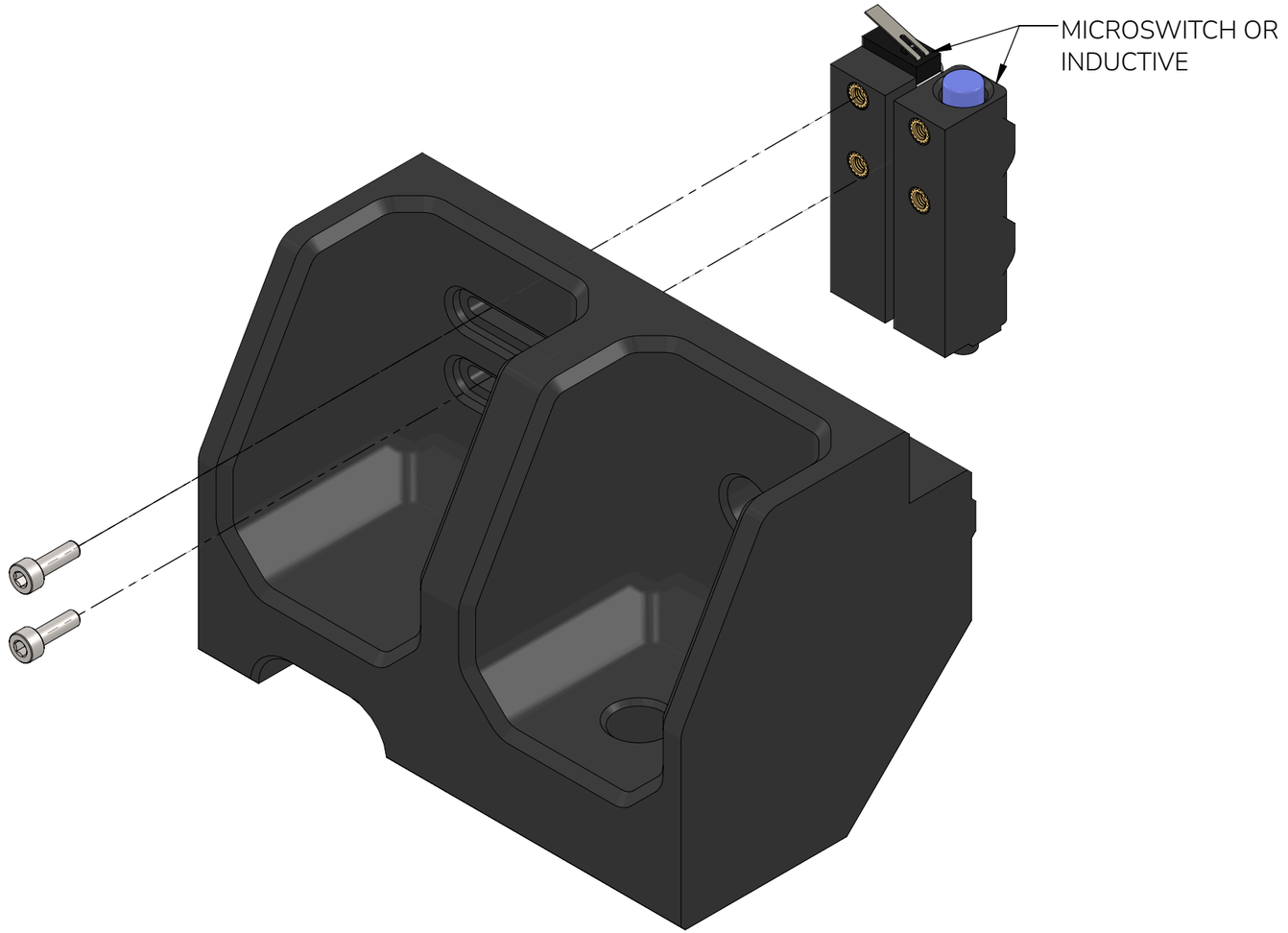

	PART NUMBER	QTY.
1	Y-AXIS-HOLD-DOWN-C2	1
2	Y-AXIS-INDUCTIVE-SWITCH-ASSEMBLY	1
3	Y-AXIS-MICROSWITCH-ASSEMBLY	1
4	ISO-4762-M3X10-SHCS	2

Y AXIS HOLD DOWNS

C2 PRE-ASSEMBLY

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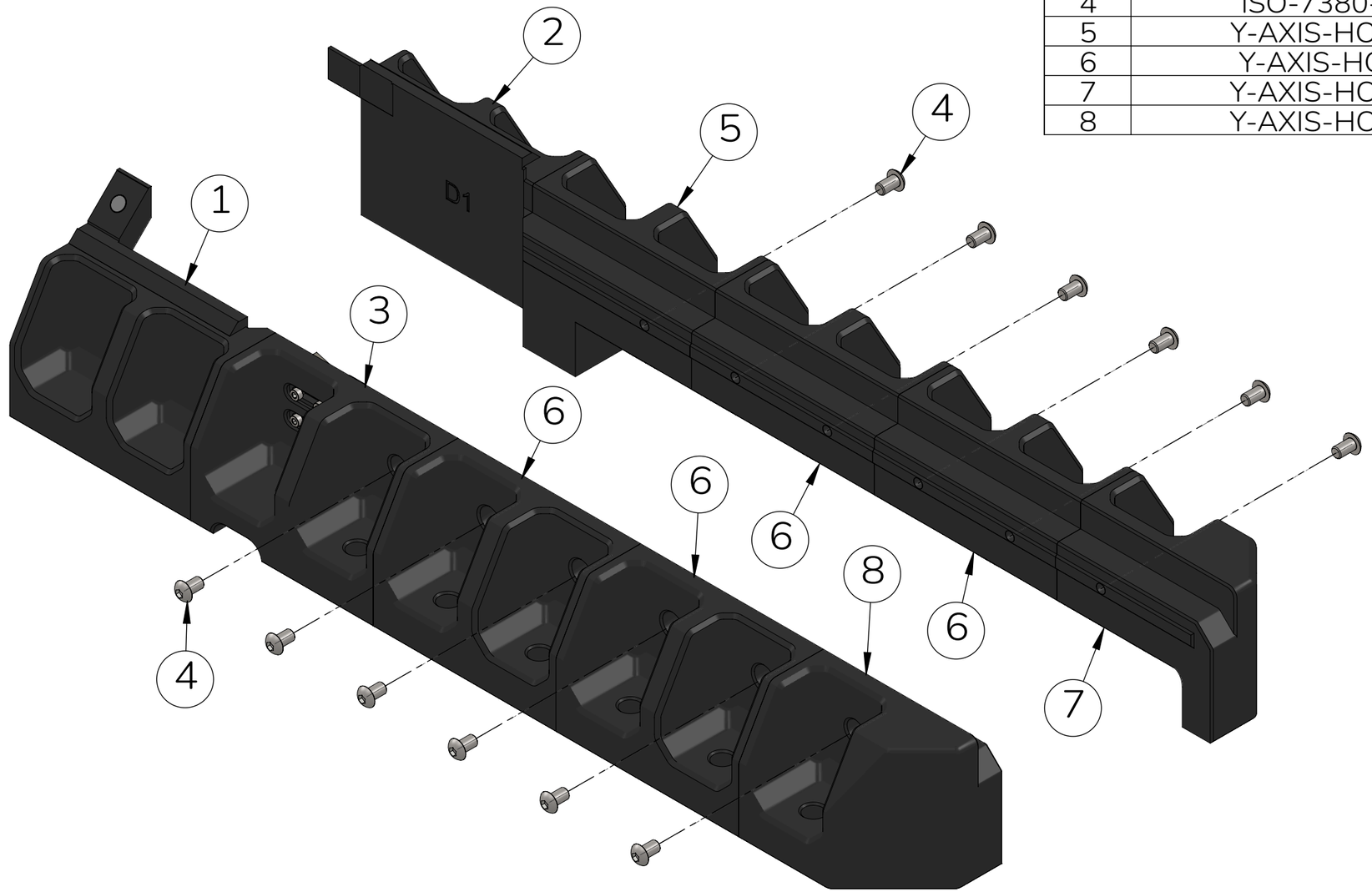
	PART NUMBER	QTY.
1	D2-HOLD-DOWN-SUBASSEMBLY	1
2	D1-HOLD-DOWN-SUBASSEMBLY	1
3	C2-HOLD-DOWN-SUBASSEMBLY	1
4	ISO-7380-M5X10 BHCS	12
5	Y-AXIS-HOLD-DOWN-C1	1
6	Y-AXIS-HOLD-DOWN-B	4
7	Y-AXIS-HOLD-DOWN-A1	1
8	Y-AXIS-HOLD-DOWN-A2	1



Y AXIS HOLD DOWNS

ASSEMBLY

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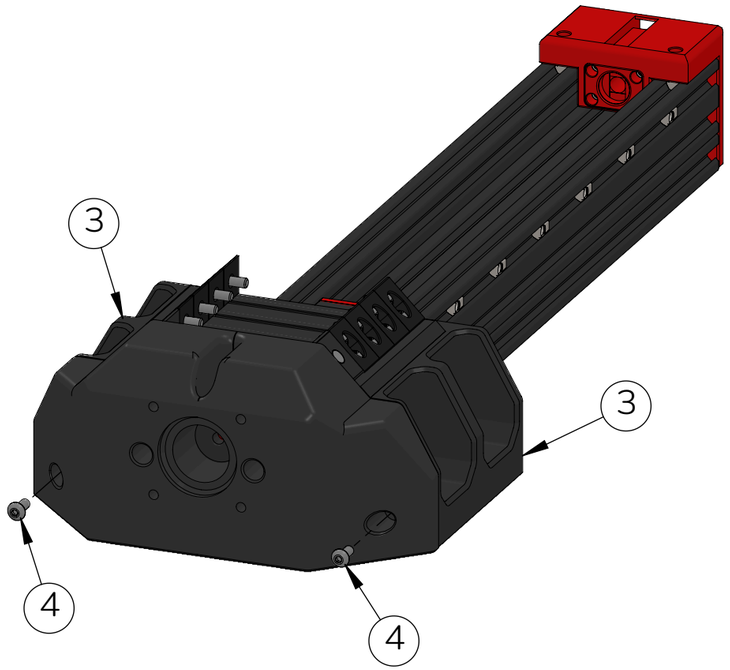
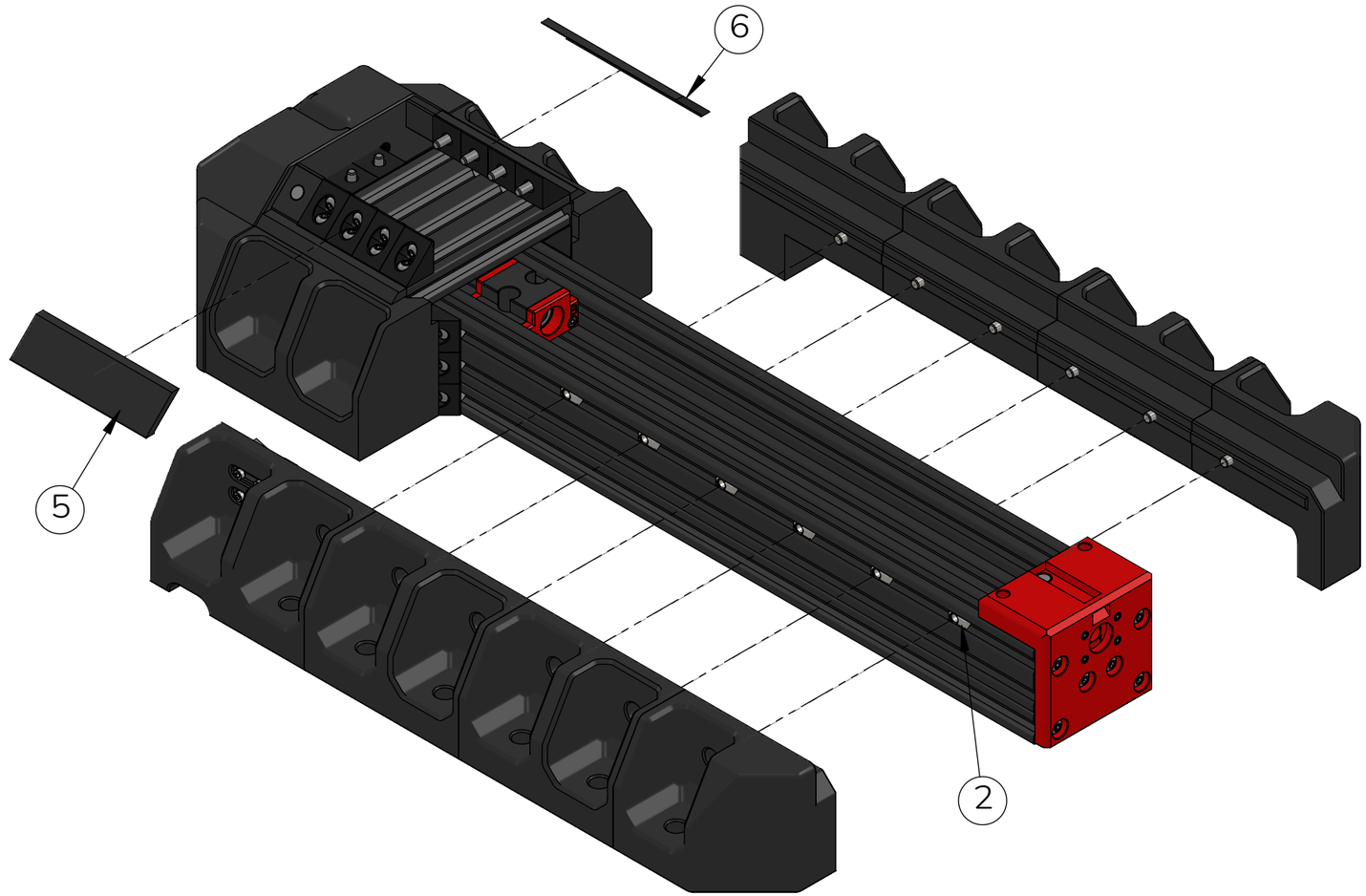
Z & X AXIS HIDDEN FOR CLARITY

	PART NUMBER	QTY.
1	MILO-SUBASSEMBLY-B	1
2	M5 DROP IN NUT	12
3	HOLD-DOWN-ASSEMBLY	1
4	ISO-7380-M5X12 BHCS	2
5	DUST-COVER-A-SUBASSEMBLY	1
6	DUST-COVER-B-SUBASSEMBLY	1





MILO SUB-ASSEMBLY B

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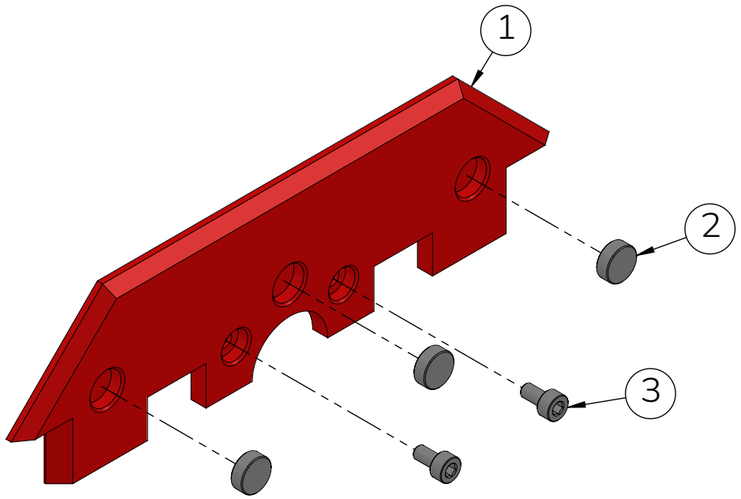
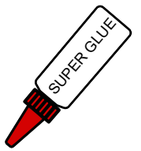
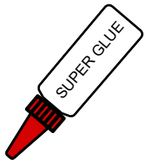


	PART NUMBER	QTY.
1	Y-AXIS-BELLOW-COLUMN-MAG-MOUNT	1
2	8X3MM NEODYMIUM MAGNET	3
3	ISO-4762-M3X6-SHCS	2

You will likely need to modify the length of the bellows. With the Y-Axis moved to the very front of your machine, the bellows should be taut. This will allow full Y-Axis travel.

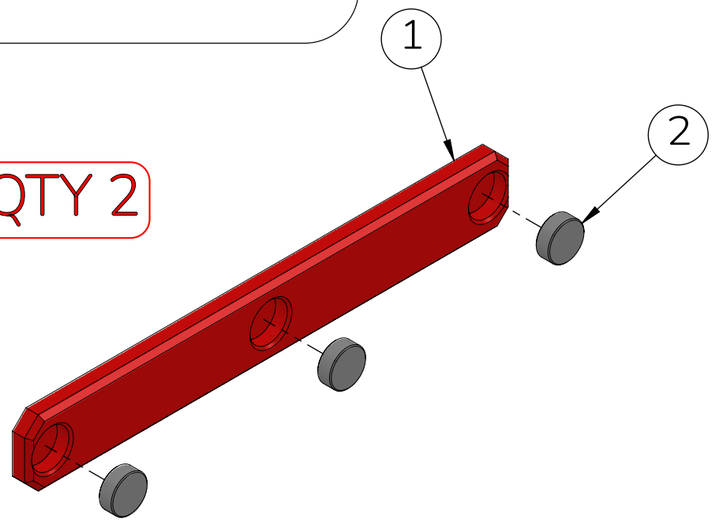


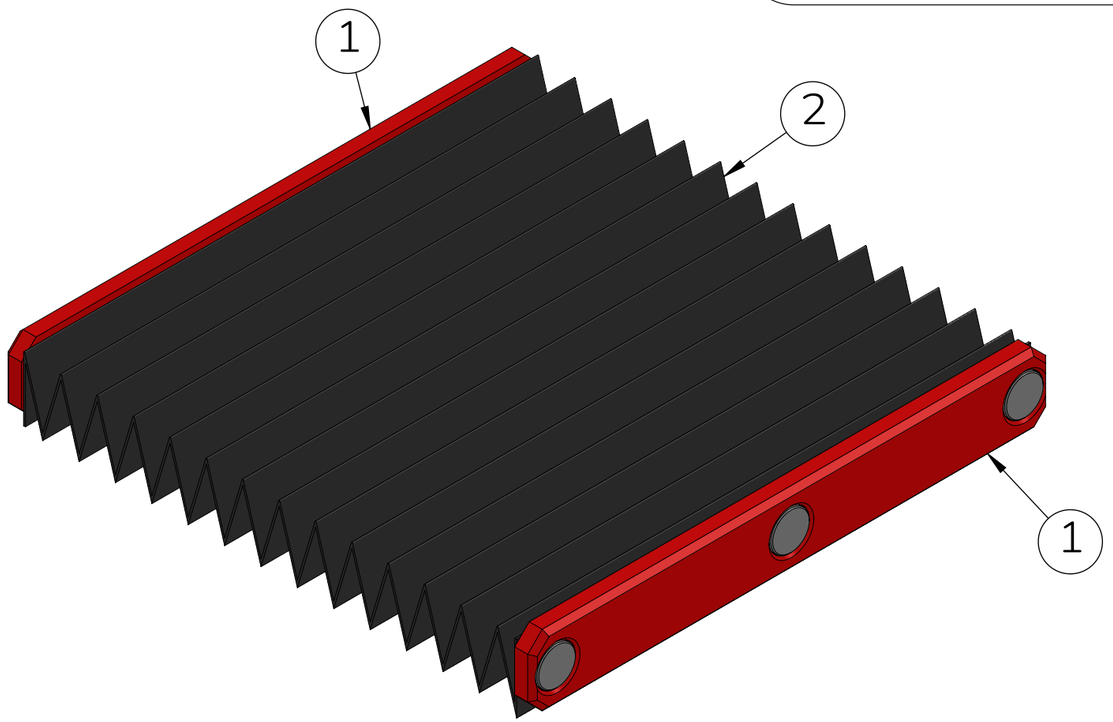
Super glue or use double sided tape for attaching the bellows to the mag mount.



If the magnets aren't tight, you can apply a dab of super glue.

QTY 2





	PART NUMBER	QTY.
1	BELLOW-ATTACHMENT-PLATE	1
2	8X3MM NEODYMIUM MAGNET	3

	PART NUMBER	QTY.
1	BELLOW-ATTACHMENT-PLATE-ASSEMBLY	2
2	BELLOWS	1

Y AXIS BELLOWS/WAYCOVER

PRE-ASSEMBLY

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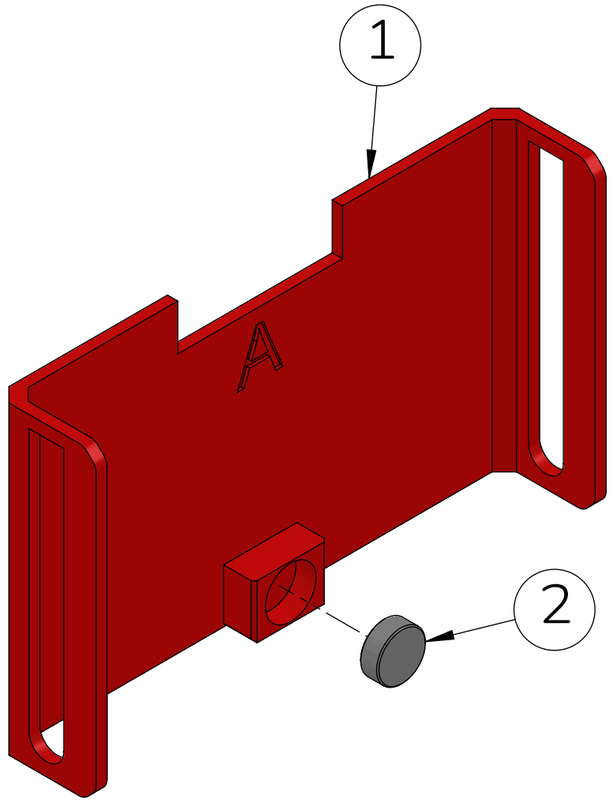
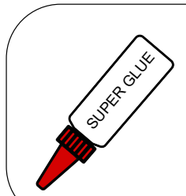
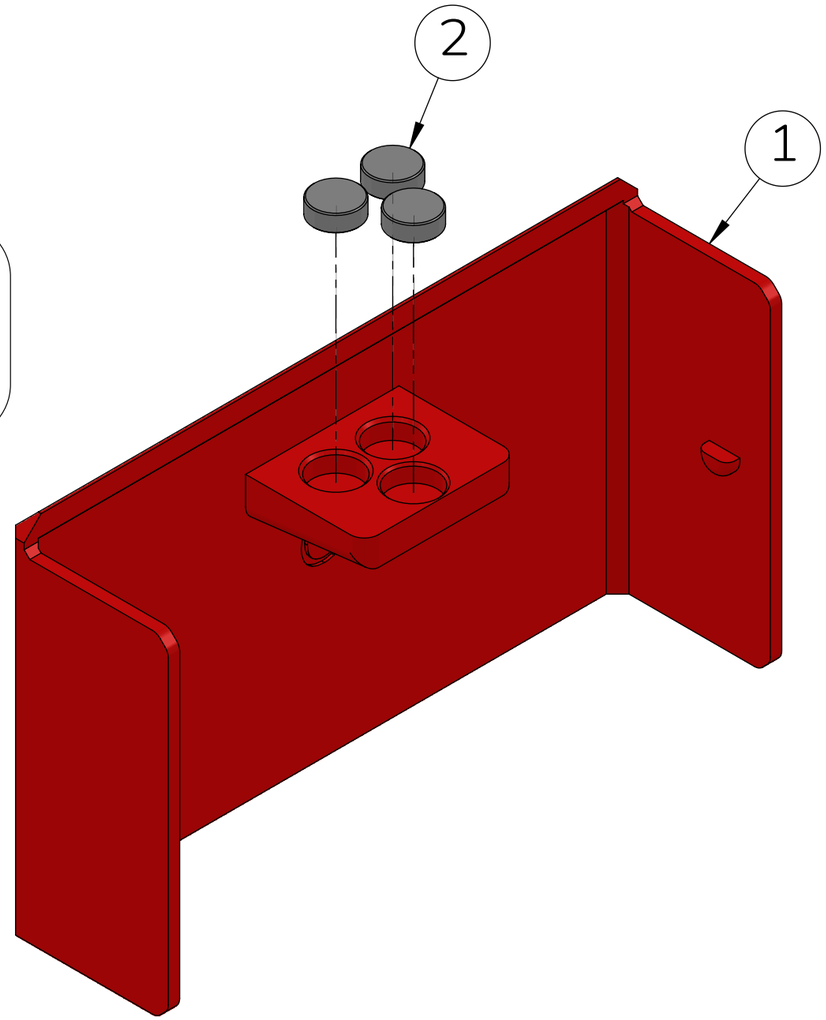




If the magnets aren't tight, you can apply a dab of super glue.



	PART NUMBER	QTY.
1	WAY-COVER-Y-FRONT-A	1
2	8X3MM NEODYMIUM MAGNET	1

	PART NUMBER	QTY.
1	WAY-COVER-Y-FRONT-G	1
2	8X3MM NEODYMIUM MAGNET	3

YY AXIS BELLOWS/WAYCOVER

WAYCOVER PRE-ASSEMBLY

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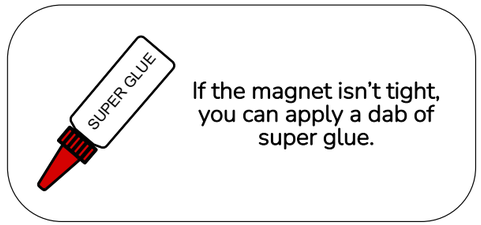
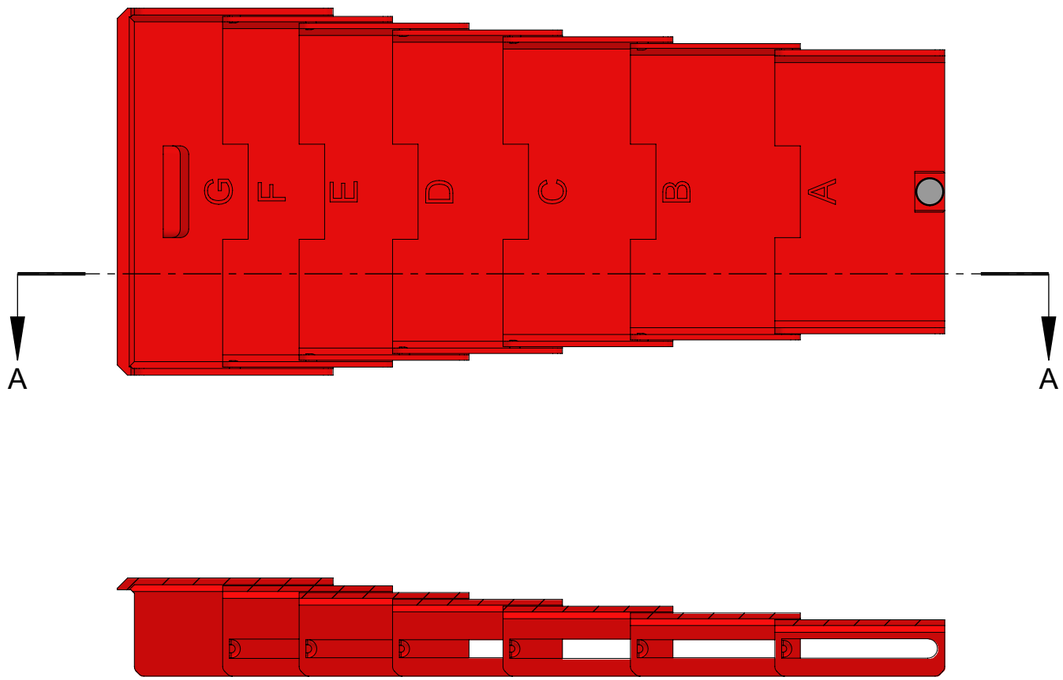
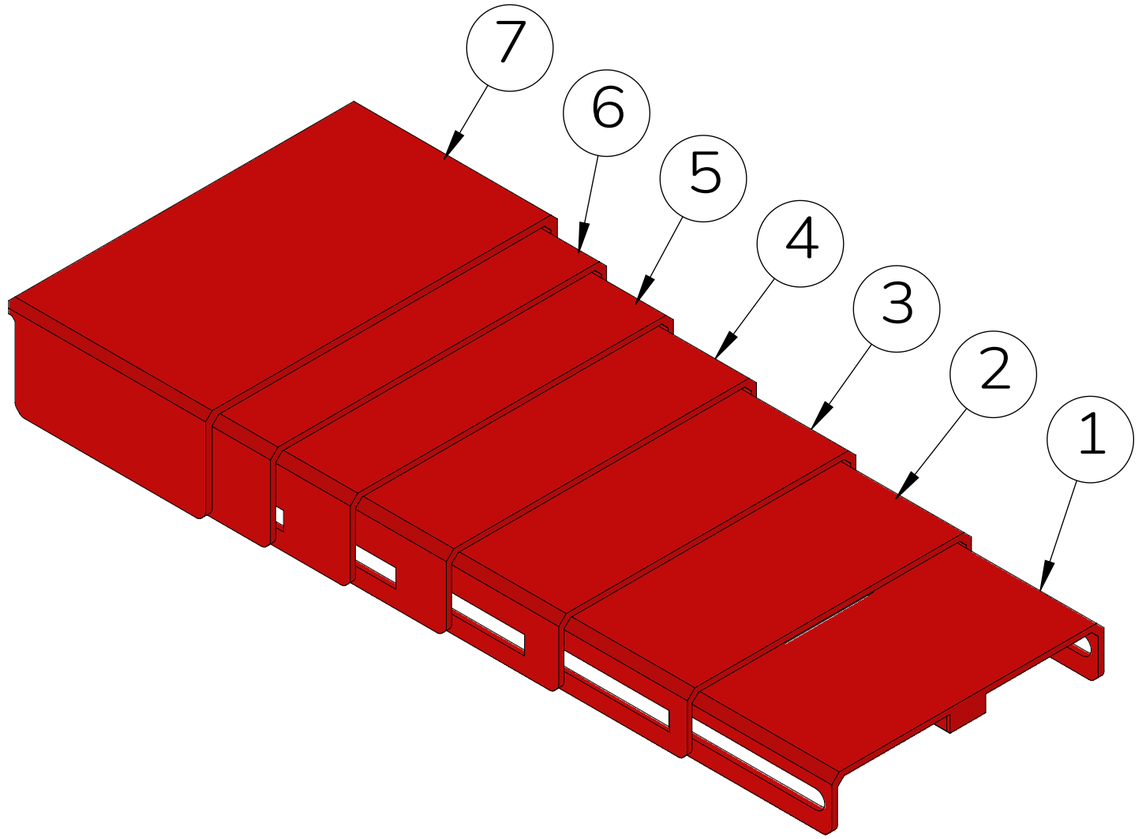
	PART NUMBER	QTY
1	WAY-COVER-A-SUBASSEMBLY	1
2	WAY-COVER-Y-FRONT-B	1
3	WAY-COVER-Y-FRONT-C	1
4	WAY-COVER-Y-FRONT-D	1
5	WAY-COVER-Y-FRONT-E	1
6	WAY-COVER-Y-FRONT-F	1
7	WAY-COVER-G-SUBASSEMBLY	1



If the magnet isn't tight, you can apply a dab of super glue.





YY AXIS BELLOWS/WAYCOVER

WAYCOVER ASSEMBLY

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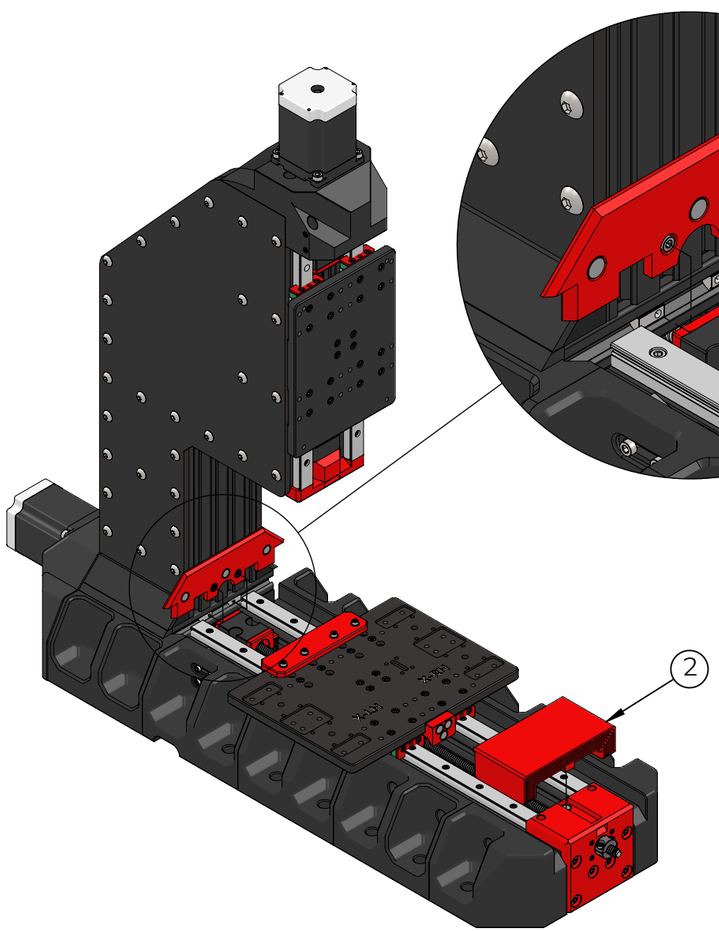
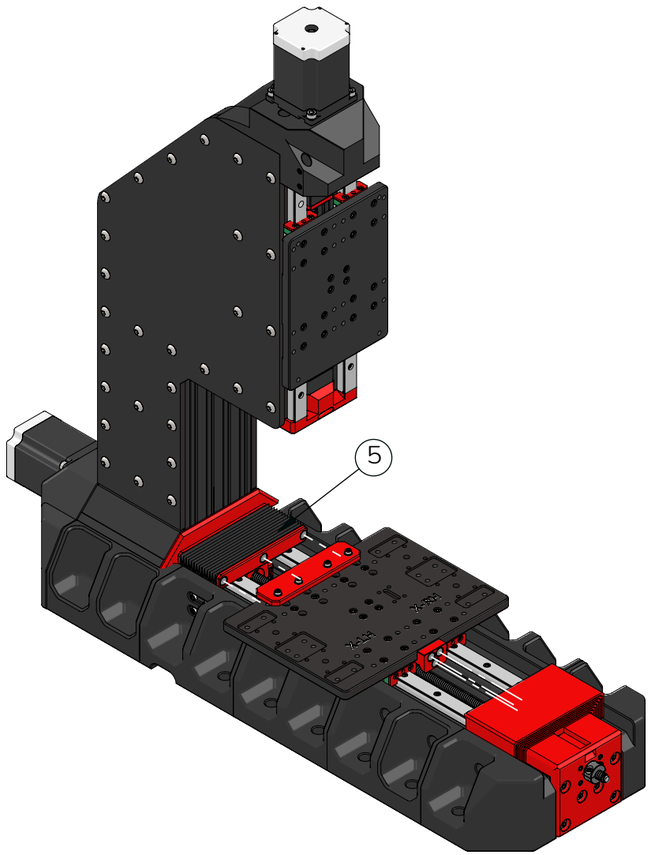
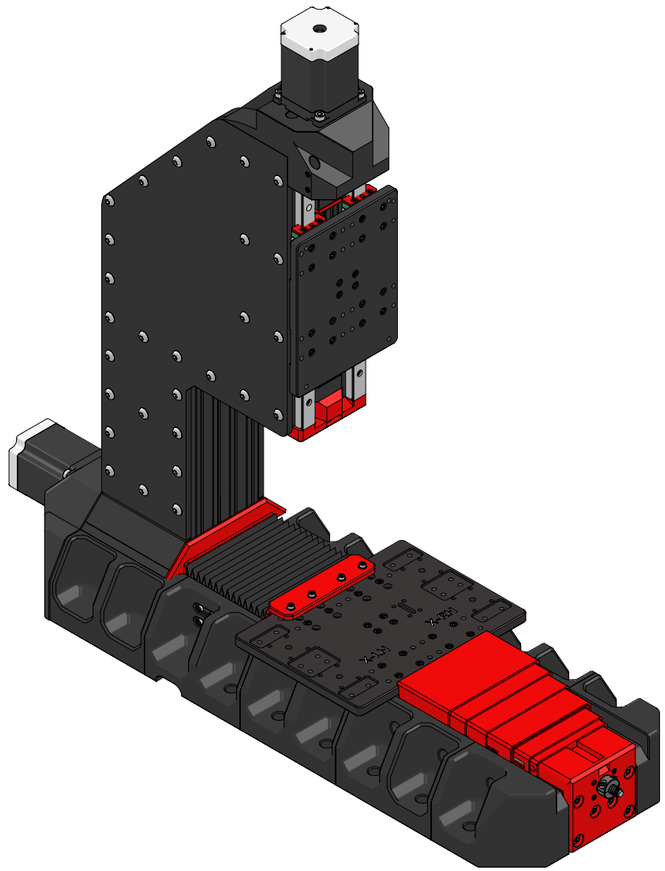
X AXIS HIDDEN FOR CLARITY

	PART NUMBER	QTY.
1	MILO-SUBASSEMBLY-C	1
2	Y-AXIS-WAY-COVER-ASSEMBLY	1
3	M3 DROP IN NUT	2
4	Y-AXIS-BELLOW-COLUMN-MAG-MOUNT-ASSEMBLY	1
5	ISO-4762-M3X6-SHCS	2







MILO SUBASSEMBLY C

WAY COVER INSTALL

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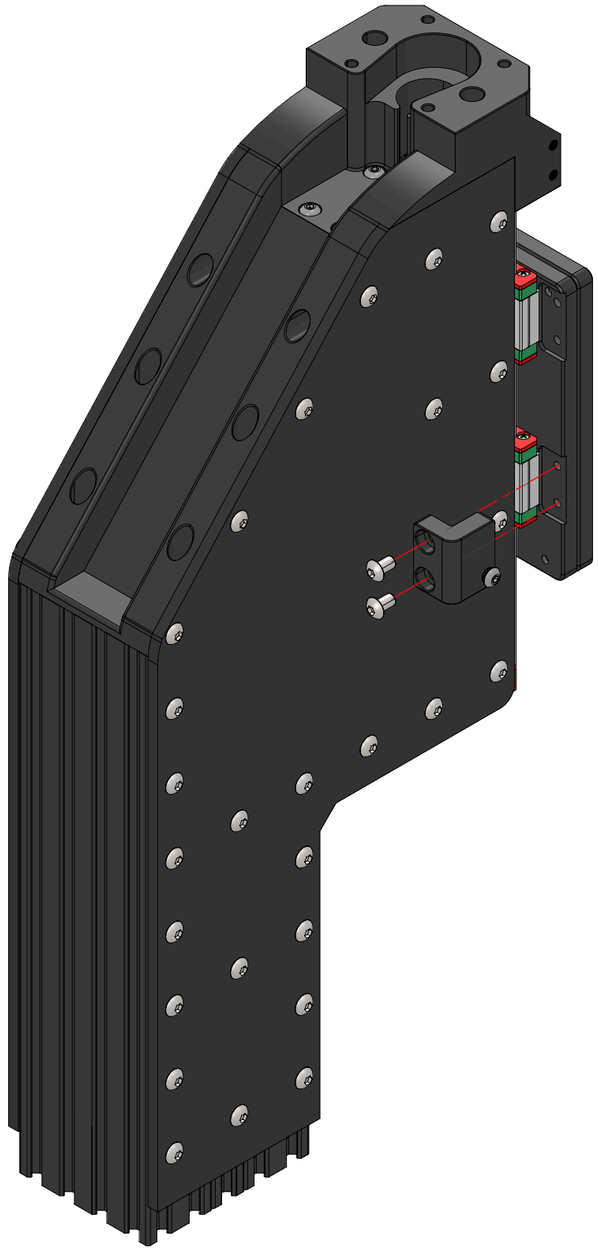
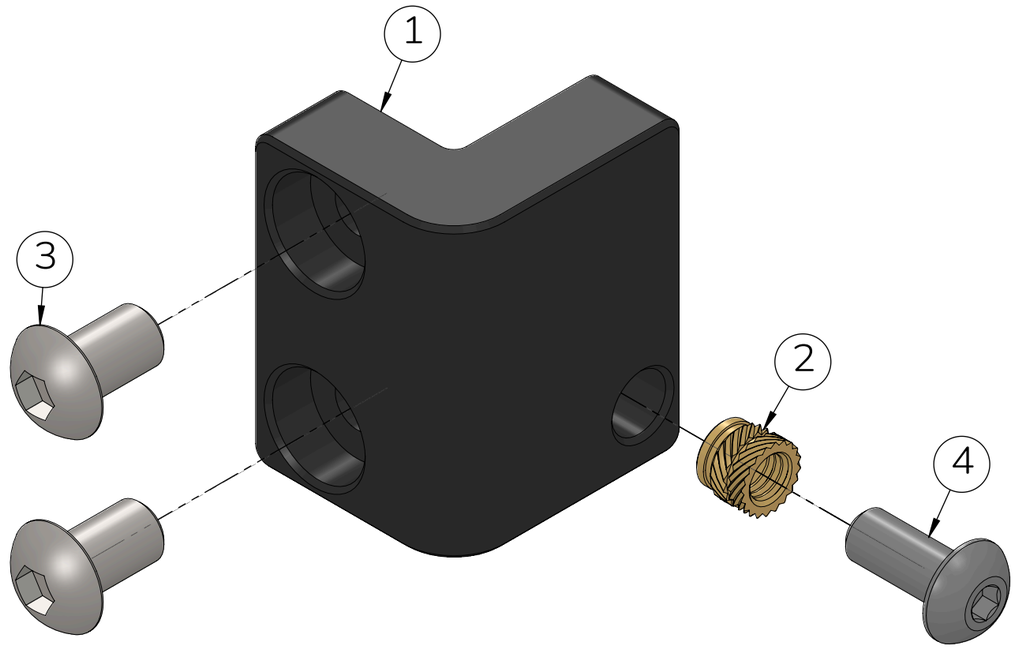
	PART NUMBER	QTY.
1	TRAM-AID	1
2	M5X5.8mm HEATSET INSERT	1
3	ISO-7380-M5X10 BHCS	2
4	ISO-7380-M5X12 BHCS	1





MILO SUBASSEMBLY C

SPINDLE TRAM AID

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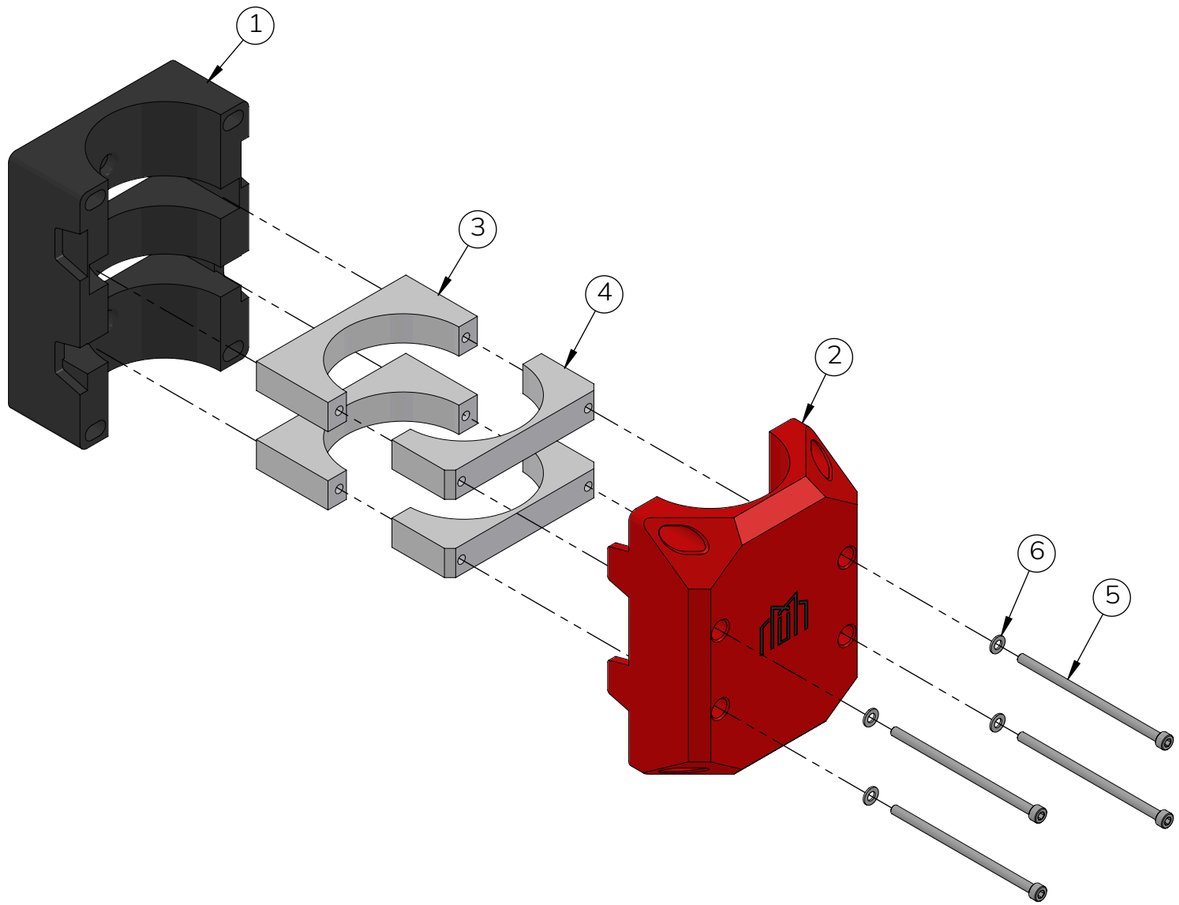


	PART NUMBER	QTY.
1	80mm-HOI-SPINDLE-MOUNT-BACK-PRINTED	1
2	80mm-HOI-SPINDLE-MOUNT-FRONT-PRINTED	1
3	80mm-HOI-SPINDLE-MOUNT-BACK	2
4	80mm-HOI-SPINDLE-MOUNT-FRONT	2
5	ISO-4762-M5X15-SHCS	4
6	ISO-7089-M5-WASHER	4
7	LOGO INSERT	1

Spindle Mount Options

80mm Printed Spindle Mount

80mm HOI Spindle Mount

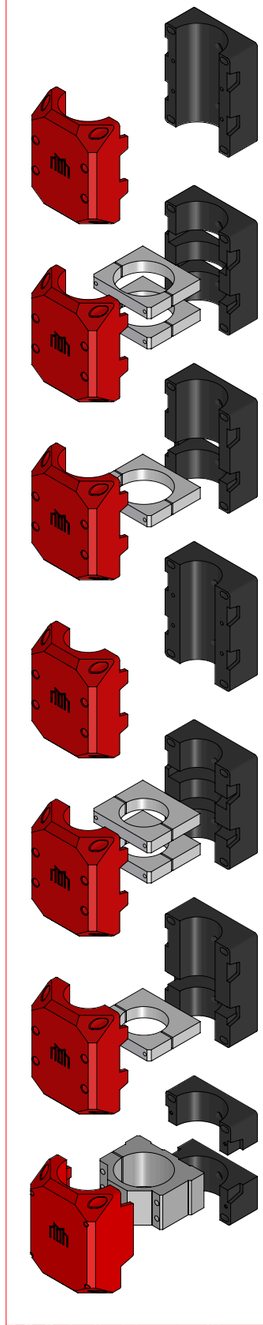
80mm Half Plate Spindle Mount

65mm Printed Spindle Mount

65mm HOI Spindle Mount

65mm Half Plate Spindle Mount

80mm LDO Spindle Mount



Note: LDO Spindle Mount will requires a LDO specific tram plate

MILO SUBASSEMBLY D

SPINDLE MOUNT REFERENCE

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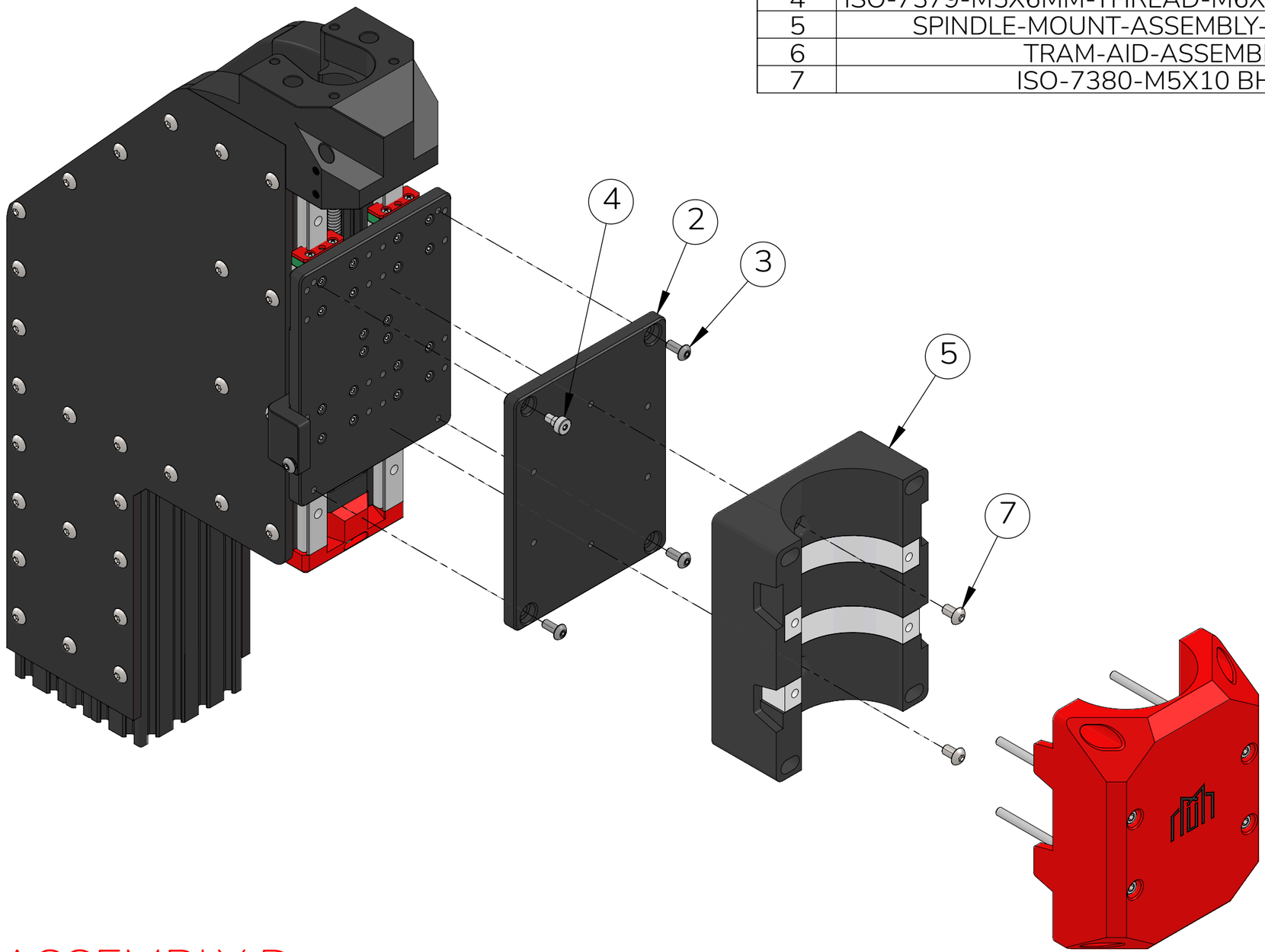
	PART NUMBER	QTY.
1	MILO-SUBASSEMBLY-D	1
2	TRAM-PLATE-FINAL	1
3	ISO-7380-M5X12 BHCS	3
4	ISO-7379-M5X6MM-THREAD-M6X4MM-SHOULDER	1
5	SPINDLE-MOUNT-ASSEMBLY-HOI-80mm	1
6	TRAM-AID-ASSEMBLY	1
7	ISO-7380-M5X10 BHCS	2



MILO SUBASSEMBLY D

SPINDLE MOUNT MOUNTING

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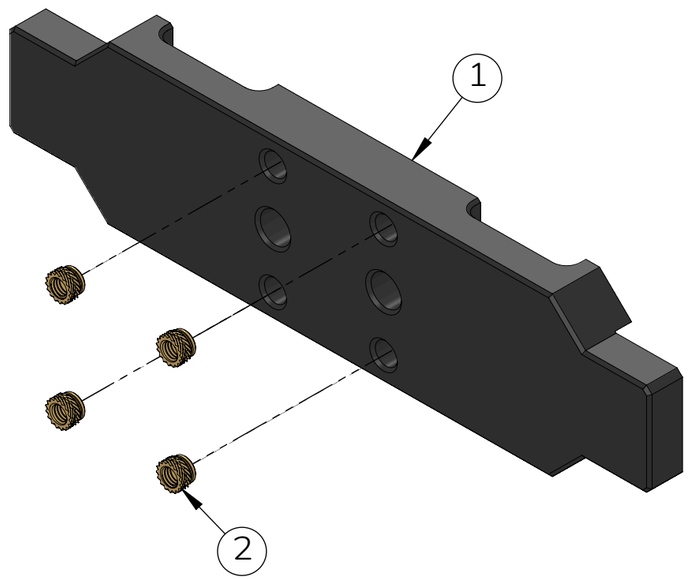
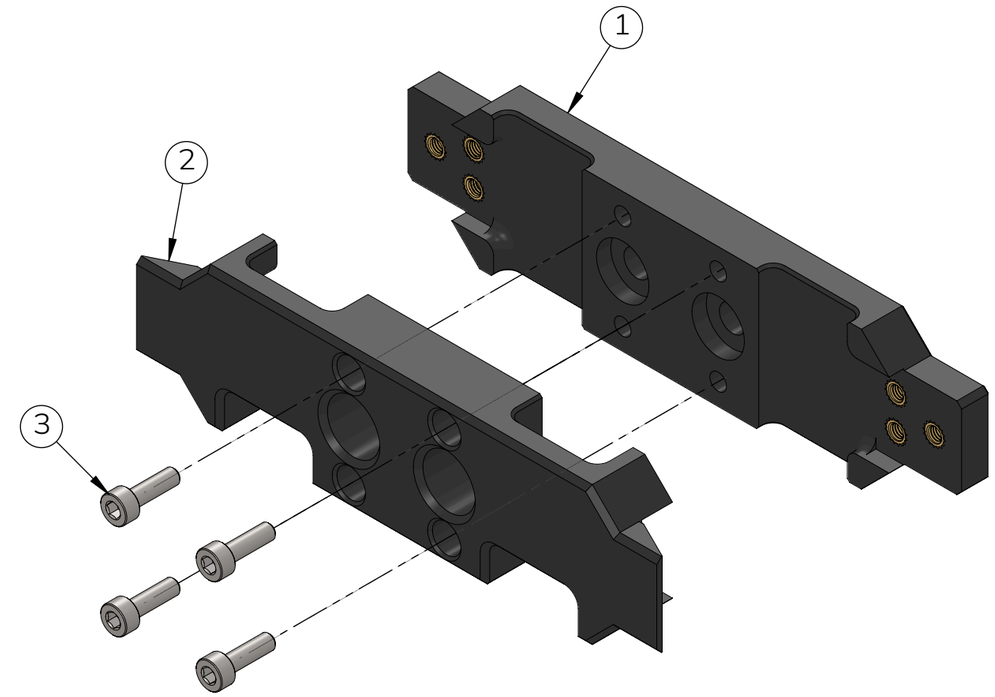
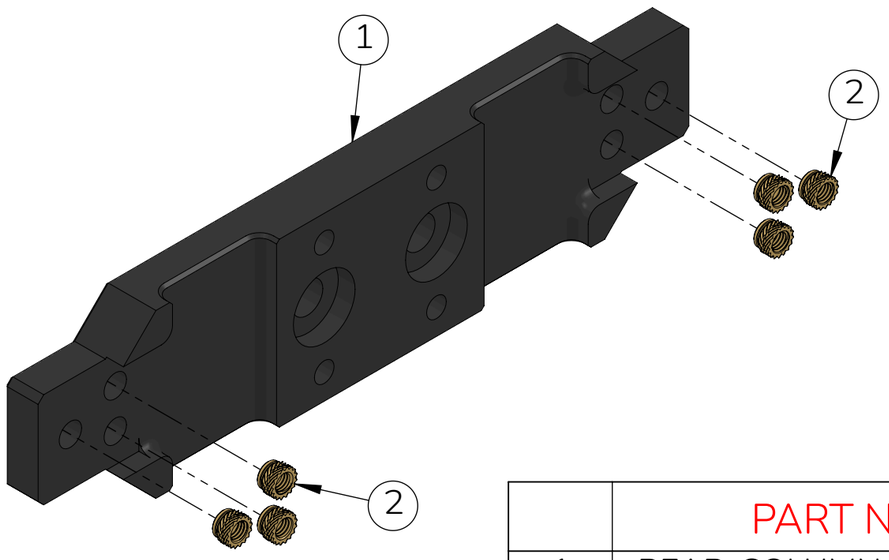
	PART NUMBER	QTY.
1	REAR-COLUMN-CHAIN-ANCHOR-PRE-ASSEMBLY	1
2	REAR-COLUMN-CHAIN -ANCHOR-FACEPLATE	1
3	ISO-4762-M3X10-SHCS	4







	PART NUMBER	QTY.
1	REAR-COLUMN-CHAIN-ANCHOR	1
2	M3X3MM HEATSET INSERT	10

MILO SUB-ASSEMBLY E

REAR COLUMN CHAIN ANCHOR PRE-ASSEMBLY

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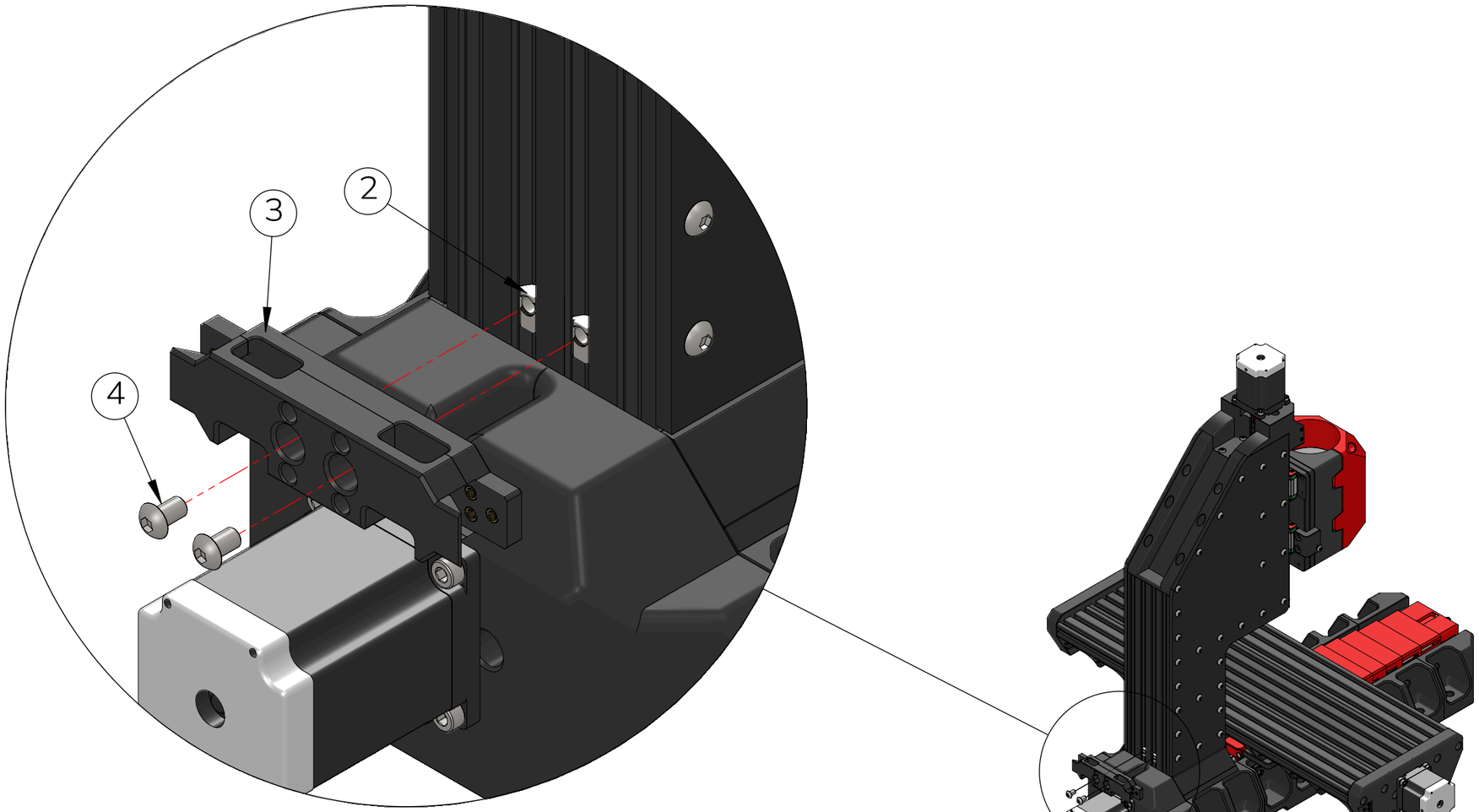
	PART NUMBER	QTY.
1	MILO-SUBASSEMBLY-E	1
2	M5 DROP IN NUT	2
3	REAR-COLUMN-CHAIN-ANCHOR-SUBASSEMBLY	1
4	ISO-7380-M5X10 BHCS	2



MILO SUB-ASSEMBLY E

ASSEMBLY STEP 1

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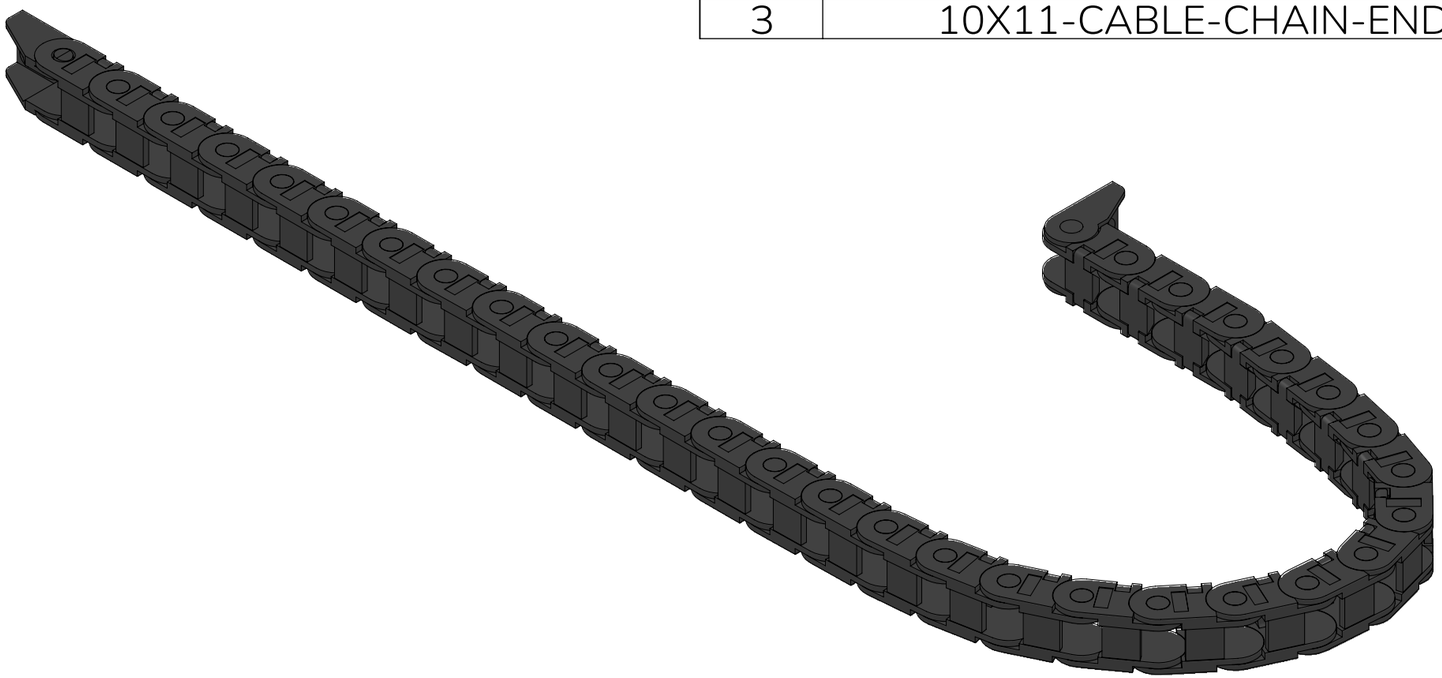
QTY 2

	PART NUMBER	QTY.
1	10X11-CABLE-CHAIN-FIXED	1
2	10X11-CABLE-CHAIN-LINK	31
3	10X11-CABLE-CHAIN-END	1



MILO SUB-ASSEMBLY E

CABLE CHAIN ASSEMBLY PRE-ASSEMBLY

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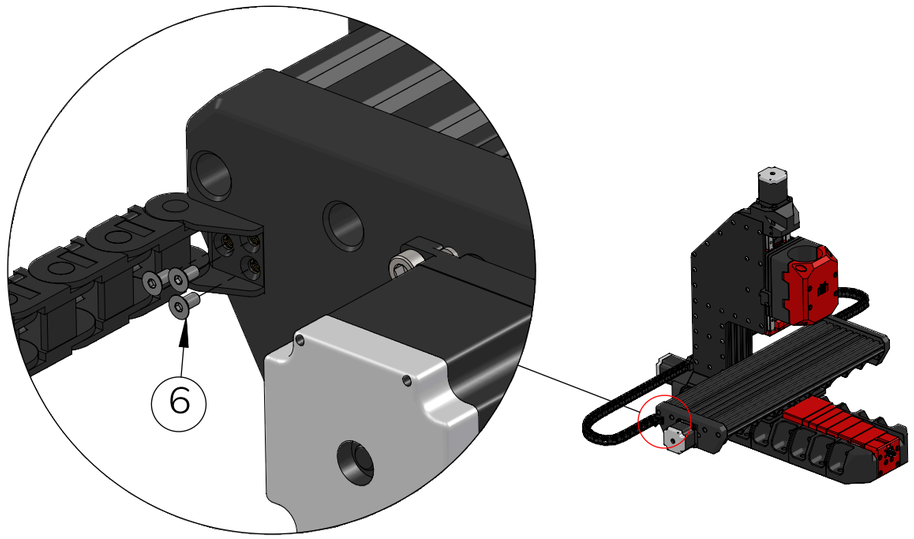

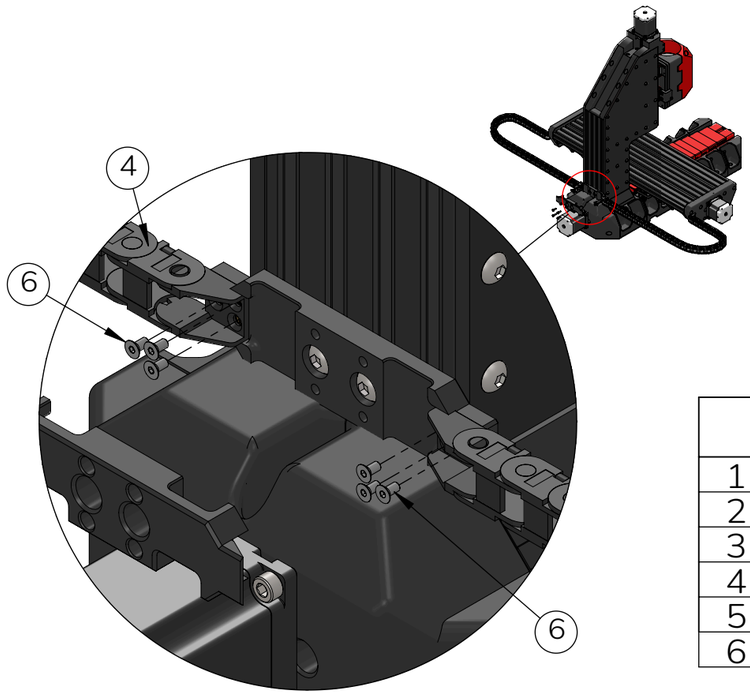

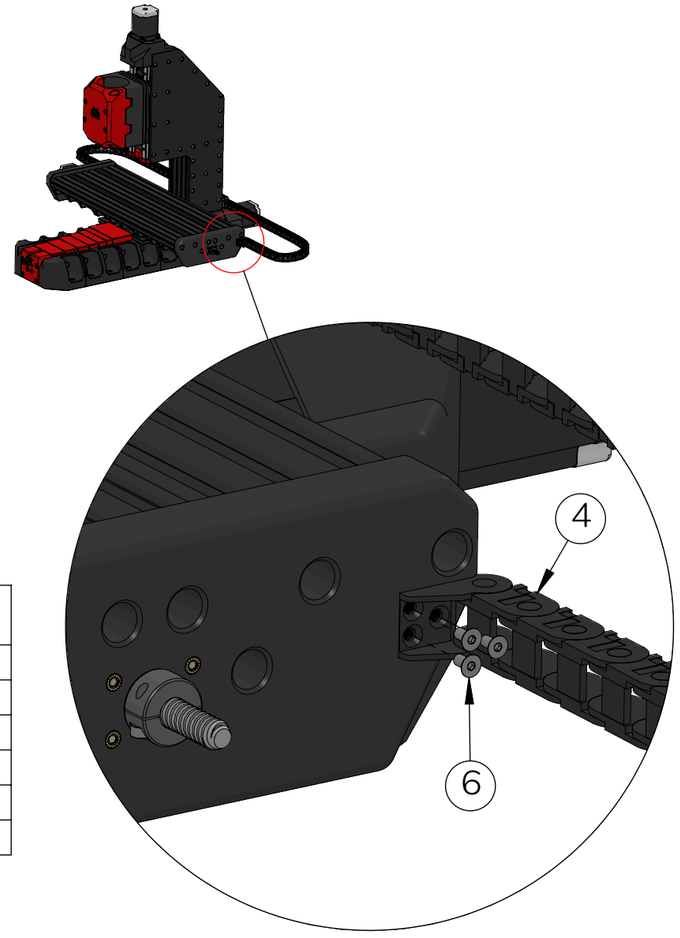

	PART NUMBER	QTY.
1	MILO-SUBASSEMBLY-E	1
2	M5 DROP IN NUT	2
3	REAR-COLUMN-CHAIN-ANCHOR-SUBASSEMBLY	1
4	CABLE-CHAIN-ASSEMBLY	2
5	ISO-7380-M5X10 BHCS	2
6	ISO-10642-M3X6MM-FHCS	12

MILO SUB-ASSEMBLY E

ASSEMBLY STEP 2

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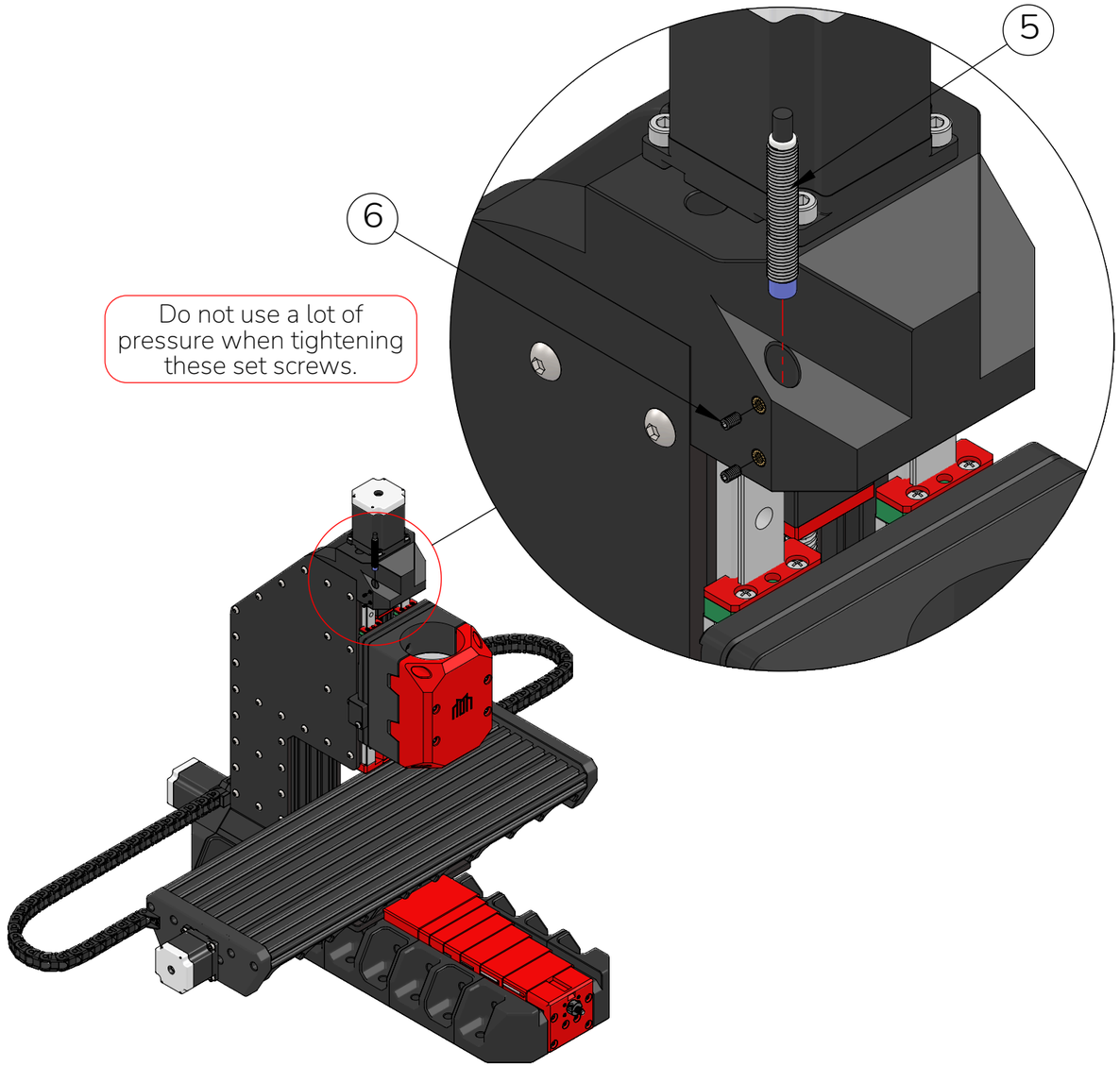
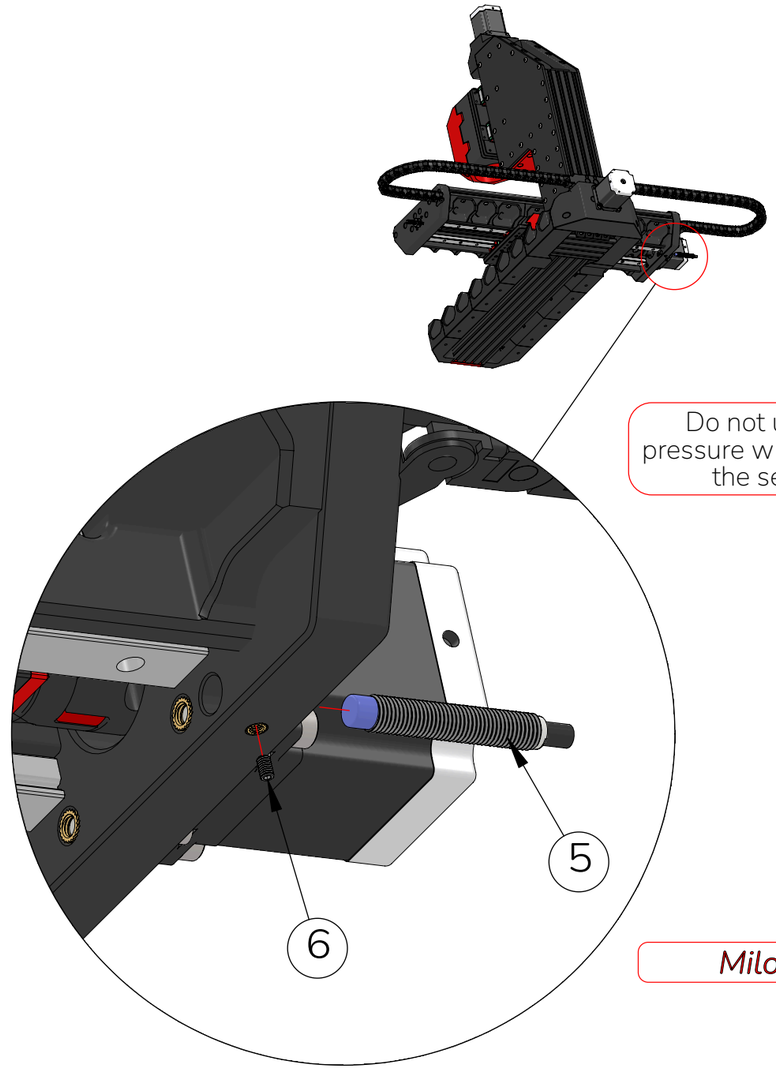
	PART NUMBER	QTY.
1	MILO-SUBASSEMBLY-E	1
5	LJ8A3-2-Z/AY	2
6	ISO-4026-M3X10-SET SCREW	3



Do not use a lot of pressure when tightening these set screws.



Do not use a lot of pressure when tightening the set screw.

Milo Only

MILO SUB-ASSEMBLY E

ASSEMBLY STEP 3

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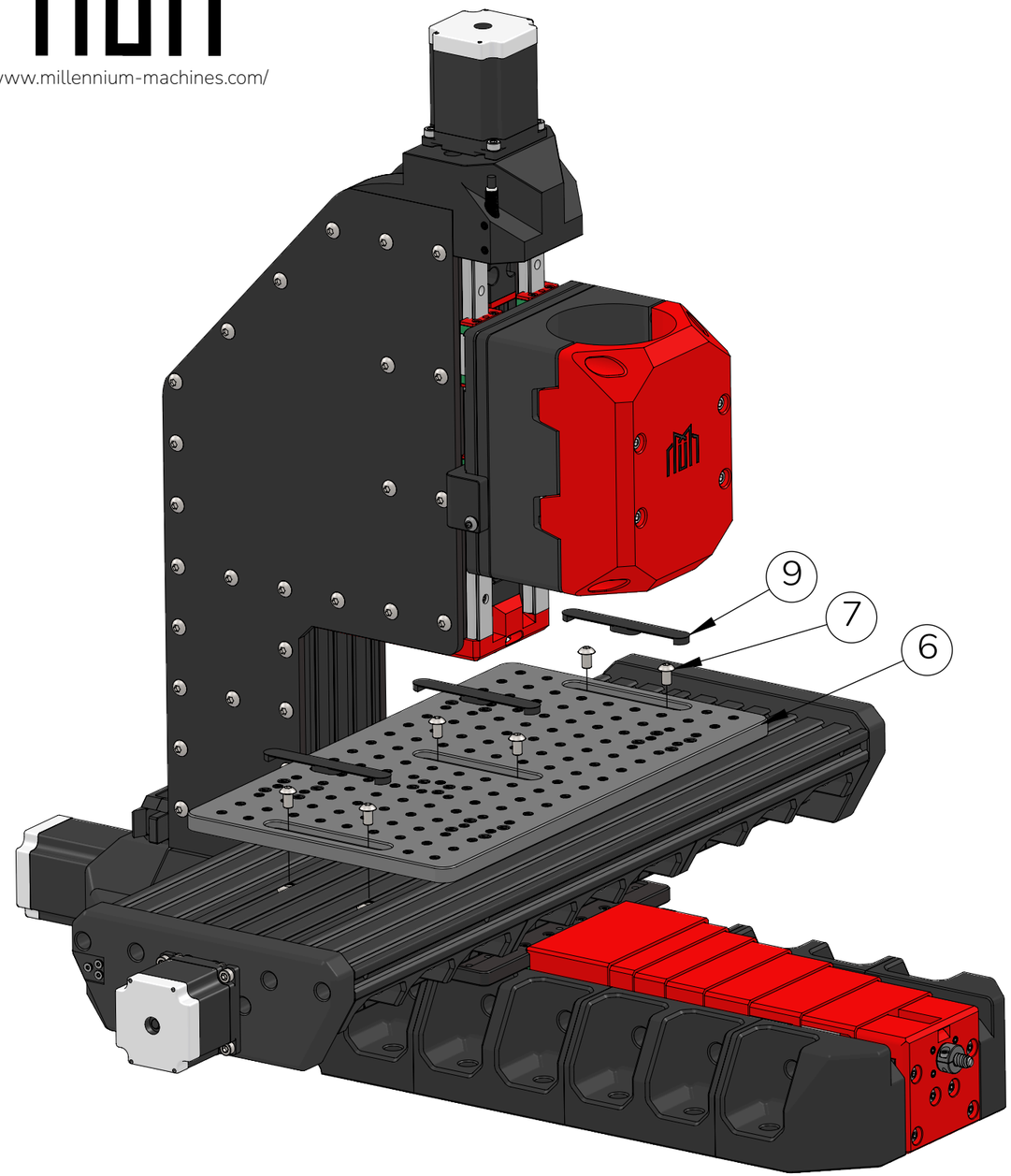
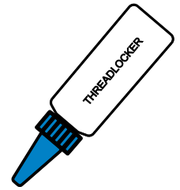
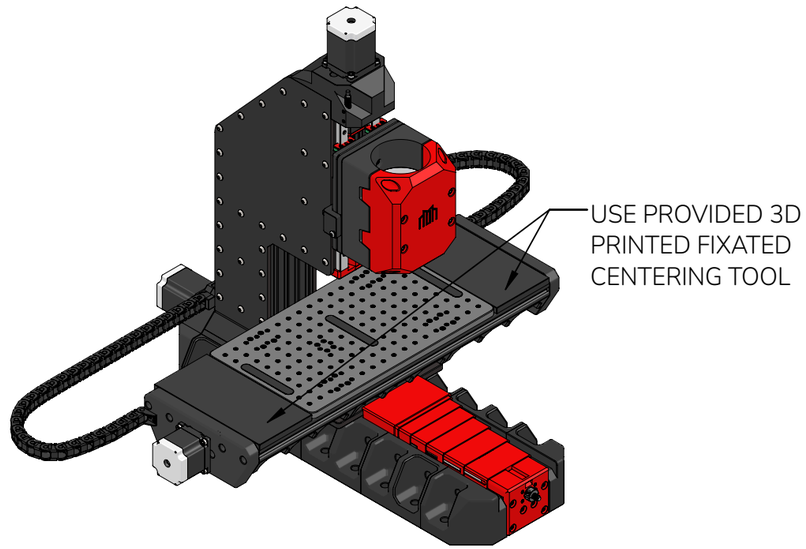
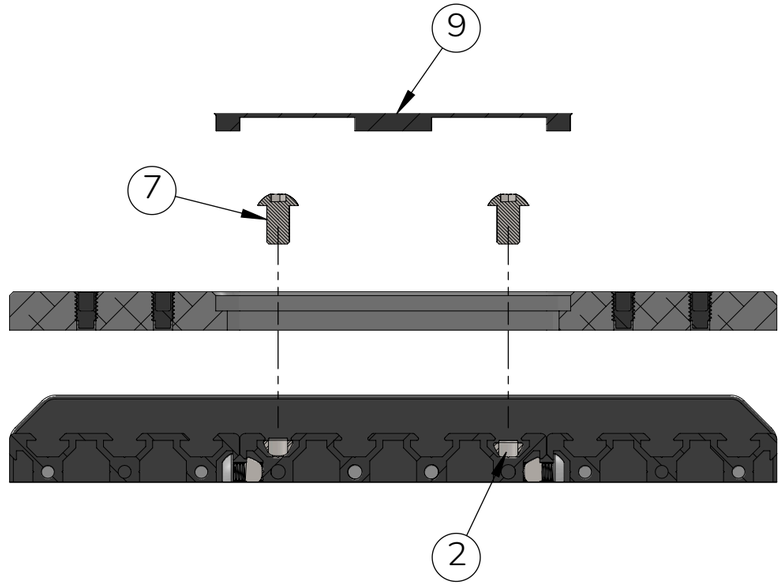




	PART NUMBER	QTY.
1	MILO-SUBASSEMBLY-E	1
2	M5 DROP IN NUT	6
6	FIXATED-V2-ASSEMBLY	1
7	ISO-7380-M5X10 BHCS	6
9	FIXATED-MOUNTING-SLOT-COVER	3





MILO SUB-ASSEMBLY E

ASSEMBLY STEP 4